

Reinforcement Learning: Theory and Algorithms

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WORKING DRAFT:

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Notation

This section collects notation and conventions that will be used throughout the book.

General Mathematical Notation

- For an integer $K \geq 1$, we slightly abuse notation and use 0-based indexing:

$$[K] := \{0, 1, 2, \dots, K - 1\}.$$

- We use $\mathbb{1}\{\mathcal{E}\}$ to denote the indicator function for an event $\mathcal{E}$, which takes value 1 if $\mathcal{E}$ is true and 0 otherwise.
- We use $\tilde{O}(\cdot)$ notation to hide polylogarithmic factors; i.e., $f(x) = \tilde{O}(g(x))$ if $f(x) = O(g(x) \cdot \text{poly}(\log x))$. Unless specified otherwise, $\log$ denotes the natural logarithm.
- For a set $\mathcal{X}$, let $\Delta(\mathcal{X})$ denote the set of probability distributions over $\mathcal{X}$.

Vectors and Norms

- For a vector v , the expressions $(v)^2$, $\sqrt{v}$, and $|v|$ denote the component-wise square, square root, and absolute value operations.
- Inequalities between vectors are understood elementwise. For vectors v, v' , we write $v \leq v'$ if $v(j) \leq v'(j)$ for all coordinates j .
- For a vector v , we refer to its j -th component as either $v(j)$ or $[v]_j$.
- For a vector x and a matrix M , unless stated otherwise, $\|x\|$ and $\|M\|$ denote the Euclidean and spectral norms, respectively. We use $\|x\|_M := \sqrt{x^\top M x}$ (with dimensions understood from context).
- We denote the ℓ_∞ -norm (max norm) by $\|x\|_\infty := \max_i |x_i|$. (Note: In the context of value functions V and Q , contraction arguments typically rely on the ℓ_∞ -norm).

Probability

- For a real-valued function f and a distribution $\mathcal{D}$, define the variance as

$$\text{Var}_{\mathcal{D}}(f) := \mathbb{E}_{x \sim \mathcal{D}}[f(x)^2] - (\mathbb{E}_{x \sim \mathcal{D}}[f(x)])^2.$$

Reinforcement Learning Context

- We write $s \in \mathcal{S}$ for states and $a \in \mathcal{A}$ for actions. Rewards are real-valued; when convenient we treat the reward function as a vector $r \in \mathbb{R}^{|\mathcal{S}||\mathcal{A}|}$ indexed by (s, a) .
- The discount factor is $\gamma \in [0, 1)$; for a policy π , the (discounted) value function is $V^\pi : \mathcal{S} \rightarrow \mathbb{R}$ and the action-value function is $Q^\pi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$.
- We overload notation so that for a distribution μ over $\mathcal{S}$,

$$V^\pi(\mu) := \mathbb{E}_{s \sim \mu} [V^\pi(s)].$$

- We overload notation and let P also denote a matrix of size $|\mathcal{S}||\mathcal{A}| \times |\mathcal{S}|$ with entries $P_{(s,a),s'} := P(s' \mid s, a)$. We also define P^π to be the transition matrix on state-action pairs induced by a deterministic policy π , namely

$$P_{(s,a),(s',a')}^\pi := \begin{cases} P(s' \mid s, a) & \text{if } a' = \pi(s'), \\ 0 & \text{if } a' \neq \pi(s'). \end{cases}$$

With this notation,

$$\begin{aligned} Q^\pi &= r + \gamma P V^\pi, \\ Q^\pi &= r + \gamma P^\pi Q^\pi, \\ Q^\pi &= (I - \gamma P^\pi)^{-1} r. \end{aligned}$$

- For a vector $Q \in \mathbb{R}^{|\mathcal{S}||\mathcal{A}|}$, define the greedy policy and its induced value as

$$\begin{aligned} \pi_Q(s) &:= \operatorname{argmax}_{a \in \mathcal{A}} Q(s, a), \\ V_Q(s) &:= \max_{a \in \mathcal{A}} Q(s, a). \end{aligned}$$

- For a vector $Q \in \mathbb{R}^{|\mathcal{S}||\mathcal{A}|}$, the *Bellman optimality operator* $\mathcal{T} : \mathbb{R}^{|\mathcal{S}||\mathcal{A}|} \rightarrow \mathbb{R}^{|\mathcal{S}||\mathcal{A}|}$ is defined by

$$\mathcal{T}Q := r + \gamma P V_Q. \tag{0.1}$$

- For a stationary policy π , the *Bellman evaluation operator* $\mathcal{T}^\pi : \mathbb{R}^{|\mathcal{S}||\mathcal{A}|} \rightarrow \mathbb{R}^{|\mathcal{S}||\mathcal{A}|}$ is defined by

$$\mathcal{T}^\pi Q := r + \gamma P^\pi Q.$$

Part I

Fundamentals

Chapter 1

Markov Decision Processes

1.1 Discounted (Infinite-Horizon) Markov Decision Processes

In reinforcement learning, the interactions between the agent and the environment are often described by an infinite-horizon, discounted Markov decision process (MDP)

$$M = (\mathcal{S}, \mathcal{A}, P, r, \gamma, \mu),$$

specified as follows:

- A state space $\mathcal{S}$, which may be finite or infinite. For mathematical convenience, we will assume that $\mathcal{S}$ is finite or countably infinite.
- An action space $\mathcal{A}$, which also may be discrete or infinite. For mathematical convenience, we will assume that $\mathcal{A}$ is finite.
- A transition function $P : \mathcal{S} \times \mathcal{A} \rightarrow \Delta(\mathcal{S})$, where $\Delta(\mathcal{S})$ is the space of probability distributions over $\mathcal{S}$ (i.e., the probability simplex). $P(s' \mid s, a)$ is the probability of transitioning into state s' upon taking action a in state s . We use $P_{s,a}$ to denote the vector $P(\cdot \mid s, a)$.
- A reward function $r : \mathcal{S} \times \mathcal{A} \rightarrow [0, 1]$. $r(s, a)$ is the immediate reward associated with taking action a in state s . More generally, $r(s, a)$ could be a random variable (with a distribution that depends on s, a). While we largely focus on the deterministic case, extensions to stochastic rewards are often straightforward.
- A discount factor $\gamma \in [0, 1)$, which defines an effective horizon for the problem.
- An initial state distribution $\mu \in \Delta(\mathcal{S})$, which specifies how the initial state s_0 is generated.

In many cases, we will assume that the initial state is fixed at s_0 , i.e. μ is a distribution supported only on s_0 .

1.1.1 The objective, policies, and values

Policies. In a given MDP $M = (\mathcal{S}, \mathcal{A}, P, r, \gamma, \mu)$, the agent interacts with the environment according to the following protocol: the agent starts at some state $s_0 \sim \mu$; at each time step $t = 0, 1, 2, \dots$, the agent takes an action $a_t \in \mathcal{A}$, obtains the immediate reward $r_t = r(s_t, a_t)$, and observes the next state s_{t+1} sampled according to $s_{t+1} \sim P(\cdot \mid$

s_t, a_t). The interaction record at time t ,

$$\tau_t = (s_0, a_0, r_0, s_1, \dots, s_t, a_t, r_t),$$

is called a *trajectory* (or history) up to time t .

In the most general setting, a policy specifies a decision-making strategy in which the agent chooses actions adaptively based on the history of observations; precisely, a policy is a (possibly randomized) mapping from a trajectory to a distribution over actions, i.e. $\pi : \mathcal{H} \rightarrow \Delta(\mathcal{A})$, where $\mathcal{H}$ is the set of all possible trajectories (of all lengths) and $\Delta(\mathcal{A})$ is the space of probability distributions over $\mathcal{A}$. A *stationary* policy $\pi : \mathcal{S} \rightarrow \Delta(\mathcal{A})$ depends only on the current state, i.e. $a_t \sim \pi(\cdot \mid s_t)$. A deterministic stationary policy is of the form $\pi : \mathcal{S} \rightarrow \mathcal{A}$.

Values. We now define values for (general) policies. For a fixed policy and a starting state $s_0 = s$, we define the value function $V_M^\pi : \mathcal{S} \rightarrow \mathbb{R}$ as

$$V_M^\pi(s) = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \mid \pi, s_0 = s \right].$$

The expectation is with respect to the randomness of the trajectory (state transitions and any stochasticity in π). Since $r(s, a) \in [0, 1]$, we have $0 \leq V_M^\pi(s) \leq 1/(1 - \gamma)$.

Similarly, the action-value (or Q-value) function $Q_M^\pi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ is defined as

$$Q_M^\pi(s, a) = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \mid \pi, s_0 = s, a_0 = a \right],$$

and $Q_M^\pi(s, a)$ is also bounded by $1/(1 - \gamma)$.

Goal. Given a state s , the goal of the agent is to find a policy π that maximizes the value:

$$\sup_{\pi} V_M^\pi(s), \tag{1.1}$$

where the supremum is over all (possibly non-stationary and randomized) policies. As we shall see, there exists a stationary deterministic policy which is simultaneously optimal for all starting states s .

We drop the dependence on M and write V^π when it is clear from context.

Example 1.1 (Navigation). Navigation is perhaps the simplest example of reinforcement learning. The state is the agent's current location. The four actions might be moving one step east, west, north, or south. In the simplest setting the transitions are deterministic. Taking the north action moves the agent one step north of its location. The agent might have a goal state g , and the reward is 0 until the agent reaches the goal, and 1 upon reaching g . Since $\gamma < 1$, there is an incentive to reach the goal earlier. As a result, the optimal behavior corresponds to finding a shortest path from the initial to the goal state. If a policy reaches the goal in d steps (and then stays there), the value from the initial state is γ^d .

Example 1.2 (Conversational agent). This is another natural reinforcement learning problem. The state can be the transcript of the conversation so far, along with additional context (e.g., task context, user profile information, or other relevant world state). Actions depend on the domain; in the most basic form, an action is the next utterance.

Sometimes, conversational agents are designed for task completion, such as a travel assistant, tech support agent, or virtual receptionist. In these cases, there may be a predefined set of *slots* that the agent must fill before it can propose a solution (e.g., travel dates, origin, destination, and mode of travel). Actions can be natural language queries aimed at filling these slots.

In task-completion settings, reward is naturally defined as a binary indicator of whether the task was completed (e.g., whether travel was successfully booked). Depending on the domain, the reward can be refined to reflect quality or cost. In more open-ended conversational settings, the reward may reflect user satisfaction.

Example 1.3 (Strategic games). This is a popular category of applications, where reinforcement learning has achieved human-level performance in Backgammon, Go, Chess, and various forms of Poker. Typically, the state is the current game position, actions are legal moves, and reward is the eventual win/loss outcome (or a more detailed score, when defined). Technically, these are multi-agent settings, yet many successful approaches use methods that do not explicitly model multiple agents.

1.1.2 Bellman Consistency Equations for Stationary Policies

Stationary policies satisfy the following consistency conditions:

Lemma 1.4. *Suppose that π is a stationary policy (possibly randomized). Then V^π and Q^π satisfy the following Bellman consistency equations: for all $s \in \mathcal{S}$ and $a \in \mathcal{A}$,*

$$\begin{aligned} V^\pi(s) &= \mathbb{E}_{a \sim \pi(\cdot|s)}[Q^\pi(s, a)], \\ Q^\pi(s, a) &= r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)}[V^\pi(s')]. \end{aligned}$$

In particular, if π is deterministic, then $V^\pi(s) = Q^\pi(s, \pi(s))$.

We leave the proof as an exercise to the reader.

It is helpful to view V^π as a vector of length $|\mathcal{S}|$, and Q^π and r as vectors of length $|\mathcal{S}| \cdot |\mathcal{A}|$. We overload notation and let P also refer to a matrix of size $(|\mathcal{S}| \cdot |\mathcal{A}|) \times |\mathcal{S}|$ where the entry $P_{(s,a),s'}$ is equal to $P(s' | s, a)$.

We also define P^π to be the transition matrix on state-action pairs induced by a stationary policy π , specifically:

$$P_{(s,a),(s',a')}^\pi := P(s' | s, a) \pi(a' | s').$$

In particular, for deterministic policies we have:

$$P_{(s,a),(s',a')}^\pi := \begin{cases} P(s' | s, a) & \text{if } a' = \pi(s'), \\ 0 & \text{if } a' \neq \pi(s'). \end{cases}$$

With this notation, it is straightforward to verify:

$$\begin{aligned} Q^\pi &= r + \gamma P V^\pi, \\ Q^\pi &= r + \gamma P^\pi Q^\pi. \end{aligned}$$

Corollary 1.5. *Suppose that π is a stationary policy. Then*

$$Q^\pi = (I - \gamma P^\pi)^{-1} r, \tag{1.2}$$

where I is the identity matrix.

Proof: To see that $I - \gamma P^\pi$ is invertible, observe that for any non-zero vector $x \in \mathbb{R}^{|\mathcal{S}||\mathcal{A}|}$,

$$\begin{aligned} \|(I - \gamma P^\pi)x\|_\infty &= \|x - \gamma P^\pi x\|_\infty \\ &\geq \|x\|_\infty - \gamma \|P^\pi x\|_\infty && \text{(triangle inequality for norms)} \\ &\geq \|x\|_\infty - \gamma \|x\|_\infty && (P^\pi \text{ is row-stochastic}) \\ &= (1 - \gamma)\|x\|_\infty > 0, && (\gamma < 1, x \neq 0) \end{aligned}$$

which implies $I - \gamma P^\pi$ is full rank. ■

The following lemma will also be useful:

Lemma 1.6. *We have that*

$$[(1 - \gamma)(I - \gamma P^\pi)^{-1}]_{(s,a),(s',a')} = (1 - \gamma) \sum_{t=0}^{\infty} \gamma^t \mathbb{P}^\pi(s_t = s', a_t = a' \mid s_0 = s, a_0 = a),$$

so we can view the (s, a) -th row of this matrix as an induced distribution over state-action pairs when following π after starting with $s_0 = s$ and $a_0 = a$.

We leave the proof as an exercise to the reader.

1.1.3 Bellman Optimality Equations

A remarkable and convenient property of discounted MDPs is that there exists a stationary and deterministic policy that simultaneously maximizes $V^\pi(s)$ for all $s \in \mathcal{S}$. This is formalized in the following theorem.

Theorem 1.7. *Let Π be the set of all (possibly non-stationary and randomized) policies. Define*

$$\begin{aligned} V^*(s) &:= \sup_{\pi \in \Pi} V^\pi(s), \\ Q^*(s, a) &:= \sup_{\pi \in \Pi} Q^\pi(s, a). \end{aligned}$$

These quantities are finite since $V^\pi(s)$ and $Q^\pi(s, a)$ are bounded between 0 and $1/(1 - \gamma)$.

There exists a stationary and deterministic policy π such that for all $s \in \mathcal{S}$ and $a \in \mathcal{A}$,

$$\begin{aligned} V^\pi(s) &= V^*(s), \\ Q^\pi(s, a) &= Q^*(s, a). \end{aligned}$$

We refer to such a π as an optimal policy.

Proof: For any $\pi \in \Pi$ and time t , π specifies a distribution over actions conditioned on the history of observations. For the purposes of this proof, it is helpful to formally let S_t , A_t , and R_t denote random variables, distinguishing them from realizations denoted by lower-case symbols.

We first show that conditioned on $(S_0, A_0, R_0, S_1) = (s, a, r, s')$, the maximum future discounted return from time 1 onwards does not depend on s, a, r . More precisely, we claim that

$$\sup_{\pi \in \Pi} \mathbb{E} \left[\sum_{t=1}^{\infty} \gamma^t r(S_t, A_t) \mid \pi, (S_0, A_0, R_0, S_1) = (s, a, r, s') \right] = \gamma V^*(s'). \quad (1.3)$$

Fix any policy π . Define the “offset” policy $\pi_{(s,a,r)}$ as follows: for all $t \geq 0$ and all histories $(s_0, a_0, r_0, \dots, s_t)$,

$$\begin{aligned} \pi_{(s,a,r)}(A_t = b \mid S_0 = s_0, A_0 = a_0, R_0 = r_0, \dots, S_t = s_t) \\ := \pi(A_{t+1} = b \mid S_0 = s, A_0 = a, R_0 = r, S_1 = s_0, A_1 = a_0, R_1 = r_0, \dots, S_{t+1} = s_t). \end{aligned}$$

By a change of variables on the time index and the definition of $\pi_{(s,a,r)}$, we have

$$\mathbb{E} \left[\sum_{t=1}^{\infty} \gamma^t r(S_t, A_t) \mid \pi, (S_0, A_0, R_0, S_1) = (s, a, r, s') \right] = \gamma \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r(S_t, A_t) \mid \pi_{(s,a,r)}, S_0 = s' \right] = \gamma V^{\pi_{(s,a,r)}}(s').$$

Moreover, for each fixed (s, a, r) , the set $\{\pi_{(s,a,r)} : \pi \in \Pi\}$ is equal to Π itself (intuitively, shifting a policy by one step is a bijection on the class of all history-dependent randomized policies). Therefore,

$$\sup_{\pi \in \Pi} \mathbb{E} \left[\sum_{t=1}^{\infty} \gamma^t r(S_t, A_t) \mid \pi, (S_0, A_0, R_0, S_1) = (s, a, r, s') \right] = \gamma \cdot \sup_{\pi \in \Pi} V^\pi(s') = \gamma V^*(s'),$$

proving (1.3).

We now define a stationary deterministic policy $\tilde{\pi}$ by choosing, for each state s , any maximizer of the one-step lookahead objective:

$$\tilde{\pi}(s) \in \operatorname{argmax}_{a \in \mathcal{A}} \mathbb{E} \left[r(s, a) + \gamma V^*(S_1) \mid S_0 = s, A_0 = a \right].$$

We claim that $V^{\tilde{\pi}}(s) = V^*(s)$ for all s .

Fix an initial state s_0 . Using the law of iterated expectations,

$$\begin{aligned} V^*(s_0) &= \sup_{\pi \in \Pi} \mathbb{E} \left[r(S_0, A_0) + \sum_{t=1}^{\infty} \gamma^t r(S_t, A_t) \mid S_0 = s_0 \right] \\ &= \sup_{\pi \in \Pi} \mathbb{E} \left[r(s_0, A_0) + \mathbb{E} \left[\sum_{t=1}^{\infty} \gamma^t r(S_t, A_t) \mid \pi, (S_0, A_0, R_0, S_1) = (s_0, A_0, r_0, S_1) \right] \mid S_0 = s_0 \right] \\ &\leq \sup_{\pi \in \Pi} \mathbb{E} \left[r(s_0, A_0) + \sup_{\pi' \in \Pi} \mathbb{E} \left[\sum_{t=1}^{\infty} \gamma^t r(S_t, A_t) \mid \pi', (S_0, A_0, R_0, S_1) = (s_0, A_0, r_0, S_1) \right] \mid S_0 = s_0 \right] \\ &\stackrel{(a)}{=} \sup_{\pi \in \Pi} \mathbb{E} \left[r(s_0, A_0) + \gamma V^*(S_1) \mid S_0 = s_0 \right] \\ &= \sup_{a_0 \in \mathcal{A}} \mathbb{E} \left[r(s_0, a_0) + \gamma V^*(S_1) \mid S_0 = s_0, A_0 = a_0 \right] \\ &\stackrel{(b)}{=} \mathbb{E} \left[r(s_0, A_0) + \gamma V^*(S_1) \mid S_0 = s_0, \tilde{\pi} \right]. \end{aligned}$$

Here, step (a) uses (1.3), and step (b) follows from the definition of $\tilde{\pi}$.

Applying the same argument recursively (conditioning on S_1 , then S_2 , etc.) yields

$$V^*(s_0) \leq \mathbb{E} \left[r(S_0, A_0) + \gamma r(S_1, A_1) + \cdots + \gamma^t V^*(S_{t+1}) \mid S_0 = s_0, \tilde{\pi} \right] \quad \text{for all } t \geq 0.$$

Taking $t \rightarrow \infty$ and using bounded convergence (since V^* is bounded), we obtain $V^*(s_0) \leq V^{\tilde{\pi}}(s_0)$. Since by definition $V^{\tilde{\pi}}(s_0) \leq \sup_{\pi \in \Pi} V^\pi(s_0) = V^*(s_0)$, we conclude $V^{\tilde{\pi}}(s_0) = V^*(s_0)$. As s_0 was arbitrary, this proves the first claim.

For the same policy $\tilde{\pi}$, an analogous argument starting from $Q^*(s, a)$ shows $Q^{\tilde{\pi}}(s, a) = Q^*(s, a)$ for all (s, a) , completing the proof. $\blacksquare$

This shows that we may restrict ourselves to stationary and deterministic policies without any loss in performance. The following theorem, also due to [Bellman, 1956], gives a precise characterization of the optimal action-value function.

Theorem 1.8 (Bellman optimality equations). *We say that a vector $Q \in \mathbb{R}^{|\mathcal{S}| \times |\mathcal{A}|}$ satisfies the Bellman optimality equations if, for all $(s, a) \in \mathcal{S} \times \mathcal{A}$,*

$$Q(s, a) = r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot \mid s, a)} \left[\max_{a' \in \mathcal{A}} Q(s', a') \right].$$

For any $Q \in \mathbb{R}^{|\mathcal{S}| \times |\mathcal{A}|}$, we have $Q = Q^$ if and only if Q satisfies the Bellman optimality equations. Furthermore, any deterministic policy defined by*

$$\pi(s) \in \operatorname{argmax}_{a \in \mathcal{A}} Q^*(s, a)$$

is an optimal policy (with ties broken in an arbitrary but fixed manner).

Before we prove this claim, we introduce a few definitions. Let π_Q denote a greedy policy with respect to a vector $Q \in \mathbb{R}^{|S||\mathcal{A}|}$:

$$\pi_Q(s) \in \operatorname{argmax}_{a \in \mathcal{A}} Q(s, a),$$

where ties are broken in some arbitrary but fixed manner. With this notation, the optimal policy can be written as

$$\pi^* = \pi_{Q^*}.$$

We also use the following notation to map a vector $Q \in \mathbb{R}^{|S||\mathcal{A}|}$ to a vector of length $|S|$:

$$V_Q(s) := \max_{a \in \mathcal{A}} Q(s, a).$$

The Bellman optimality operator $\mathcal{T}_M : \mathbb{R}^{|S||\mathcal{A}|} \rightarrow \mathbb{R}^{|S||\mathcal{A}|}$ is defined as

$$\mathcal{T}Q := r + \gamma P V_Q. \quad (1.4)$$

With this notation, the Bellman optimality equations can be written concisely as

$$Q = \mathcal{T}Q.$$

Thus, the previous theorem states that $Q = Q^*$ if and only if Q is a fixed point of the operator $\mathcal{T}$.

Proof: We begin by showing that

$$V^*(s) = \max_{a \in \mathcal{A}} Q^*(s, a). \quad (1.5)$$

Let π^* be an optimal stationary deterministic policy, which exists by Theorem 1.7. Consider the (non-stationary) policy that takes action a in state s at time 0 and then follows π^* thereafter. By definition of V^* as the supremum over all (possibly non-stationary) policies,

$$V^*(s) \geq Q^{\pi^*}(s, a) = Q^*(s, a),$$

where the last equality uses that π^* is optimal for all starting states. Since a is arbitrary, this gives $V^*(s) \geq \max_a Q^*(s, a)$. Conversely, since π^* is deterministic, Lemma 1.4 yields

$$V^*(s) = V^{\pi^*}(s) = Q^{\pi^*}(s, \pi^*(s)) \leq \max_a Q^{\pi^*}(s, a) = \max_a Q^*(s, a),$$

proving (1.5).

First, we show the forward direction, i.e. that Q^* satisfies $Q^* = \mathcal{T}Q^*$. For any (s, a) ,

$$\begin{aligned} Q^*(s, a) &= \sup_{\pi \in \Pi} Q^\pi(s, a) \\ &= r(s, a) + \gamma \sup_{\pi \in \Pi} \mathbb{E}_{s' \sim P(\cdot | s, a)} [V^\pi(s')] \\ &= r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} \left[\sup_{\pi \in \Pi} V^\pi(s') \right] \\ &= r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} [V^*(s')] \\ &= r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} \left[\max_{a'} Q^*(s', a') \right], \end{aligned}$$

where the third line uses that the transition distribution does not depend on π , and the last line uses (1.5). This proves sufficiency.

For the converse, suppose $Q = \mathcal{T}Q$ for some Q . We now show that $Q = Q^*$. Let $\pi = \pi_Q$ be a greedy policy with respect to Q . The identity $Q = \mathcal{T}Q$ means

$$Q(s, a) = r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)}[V_Q(s')] = r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)}[Q(s', \pi(s'))],$$

which in matrix form is $Q = r + \gamma P^\pi Q$. Thus,

$$Q = (I - \gamma P^\pi)^{-1} r = Q^\pi,$$

using Corollary 1.5.

It remains to show that π is optimal. Let π' be any other deterministic stationary policy. First note that

$$[(P^\pi - P^{\pi'})Q^\pi]_{s,a} = \mathbb{E}_{s' \sim P(\cdot | s, a)}[Q^\pi(s', \pi(s')) - Q^\pi(s', \pi'(s'))] \geq 0,$$

since π is greedy with respect to $Q = Q^\pi$. Using $Q^\pi = r + \gamma P^\pi Q^\pi$ and $Q^{\pi'} = (I - \gamma P^{\pi'})^{-1} r$, we have

$$\begin{aligned} Q^\pi - Q^{\pi'} &= (I - \gamma P^{\pi'})^{-1} \left((I - \gamma P^{\pi'}) - (I - \gamma P^\pi) \right) Q^\pi \\ &= \gamma (I - \gamma P^{\pi'})^{-1} (P^\pi - P^{\pi'}) Q^\pi \\ &\geq 0, \end{aligned}$$

where the last step uses that $(I - \gamma P^{\pi'})^{-1}$ has nonnegative entries (indeed, $(I - \gamma P^{\pi'})^{-1} = \sum_{t=0}^{\infty} \gamma^t (P^{\pi'})^t$). Thus $Q^\pi \geq Q^{\pi'}$ for all deterministic stationary π' . Since Theorem 1.7 establishes that an optimal policy exists within the set of stationary deterministic policies, and π dominates all such policies, π must be globally optimal. Therefore $Q = Q^\pi = Q^*$, completing the proof. ■

1.2 Finite-Horizon Markov Decision Processes

In some settings, it is more natural to work with finite-horizon (and hence time-dependent) Markov Decision Processes; see the discussion below. A finite-horizon, time-dependent MDP $M = (\mathcal{S}, \mathcal{A}, \{P_h\}_{h=0}^{H-1}, \{r_h\}_{h=0}^{H-1}, H, \mu)$ is specified by:

- A state space $\mathcal{S}$, which may be finite or infinite.
- An action space $\mathcal{A}$, which may be finite or infinite.
- A time-dependent transition kernel $P_h : \mathcal{S} \times \mathcal{A} \rightarrow \Delta(\mathcal{S})$, where $\Delta(\mathcal{S})$ is the set of probability distributions over $\mathcal{S}$. Here $P_h(s' | s, a)$ is the probability of transitioning to state s' after taking action a in state s at time step h . (The stationary setting is recovered when P_h does not depend on h .)
- A time-dependent reward function $r_h : \mathcal{S} \times \mathcal{A} \rightarrow [0, 1]$, where $r_h(s, a)$ is the immediate reward obtained by taking action a in state s at time step h . More generally, one may allow $r_h(s, a)$ to be stochastic, but we focus on deterministic rewards for simplicity.
- A positive integer horizon H .
- An initial state distribution $\mu \in \Delta(\mathcal{S})$, which specifies how s_0 is generated.

A (possibly non-stationary and randomized) policy π specifies, at each time h , a distribution over actions conditioned on the observed history up to time h . (Equivalently, one may restrict attention to Markov policies $\pi(s, h)$ without loss of optimality in finite-horizon MDPs; see the theorem below.)

For a policy π , state s , and time $h \in \{0, \dots, H\}$, define the value function $V_h^\pi : \mathcal{S} \rightarrow \mathbb{R}$ as the expected cumulative reward from time h onward:

$$V_h^\pi(s) = \mathbb{E} \left[\sum_{t=h}^{H-1} r_t(S_t, A_t) \mid \pi, S_h = s \right],$$

where the expectation is over the trajectory induced by π and the transition kernels. Similarly, define the state-action value (Q-value) function $Q_h^\pi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ by

$$Q_h^\pi(s, a) = \mathbb{E} \left[\sum_{t=h}^{H-1} r_t(S_t, A_t) \mid \pi, S_h = s, A_h = a \right].$$

We also write $V^\pi(s) = V_0^\pi(s)$.

Given a starting state s , the goal is to find a policy maximizing the value, i.e.,

$$\max_{\pi} V^\pi(s), \tag{1.6}$$

where the maximum is over all (possibly non-stationary and randomized) policies.

Theorem 1.9 (Bellman optimality equations: finite horizon). *For $h \in \{0, \dots, H\}$, define*

$$Q_h^*(s, a) = \sup_{\pi} Q_h^\pi(s, a),$$

where the sup is over all (possibly non-stationary and randomized) policies, and define $Q_H^*(s, a) \equiv 0$ (terminal condition) for all (s, a) .

Then $\{Q_h^*\}_{h=0}^H$ is the unique collection of functions satisfying, for all $h \in \{0, \dots, H-1\}$ and $(s, a) \in \mathcal{S} \times \mathcal{A}$,

$$Q_h(s, a) = r_h(s, a) + \mathbb{E}_{s' \sim P_h(\cdot | s, a)} \left[\max_{a' \in \mathcal{A}} Q_{h+1}(s', a') \right]. \tag{1.7}$$

Moreover, any policy π^* defined by

$$\pi^*(s, h) \in \operatorname{argmax}_{a \in \mathcal{A}} Q_h^*(s, a), \quad h = 0, \dots, H-1,$$

is an optimal policy.

We leave the proof as an exercise to the reader. (It follows by backward induction.)

Discussion: stationary vs. time-dependent models. For the purposes of this book, it is natural to study both models: we typically assume stationary dynamics in the infinite-horizon discounted setting and time-dependent dynamics in the finite-horizon setting.

From a theoretical perspective, the finite-horizon *time-dependent* formulation is often more convenient: backward induction avoids contraction arguments, and many sharp statistical rates admit simpler proofs in this setting.

From a practical perspective, explicitly time-dependent models are less common because they lead to policies and value functions whose tabular representations scale as $O(H|\mathcal{S}||\mathcal{A}|)$ and $O(H|\mathcal{S}|)$, respectively, i.e., a factor $O(H)$ larger than in the stationary case. In practice, one often folds temporal information into the state (e.g., augmenting the state with the time index), which recovers a stationary MDP on an expanded state space and can be combined naturally with function approximation to represent policies and value functions compactly.

	Value Iteration	Policy Iteration	LP Algorithms
Poly?	$ \mathcal{S} ^2 \mathcal{A} \frac{L(P, r, \gamma) \log \frac{1}{1-\gamma}}{1-\gamma}$	$(\mathcal{S} ^3 + \mathcal{S} ^2 \mathcal{A}) \frac{L(P, r, \gamma) \log \frac{1}{1-\gamma}}{1-\gamma}$	$ \mathcal{S} ^3 \mathcal{A} L(P, r, \gamma)$
Strongly Poly?	X	$(\mathcal{S} ^3 + \mathcal{S} ^2 \mathcal{A}) \cdot \min \left\{ \frac{ \mathcal{A} ^{ \mathcal{S} }}{ \mathcal{S} }, \frac{ \mathcal{S} ^2 \mathcal{A} \log \frac{ \mathcal{S} ^2}{1-\gamma}}{1-\gamma} \right\}$	$ \mathcal{S} ^4 \mathcal{A} ^4 \log \frac{ \mathcal{S} }{1-\gamma}$

Table 1.1: Computational complexities of various approaches (universal constants are omitted). Polynomial-time algorithms may depend on the bit complexity $L(P, r, \gamma)$, while strongly polynomial algorithms do not. Note that value iteration and policy iteration have iteration complexity scaling as $1/(1-\gamma)$; thus, if γ is treated as part of the input (encoded in $L(P, r, \gamma)$), these methods are not polynomial-time in the bit-length model. Similarly, policy iteration is strongly polynomial only when γ is treated as a fixed constant. In contrast, the linear programming approach yields both polynomial-time (in the bit-length model) and strongly polynomial algorithms; the latter can be obtained via interior point methods. See the text and Section 1.6 for references. Here, $|\mathcal{S}|^2|\mathcal{A}|$ is the assumed runtime per iteration of value iteration, and $|\mathcal{S}|^3 + |\mathcal{S}|^2|\mathcal{A}|$ is the assumed runtime per iteration of policy iteration; these are consistent with cubic-time linear system solves.

1.3 Computational Complexity

This section concerns the problem of computing an optimal policy when the MDP $M = (\mathcal{S}, \mathcal{A}, P, r, \gamma)$ is known; this is the classical *planning* problem in reinforcement learning. While much of this book is concerned with statistical limits, understanding computational limits is also informative. We will consider algorithms that return both exact and approximately optimal policies. In particular, we will distinguish polynomial-time (and strongly polynomial-time) algorithms.

Assume that (P, r, γ) are specified with rational entries. Let $L(P, r, \gamma)$ denote the total bit-length of the description of M . We measure running time in the standard bit-complexity model in which basic arithmetic operations $+$, $-$, $\times$, $\div$ on numbers with $O(L(P, r, \gamma))$ bits take unit time.¹ In this setting, one may hope for an algorithm that returns an *exact* optimal policy with running time polynomial in $|\mathcal{S}|$, $|\mathcal{A}|$, and $L(P, r, \gamma)$.

It is also useful to understand which algorithms are *strongly* polynomial. Here, one does not explicitly restrict (P, r, γ) to be rational. An algorithm is strongly polynomial if it returns an optimal policy with running time polynomial in $|\mathcal{S}|$ and $|\mathcal{A}|$ only (with no dependence on $L(P, r, \gamma)$).

The next two subsections cover classical *iterative* algorithms that compute Q^* . We then discuss the linear programming approach.

1.3.1 Value Iteration

Perhaps the simplest algorithm for discounted MDPs is to iteratively apply the Bellman optimality operator $\mathcal{T}$. Starting from some initialization Q , we repeat

$$Q \leftarrow \mathcal{T}Q.$$

This algorithm is referred to as *Q-value iteration*.

Lemma 1.10 (Contraction). *For any $Q, Q' \in \mathbb{R}^{|\mathcal{S}||\mathcal{A}|}$,*

$$\|\mathcal{T}Q - \mathcal{T}Q'\|_\infty \leq \gamma \|Q - Q'\|_\infty.$$

¹Equivalently, one can state bounds that scale polynomially with the bit-length of the numbers manipulated by the algorithm. We use the

Proof: First, for any $s \in \mathcal{S}$, we claim

$$|V_Q(s) - V_{Q'}(s)| \leq \max_{a \in \mathcal{A}} |Q(s, a) - Q'(s, a)|.$$

Assume $V_Q(s) \geq V_{Q'}(s)$ (the other case is symmetric), and let $a \in \operatorname{argmax}_{a \in \mathcal{A}} Q(s, a)$ be a greedy action for Q at s . Then

$$\begin{aligned} V_Q(s) - V_{Q'}(s) &= Q(s, a) - \max_{a' \in \mathcal{A}} Q'(s, a') \\ &\leq Q(s, a) - Q'(s, a) \leq \max_{a \in \mathcal{A}} |Q(s, a) - Q'(s, a)|. \end{aligned}$$

Using this bound,

$$\begin{aligned} \|\mathcal{T}Q - \mathcal{T}Q'\|_\infty &= \gamma \|PV_Q - PV_{Q'}\|_\infty = \gamma \|P(V_Q - V_{Q'})\|_\infty \\ &\leq \gamma \|V_Q - V_{Q'}\|_\infty = \gamma \max_s |V_Q(s) - V_{Q'}(s)| \\ &\leq \gamma \max_s \max_a |Q(s, a) - Q'(s, a)| = \gamma \|Q - Q'\|_\infty, \end{aligned}$$

where the inequality $\|Px\|_\infty \leq \|x\|_\infty$ uses that each component of Px is a convex combination of the entries of x . ■

The following result bounds the suboptimality of the greedy policy induced by an approximate Q -function.

Lemma 1.11. (*Q-error amplification*) For any $Q \in \mathbb{R}^{|\mathcal{S}||\mathcal{A}|}$,

$$V^{\pi_Q} \geq V^* - \frac{2\|Q - Q^*\|_\infty}{1 - \gamma} \mathbf{1},$$

where $\mathbf{1}$ denotes the vector of all ones.

Proof: Fix a state s and let $a = \pi_Q(s)$. Then

$$\begin{aligned} V^*(s) - V^{\pi_Q}(s) &= Q^*(s, \pi^*(s)) - Q^{\pi_Q}(s, a) \\ &= (Q^*(s, \pi^*(s)) - Q^*(s, a)) + (Q^*(s, a) - Q^{\pi_Q}(s, a)). \end{aligned}$$

For the first term, using that a is greedy for Q ,

$$Q(s, \pi^*(s)) \leq Q(s, a) \implies Q^*(s, \pi^*(s)) - Q^*(s, a) \leq (Q^*(s, \pi^*(s)) - Q(s, \pi^*(s))) + (Q(s, a) - Q^*(s, a)) \leq 2\|Q - Q^*\|_\infty.$$

For the second term, Bellman consistency for π_Q gives

$$Q^*(s, a) - Q^{\pi_Q}(s, a) = \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)} [V^*(s') - V^{\pi_Q}(s')] \leq \gamma \|V^* - V^{\pi_Q}\|_\infty.$$

Combining,

$$V^*(s) - V^{\pi_Q}(s) \leq 2\|Q - Q^*\|_\infty + \gamma \|V^* - V^{\pi_Q}\|_\infty.$$

Taking the supremum over s on the left-hand side yields

$$\|V^* - V^{\pi_Q}\|_\infty \leq 2\|Q - Q^*\|_\infty + \gamma \|V^* - V^{\pi_Q}\|_\infty,$$

and rearranging gives

$$\|V^* - V^{\pi_Q}\|_\infty \leq \frac{2}{1 - \gamma} \|Q - Q^*\|_\infty.$$

Equivalently, $V^{\pi_Q} \geq V^* - \frac{2}{1 - \gamma} \|Q - Q^*\|_\infty \mathbf{1}$. ■

Theorem 1.12. (*Q-value iteration convergence*) Set $Q^{(0)} = 0$, and for $k = 0, 1, \dots$ define

$$Q^{(k+1)} = \mathcal{T}Q^{(k)}.$$

Let $\pi^{(k)} = \pi_{Q^{(k)}}$. For

$$k \geq \frac{1}{1-\gamma} \log\left(\frac{2}{(1-\gamma)^2 \epsilon}\right),$$

we have

$$V^{\pi^{(k)}} \geq V^* - \epsilon \mathbb{1}.$$

Proof: Since $\|Q^*\|_\infty \leq 1/(1-\gamma)$, and $Q^{(k)} = \mathcal{T}^k Q^{(0)}$ with $Q^* = \mathcal{T}Q^*$, the contraction property (Lemma 1.10) gives

$$\begin{aligned} \|Q^{(k)} - Q^*\|_\infty &= \|\mathcal{T}^k Q^{(0)} - \mathcal{T}^k Q^*\|_\infty \leq \gamma^k \|Q^{(0)} - Q^*\|_\infty \\ &= \gamma^k \|Q^*\|_\infty \leq \frac{\gamma^k}{1-\gamma} \leq \frac{\exp(-(1-\gamma)k)}{1-\gamma}, \end{aligned}$$

where we used $\gamma^k \leq \exp(-(1-\gamma)k)$. Choosing k as stated ensures $\|Q^{(k)} - Q^*\|_\infty \leq \frac{(1-\gamma)\epsilon}{2}$, and applying Lemma 1.11 yields $V^{\pi^{(k)}} \geq V^* - \epsilon \mathbb{1}$. ■

Iteration complexity for an exact solution. The previous results guarantee that Q -value iteration converges geometrically to Q^* , and hence yields an ϵ -optimal policy after $O\left(\frac{1}{1-\gamma} \log \frac{1}{\epsilon}\right)$ iterations. To obtain an *exact* optimal policy in the setting where (P, r, γ) are rational, one can combine this convergence with a standard bit-complexity separation argument. In particular, when the data (P, r, γ) have total description length $L(P, r, \gamma)$, the value of any *fixed* deterministic stationary policy can be written as a rational function of these inputs whose numerator and denominator have bit-length polynomial in $L(P, r, \gamma)$. Consequently, among the finite set of deterministic stationary policies, either two policies have exactly the same value (a true tie) or their values differ by at least $2^{-\text{poly}(L(P, r, \gamma))}$. Thus, once $\|Q^{(k)} - Q^*\|_\infty$ is smaller than a corresponding threshold, any greedy policy with respect to $Q^{(k)}$ must be optimal (up to arbitrary tie-breaking). This yields the polynomial-time bound in Table 1.1 when γ is treated as a fixed constant. Value iteration is not strongly polynomial: as $\gamma \uparrow 1$, the number of iterations required to reach the separation scale can grow arbitrarily large, and moreover the algorithm may never certify exact optimality in finite time without explicitly appealing to such bit-length arguments.

1.3.2 Policy Iteration

Policy iteration for discounted MDPs starts from an arbitrary deterministic stationary policy π_0 and repeats, for $k = 0, 1, 2, \dots$:

1. *Policy evaluation.* Compute the value of the current policy, e.g., compute V^{π_k} by solving the linear system $V^{\pi_k} = r(\cdot, \pi_k(\cdot)) + \gamma P^{\pi_k} V^{\pi_k}$, and then obtain Q^{π_k} via $Q^{\pi_k}(s, a) = r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)}[V^{\pi_k}(s')]$. (Equivalently, one may compute Q^{π_k} directly using Equation 1.2.)
2. *Policy improvement.* Update the policy to be greedy w.r.t. Q^{π_k} :

$$\pi_{k+1} = \pi_{Q^{\pi_k}}.$$

The first step is called *policy evaluation* and the second *policy improvement*.

Lemma 1.13 (Monotonicity and contraction of policy iteration). *For all $k \geq 0$,*

1. (*Sandwich bound*) $Q^{\pi_{k+1}} \geq \mathcal{T}Q^{\pi_k} \geq Q^{\pi_k}$ (elementwise).
2. (*Geometric contraction to optimality*)

$$\|Q^* - Q^{\pi_{k+1}}\|_\infty \leq \gamma \|Q^* - Q^{\pi_k}\|_\infty.$$

Proof: We repeatedly use that $\mathcal{T}$ is monotone and a γ -contraction in $\|\cdot\|_\infty$ (Lemma 1.10), and that $Q^* \geq Q^\pi$ elementwise for any policy π .

Step 1: $\mathcal{T}Q^{\pi_k} \geq Q^{\pi_k}$. For deterministic π_k , we have $V^{\pi_k}(s) = Q^{\pi_k}(s, \pi_k(s))$. Moreover, by definition, $V_{Q^{\pi_k}}(s) = \max_a Q^{\pi_k}(s, a) \geq Q^{\pi_k}(s, \pi_k(s)) = V^{\pi_k}(s)$. Therefore, for every (s, a) ,

$$\mathcal{T}Q^{\pi_k}(s, a) = r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)}[V_{Q^{\pi_k}}(s')] \geq r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)}[V^{\pi_k}(s')] = Q^{\pi_k}(s, a).$$

Step 2: $Q^{\pi_{k+1}} \geq \mathcal{T}Q^{\pi_k}$. Let $\pi' = \pi_{k+1} = \pi_{Q^{\pi_k}}$. Define the (policy-specific) Bellman evaluation operator on state values,

$$(\mathcal{T}^{\pi'} V)(s) := r(s, \pi'(s)) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, \pi'(s))}[V(s')].$$

This operator is monotone and a γ -contraction in $\|\cdot\|_\infty$, with unique fixed point $V^{\pi'}$.

By greediness of π' w.r.t. Q^{π_k} ,

$$(\mathcal{T}^{\pi'} V^{\pi_k})(s) = r(s, \pi'(s)) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, \pi'(s))}[V^{\pi_k}(s')] = Q^{\pi_k}(s, \pi'(s)) = \max_a Q^{\pi_k}(s, a) = V_{Q^{\pi_k}}(s) \geq V^{\pi_k}(s).$$

Applying $\mathcal{T}^{\pi'}$ repeatedly and using monotonicity yields an increasing sequence $V^{\pi_k} \leq (\mathcal{T}^{\pi'})V^{\pi_k} \leq (\mathcal{T}^{\pi'})^2 V^{\pi_k} \leq \dots$, which converges to the fixed point $V^{\pi'}$. Hence,

$$V^{\pi_{k+1}} = V^{\pi'} \geq V_{Q^{\pi_k}} \quad (\text{elementwise}).$$

Now, for any (s, a) ,

$$Q^{\pi_{k+1}}(s, a) = r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)}[V^{\pi_{k+1}}(s')] \geq r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)}[V_{Q^{\pi_k}}(s')] = \mathcal{T}Q^{\pi_k}(s, a),$$

which proves $Q^{\pi_{k+1}} \geq \mathcal{T}Q^{\pi_k}$ and completes the sandwich bound.

Step 3: contraction. Since $Q^* = \mathcal{T}Q^*$ and $Q^* \geq Q^{\pi_{k+1}} \geq \mathcal{T}Q^{\pi_k}$ elementwise,

$$0 \leq Q^* - Q^{\pi_{k+1}} \leq Q^* - \mathcal{T}Q^{\pi_k} = \mathcal{T}Q^* - \mathcal{T}Q^{\pi_k}.$$

Taking $\|\cdot\|_\infty$ and applying Lemma 1.10 gives

$$\|Q^* - Q^{\pi_{k+1}}\|_\infty \leq \|\mathcal{T}Q^* - \mathcal{T}Q^{\pi_k}\|_\infty \leq \gamma \|Q^* - Q^{\pi_k}\|_\infty,$$

as claimed. ■

With Lemma 1.13, a geometric convergence bound follows immediately.

Theorem 1.14 (Policy iteration convergence). *Let π_0 be any initial deterministic stationary policy, and let $\{\pi_k\}$ be simplified unit-cost model to keep the dependence on $L(P, r, \gamma)$ explicit.*

generated by policy iteration. Then for all $k \geq 0$,

$$\|Q^* - Q^{\pi_k}\|_\infty \leq \gamma^k \|Q^* - Q^{\pi_0}\|_\infty \leq \frac{\gamma^k}{1 - \gamma}.$$

In particular, for

$$k \geq \frac{\log\left(\frac{1}{(1-\gamma)\epsilon}\right)}{1 - \gamma},$$

we have the elementwise performance guarantee

$$Q^{\pi_k} \geq Q^* - \epsilon \mathbf{1}.$$

Iteration complexity for an exact solution. Policy iteration improves the value monotonically and converges geometrically, but unlike value iteration it can terminate with an *exact* optimal policy in finitely many iterations because the policy changes only when the greedy action changes. A naive worst-case bound on the number of iterations is the number of deterministic stationary policies, $|\mathcal{A}|^{|\mathcal{S}|}$, since policy iteration never revisits a policy. A modest refinement improves this to $\frac{|\mathcal{A}|^{|\mathcal{S}|}}{|\mathcal{S}|}$. More strongly, when one measures complexity in the *strongly polynomial* model (unit-cost real arithmetic) and treats γ as fixed, substantially better bounds are known: policy iteration finds an optimal policy in a number of iterations polynomial in $|\mathcal{S}|$ and $|\mathcal{A}|$ (see Table 1.1 and Section 1.6 for references and precise statements).

1.3.3 Value Iteration for Finite-Horizon MDPs

We now specify the value iteration algorithm for finite-horizon MDPs. In the finite-horizon (time-dependent) setting, dynamic programming via backward induction computes the optimal value functions exactly, and the natural analogues of value iteration and policy iteration coincide.

Let $Q_H(s, a) \equiv 0$ for all (s, a) . For $h = H - 1, H - 2, \dots, 0$, define

$$Q_h(s, a) = r_h(s, a) + \mathbb{E}_{s' \sim P_h(\cdot|s, a)} \left[\max_{a' \in \mathcal{A}} Q_{h+1}(s', a') \right]. \quad (1.8)$$

Equivalently, one may initialize $Q_{H-1}(s, a) = r_{H-1}(s, a)$ and then run the same recursion for $h = H - 2, \dots, 0$.

By Theorem 1.9, the resulting functions satisfy $Q_h = Q_h^*$ for all $h \in \{0, \dots, H\}$, and the policy

$$\pi(s, h) \in \operatorname{argmax}_{a \in \mathcal{A}} Q_h^*(s, a)$$

(with arbitrary tie-breaking) is optimal.

1.3.4 The Linear Programming Approach

It is helpful to understand an alternative approach to computing an optimal policy when the MDP is known. In particular, consider a discounted MDP $M = (\mathcal{S}, \mathcal{A}, P, r, \gamma, \mu)$ where P , r , and γ are specified by rational numbers. From a bit-complexity perspective, the iterative algorithms above are not, in general, polynomial-time in the description length of the input because their iteration complexity typically depends polynomially on $1/(1 - \gamma)$, while a rational value of $1 - \gamma$ may be represented with only $O\left(\log \frac{1}{1-\gamma}\right)$ bits. In contrast, the LP approach yields polynomial-time algorithms in the standard bit model.

The Primal LP

Consider the following linear program in variables $V \in \mathbb{R}^{|\mathcal{S}|}$:

$$\begin{aligned} \min \quad & \sum_{s \in \mathcal{S}} \mu(s) V(s) \\ \text{subject to} \quad & V(s) \geq r(s, a) + \gamma \sum_{s' \in \mathcal{S}} P(s' | s, a) V(s') \quad \forall s \in \mathcal{S}, a \in \mathcal{A}. \end{aligned}$$

The feasible region is exactly the set of value functions that dominate their Bellman optimality backup. The optimal value function V^* is the (unique) minimal feasible vector (componentwise). If μ has full support, then V^* is also the unique optimizer of the above objective (since increasing any coordinate increases the objective).

Computational complexity for an exact solution. Table 1.1 summarizes standard runtime bounds for solving this LP when the coefficients are rational and we measure complexity in the bit model. There are also specialized interior-point methods for discounted MDP LPs with strong guarantees; see Section 1.6 for references and further discussion.

Policy iteration and simplex. Policy iteration can be interpreted as a pivoting method for closely related LP formulations (and, in particular, has a well-known connection to simplex-type methods under appropriate pivot rules). While the simplex method is not strongly polynomial for general LPs, policy-iteration/simplex variants admit strong polynomial-time guarantees for discounted MDPs when the discount factor is treated as fixed; see [Ye, 2011].

The Dual LP and the State–Action Polytope

We next introduce the discounted occupancy measure, which yields a convenient dual view. For a (possibly randomized) stationary policy π and a fixed starting state s_0 , define the distribution over state–action pairs

$$d_{s_0}^\pi(s, a) := (1 - \gamma) \sum_{t=0}^{\infty} \gamma^t \Pr^\pi(s_t = s, a_t = a | s_0). \quad (1.9)$$

It is straightforward to verify that $d_{s_0}^\pi$ is a probability distribution over $\mathcal{S} \times \mathcal{A}$ (since $(1 - \gamma) \sum_{t \geq 0} \gamma^t = 1$). For a starting distribution μ , we also define the averaged occupancy measure

$$d_\mu^\pi(s, a) := \mathbb{E}_{s_0 \sim \mu} [d_{s_0}^\pi(s, a)].$$

Flow (consistency) constraints. The discounted occupancy satisfies a linear recursion. For a fixed start state s_0 , one has for every $s \in \mathcal{S}$,

$$\sum_{a \in \mathcal{A}} d_{s_0}^\pi(s, a) = (1 - \gamma) \mathbb{1}\{s = s_0\} + \gamma \sum_{s' \in \mathcal{S}} \sum_{a' \in \mathcal{A}} P(s | s', a') d_{s_0}^\pi(s', a'). \quad (1.10)$$

Averaging (1.10) over $s_0 \sim \mu$ yields the corresponding relation for d_μ^π :

$$\sum_{a \in \mathcal{A}} d_\mu^\pi(s, a) = (1 - \gamma) \mu(s) + \gamma \sum_{s' \in \mathcal{S}} \sum_{a' \in \mathcal{A}} P(s | s', a') d_\mu^\pi(s', a') \quad \forall s \in \mathcal{S}. \quad (1.11)$$

Motivated by (1.11), define the *state–action polytope*

$$\mathcal{K}_\mu := \left\{ d \in \mathbb{R}^{|\mathcal{S}| \cdot |\mathcal{A}|} \left| \begin{aligned} & d(s, a) \geq 0 \quad \forall (s, a), \\ & \sum_{a \in \mathcal{A}} d(s, a) = (1 - \gamma) \mu(s) + \gamma \sum_{s', a'} P(s | s', a') d(s', a') \quad \forall s \in \mathcal{S} \end{aligned} \right. \right\}.$$

Proposition 1.15. $\mathcal{K}_\mu$ is exactly the set of discounted occupancy measures induced by stationary policies: $d \in \mathcal{K}_\mu$ if and only if there exists a stationary (possibly randomized) policy π such that $d_\mu^\pi = d$.

The dual LP. Using the identity

$$V^\pi(\mu) = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \right] = \frac{1}{1-\gamma} \sum_{s \in \mathcal{S}} \sum_{a \in \mathcal{A}} d_\mu^\pi(s, a) r(s, a),$$

we can write a dual LP in the variables $d \in \mathbb{R}^{|\mathcal{S}| \cdot |\mathcal{A}|}$:

$$\begin{aligned} \max \quad & \frac{1}{1-\gamma} \sum_{s \in \mathcal{S}} \sum_{a \in \mathcal{A}} d(s, a) r(s, a) \\ \text{subject to} \quad & d \in \mathcal{K}_\mu. \end{aligned}$$

One can verify that this is dual to the primal LP above (and strong duality holds under standard conditions). This dual perspective provides an alternative route to computing an optimal policy.

If d^* is an optimal solution and μ has full support, then an optimal policy can be recovered via

$$\pi^*(a \mid s) = \frac{d^*(s, a)}{\sum_{a'} d^*(s, a')} \quad \text{whenever } \sum_{a'} d^*(s, a') > 0,$$

with $\pi^*(\cdot \mid s)$ defined arbitrarily on states with zero occupancy. An alternative optimal policy is $\arg\max_a d^*(s, a)$; these coincide when the optimizer is unique.

1.4 Sample Complexity and Sampling Models

Much of reinforcement learning is concerned with finding a near-optimal policy (or, equivalently, achieving near-optimal return) in settings where the MDP is *not* known to the learner. We will study these questions under several different *sampling models*, which specify how the agent obtains information about the unknown underlying MDP. In each model, we are interested in understanding the number of samples required to compute a near-optimal policy, i.e. the *sample complexity*. Ultimately, we are interested in results that remain meaningful when the number of states and actions is large (or even when $\mathcal{S}$ is countably or uncountably infinite). This is analogous in spirit to generalization in supervised learning, but, as we shall see, it is fundamentally more challenging in reinforcement learning because the data distribution depends on the learner's behavior.

The episodic setting. In the episodic setting, the learner interacts with the environment in *episodes*. In each episode, the learner starts from an initial state $s_0 \sim \mu$, acts for a finite number of steps, observes the resulting trajectory, and then the environment resets and a new episode begins with a fresh draw $s_0 \sim \mu$. This episodic model of feedback is applicable to both finite-horizon and infinite-horizon MDPs:

- **Finite-horizon MDPs.** Each episode lasts exactly H steps, after which the environment resets to $s_0 \sim \mu$.
- **Infinite-horizon MDPs.** It is often natural to impose an episodic protocol even in the infinite-horizon discounted setting by ending each episode after a random (finite) number of steps. A standard model is to terminate the episode at each time step independently with probability $1 - \gamma$ (equivalently, the episode length has a geometric distribution). After termination, the state resets to $s_0 \sim \mu$. Under this protocol, the undiscounted cumulative reward observed in one episode is an unbiased estimate of the infinite-horizon discounted value $V^\pi(\mu)$.

In this setting, we are often interested either in the number of episodes needed to find a near-optimal policy (a *PAC* guarantee), or in *regret* guarantees (which we study in Chapter 6). Both are statements about the statistical (sample) complexity of learning.

The episodic setting is challenging because the agent must explore to obtain information about relevant state–action pairs. As we shall see in Chapter 6, this exploration must be strategic: simply behaving randomly can be far too inefficient. It is therefore often helpful to also study the statistical complexity of RL in an abstract sampling model that bypasses exploration concerns.

The generative model setting. A *generative model* (or simulator oracle) takes as input a state–action pair (s, a) and returns an independent sample $s' \sim P(\cdot \mid s, a)$ together with the reward $r(s, a)$ (or, in the stochastic-reward setting, an independent sample of the reward). This model allows the learner to sample transitions from any chosen (s, a) directly, and is a convenient abstraction for isolating statistical questions from exploration dynamics.

The offline RL setting. In *offline* RL, the agent is given a fixed dataset collected in advance, typically generated by some (unknown) behavior policy or a mixture of policies. In the simplest model, the dataset consists of tuples $\{(s, a, r, s')\}$, where r corresponds to $r(s, a)$ in the deterministic-reward case (or a reward sample otherwise) and $s' \sim P(\cdot \mid s, a)$. For simplicity, it is often helpful to assume that the state–action pairs (s, a) in the dataset are sampled i.i.d. from some distribution ν over $\mathcal{S} \times \mathcal{A}$, after which (r, s') are drawn from the conditional dynamics given (s, a) . This i.i.d. model captures the core statistical difficulties associated with distribution shift and coverage.

1.5 Bonus: Advantages and the Performance Difference Lemma

Throughout, we will overload notation so that for a distribution μ over $\mathcal{S}$,

$$V^\pi(\mu) := \mathbb{E}_{s \sim \mu}[V^\pi(s)].$$

The *advantage* of a policy π is defined as

$$A^\pi(s, a) := Q^\pi(s, a) - V^\pi(s).$$

In particular, for an optimal policy π^* , we have

$$A^{\pi^*}(s, a) = Q^*(s, a) - V^*(s) \leq 0 \quad \forall (s, a),$$

with equality holding for any action $a \in \operatorname{argmax}_{a'} Q^*(s, a')$.

Analogous to the discounted state–action occupancy measure in Equation 1.9, we define a discounted *state* occupancy measure. To avoid introducing new notation, we use the same symbol $d_{s_0}^\pi$; its meaning will be clear from whether it is evaluated on s or on (s, a) :

$$d_{s_0}^\pi(s) := (1 - \gamma) \sum_{t=0}^{\infty} \gamma^t \Pr^\pi(s_t = s \mid s_0). \quad (1.12)$$

Here, $\Pr^\pi(s_t = s \mid s_0)$ denotes the state visitation probability at time t under π starting from s_0 . As before, for a distribution μ over $\mathcal{S}$, we write

$$d_\mu^\pi(s) := \mathbb{E}_{s_0 \sim \mu}[d_{s_0}^\pi(s)].$$

The following lemma is a standard and extremely useful identity.

Lemma 1.16. (The performance difference lemma) For all policies π, π' and all distributions μ over $\mathcal{S}$,

$$V^\pi(\mu) - V^{\pi'}(\mu) = \frac{1}{1-\gamma} \mathbb{E}_{s \sim d_\mu^\pi} \mathbb{E}_{a \sim \pi(\cdot|s)} [A^{\pi'}(s, a)].$$

Proof: We first prove the identity for a fixed start state $s_0 = s$, and then average over $s \sim \mu$.

Fix $s_0 = s$ and let $\tau = (s_0, a_0, s_1, a_1, \dots)$ denote the (infinite) trajectory generated by following π . For any function $f : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$, by the definition (1.12) we have

$$\mathbb{E}_{\tau \sim \text{Pr}^\pi(\cdot|s_0=s)} \left[\sum_{t=0}^{\infty} \gamma^t f(s_t, a_t) \right] = \frac{1}{1-\gamma} \mathbb{E}_{s_t \sim d_s^\pi} \mathbb{E}_{a \sim \pi(\cdot|s_t)} [f(s_t, a)]. \quad (1.13)$$

Next, we use a telescoping decomposition. Since $V^{\pi'}$ is bounded and $\gamma \in [0, 1)$,

$$\sum_{t=0}^{\infty} \gamma^t (\gamma V^{\pi'}(s_{t+1}) - V^{\pi'}(s_t)) = -V^{\pi'}(s_0) \quad (\text{almost surely}).$$

Therefore,

$$\begin{aligned} V^\pi(s) - V^{\pi'}(s) &= \mathbb{E}_{\tau \sim \text{Pr}^\pi(\cdot|s_0=s)} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \right] - V^{\pi'}(s) \\ &= \mathbb{E}_{\tau \sim \text{Pr}^\pi(\cdot|s_0=s)} \left[\sum_{t=0}^{\infty} \gamma^t (r(s_t, a_t) + \gamma V^{\pi'}(s_{t+1}) - V^{\pi'}(s_t)) \right] \\ &= \mathbb{E}_{\tau \sim \text{Pr}^\pi(\cdot|s_0=s)} \left[\sum_{t=0}^{\infty} \gamma^t (r(s_t, a_t) + \gamma \mathbb{E}[V^{\pi'}(s_{t+1}) | s_t, a_t] - V^{\pi'}(s_t)) \right] \\ &= \mathbb{E}_{\tau \sim \text{Pr}^\pi(\cdot|s_0=s)} \left[\sum_{t=0}^{\infty} \gamma^t (Q^{\pi'}(s_t, a_t) - V^{\pi'}(s_t)) \right] \\ &= \mathbb{E}_{\tau \sim \text{Pr}^\pi(\cdot|s_0=s)} \left[\sum_{t=0}^{\infty} \gamma^t A^{\pi'}(s_t, a_t) \right]. \end{aligned}$$

Applying (1.13) with $f = A^{\pi'}$ yields

$$V^\pi(s) - V^{\pi'}(s) = \frac{1}{1-\gamma} \mathbb{E}_{s' \sim d_s^\pi} \mathbb{E}_{a \sim \pi(\cdot|s')} [A^{\pi'}(s', a)].$$

Finally, averaging over $s \sim \mu$ gives the stated result. ■

1.6 Bibliographic Remarks and Further Reading

We refer the reader to Puterman [1994] for a comprehensive treatment of dynamic programming and MDPs; this text also contains a thorough treatment of the dual LP formulation and a proof of Lemma 1.15.

Regarding the computational complexity of planning, Ye [2011] provides an excellent summary of various approaches. Notably, Ye [2011] showed that policy iteration is a strongly polynomial-time algorithm for a fixed discount rate.²

²The strongly polynomial runtime stated in Table 1.1 differs slightly from that in [Ye, 2011] because we assume the cost per iteration of policy

Mansour and Singh [1999] proved that the number of iterations of policy iteration is bounded by roughly $|\mathcal{A}|^{|\mathcal{S}|}/|\mathcal{S}|$ (improving on the trivial $|\mathcal{A}|^{|\mathcal{S}|}$ bound). For general strongly polynomial algorithms (independent of γ), the CIPA algorithm proposed by Ye [2005] is an interior-point method achieving the runtime claimed in Table 1.1.

The error amplification bound (Lemma 1.11) is due to Singh and Yee [1994].

The performance difference lemma (Lemma 1.16) was explicitly stated and utilized by Kakade and Langford [2002], Kakade [2003], though the result (based on a telescoping expansion of advantages) was implicit in the analysis of several prior works.

iteration is $O(|\mathcal{S}|^3 + |\mathcal{S}|^2|\mathcal{A}|)$ (dominated by solving the linear system).

Chapter 2

Sample Complexity with a Generative Model

This chapter begins our study of the sample complexity, where we focus on the (minmax) number of transitions we need to observe in order to accurately estimate Q^* or in order to find a near optimal policy. We assume that we have access to a generative model (as defined in Section 1.4) and that the reward function is deterministic (the latter is often a mild assumption, due to that much of the difficulty in RL is due to the uncertainty in the transition model P).

This chapter follows the results due to [Azar et al., 2013], along with some improved rates due to [Agarwal et al., 2020c]. One of the key observations in this chapter is that we can find a near optimal policy using a number of observed transitions that is *sublinear* in the model size, i.e. use a number of samples that is smaller than $O(|\mathcal{S}|^2|\mathcal{A}|)$. In other words, we do not need to learn an accurate model of the world in order to learn to act near optimally.

Notation. We define $\widehat{M}$ to be the empirical MDP that is identical to the original M , except that it uses $\widehat{P}$ instead of P for the transition model. When clear from context, we drop the subscript on M on the values, action values (and one-step variances and variances which we define later). We let $\widehat{V}^\pi$, $\widehat{Q}^\pi$, $\widehat{Q}^*$, and $\widehat{\pi}^*$ denote the value function, state-action value function, optimal state-action value, and optimal policy in $\widehat{M}$, respectively.

2.1 Warmup: a naive model-based approach

A central question in this chapter is: *Do we require an accurate model of the world in order to find a near optimal policy?* Recall that a generative model takes as input a state action pair (s, a) and returns a sample $s' \sim P(\cdot|s, a)$ and the reward $r(s, a)$ (or a sample of the reward if the rewards are stochastic). Let us consider the most naive approach to learning (when we have access to a generative model): suppose we call our simulator N times at each state action pair. Let $\widehat{P}$ be our empirical model, defined as follows:

$$\widehat{P}(s'|s, a) = \frac{\text{count}(s', s, a)}{N}$$

where $\text{count}(s', s, a)$ is the number of times the state-action pair (s, a) transitions to state s' . As the N is the number of calls for each state action pair, the total number of calls to our generative model is $|\mathcal{S}||\mathcal{A}|N$. As before, we can view $\widehat{P}$ as a matrix of size $|\mathcal{S}||\mathcal{A}| \times |\mathcal{S}|$.

Note that since P has a $|\mathcal{S}|^2|\mathcal{A}|$ parameters, we would expect that observing $O(|\mathcal{S}|^2|\mathcal{A}|)$ transitions is sufficient to provide us with an accurate model. The following proposition shows that this is the case.

Proposition 2.1. *There exists an absolute constant c such that the following holds. Suppose $\epsilon \in (0, \frac{1}{1-\gamma})$ and that we obtain*

$$\# \text{ samples from generative model} = |\mathcal{S}||\mathcal{A}|N \geq \frac{\gamma}{(1-\gamma)^4} \frac{|\mathcal{S}|^2|\mathcal{A}| \log(c|\mathcal{S}||\mathcal{A}|/\delta)}{\epsilon^2}$$

where we uniformly sample every state action pair. Then, with probability greater than $1 - \delta$, we have:

- (Model accuracy) The transition model has error bounded as:

$$\max_{s,a} \|P(\cdot|s,a) - \hat{P}(\cdot|s,a)\|_1 \leq (1-\gamma)^2 \epsilon.$$

- (Uniform value accuracy) For all policies π ,

$$\|Q^\pi - \hat{Q}^\pi\|_\infty \leq \epsilon$$

- (Near optimal planning) Suppose that $\hat{\pi}$ is the optimal policy in $\hat{M}$. We have that:

$$\|\hat{Q}^* - Q^*\|_\infty \leq \epsilon, \text{ and } \|Q^{\hat{\pi}} - Q^*\|_\infty \leq 2\epsilon.$$

Before we provide the proof, the following lemmas will be helpful throughout:

Lemma 2.2. (Simulation Lemma) For all π we have that:

$$Q^\pi - \hat{Q}^\pi = \gamma(I - \gamma\hat{P}^\pi)^{-1}(P - \hat{P})V^\pi$$

Proof: Using our matrix equality for Q^π (see Equation 1.2), we have:

$$\begin{aligned} Q^\pi - \hat{Q}^\pi &= (I - \gamma P^\pi)^{-1}r - (I - \gamma\hat{P}^\pi)^{-1}r \\ &= (I - \gamma\hat{P}^\pi)^{-1}((I - \gamma\hat{P}^\pi) - (I - \gamma P^\pi))Q^\pi \\ &= \gamma(I - \gamma\hat{P}^\pi)^{-1}(P^\pi - \hat{P}^\pi)Q^\pi \\ &= \gamma(I - \gamma\hat{P}^\pi)^{-1}(P - \hat{P})V^\pi \end{aligned}$$

which proves the claim. ■

Lemma 2.3. For any policy π , MDP M and vector $v \in \mathbb{R}^{|\mathcal{S}| \times |\mathcal{A}|}$, we have $\|(I - \gamma P^\pi)^{-1}v\|_\infty \leq \|v\|_\infty / (1 - \gamma)$.

Proof: Note that $v = (I - \gamma P^\pi)(I - \gamma P^\pi)^{-1}v = (I - \gamma P^\pi)w$, where $w = (I - \gamma P^\pi)^{-1}v$. By triangle inequality, we have

$$\|v\|_\infty = \|(I - \gamma P^\pi)w\|_\infty \geq \|w\|_\infty - \gamma \|P^\pi w\|_\infty \geq \|w\|_\infty - \gamma \|w\|_\infty,$$

where the final inequality follows since $P^\pi w$ is an average of the elements of w by the definition of P^π so that $\|P^\pi w\|_\infty \leq \|w\|_\infty$. Rearranging terms completes the proof. ■

Now we are ready to complete the proof of our proposition.

Proof: Using the concentration of a distribution in the ℓ_1 norm (Lemma A.8), we have that for a fixed s, a that, with probability greater than $1 - \delta$, we have:

$$\|P(\cdot|s,a) - \hat{P}(\cdot|s,a)\|_1 \leq c \sqrt{\frac{|\mathcal{S}| \log(1/\delta)}{m}}$$

where m is the number of samples used to estimate $\hat{P}(\cdot|s, a)$. The first claim now follows by the union bound (and redefining δ and c appropriately).

For the second claim, we have that:

$$\begin{aligned}\|Q^\pi - \hat{Q}^\pi\|_\infty &= \|\gamma(I - \gamma\hat{P}^\pi)^{-1}(P - \hat{P})V^\pi\|_\infty \leq \frac{\gamma}{1-\gamma}\|(P - \hat{P})V^\pi\|_\infty \\ &\leq \frac{\gamma}{1-\gamma} \left(\max_{s,a} \|P(\cdot|s, a) - \hat{P}(\cdot|s, a)\|_1 \right) \|V^\pi\|_\infty \leq \frac{\gamma}{(1-\gamma)^2} \max_{s,a} \|P(\cdot|s, a) - \hat{P}(\cdot|s, a)\|_1\end{aligned}$$

where the penultimate step uses Holder's inequality. The second claim now follows.

For the final claim, first observe that $|\sup_x f(x) - \sup_x g(x)| \leq \sup_x |f(x) - g(x)|$, where f and g are real valued functions. This implies:

$$|\hat{Q}^*(s, a) - Q^*(s, a)| = \left| \sup_{\pi} \hat{Q}^\pi(s, a) - \sup_{\pi} Q^\pi(s, a) \right| \leq \sup_{\pi} |\hat{Q}^\pi(s, a) - Q^\pi(s, a)| \leq \epsilon$$

which proves the first inequality. The second inequality is left as an exercise to the reader. $\blacksquare$

2.2 Sublinear Sample Complexity

In the previous approach, we are able to accurately estimate the value of *every* policy in the unknown MDP M . However, with regards to planning, we only need an accurate estimate $\hat{Q}^*$ of Q^* , which we may hope would require less samples. Let us now see that the model based approach can be refined to obtain minmax optimal sample complexity, which we will see is sublinear in the model size.

We will state our results in terms of N , and recall that N is the # of calls to the generative models per state-action pair, so that:

$$\# \text{ samples from generative model} = |\mathcal{S}||\mathcal{A}|N.$$

Let us start with a crude bound on the optimal action-values, which provides a sublinear rate. In the next section, we will improve upon this to obtain the minmax optimal rate.

Proposition 2.4. (*Crude Value Bounds*) Let $\delta \geq 0$. With probability greater than $1 - \delta$,

$$\begin{aligned}\|Q^* - \hat{Q}^*\|_\infty &\leq \Delta_{\delta, N} \\ \|Q^* - \hat{Q}^{\pi^*}\|_\infty &\leq \Delta_{\delta, N},\end{aligned}$$

where:

$$\Delta_{\delta, N} := \frac{\gamma}{(1-\gamma)^2} \sqrt{\frac{2 \log(2|\mathcal{S}||\mathcal{A}|/\delta)}{N}}$$

Note that the first inequality above shows a sublinear rate on estimating the value function. Ultimately, we are interested in the value $V^{\hat{\pi}^*}$ when we execute $\hat{\pi}^*$, not just an estimate $\hat{Q}^*$ of Q^* . Here, by Lemma 1.11, we lose an additional horizon factor and have:

$$\|Q^* - \hat{Q}^{\hat{\pi}^*}\|_\infty \leq \frac{1}{1-\gamma} \Delta_{\delta, N}.$$

As we see in Theorem 2.6, this is improvable.

Before we provide the proof, the following lemma will be helpful throughout.

Lemma 2.5. (Component-wise Bounds) We have that:

$$\begin{aligned} Q^* - \hat{Q}^* &\leq \gamma(I - \gamma\hat{P}^{\pi^*})^{-1}(P - \hat{P})V^* \\ Q^* - \hat{Q}^* &\geq \gamma(I - \gamma\hat{P}^{\hat{\pi}^*})^{-1}(P - \hat{P})V^* \end{aligned}$$

Proof: For the first claim, the optimality of π^* in M implies:

$$Q^* - \hat{Q}^* = Q^{\pi^*} - \hat{Q}^{\hat{\pi}^*} \leq Q^{\pi^*} - \hat{Q}^{\pi^*} = \gamma(I - \gamma\hat{P}^{\pi^*})^{-1}(P - \hat{P})V^*,$$

where we have used Lemma 2.2 in the final step. This proves the first claim.

For the second claim,

$$\begin{aligned} Q^* - \hat{Q}^* &= Q^{\pi^*} - \hat{Q}^{\hat{\pi}^*} \\ &= (1 - \gamma) \left((I - \gamma P^{\pi^*})^{-1}r - (I - \gamma\hat{P}^{\hat{\pi}^*})^{-1}r \right) \\ &= (I - \gamma\hat{P}^{\pi^*})^{-1}((I - \gamma\hat{P}^{\hat{\pi}^*}) - (I - \gamma P^{\pi^*}))Q^* \\ &= \gamma(I - \gamma\hat{P}^{\pi^*})^{-1}(P^{\pi^*} - \hat{P}^{\hat{\pi}^*})Q^* \\ &\geq \gamma(I - \gamma\hat{P}^{\pi^*})^{-1}(P^{\pi^*} - \hat{P}^{\pi^*})Q^* \\ &= \gamma(I - \gamma\hat{P}^{\pi^*})^{-1}(P - \hat{P})V^*, \end{aligned}$$

where the inequality follows from $\hat{P}^{\hat{\pi}^*}Q^* \leq \hat{P}^{\pi^*}Q^*$, due to the optimality of π^* . This proves the second claim. ■

Proof: Following from the simulation lemma (Lemma 2.2) and Lemma 2.3, we have:

$$\|Q^* - \hat{Q}^{\pi^*}\|_{\infty} \leq \frac{\gamma}{1 - \gamma} \|(P - \hat{P})V^*\|_{\infty}.$$

Also, the previous lemma, implies that:

$$\|Q^* - \hat{Q}^*\|_{\infty} \leq \frac{\gamma}{1 - \gamma} \|(P - \hat{P})V^*\|_{\infty}$$

By applying Hoeffding's inequality and the union bound,

$$\|(P - \hat{P})V^*\|_{\infty} = \max_{s,a} |\mathbb{E}_{s' \sim P(\cdot|s,a)}[V^*(s')] - \mathbb{E}_{s' \sim \hat{P}(\cdot|s,a)}[V^*(s')]| \leq \frac{1}{1 - \gamma} \sqrt{\frac{2 \log(2|S||\mathcal{A}|/\delta)}{N}}$$

which holds with probability greater than $1 - \delta$. This completes the proof. ■

2.3 Minmax Optimal Sample Complexity (and the Model Based Approach)

We now see that the model based approach is minmax optimal, for both the discounted case and the finite horizon setting.

2.3.1 The Discounted Case

Upper bounds. The following theorem refines our crude bound on $\hat{Q}^*$.

Theorem 2.6. For $\delta \geq 0$ and for an appropriately chosen absolute constant c , we have that:

- (Value estimation) With probability greater than $1 - \delta$,

$$\|Q^* - \hat{Q}^*\|_\infty \leq \gamma \sqrt{\frac{c}{(1-\gamma)^3} \frac{\log(c|\mathcal{S}||\mathcal{A}|/\delta)}{N}} + \frac{c\gamma}{(1-\gamma)^3} \frac{\log(c|\mathcal{S}||\mathcal{A}|/\delta)}{N}.$$

- (Sub-optimality) If $N \geq \frac{1}{(1-\gamma)^2}$, then with probability greater than $1 - \delta$,

$$\|Q^* - Q^{\hat{\pi}^*}\|_\infty \leq \gamma \sqrt{\frac{c}{(1-\gamma)^3} \frac{\log(c|\mathcal{S}||\mathcal{A}|/\delta)}{N}}.$$

This immediately provides the following corollary.

Corollary 2.7. *Provided that $\epsilon \leq 1$ and that*

$$\# \text{ samples from generative model} = |\mathcal{S}||\mathcal{A}|N \geq \frac{c|\mathcal{S}||\mathcal{A}|}{(1-\gamma)^3} \frac{\log(c|\mathcal{S}||\mathcal{A}|/\delta)}{\epsilon^2},$$

then with probability greater than $1 - \delta$,

$$\|Q^* - \hat{Q}^*\|_\infty \leq \epsilon.$$

Furthermore, provided that $\epsilon \leq \sqrt{\frac{1}{1-\gamma}}$ and that

$$\# \text{ samples from generative model} = |\mathcal{S}||\mathcal{A}|N \geq \frac{c|\mathcal{S}||\mathcal{A}|}{(1-\gamma)^3} \frac{\log(c|\mathcal{S}||\mathcal{A}|/\delta)}{\epsilon^2},$$

then with probability greater than $1 - \delta$,

$$\|Q^* - Q^{\hat{\pi}^*}\|_\infty \leq \epsilon.$$

We only prove the first claim in Theorem 2.6 on the estimation accuracy. With regards to the sub-optimality, note that Theorem 1.11 already implies a sub-optimality gap, though with an amplification of the estimation error by $2/(1-\gamma)$. The argument for the improvement provided in the second claim is more involved (See Section 2.6 for further discussion).

Lower Bounds. Let us say that an estimation algorithm $\mathcal{A}$, which is a map from samples to an estimate $\hat{Q}^*$, is (ϵ, δ) -good on MDP M if $\|Q^* - \hat{Q}^*\|_\infty \leq \epsilon$ holds with probability greater than $1 - \delta$.

Theorem 2.8. *There exists ϵ_0, δ_0, c and a set of MDPs $\mathcal{M}$ such that for $\epsilon \in (0, \epsilon_0)$ and $\delta \in (0, \delta_0)$ if algorithm $\mathcal{A}$ is (ϵ, δ) -good on all $M \in \mathcal{M}$, then $\mathcal{A}$ must use a number of samples that is lower bounded as follows*

$$\# \text{ samples from generative model} \geq \frac{c}{(1-\gamma)^3} \frac{|\mathcal{S}||\mathcal{A}| \log(c|\mathcal{S}||\mathcal{A}|/\delta)}{\epsilon^2}.$$

In other words, this theorem shows that the model based approach minmax optimal.

2.3.2 Finite Horizon Setting

Recall the setting of finite horizon MDPs defined in Section 1.2. Again, we can consider the most naive approach to learning (when we have access to a generative model): suppose we call our simulator N times for every $(s, a, h) \in \mathcal{S} \times \mathcal{A} \times [H]$, i.e. we obtain N i.i.d. samples where $s' \sim P_h(\cdot | s, a)$, for every $(s, a, h) \in \mathcal{S} \times \mathcal{A} \times [H]$. Note that the total number of observed transitions is $H|\mathcal{S}||\mathcal{A}|N$.

Upper bounds. The following theorem provides an upper bound on the model based approach.

Theorem 2.9. *For $\delta \geq 0$ and with probability greater than $1 - \delta$, we have that:*

- (Value estimation)

$$\|Q_0^* - \hat{Q}_0^*\|_\infty \leq cH \sqrt{\frac{\log(c|\mathcal{S}||\mathcal{A}|/\delta)}{N}} + cH \frac{\log(c|\mathcal{S}||\mathcal{A}|/\delta)}{N},$$

- (Sub-optimality)

$$\|Q_0^* - Q_0^{\hat{\pi}^*}\|_\infty \leq cH \sqrt{\frac{\log(c|\mathcal{S}||\mathcal{A}|/\delta)}{N}} + cH \frac{\log(c|\mathcal{S}||\mathcal{A}|/\delta)}{N},$$

where c is an absolute constant.

Note that the above bound requires N to be $O(H^2)$ in order to achieve an ϵ -optimal policy, while in the discounted case, we require N to be $O(1/(1 - \gamma)^3)$ for the same guarantee. While this may seem like an improvement by a horizon factor, recall that for the finite horizon case, N corresponds to observing $O(H)$ more transitions than in the discounted case.

Lower Bounds. In the minmax sense of Theorem 2.8, the previous upper bound provided by the model based approach for the finite horizon setting achieves the minmax optimal sample complexity.

2.4 Analysis

We now prove (the first claim in) Theorem 2.6.

2.4.1 Variance Lemmas

The key to the sharper analysis is to more sharply characterize the variance in our estimates.

Denote the variance of any real valued f under a distribution $\mathcal{D}$ as:

$$\text{Var}_{\mathcal{D}}(f) := E_{x \sim \mathcal{D}}[f(x)^2] - (E_{x \sim \mathcal{D}}[f(x)])^2$$

Slightly abusing the notation, for $V \in R^{|\mathcal{S}|}$, we define the vector $\text{Var}_P(V) \in R^{|\mathcal{S}||\mathcal{A}|}$ as:

$$\text{Var}_P(V)(s, a) := \text{Var}_{P(\cdot|s,a)}(V)$$

Equivalently,

$$\text{Var}_P(V) = P(V)^2 - (PV)^2.$$

Now we characterize a relevant deviation in terms of its variance.

Lemma 2.10. *Let $\delta > 0$. With probability greater than $1 - \delta$,*

$$|(P - \hat{P})V^*| \leq \sqrt{\frac{2\log(2|\mathcal{S}||\mathcal{A}|/\delta)}{N}} \sqrt{\text{Var}_P(V^*)} + \frac{1}{1 - \gamma} \frac{2\log(2|\mathcal{S}||\mathcal{A}|/\delta)}{3N} \mathbf{1}.$$

Proof: The claims follows from Bernstein's inequality along with a union bound over all state-action pairs. $\blacksquare$

The key ideas in the proof are in how we bound $\|(I - \gamma \hat{P}^{\pi^*})^{-1} \sqrt{\text{Var}_P(V^*)}\|_\infty$ and $\|(I - \gamma \hat{P}^{\pi^*})^{-1} \sqrt{\text{Var}_P(V^*)}\|_\infty$.

It is helpful to define Σ_M^π as the variance of the discounted reward, i.e.

$$\Sigma_M^\pi(s, a) := \mathbb{E} \left[\left(\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) - Q_M^\pi(s, a) \right)^2 \middle| s_0 = s, a_0 = a \right]$$

where the expectation is induced under the trajectories induced by π in M . It is straightforward to verify that $\|\Sigma_M^\pi\|_\infty \leq \gamma^2/(1-\gamma)^2$.

The following lemma shows that Σ_M^π satisfies a Bellman consistency condition.

Lemma 2.11. (Bellman consistency of Σ) For any MDP M ,

$$\Sigma_M^\pi = \gamma^2 \text{Var}_P(V_M^\pi) + \gamma^2 P^\pi \Sigma_M^\pi \quad (2.1)$$

where P is the transition model in MDP M .

The proof is left as an exercise to the reader.

Lemma 2.12. (Weighted Sum of Deviations) For any policy π and MDP M ,

$$\|(I - \gamma P^\pi)^{-1} \sqrt{\text{Var}_P(V_M^\pi)}\|_\infty \leq \sqrt{\frac{2}{(1-\gamma)^3}},$$

where P is the transition model of M .

Proof: Note that $(1-\gamma)(I - \gamma P^\pi)^{-1}$ is matrix whose rows are a probability distribution. For a positive vector v and a distribution ν (where ν is vector of the same dimension of v), Jensen's inequality implies that $\nu \cdot \sqrt{v} \leq \sqrt{\nu \cdot v}$. This implies:

$$\begin{aligned} \|(I - \gamma P^\pi)^{-1} \sqrt{v}\|_\infty &= \frac{1}{1-\gamma} \|(1-\gamma)(I - \gamma P^\pi)^{-1} \sqrt{v}\|_\infty \\ &\leq \sqrt{\left\| \frac{1}{1-\gamma} (I - \gamma P^\pi)^{-1} v \right\|_\infty} \\ &\leq \sqrt{\left\| \frac{2}{1-\gamma} (I - \gamma^2 P^\pi)^{-1} v \right\|_\infty}. \end{aligned}$$

where we have used that $\|(I - \gamma P^\pi)^{-1} v\|_\infty \leq 2\|(I - \gamma^2 P^\pi)^{-1} v\|_\infty$ (which we will prove shortly). The proof is completed as follows: by Equation 2.1, $\Sigma_M^\pi = \gamma^2(I - \gamma^2 P^\pi)^{-1} \text{Var}_P(V_M^\pi)$, so taking $v = \text{Var}_P(V_M^\pi)$ and using that $\|\Sigma_M^\pi\|_\infty \leq \gamma^2/(1-\gamma)^2$ completes the proof.

Finally, to see that $\|(I - \gamma P^\pi)^{-1} v\|_\infty \leq 2\|(I - \gamma^2 P^\pi)^{-1} v\|_\infty$, observe:

$$\begin{aligned} \|(I - \gamma P^\pi)^{-1} v\|_\infty &= \|(I - \gamma P^\pi)^{-1} (I - \gamma^2 P^\pi) (I - \gamma^2 P^\pi)^{-1} v\|_\infty \\ &= \|(I - \gamma P^\pi)^{-1} ((1-\gamma)I + \gamma(I - \gamma P^\pi)) (I - \gamma^2 P^\pi)^{-1} v\|_\infty \\ &= \|(1-\gamma)(I - \gamma P^\pi)^{-1} + \gamma I\| (I - \gamma^2 P^\pi)^{-1} v\|_\infty \\ &\leq (1-\gamma)\|(I - \gamma P^\pi)^{-1} (I - \gamma^2 P^\pi)^{-1} v\|_\infty + \gamma\|(I - \gamma^2 P^\pi)^{-1} v\|_\infty \\ &\leq \frac{1-\gamma}{1-\gamma} \|(I - \gamma^2 P^\pi)^{-1} v\|_\infty + \gamma\|(I - \gamma^2 P^\pi)^{-1} v\|_\infty \\ &\leq 2\|(I - \gamma^2 P^\pi)^{-1} v\|_\infty \end{aligned}$$

which proves the claim. $\blacksquare$

2.4.2 Completing the proof

Lemma 2.13. *Let $\delta \geq 0$. With probability greater than $1 - \delta$, we have:*

$$\begin{aligned}\text{Var}_P(V^*) &\leq 2\text{Var}_{\hat{P}}(\hat{V}^{\pi^*}) + \Delta'_{\delta,N}\mathbb{1} \\ \text{Var}_P(V^*) &\leq 2\text{Var}_{\hat{P}}(\hat{V}^*) + \Delta'_{\delta,N}\mathbb{1}\end{aligned}$$

where

$$\Delta'_{\delta,N} := \frac{1}{(1-\gamma)^2} \sqrt{\frac{18 \log(6|\mathcal{S}||\mathcal{A}|/\delta)}{N}} + \frac{1}{(1-\gamma)^4} \frac{4 \log(6|\mathcal{S}||\mathcal{A}|/\delta)}{N}.$$

Proof: By definition,

$$\begin{aligned}\text{Var}_P(V^*) &= \text{Var}_P(V^*) - \text{Var}_{\hat{P}}(V^*) + \text{Var}_{\hat{P}}(V^*) \\ &= P(V^*)^2 - (PV^*)^2 - \hat{P}(V^*)^2 + (\hat{P}V^*)^2 + \text{Var}_{\hat{P}}(V^*) \\ &= (P - \hat{P})(V^*)^2 - \left((PV^*)^2 - (\hat{P}V^*)^2\right) + \text{Var}_{\hat{P}}(V^*)\end{aligned}$$

Now we bound each of these terms with Hoeffding's inequality and the union bound. For the first term, with probability greater than $1 - \delta$,

$$\|(P - \hat{P})(V^*)^2\|_\infty \leq \frac{1}{(1-\gamma)^2} \sqrt{\frac{2 \log(2|\mathcal{S}||\mathcal{A}|/\delta)}{N}}.$$

For the second term, again with probability greater than $1 - \delta$,

$$\begin{aligned}\|(PV^*)^2 - (\hat{P}V^*)^2\|_\infty &\leq \|PV^* + \hat{P}V^*\|_\infty \|PV^* - \hat{P}V^*\|_\infty \\ &\leq \frac{2}{1-\gamma} \|(P - \hat{P})V^*\|_\infty \leq \frac{2}{(1-\gamma)^2} \sqrt{\frac{2 \log(2|\mathcal{S}||\mathcal{A}|/\delta)}{N}}.\end{aligned}$$

where we have used that $(\cdot)^2$ is a component-wise operation in the second step. For the last term:

$$\begin{aligned}\text{Var}_{\hat{P}}(V^*) &= \text{Var}_{\hat{P}}(V^* - \hat{V}^{\pi^*} + \hat{V}^{\pi^*}) \\ &\leq 2\text{Var}_{\hat{P}}(V^* - \hat{V}^{\pi^*}) + 2\text{Var}_{\hat{P}}(\hat{V}^{\pi^*}) \\ &\leq 2\|V^* - \hat{V}^{\pi^*}\|_\infty^2 + 2\text{Var}_{\hat{P}}(\hat{V}^{\pi^*}) \\ &= 2\Delta_{\delta,N}^2 + 2\text{Var}_{\hat{P}}(\hat{V}^{\pi^*}).\end{aligned}$$

where $\Delta_{\delta,N}$ is defined in Proposition 2.4. To obtain a cumulative probability of error less than δ , we replace δ in the above claims with $\delta/3$. Combining these bounds completes the proof of the first claim. The argument in the above display also implies that $\text{Var}_{\hat{P}}(V^*) \leq 2\Delta_{\delta,N}^2 + 2\text{Var}_{\hat{P}}(\hat{V}^*)$ which proves the second claim. $\blacksquare$

Using Lemma 2.10 and 2.13, we have the following corollary.

Corollary 2.14. *Let $\delta \geq 0$. With probability greater than $1 - \delta$, we have:*

$$\begin{aligned}|(P - \hat{P})V^*| &\leq c \sqrt{\frac{\text{Var}_{\hat{P}}(\hat{V}^{\pi^*}) \log(c|\mathcal{S}||\mathcal{A}|/\delta)}{N}} + \Delta''_{\delta,N}\mathbb{1} \\ |(P - \hat{P})V^*| &\leq c \sqrt{\frac{\text{Var}_{\hat{P}}(\hat{V}^*) \log(c|\mathcal{S}||\mathcal{A}|/\delta)}{N}} + \Delta''_{\delta,N}\mathbb{1},\end{aligned}$$

where

$$\Delta''_{\delta,N} := c \frac{1}{1-\gamma} \left(\frac{\log(c|\mathcal{S}||\mathcal{A}|/\delta)}{N} \right)^{3/4} + \frac{c}{(1-\gamma)^2} \frac{\log(c|\mathcal{S}||\mathcal{A}|/\delta)}{N},$$

and where c is an absolute constant.

Proof:(of Theorem 2.6) The proof consists of bounding the terms in Lemma 2.5. We have:

$$\begin{aligned}
Q^* - \hat{Q}^* &\leq \gamma \|(I - \gamma \hat{P}^{\pi^*})^{-1} (P - \hat{P}) V^*\|_\infty \\
&\leq c\gamma \sqrt{\frac{\log(c|\mathcal{S}||\mathcal{A}|/\delta)}{N}} \|(I - \gamma \hat{P}^{\pi^*})^{-1} \sqrt{\text{Var}_{\hat{P}}(\hat{V}^{\pi^*})}\|_\infty + \frac{c\gamma}{(1-\gamma)^2} \left(\frac{\log(c|\mathcal{S}||\mathcal{A}|/\delta)}{N} \right)^{3/4} \\
&\quad + \frac{c\gamma}{(1-\gamma)^3} \frac{\log(c|\mathcal{S}||\mathcal{A}|/\delta)}{N} \\
&\leq \gamma \sqrt{\frac{2}{(1-\gamma)^3}} \sqrt{\frac{\log(c|\mathcal{S}||\mathcal{A}|/\delta)}{N}} + \frac{c\gamma}{(1-\gamma)^2} \left(\frac{\log(c|\mathcal{S}||\mathcal{A}|/\delta)}{N} \right)^{3/4} + \frac{c\gamma}{(1-\gamma)^3} \frac{\log(c|\mathcal{S}||\mathcal{A}|/\delta)}{N} \\
&\leq 3\gamma \sqrt{\frac{1}{(1-\gamma)^3}} c \sqrt{\frac{\log(c|\mathcal{S}||\mathcal{A}|/\delta)}{N}} + 2 \frac{c\gamma}{(1-\gamma)^3} \frac{\log(c|\mathcal{S}||\mathcal{A}|/\delta)}{N},
\end{aligned}$$

where the first step uses Corollary 2.14; the second uses Lemma 2.12; and the last step uses that $2ab \leq a^2 + b^2$ (and choosing a, b appropriately). The proof of the lower bound is analogous. Taking a different absolute constant completes the proof. $\blacksquare$

2.5 Scalings and Effective Horizon Dependencies

It will be helpful to more intuitively understand why $1/(1-\gamma)^3$ is the effective horizon dependency one might hope to expect, from a dimensional analysis viewpoint. Due to that Q^* is a quantity that is as large as $1/(1-\gamma)$, to account for this scaling, it is natural to look at obtaining relative accuracy.

In particular, if

$$N \geq \frac{c}{1-\gamma} \frac{|\mathcal{S}||\mathcal{A}| \log(c|\mathcal{S}||\mathcal{A}|/\delta)}{\epsilon^2},$$

then with probability greater than $1 - \delta$, then

$$\|Q^* - Q^{\hat{\pi}^*}\|_\infty \leq \frac{\epsilon}{1-\gamma}, \text{ and } \|Q^* - \hat{Q}^*\|_\infty \leq \frac{\epsilon}{1-\gamma}.$$

(provided that $\epsilon \leq \sqrt{1-\gamma}$ using Theorem 2.6). In other words, if we had normalized the value functions ¹, then for additive accuracy (on our normalized value functions) our sample size would scale linearly with the effective horizon.

2.6 Bibliographic Remarks and Further Readings

The notion of a generative model was first introduced in [Kearns and Singh, 1999], which made the argument that, up to horizon factors and logarithmic factors, both model based methods and model free methods are comparable. [Kakade, 2003] gave an improved version of this rate (analogous to the crude bounds seen here).

The first claim in Theorem 2.6 is due to [Azar et al., 2013], and the proof in this section largely follows this work. Improvements are possible with regards to bounding the quality of $\hat{\pi}^*$; here, Theorem 2.6 shows that the model based approach is near optimal even for policy itself; showing that the quality of $\hat{\pi}^*$ does suffer any amplification factor of $1/(1-\gamma)$. [Sidford et al., 2018] provides the first proof of this improvement using a variance reduction algorithm with value iteration. The second claim in Theorem 2.6 is due to [Agarwal et al., 2020c], which shows that the naive model based approach is sufficient. The lower bound in Theorem 2.8 is due to [Azar et al., 2013].

¹Rescaling the value functions by multiplying by $(1-\gamma)$, i.e. $Q^\pi \leftarrow (1-\gamma)Q^\pi$, would keep the values bounded between 0 and 1. Throughout,

We also remark that we may hope for the sub-optimality bounds (on the value of the argmax policy) to hold up to for “large” ϵ , i.e. up to $\epsilon \leq 1/(1 - \gamma)$ (see the second claim in Theorem 2.6). Here, the work in [Li et al., 2020] shows this limit is achievable, albeit with a slightly different algorithm where they introduce perturbations. It is currently an open question if the naive model based approach also achieves the non-asymptotic statistical limit.

This chapter also provided results, without proof, on the optimal sample complexity in the finite horizon setting (see Section 2.3.2). The proof of this claim would also follow from the line of reasoning in [Azar et al., 2013], with the added simplification that the sub-optimality analysis is simpler in the finite-horizon setting with time-dependent transition matrices (e.g. see [Yin et al., 2021]).

this book it is helpful to understand sample size with regards to normalized quantities.

Chapter 3

Linear Bellman Completeness

Up to now we have focussed on “tabular” MDPs, where the sample complexity (and the computational complexity) scaled polynomially in the size of the state and action spaces. Ultimately, we seek to obtain methods which are applicable to cases where number of states and actions is large (or, possibly, countably or uncountably infinite).

In this chapter, we will consider one of the simplest conditions, where such a result is possible. In particular, we will consider a setting where we have a set of features which satisfy the *linear Bellman completion* condition, and we show that, with access to a generative model, there is a simple algorithm that learns a near optimal policy with polynomial sample complexity in the dimension of these features.

The linear Bellman completeness concept and first analysis was due to [Munos, 2005]. This chapter studies this concept in the setting of finite horizon MDPs, where we have access to a generative model.

3.1 The Linear Bellman Completeness Condition

We will work with given a feature mapping $\phi : \mathcal{S} \times \mathcal{A} \mapsto \mathbb{R}^d$, where we assume the following property is true: given any linear function $f(s, a) := \theta^\top \phi(s, a)$, we have that the Bellman operator applied to $f(s, a)$ also returns a linear function with respect to ϕ . Precisely, we have:

Definition 3.1 (Linear Bellman Completeness). We say the features ϕ satisfy the *linear Bellman completeness* property if for all $\theta \in \mathbb{R}^d$ and $(s, a, h) \in \mathcal{S} \times \mathcal{A} \times [H]$, there exists $w \in \mathbb{R}^d$ such that:

$$w^\top \phi(s, a) = r(s, a) + \mathbb{E}_{s' \sim P_h(s, a)} \max_{a'} \theta^\top \phi(s', a').$$

As w depends on θ , we use the notation $\mathcal{T}_h : \mathbb{R}^d \mapsto \mathbb{R}^d$ to represent such a w , i.e., $w := \mathcal{T}_h(\theta)$ in the above equation.

Note that the above implies that $r(s, a)$ is in the span of ϕ (to see this, take $\theta = 0$). Furthermore, it also implies that $Q_h^*(s, a)$ is linear in ϕ , i.e., there exists θ_h^* such that $Q_h^*(s, a) = (\theta_h^*)^\top \phi(s, a)$. We can easily see that tabular MDP is captured by the linear Bellman completeness with $\phi(s, a) \in \mathbb{R}^{|\mathcal{S}| \times |\mathcal{A}|}$ being a one-hot encoding vector with zeros everywhere except one at the entry corresponding to state-action (s, a) . In Chapters 7 and 8, we will provide precise examples of models which satisfy the linear Bellman completeness condition.

We focus on the generative model setting, i.e., where we can input any (s, a) pair to obtain a next state sample $s' \sim P(s, a)$ and reward $r(s, a)$. Our goal here is to design an algorithm that finds a policy $\hat{\pi}$ such that $V^{\hat{\pi}} \geq V^* - \epsilon$, with number of samples scaling polynomially with respect to all relevant parameters $d, H, 1/\epsilon$.

Algorithm 1 Least Squares Value Iteration

1: **Input:** $\mathcal{D}_0, \dots, \mathcal{D}_{H-1}$
2: Set $V_H(s) = 0$ for all $s \in \mathcal{S}$
3: **for** $h = H - 1 \rightarrow 0$ **do**
4: Solve least squares

$$\theta_h = \arg \min_{\theta} \sum_{s,a,r,s' \in \mathcal{D}_h} (\theta^\top \phi(s,a) - r - V_{h+1}(s'))^2 \quad (3.1)$$

5: Set $V_h(s) = \max_{a \in \mathcal{A}} \theta_h^\top \phi(s,a), \forall s$
6: **end for**
7: **Return:** $\{\hat{\pi}_h(s)\}_{h=0}^{H-1}$ where $\hat{\pi}_h(s) := \arg \max_a \theta_h^\top \phi(s,a), \forall s, h$.

3.2 The LSVI Algorithm

We first present the least square value iteration (LSVI) algorithm (Alg. 1) here. The algorithm takes H many datasets $\mathcal{D}_0, \dots, \mathcal{D}_{H-1}$ as inputs, where each $\mathcal{D}_h = \{s, a, r, s'\}$.

For each time step, LSVI estimate $Q_h^*(s, a)$ via $\theta_h^\top \phi(s, a)$ with θ_h computed from a least square problem.

To make sure the algorithm succeeds in terms of finding a near optimal policy, we need to design the datasets $\mathcal{D}_0, \dots, \mathcal{D}_{H-1}$. Intuitively, we may want to make sure that in each $\mathcal{D}_h$, we have $\sum_{s,a \in \mathcal{D}_h} \phi(s,a) \phi(s,a)^\top$ being full rank, so that least square has closed-form solution, and θ_h has a good generalization bound. In the next section, we will leverage the generative model setting and the D-optimal design to construct such datasets.

3.3 LSVI with D-Optimal Design

In this section, we start by introducing the D-optimal design and its properties. We then use the D-optimal design to construct a dataset and give a generalization bound for the solution to the least squares problem on the constructed dataset. Finally, we present the sample complexity bound for LSVI and its analysis.

We will use the notation $\|x\|_M^2 = x^\top M x$ for a matrix M and vector x of appropriate dimensions.

3.3.1 D-Optimal Design

We now specify a sampling distribution which, roughly speaking, ensures good coverage (in a spectral sense) over our feature set.

Theorem 3.2. *Suppose $\mathcal{X} \subset \mathbb{R}^d$ is a compact set (and full dimensional). There exists a distribution ρ on $\mathcal{X}$ such that:*

- ρ is supported on at most $d(d+1)/2$ points (all of which are in $\mathcal{X}$).
- Define

$$\Sigma = \mathbb{E}_{x \sim \rho} [xx^\top]$$

We have that for all $x \in \mathcal{X}$,

$$\|x\|_{\Sigma^{-1}}^2 \leq d.$$

The distribution ρ is referred to as the D-optimal design.

With these two properties, one can show that for any $x \in \mathcal{X}$, it can be written as a linear combination of the points on the support of ρ , i.e., $x = \sum_{x_i \in \text{support}(\rho)} \alpha_i \rho(x_i) x_i$, where $\|\alpha\|_2^2 \leq d$.

The following specifies an explicit construction: ρ is the following maximizer:

$$\rho \in \operatorname{argmax}_{\nu \in \Delta(\mathcal{X})} \ln \det (\mathbb{E}_{x \sim \nu} x x^\top). \quad (3.2)$$

(and there exists an optimizer which is supported on at most $d(d+1)/2$ points). The design ρ has a geometric interpretation: the centered ellipsoid $\mathcal{E} = \{v : \|v\|_{\Sigma^{-1}}^2 \leq d\}$ is the unique minimum volume centered ellipsoid containing $\mathcal{X}$. We do not provide the proof here (See Section 3.6).

In our setting, we will utilize the D-optimal design on the set $\Phi := \{\phi(s, a) : s, a \in \mathcal{S} \times \mathcal{A}\}$.

Change of basis interpretation with D-optimal design. Consider the coordinate transformation:

$$\tilde{x} := \Sigma^{-1/2} x.$$

Here we have that:

$$\tilde{\Sigma} := \mathbb{E}_{x \sim \rho} [\tilde{x} \tilde{x}^\top] = I.$$

Furthermore, we still have that $\|\tilde{x}\|_{\tilde{\Sigma}^{-1}}^2 \leq d$. In this sense, we can interpret the D-optimal design as providing a well-conditioned sampling distribution for the set $\mathcal{X}$.

3.3.2 Performance Guarantees

Now we explain how we can construct a dataset $\mathcal{D}_h$ for $h = 0, \dots, H-1$ using the D-optimal design ρ and the generative model:

- At h , for each $s, a \in \text{support}(\rho)$, we sample $n_{s,a} := \lceil \rho(s, a) N \rceil$ many next states independently from $P_h(\cdot | s, a)$; combine all $(s, a, r(s, a), s')$ and denote the whole dataset as $\mathcal{D}_h$
- Repeat for all $h \in [0, \dots, H-1]$ to construct $\mathcal{D}_0, \dots, \mathcal{D}_{H-1}$.

The key property of the dataset $\mathcal{D}_h$ is that the empirical covariance matrix $\sum_{s,a \in \mathcal{D}_h} \phi(s, a) \phi(s, a)^\top$ is full rank. Denote $\Lambda_h = \sum_{s,a \in \mathcal{D}_h} \phi(s, a) \phi(s, a)^\top$, we can verify that

$$\Lambda_h = \sum_{s,a \in \mathcal{D}} \phi(s, a) \phi(s, a)^\top \succcurlyeq N \Sigma \quad (3.3)$$

where the inequality comes from the fact that in $\mathcal{D}_h$, for each $s, a \in \text{support}(\rho)$, we have $\lceil \rho(s, a) N \rceil$ many copies of it. The D-optimal design ensures that Λ covers all $\phi(s, a)$ in the sense that $\max_{s,a} \phi(s, a)^\top \Lambda_h^{-1} \phi(s, a) \leq d$ due to the property of the D-optimal design. This is the key property that ensures the solution of the least squares on $\mathcal{D}_h$ is accurate globally.

Now we state the main sample complexity bound theorem for LSVI.

Theorem 3.3 (Sample Complexity of LSVI). *Suppose our features satisfy the linear Bellman completion property. Fix $\delta \in (0, 1)$ and $\epsilon \in (0, 1)$. Set parameter $N := \left\lceil \frac{64H^6 d^2 \ln(1/\delta)}{\epsilon^2} \right\rceil$. With probability at least $1 - \delta$, Algorithm 1 outputs $\hat{\pi}$ such that:*

$$\mathbb{E}_{s \sim \mu} V_0^*(s) - \mathbb{E}_{s \sim \mu} V_0^{\hat{\pi}}(s) \leq \epsilon,$$

with total number of samples $H \left(d^2 + \frac{64H^6 d^2 \ln(1/\delta)}{\epsilon^2} \right)$.

3.3.3 Analysis

To prove the theorem, we first show that if we had $\{\widehat{Q}_h\}_h$ such that the Bellman residual $\|\widehat{Q}_h - \mathcal{T}_h \widehat{Q}_{h+1}\|_\infty$ is small, then the performance of the greedy policies with respect to $\widehat{Q}_h$ is already near optimal. Note that this result is general and has nothing to do with the LSVI algorithm.

Lemma 3.4. *Assume that for all h , we have $\|\widehat{Q}_h - \mathcal{T}_{h+1} \widehat{Q}_{h+1}\|_\infty \leq \epsilon$. Then we have:*

1. *Accuracy of $\widehat{Q}_h$: for all h : $\|\widehat{Q}_h - Q_h^*\|_\infty \leq (H - h)\epsilon$,*
2. *Policy performance: for $\widehat{\pi}_h(s) := \operatorname{argmax}_a \widehat{Q}_h(s, a)$, we have $|V^{\widehat{\pi}} - V^*| \leq 2H^2\epsilon$.*

Proof: The proof of this lemma is elementary, which is analog to what we have proved in Value Iteration in the discounted setting. For completeness, we provide a proof here.

Starting from $Q_H(s, a) = 0, \forall s, a$, we have $\mathcal{T}_{H-1} Q_H = r$ by definition. The condition implies that $\|\widehat{Q}_{H-1} - r\|_\infty \leq \epsilon$, which means that $\|\widehat{Q}_{H-1} - Q_{H-1}^*\|_\infty \leq \epsilon$. This proves the base case.

Our inductive hypothesis is that $\|\widehat{Q}_{h+1} - Q_{h+1}^*\|_\infty \leq (H - h - 1)\epsilon$. Note that:

$$\begin{aligned} |\widehat{Q}_h(s, a) - Q_h^*(s, a)| &\leq |\widehat{Q}_h(s, a) - \mathcal{T}_h \widehat{Q}_{h+1}(s, a)| + |\mathcal{T}_h \widehat{Q}_{h+1}(s, a) - Q_h^*(s, a)| \\ &\leq \epsilon + |\mathcal{T}_h \widehat{Q}_{h+1}(s, a) - \mathcal{T}_h Q_{h+1}^*(s, a)| \leq \epsilon + (H - h - 1)\epsilon = (H - h)\epsilon, \end{aligned}$$

which concludes the proof for the first claim.

For the second claim, Starting from time step $H - 1$, for any s , we have:

$$\begin{aligned} V_{H-1}^{\widehat{\pi}}(s) - V_{H-1}^*(s) &= Q_{H-1}^{\widehat{\pi}}(s, \widehat{\pi}_{H-1}(s)) - Q_{H-1}^*(s, \pi^*(s)) \\ &= Q_{H-1}^{\widehat{\pi}}(s, \widehat{\pi}_{H-1}(s)) - Q_{H-1}^*(s, \widehat{\pi}_{H-1}(s)) + Q_{H-1}^*(s, \widehat{\pi}_{H-1}(s)) - Q_{H-1}^*(s, \pi^*(s)) \\ &= Q_{H-1}^*(s, \widehat{\pi}_{H-1}(s)) - Q_{H-1}^*(s, \pi^*(s)) \\ &\geq Q_{H-1}^*(s, \widehat{\pi}_{H-1}(s)) - \widehat{Q}_{H-1}(s, \widehat{\pi}_{H-1}(s)) + \widehat{Q}_{H-1}(s, \pi^*(s)) - Q_{H-1}^*(s, \pi^*(s)) \\ &\geq 2\epsilon, \end{aligned}$$

where the third equality uses the fact that $Q_{H-1}^{\widehat{\pi}}(s, a) = Q_{H-1}^*(s, a) = r(s, a)$, the first inequality uses the definition of $\widehat{\pi}_{H-1}$ and $\widehat{Q}_{H-1}(s, \widehat{\pi}_{H-1}(s)) \geq \widehat{Q}_{H-1}(s, a), \forall a$. This concludes the base case. For time step $h + 1$, our inductive hypothesis is defined as $V_{h+1}^{\widehat{\pi}}(s) - V_{h+1}^*(s) \geq -2(H - h - 1)H\epsilon$. For time step h , we have:

$$\begin{aligned} V_h^{\widehat{\pi}}(s) - V_h^*(s) &= Q_h^{\widehat{\pi}}(s, \widehat{\pi}_h(s)) - Q_h^*(s, \pi^*(s)) \\ &= Q_h^{\widehat{\pi}}(s, \widehat{\pi}_h(s)) - Q_h^*(s, \widehat{\pi}_h(s)) + Q_h^*(s, \widehat{\pi}_h(s)) - Q_h^*(s, \pi^*(s)) \\ &= \mathbb{E}_{s' \sim P_h(s, \widehat{\pi}_h(s))} [V_{h+1}^{\widehat{\pi}}(s') - V_{h+1}^*(s')] + Q_h^*(s, \widehat{\pi}_h(s)) - Q_h^*(s, \pi^*(s)) \\ &\geq -2(H - h - 1)\epsilon + Q_h^*(s, \widehat{\pi}_h(s)) - \widehat{Q}_h(s, \widehat{\pi}_h(s)) + \widehat{Q}_h(s, \pi^*(s)) - Q_h^*(s, \pi^*(s)) \\ &\geq -2(H - h - 1)H\epsilon - 2(H - h)\epsilon \geq -2(H - h)H\epsilon. \end{aligned}$$

This concludes that for any s , at time step $h = 0$, we have $V_0^{\widehat{\pi}}(s) - V_0^*(s) \geq -2H^2\epsilon$. ■

To conclude the proof, what we left to show is that the $\widehat{Q}_h$ that is returned by LSVI satisfies the condition in the above lemma. Note that $\widehat{Q}_h(s, a) = \widehat{\theta}_h^\top \phi(s, a)$, and $\mathcal{T}_h \widehat{Q}_{h+1}$ is indeed the Bayes optimal solution of the least squares in Eq. 3.1, and via the Bellman completion assumption, we have $\mathcal{T}_h \widehat{Q}_{h+1}(s, a) = \mathcal{T}_h(\widehat{\theta}_{h+1})^\top \phi(s, a), \forall s, a$. With the constructed $\mathcal{D}_h$ based on the D-optimal design, Linear regression immediately gives us a generalization bound on $\|\widehat{Q}_h - \mathcal{T}_h \widehat{Q}_{h+1}\|_\infty$. The following theorem formalize the above statement.

Lemma 3.5. Fix $\delta \in (0, 1)$ and $\epsilon \in (0, 1)$, set $N = \left\lceil \frac{16H^2 d^2 \ln(H/\delta)}{\epsilon^2} \right\rceil$. With probability at least $1 - \delta$, for all h , we have:

$$\forall h : \quad \left\| \hat{Q}_h - \mathcal{T}_h \hat{Q}_{h+1} \right\|_\infty \leq \epsilon.$$

Proof: The proof uses a standard result for ordinary least squares (see Theorem A.10 in the Appendix). Consider a time step h . In time step h , we perform linear regression from $\phi(s, a)$ to $r(s, a) + \max_{a'} \hat{\theta}_{h+1}^\top \phi(s', a')$. Note that the Bayes optimal here is

$$\mathbb{E}_{s' \sim P_h(s, a)} \left[r(s, a) + \max_{a'} \hat{\theta}_{h+1}^\top \phi(s', a') \right] = \mathcal{T}_h(\hat{\theta}_{h+1})^\top \phi(s, a).$$

Also we can show that the noise $\epsilon_{s, a} := \left(r(s, a) + \max_{a'} \hat{Q}_{h+1}(s', a') - \mathcal{T}_h \hat{Q}_h(s, a) \right)$ is bounded by $2H$, and also note that the noises are independent, and has mean zero: $\mathbb{E}_{s' \sim P_h(s, a)} \left[r(s, a) + \max_{a'} \hat{Q}_{h+1}(s', a') - \mathcal{T}_h \hat{Q}_h(s, a) \right] = 0$. Using Theorem A.10, we have that with probability at least $1 - \delta$,

$$\left\| \hat{\theta}_h - \mathcal{T}_h(\hat{\theta}_{h+1}) \right\|_{\Lambda_h}^2 \leq 4H^2 \left(d + \sqrt{d \ln(1/\delta)} + 2 \ln(1/\delta) \right).$$

Now for any s, a , we must have:

$$\begin{aligned} \left| (\hat{\theta}_h - \mathcal{T}_h(\hat{\theta}_{h+1}))^\top \phi(s, a) \right|^2 &\leq \left\| \hat{\theta}_h - \mathcal{T}_h(\hat{\theta}_{h+1}) \right\|_{\Lambda_h}^2 \left\| \phi(s, a) \right\|_{\Lambda_h^{-1}}^2 \leq \left\| \hat{\theta}_h - \mathcal{T}_h(\hat{\theta}_{h+1}) \right\|_{\Lambda_h}^2 \frac{d}{N} \\ &\leq 4H^2 \frac{d^2 + d^{1.5} \sqrt{\ln(1/\delta)} + 2d \ln(1/\delta)}{N} \leq \frac{16H^2 d^2 \ln(1/\delta)}{N}, \end{aligned}$$

which implies that:

$$\left\| \hat{Q}_h - \mathcal{T}_h \hat{Q}_{h+1} \right\|_\infty = \max_{s, a} \left| (\hat{\theta}_h - \mathcal{T}_h(\hat{\theta}_{h+1}))^\top \phi(s, a) \right| \leq \frac{4Hd \sqrt{\ln(1/\delta)}}{\sqrt{N}}.$$

Set the right hand side to be ϵ , we have $N = \frac{16H^2 d^2 \ln(1/\delta)}{\epsilon^2}$.

Finally, apply union bound over all $h = 0, \dots, H - 1$, we conclude the proof. $\blacksquare$

Now we are ready to conclude the proof for the main theorem.

Proof:[Proof of Theorem 3.3] Using Lemma 3.5, with $N = \left\lceil \frac{64H^6 d^2 \ln(H/\delta)}{\epsilon^2} \right\rceil$, we have that $\left\| \hat{Q}_h - \mathcal{T}_h \hat{Q}_{h+1} \right\|_\infty \leq \epsilon/(2H^2)$ for all h , with probability at least $1 - \delta$. Now apply Lemma 3.4, we immediately have that $|V^{\hat{\pi}} - V^*| \leq \epsilon$.

Regarding sample complexity, we have:

$$H \sum_{s, a} [\rho(s, a) N] \leq H \sum_{s, a \in \text{support}(\rho)} (1 + \rho(s, a) N) \leq H \left(d^2 + \frac{64H^6 d^2 \ln(H/\delta)}{\epsilon^2} \right),$$

where the second inequality use the support size upper bound of the D-optimal design ρ (see Lemma 3.2). This completes the proof. $\blacksquare$

3.4 How Strong is Bellman Completion as a Modeling?

To be added.

3.5 Offline Reinforcement Learning

One notable observation about LSVI is that the algorithm is non-adaptive, in the sense that it works using a fixed set of observed transitions and rewards. Hence, it is applicable to the offline setting discussed in Chapter 1.

We now discuss two offline objectives: that of learning a near optimal policy and that of policy evaluation (i.e. evaluating the quality of a given policy).

We consider the setting where we have a H datasets of the form $D_h = \{(s_i, a_i, s'_i, r(s_i, a_i))\}_{i=1}^N$, where we assume for each i and h , that we have independent samples $s'_i \sim P_h(\cdot | s_i, a_i)$. In other words, D_h is a dataset of observed transitions corresponding to stage h , where each next state has been sampled independently. Note that here we have not made explicit distributional assumption about how the (s_i, a_i) 's have been generated. The results we present can also be extended to where we have a fixed data generation distribution over these quadruples.

It is straight forward to extend the LSVI guarantees to these setting provided our dataset has coverage in the following sense:

Assumption 3.6. (Coverage) Suppose that for each $h \in [H]$, we have that:

$$\frac{1}{N} \sum_{(s_i, a_i) \in D_h} \phi(s_i, a_i) \phi(s_i, a_i)^\top \succeq \frac{1}{\kappa} \Sigma$$

where Σ is the D-optimal design covariance (see Equation 3.2).

3.5.1 Offline Learning

A minor modification of the proof in Theorem 3.3 leads to the following guarantee:

Theorem 3.7 (Sample Complexity of LSVI). *Suppose that Assumption 3.6 holds and that our features satisfy the linear Bellman completion property. Fix $\delta \in (0, 1)$ and $\epsilon \in (0, 1)$. Set parameter $N := \left\lceil \frac{c\kappa H^6 d^2 \ln(1/\delta)}{\epsilon^2} \right\rceil$, where c is an appropriately chosen absolute constant. With probability at least $1 - \delta$, Algorithm 1 outputs $\hat{\pi}$ such that:*

$$\mathbb{E}_{s \sim \mu}[V_0^*(s)] - \mathbb{E}_{s \sim \mu}[V_0^{\hat{\pi}}(s)] \leq \epsilon.$$

3.5.2 Offline Policy Evaluation

Here we are interested in question of evaluating some given policy π using the offline data. For this, we will make a completeness assumption with respect to π as follows:

Definition 3.8 (Linear Completeness for π). We say the features ϕ satisfy the *policy completeness property* for π if for all $\theta \in \mathbb{R}^d$ and $(s, a, h) \in \mathcal{S} \times \mathcal{A} \times [H]$, there exists $w \in \mathbb{R}^d$ such that:

$$w^\top \phi(s, a) = r(s, a) + \mathbb{E}_{s' \sim P_h(s, a)} \theta^\top \phi(s', \pi(s')).$$

For this case, Algorithm 2, Least Squares Policy Evaluation (LSPE), is a modified version of LSVI for the purposes of estimation; note the algorithm no longer estimates $V_h(s)$ using the greedy policy. Again, a nearly identical argument to the proof in Theorem 3.3 leads to the following guarantee. Here, we set:

$$\hat{V}(\mu) = \sum_{s_i \in D_0} \theta_0 \cdot \phi(s_i, \pi(s_i))$$

where θ_0 is the parameter returned by LSVI.

Algorithm 2 Least Squares Policy Evaluation

1: **Input:** $\pi, \mathcal{D}_0, \dots, \mathcal{D}_{H-1}$
2: Set $V_H(s) = 0$ for all $s \in \mathcal{S}$
3: **for** $h = H - 1 \rightarrow 0$ **do**
4: Solve least squares

$$\hat{\theta}_h = \arg \min_{\theta} \sum_{s,a,r,s' \in \mathcal{D}_h} (\theta^\top \phi(s,a) - r - V_{h+1}(s'))^2 \quad (3.4)$$

5: Set $V_h(s) = \hat{\theta}_h^\top \phi(s, \pi(s)), \forall s$
6: **end for**
7: **Return:** $\{\hat{\theta}_h\}_{h=0}^{H-1}$.

Theorem 3.9 (Sample Complexity of LSPE). *Suppose that Assumption 3.6 holds and that our features satisfy the linear policy completion property (with respect to π). Fix $\delta \in (0, 1)$ and $\epsilon \in (0, 1)$. Set parameter $N := \left\lceil \frac{c\kappa H^4 d^2 \ln(1/\delta)}{\epsilon^2} \right\rceil$, where c is an appropriately chosen absolute constant. With probability at least $1 - \delta$, Algorithm 2 outputs $\hat{\theta}_0$ such that for all s ,*

$$|V_0^\pi(s) - \hat{\theta}_0^\top \phi(s, \pi(s))| \leq \epsilon.$$

Note that the above theorem has a sample complexity improvement by a factor of H . This is due to that the analysis is improvable, as we only care about value accuracy (the first claim in Lemma 3.4, rather than the second, is what is relevant here). We should note that the sample size bounds presented in this chapter have not been optimized with regards to their H dependencies.

3.6 Bibliographic Remarks and Further Readings

The idea of Bellman completion under general function class was introduced in [Munos, 2005] under the setting of batch RL. For the episodic online learning setting, Zanette et al. [2020] provided a statistically efficient algorithm under the linear Bellman completion condition, and [Jin et al., 2021] proposes a statistically efficient algorithms under the Bellman completion condition with general function approximation.

We refer readers to [Lattimore and Szepesvári, 2020] for a proof of the D-optimal design; the idea directly follows from John's theorem (e.g. see [Ball, 1997, Todd, 2016]).

Chapter 4

Fitted Dynamic Programming Methods

Let us again consider the question of learning in large MDPs, when the underlying MDPs is unknown. In the previous chapter, we relied on the linear Bellman completeness assumption to provide strong guarantees with only polynomial dependence on the dimension of the feature space and the horizon H . This chapter considers the approach of using *function approximation* methods with iterative dynamic programming approaches.

In particular, we now consider approaches which rely on (average-case) supervised learning methods (namely regression), where we use regression to approximate the target functions in both the value iteration and policy iteration algorithms. We refer to these algorithms as *fitted Q -iteration* (FQI) and *fitted policy iteration* (FPI). The FQI algorithm can be implemented in an offline RL sampling model while the FPI algorithm requires the sampling access to an episodic model (see Chapter 1 for review of these sampling models).

This chapter focuses on obtaining of *average case function approximation error* bounds, provided we have a somewhat stringent condition on how the underlying MDP behaves, quantified by the *concentrability coefficient*. This notion was introduced in [Munos, 2003, 2005]. While the notion is somewhat stringent, we will see that it is not avoidable without further assumptions. The next chapter more explicitly considers lower bounds, while Chapters 12 and 13 seek to relax the concentrability notion.

4.1 Fitted Q -Iteration (FQI) and Offline RL

We consider infinite horizon discounted MDP $\mathcal{M} = \{\mathcal{S}, \mathcal{A}, \gamma, P, r, \mu\}$ where μ is the initial state distribution. We assume reward is bounded, i.e., $\sup_{s,a} r(s, a) \in [0, 1]$. For notation simplicity, we denote $V_{\max} := 1/(1 - \gamma)$.

Given any $f : \mathcal{S} \times \mathcal{A} \mapsto \mathbb{R}$, we denote the Bellman operator $\mathcal{T}f : \mathcal{S} \times \mathcal{A} \mapsto \mathbb{R}$ as follows. For all $s, a \in \mathcal{S} \times \mathcal{A}$,

$$\mathcal{T}f(s, a) := r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} \max_{a' \in \mathcal{A}} f(s', a').$$

We assume that we have a distribution $\nu \in \Delta(\mathcal{S} \times \mathcal{A})$. We collect a dataset $\mathcal{D} := \{(s_i, a_i, r_i, s'_i)\}_{i=1}^n$ where $s_i, a_i \sim \nu, r_i = r(s_i, a_i), s'_i \sim P(\cdot | s_i, a_i)$. Given $\mathcal{D}$, our goal is to output a near optimal policy for the MDP, that is we would like our algorithm to produce a policy $\hat{\pi}$ such that, with probability at least $1 - \delta$, $V(\hat{\pi}) \geq V^* - \epsilon$, for some (ϵ, δ) pair. As usual, the number of samples n will depend on the accuracy parameters (ϵ, δ) and we would like n to scale favorably with these. Given any distribution $\nu \in \mathcal{S} \times \mathcal{A}$, and any function $f : \mathcal{S} \times \mathcal{A} \mapsto \mathbb{R}$, we write $\|f\|_{2,\nu}^2 := \mathbb{E}_{s,a \sim \nu} f^2(s, a)$

Denote a function class $\mathcal{F} = \{f : \mathcal{S} \times \mathcal{A} \mapsto [0, V_{\max}]\}$.

We require the data distribution ν is exploratory enough.

Assumption 4.1 (Concentrability). There exists a constant C such that for any policy π (including non-stationary policies), we have:

$$\forall \pi, s, a : \frac{d^\pi(s, a)}{\nu(s, a)} \leq C.$$

Note that concentrability does not require that the state space is finite, but it does place some constraints on the system dynamics. Note that the above assumption requires that ν to cover all possible policies's state-action distribution, even including non-stationary policies.

In addition to the above two assumptions, we need an assumption on the representational condition of class $\mathcal{F}$.

Assumption 4.2 (Inherent Bellman Error). We assume the following error bound holds:

$$\epsilon_{approx, \nu} := \max_{f \in \mathcal{F}} \min_{f' \in \mathcal{F}} \|f' - \mathcal{T}f\|_{2, \nu}^2.$$

We refer to $\epsilon_{approx, \nu}$ as the *inherent Bellman error* with respect to the distribution ν .

4.1.1 The FQI Algorithm

Fitted Q Iteration (FQI) simply performs the following iteration. Start with some $f_0 \in \mathcal{F}$, FQI iterates:

$$\text{FQI: } f_t \in \operatorname{argmin}_{f \in \mathcal{F}} \sum_{i=1}^n \left(f(s'_i, a_i) - r_i - \gamma \max_{a' \in \mathcal{A}} f_{t-1}(s_i, a_i) \right)^2. \quad (4.1)$$

After k many iterations, we output a policy $\pi^k(s) := \operatorname{argmax}_a f_k(s, a), \forall s$.

To get an intuition why this approach can work, let us assume the inherent Bellman error $\epsilon_{approx} = 0$ which is saying that for any $f \in \mathcal{F}$, we have $\mathcal{T}f \in \mathcal{F}$ (i.e., Bellman completion). Note that the Bayes optimal solution is $\mathcal{T}f_{t-1}$. Due to the Bellman completion assumption, the Bayes optimal solution $\mathcal{T}f_{t-1} \in \mathcal{F}$. Thus, we should expect that f_t is close to the Bayes optimal $\mathcal{T}f_{t-1}$ under the distribution ν , i.e., with a uniform convergence argument, for the generalization bound, we should expect that:

$$\mathbb{E}_{s, a \sim \nu} (f_t(s, a) - \mathcal{T}f_{t-1}(s, a))^2 \approx \sqrt{1/n}.$$

Thus in high level, $f_t \approx \mathcal{T}f_{t-1}$ as our data distribution ν is exploratory, and we know that based on value iteration, $\mathcal{T}f_{k-1}$ is a better approximation of Q^* than f_k , i.e., $\|\mathcal{T}f_{t-1} - Q^*\|_\infty \leq \gamma \|f_{t-1} - Q^*\|_\infty$, we can expect FQI to converge to the optimal solution when $n \rightarrow \infty, t \rightarrow \infty$. We formalize the above intuition below.

4.1.2 Performance Guarantees of FQI

We first state the performance guarantee of FQI.

Theorem 4.3 (FQI guarantee). Fix $K \in \mathbb{N}^+$. Fitted Q Iteration guarantees that with probability $1 - \delta$,

$$V^* - V^{\pi^K} \leq \frac{1}{(1-\gamma)^2} \left(\sqrt{\frac{22CV_{\max}^2 \ln(|\mathcal{F}|^2 K/\delta)}{n}} + \sqrt{20C\epsilon_{approx, \nu}} \right) + \frac{\gamma^K V_{\max}}{(1-\gamma)}$$

We start from the following lemma which shows that if for all t , $\|f_{t+1} - \mathcal{T}f_t\|_{2,\nu}^2 \leq \varepsilon$, then the greedy policy with respect to f_t is approaching to π^* .

Lemma 4.4. *Assume that for all t , we have $\|f_{t+1} - \mathcal{T}f_t\|_{2,\nu} \leq \varepsilon$, then for any k , we have:*

$$V^{\pi^k} - V^* \leq \frac{\sqrt{C}\varepsilon}{(1-\gamma)^2} + \frac{\gamma^k V_{\max}}{1-\gamma}.$$

Proof: We start from the Performance Difference Lemma:

$$\begin{aligned} (1-\gamma)(V^* - V^{\pi^k}) &= \mathbb{E}_{s \sim d^{\pi^k}} [-A^*(s, \pi_k(s))] \\ &= \mathbb{E}_{s \sim d^{\pi^k}} [Q^*(s, \pi^*(s)) - Q^*(s, \pi_k(s))] \\ &\leq \mathbb{E}_{s \sim d^{\pi^k}} [Q^*(s, \pi^*(s)) - f_k(s, \pi^*(s)) + f_k(s, \pi_k(s)) - Q^*(s, \pi_k(s))] \\ &\leq \|Q^* - f_k\|_{1, d^{\pi^k} \circ \pi^*} + \|Q^* - f_k\|_{1, d^{\pi^k} \circ \pi_k} \\ &\leq \|Q^* - f_k\|_{2, d^{\pi^k} \circ \pi^*} + \|Q^* - f_k\|_{2, d^{\pi^k} \circ \pi_k}, \end{aligned}$$

where the first inequality comes from the fact that π_k is a greedy policy of f_k , i.e., $f_k(s, \pi_k(s)) \geq f_k(s, a)$ for any other a including $\pi^*(s)$. Now we bound each term on the RHS of the above inequality. We do this by consider a state-action distribution β that is induced by some policy. We have:

$$\begin{aligned} \|Q^* - f_k\|_{2,\beta} &\leq \|Q^* - \mathcal{T}f_{k-1}\|_{2,\beta} + \|f_k - \mathcal{T}f_{k-1}\|_{2,\beta} \\ &\leq \gamma \sqrt{\mathbb{E}_{s,a \sim \beta} \left[\left(\mathbb{E}_{s' \sim P(\cdot|s,a)} \max_a Q^*(s', a) - \max_a f_{k-1}(s', a) \right)^2 \right]} + \|f_k - \mathcal{T}f_{k-1}\|_{2,\beta} \\ &\leq \gamma \sqrt{\mathbb{E}_{s,a \sim \beta, s' \sim P(\cdot|s,a)} \max_a (Q^*(s', a) - f_{k-1}(s', a))^2} + \sqrt{C} \|f_k - \mathcal{T}f_{k-1}\|_{2,\nu}, \end{aligned}$$

where in the last inequality, we use the fact that $(\mathbb{E}[x])^2 \leq \mathbb{E}[x^2]$, $(\max_x f(x) - \max_x g(x))^2 \leq \max_x (f(x) - g(x))^2$ for any two functions f and g , and assumption 4.1.

Denote $\beta'(s', a') = \sum_{s,a} \beta(s, a) P(s'|s, a) \mathbf{1}\{a' = \arg\max_a (Q^*(s', a) - f_{k-1}(s', a))^2\}$, the above inequality becomes:

$$\|Q^* - f_k\|_{2,\beta} \leq \gamma \|Q^* - f_{k-1}\|_{2,\beta'} + \sqrt{C} \|f_k - \mathcal{T}f_{k-1}\|_{2,\nu}.$$

We can recursively repeat the same process for $\|Q^* - f_{k-1}\|_{2,\nu'}$ till step $k = 0$:

$$\|Q^* - f_k\|_{2,\beta} \leq \sqrt{C} \sum_{t=0}^{k-1} \gamma^t \|f_{k-t} - \mathcal{T}f_{k-t-1}\|_{2,\nu} + \gamma^k \|Q^* - f_0\|_{2,\tilde{\nu}},$$

where $\tilde{\nu}$ is some valid state-action distribution.

Note that for the first term on the RHS of the above inequality, we can bound it as follow:

$$\sqrt{C} \sum_{t=0}^{k-1} \gamma^t \|f_{k-t} - \mathcal{T}f_{k-t-1}\|_{2,\mu} \leq \sqrt{C} \sum_{t=0}^{k-1} \gamma^t \varepsilon \leq \sqrt{C} \varepsilon \frac{1}{1-\gamma}.$$

For the second term, we have:

$$\gamma \|Q^* - f_0\|_{2,\tilde{\nu}} \leq \gamma^k V_{\max}.$$

Thus, we have:

$$\|Q^* - f_k\|_{2,\beta} = \frac{\sqrt{C}\varepsilon}{1-\gamma} + \gamma^k V_{\max},$$

for all β , including $\beta = d^{\pi_k} \circ \pi^*$, and $\beta = d^{\pi_k} \circ \pi_k$. This concludes the proof. $\blacksquare$

What left is to show that least squares control the error ε . We will use the generalization bound for least squares shown in Lemma A.11.

Lemma 4.5 (Bellman error). *With probability at least $1 - \delta$, for all $t = 0, \dots, K$, we have:*

$$\|f_{t+1} - \mathcal{T}f_t\|_{2,\nu}^2 \leq \frac{22V_{\max}^2 \ln(|\mathcal{F}|^2 K/\delta)}{n} + 20\epsilon_{\text{approx},\nu}.$$

Proof:

Let us fix a function $g \in \mathcal{F}$, and consider the regression problem on dataset $\{\phi(s, a), r(s, a) + \gamma \max_{a'} g(s', a')\}$. Denote $\hat{f}_g = \operatorname{argmin}_{f \in \mathcal{F}} \sum_{i=1}^n (f(s_i, a_i) - r_i - \gamma \max_{a'} g(s', a'))$, as the least square solution.

Note that in this regression problem, we have $|r_i + \gamma \max_{a'} g(s', a')| \leq 1 + \gamma V_{\max} \leq 2V_{\max}$, and for our Bayes optimal solution, we have $|\mathcal{T}g(s, a)| = |r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s,a)} \max_{a'} g(s', a')| \leq 1 + \gamma V_{\max} \leq 2V_{\max}$. Also note that our inherent Bellman error condition implies that $\min_{f \in \mathcal{F}} \mathbb{E}_{s,a \sim \nu} (f(s, a) - \mathcal{T}g(s, a))^2 \leq \epsilon_{\text{approx},\nu}$. Thus, we can apply Lemma A.11 directly here. With probability at least $1 - \delta$, we get:

$$\|\hat{f}_g - \mathcal{T}g\|_{2,\nu}^2 \leq \frac{22V_{\max}^2 \ln(|\mathcal{F}|/\delta)}{n} + 20\epsilon_{\text{approx},\nu}.$$

Note that the above inequality holds for the fixed g . We apply a union bound over all possible $g \in \mathcal{F}$, we get that with probability at least $1 - \delta$:

$$\forall g \in \mathcal{F} : \|\hat{f}_g - \mathcal{T}g\|_{2,\nu}^2 \leq \frac{22V_{\max}^2 \ln(|\mathcal{F}|^2/\delta)}{n} + 20\epsilon_{\text{approx},\nu}. \quad (4.2)$$

Consider iteration t , set $g = f_{t+1}$ in Eq. 4.2, and by the definition of f_t , we have $\hat{f}_g = f_t$ in this case, which means that above inequality in Eq. 4.2 holds for the pair $(\hat{f}_g, g) = (f_t, f_{t+1})$. Now apply a union bound over all $t = 0, \dots, K$, we conclude the proof. $\blacksquare$

Now we are ready to prove the main theorem.

Proof:[Proof of Theorem 4.3]

From Lemma 4.5, we see that with probability at least $1 - \delta$, we have for all $t \leq K$:

$$\|f_{t+1} - \mathcal{T}f_t\|_{2,\nu} \leq \sqrt{\frac{22V_{\max}^2 \ln(|\mathcal{F}|^2 K/\delta)}{n}} + \sqrt{20\epsilon_{\text{approx},\nu}}.$$

Set ε in Lemma 4.4 to be $\sqrt{\frac{22V_{\max}^2 \ln(|\mathcal{F}|^2 K/\delta)}{n}} + \sqrt{20\epsilon_{\text{approx},\nu}}$, we conclude the proof. $\blacksquare$

4.2 Fitted Policy-Iteration (FPI)

to be added..

4.3 Failure Cases Without Assumption 4.1

4.4 FQI for Policy Evaluation

4.5 Bibliographic Remarks and Further Readings

The notion of concentrability was developed in [Munos, 2003, 2005] in order to permitting sharper bounds in terms of average case function approximation error, provided that the concentrability coefficient is bounded. These methods also permit sample based fitting methods, with sample size and error bounds, provided there is a data collection policy that induces a bounded concentrability coefficient [Munos, 2005, Szepesvári and Munos, 2005, Antos et al., 2008, Lazaric et al., 2016]. Chen and Jiang [2019] provide a more detailed discussion on this quantity.

Part II

Strategic Exploration

Chapter 5

Multi-Armed & Linear Bandits

For the case, where $\gamma = 0$ (or $H = 1$ in the undiscounted case), the problem of learning in an unknown MDP reduce to the multi-armed bandit problem. The basic algorithms and proof methodologies here are important to understand in their own right, due to that we will have to extend these with more sophisticated variants to handle the exploration-exploitation tradeoff in the more challenging reinforcement learning problem.

This chapter follows analysis of the LinUCB algorithm from the original proof in [Dani et al., 2008], with a simplified concentration analysis due to [Abbasi-Yadkori et al., 2011].

Throughout this chapter, we assume reward is stochastic.

5.1 The K -Armed Bandit Problem

The setting is where we have K decisions (the “arms”), where when we play arm $a \in \{1, 2, \dots, K\}$ we obtain a random reward $r_a \in [-1, 1]$ from $R(a) \in \Delta([-1, 1])$ which has mean reward:

$$\mathbb{E}_{r_a \sim R(a)}[r_a] = \mu_a$$

where it is easy to see that $\mu_a \in [-1, 1]$.

Every iteration t , the learner will pick an arm $I_t \in [1, 2, \dots, K]$. Our cumulative regret is defined as:

$$R_T = T \cdot \max_i \mu_i - \sum_{t=0}^{T-1} \mu_{I_t}$$

We denote $a^* = \operatorname{argmax}_i \mu_i$ as the optimal arm. We define gap $\Delta_a = \mu_{a^*} - \mu_a$ for any arm a .

Theorem 5.1. *There exists an algorithm such that with probability at least $1 - \delta$, we have:*

$$R_T = O \left(\min \left\{ \sqrt{KT \cdot \ln(TK/\delta)}, \sum_{a \neq a^*} \frac{\ln(TK/\delta)}{\Delta_a} \right\} + K \right).$$

5.1.1 The Upper Confidence Bound (UCB) Algorithm

We summarize the upper confidence bound (UCB) algorithm in Alg. 3. For simplicity, we allocate the first K rounds to pull each arm once.

Algorithm 3 UCB

```
1: Play each arm once and denote received reward as  $r_a$  for all  $a \in \{1, 2, \dots, K\}$ 
2: for  $t = 1 \rightarrow T - K$  do
3:   Execute arm  $I_t = \arg \max_{i \in [K]} \left( \hat{\mu}_i^t + \sqrt{\frac{\log(TK/\delta)}{N_i^t}} \right)$ 
4:   Observe  $r_t := r_{I_t}$ 
5: end for
```

where every iteration t , we maintain counts of each arm:

$$N_a^t = 1 + \sum_{i=1}^{t-1} \mathbf{1}\{I_i = a\},$$

where I_t is the index of the arm that is picked by the algorithm at iteration t . We main the empirical mean for each arm as follows:

$$\hat{\mu}_a^t = \frac{1}{N_a^t} \left(r_a + \sum_{i=1}^{t-1} \mathbf{1}\{I_i = a\} r_i \right).$$

Recall that r_a is the reward of arm a we got during the first K rounds.

We also main the upper confidence bound for each arm as follows:

$$\hat{\mu}_a^t + 2\sqrt{\frac{\ln(TK/\delta)}{N_a^t}}.$$

The following lemma shows that this is a valid upper confidence bound with high probability.

Lemma 5.2 (Upper Confidence Bound). *For all $t \in [1, \dots, T]$ and $a \in [1, 2, \dots, K]$, we have that with probability at least $1 - \delta$,*

$$|\hat{\mu}_a^t - \mu_a| \leq 2\sqrt{\frac{\ln(TK/\delta)}{N_a^t}}. \quad (5.1)$$

The proof of the above lemma uses Azuma-Hoeffding's inequality (Theorem A.5) for each arm a and iteration t and then apply a union bound over all T iterations and K arms. The reason one needs to use Azuma-Hoeffding's inequality is that here the number of trials of arm a , i.e. N_a^t , itself is a random variable while the classic Hoeffding's inequality only applies to the setting where the number of trials is a fixed number.

Proof: We first consider a fixed arm a . Let us define the following random variables, $X_0 = r_a - \mu_a$, $X_1 = \mathbf{1}\{I_1 = a\}(r_1 - \mu_a)$, $X_2 = \mathbf{1}\{I_2 = a\}(r_2 - \mu_a)$, $\dots$, $X_i = \mathbf{1}\{I_i = a\}(r_i - \mu_a)$, $\dots$.

Regarding the boundedness on these random variables, we have that for i where $\mathbf{1}\{I_i = a\} = 1$, we have $|X_i| \leq 1$, and for i where $\mathbf{1}\{I_i = a\} = 0$, we have $|X_i| = 0$. Now, consider $\mathbb{E}[X_i | \mathcal{H}_{<i}]$, where $\mathcal{H}_{<i}$ is all history up to but not including iteration i . Note that we have $\mathbb{E}[X_i | \mathcal{H}_{<i}] = 0$ since condition on history $\mathcal{H}_{<i}$, $\mathbf{1}\{I_i = a\}$ is a deterministic quantity (i.e., UCB algorithm determines whether or not to pull arm a at iteration i only based on the information from $\mathcal{H}_{<i}$). Thus, we can conclude that $\{X_i\}_{i=0}$ is a Martingale difference sequence. Via Azuma-Hoeffding's inequality, we have that with probability at least $1 - \delta$, for any fixed t , we have:

$$\left| \sum_{i=0}^{t-1} X_i \right| = |N_a^t \hat{\mu}_a^t - N_a^t \mu_a| \leq 2\sqrt{\ln(1/\delta) N_a^t},$$

where we use the fact that $\sum_{i=0}^{t-1} |X_i|^2 \leq N_a^t$. Divide N_a^t on both side, we have:

$$|\hat{\mu}_a^t - \mu_a| \leq 2\sqrt{\ln(1/\delta)/N_a^t}$$

Apply union bound over all $0 \leq t \leq T$ and $a \in \{1, \dots, K\}$, we have that with probability $1 - \delta$, for all t, a :

$$|\hat{\mu}_a^t - \mu_a| \leq 2\sqrt{\ln(1/\delta)/N_a^t} \leq 2\sqrt{\ln(TK/\delta)/N_a^t}$$

This concludes the proof. ■

Now we can conclude the proof of the main theorem.

Proof: Below we conditioned on the above inequality 5.1 holds. This gives us the following optimism:

$$\mu_a \leq \hat{\mu}_a^t + 2\sqrt{\frac{\ln(TK/\delta)}{N_a^t}}, \forall a, t.$$

Thus, we can upper bound the regret as follows:

$$\mu_{a^*} - \mu_{I_t} \leq \hat{\mu}_{I_t}^t + 2\sqrt{\frac{\ln(TK/\delta)}{N_{I_t}^t}} - \mu_{I_t} \leq 4\sqrt{\frac{\ln(TK/\delta)}{N_{I_t}^t}}.$$

Sum over all iterations, we get:

$$\begin{aligned} \sum_{t=0}^{T-1} \mu_{a^*} - \mu_{I_t} &\leq 4\sqrt{\ln(TK/\delta)} \sum_{t=0}^{T-1} \sqrt{\frac{1}{N_{I_t}^t}} \\ &= 4\sqrt{\ln(TK/\delta)} \sum_a \sum_{i=1}^{N_a^T} \frac{1}{\sqrt{i}} \leq 8\sqrt{\ln(TK/\delta)} \sum_a \sqrt{N_a^T} \leq 8\sqrt{\ln(TK/\delta)} \sqrt{K \sum_a N_a^T} \\ &\leq 8\sqrt{\ln(TK/\delta)} \sqrt{KT}. \end{aligned}$$

Note that our algorithm has regret K at the first K rounds.

On the other hand, if for each arm a , the gap $\Delta_a > 0$, then, we must have:

$$N_a^T \leq \frac{4\ln(TK/\delta)}{\Delta_a^2}.$$

which is because after the UCB of an arm a is below μ_{a^*} , UCB algorithm will never pull this arm a again (the UCB of a^* is no smaller than μ_{a^*}).

Thus for the regret calculation, we get:

$$\sum_{t=0}^{T-1} \mu_{a^*} - \mu_{I_t} \leq \sum_{a \neq a^*} N_a^T \Delta_a \leq \sum_{a \neq a^*} \frac{4\ln(TK/\delta)}{\Delta_a}.$$

Together with the fact that Inequality 5.1 holds with probability at least $1 - \delta$, we conclude the proof. ■

5.2 Linear Bandits: Handling Large Action Spaces

Let $D \subset \mathbb{R}^d$ be a compact (but otherwise arbitrary) set of decisions. On each round, we must choose a decision $x_t \in D$. Each such choice results in a reward $r_t \in [-1, 1]$.

Algorithm 4 The Linear UCB algorithm

Input: λ, β_t

- 1: **for** $t = 0, 1 \dots$ **do**
- 2: Execute

$$x_t = \operatorname{argmax}_{x \in D} \max_{\mu \in \text{BALL}_t} \mu \cdot x$$

and observe the reward r_t .

- 3: Update BALL_{t+1} (as specified in Equation 5.2).
 - 4: **end for**
-

We assume that, regardless of the history $\mathcal{H}$ of decisions and observed rewards, the conditional expectation of r_t is a fixed linear function, i.e. for all $x \in D$,

$$\mathbb{E}[r_t | x_t = x] = \mu^* \cdot x \in [-1, 1],$$

where $x \in D$ is arbitrary. Here, observe that we have assumed the mean reward for any decision is bounded in $[-1, 1]$. Under these assumptions, the *noise sequence*,

$$\eta_t = r_t - \mu^* \cdot x_t$$

is a martingale difference sequence.

The problem is essentially a bandit version of a fundamental geometric optimization problem, in which the agent's feedback on each round t is only the observed reward r_t and where the agent does not know μ^* apriori.

If $x_0, \dots, x_{T-1}$ are the decisions made in the game, then define the *cumulative regret* by

$$R_T = T\mu^* \cdot x^* - \sum_{t=0}^{T-1} \mu^* \cdot x_t$$

where $x^* \in D$ is an optimal decision for μ^* , i.e.

$$x^* \in \operatorname{argmax}_{x \in D} \mu^* \cdot x$$

which exists since D is compact. Observe that if the mean μ^* were known, then the optimal strategy would be to play x^* every round. Since the expected loss for each decision x equals $\mu^* \cdot x$, the cumulative regret is just the difference between the expected loss of the optimal algorithm and the expected loss for the actual decisions x_t . By the Hoeffding-Azuma inequality (see Lemma A.5), the observed reward $\sum_{t=0}^{T-1} r_t$ will be close to their (conditional) expectations $\sum_{t=0}^{T-1} \mu^* \cdot x_t$.

Since the sequence of decisions $x_1, \dots, x_{T-1}$ may depend on the particular sequence of random noise encountered, R_T is a random variable. Our goal in designing an algorithm is to keep R_T as small as possible.

5.2.1 The LinUCB algorithm

LinUCB is based on “optimism in the face of uncertainty,” which is described in Algorithm 4. At episode t , we use all previous experience to define an uncertainty region (an ellipse) BALL_t . The center of this region, $\hat{\mu}_t$, is the solution of the following regularized least squares problem:

$$\begin{aligned} \hat{\mu}_t &= \arg \min_{\mu} \sum_{\tau=0}^{t-1} \|\mu \cdot x_{\tau} - r_{\tau}\|_2^2 + \lambda \|\mu\|_2^2 \\ &= \Sigma_t^{-1} \sum_{\tau=0}^{t-1} r_{\tau} x_{\tau}, \end{aligned}$$

where λ is a parameter and where

$$\Sigma_t = \lambda I + \sum_{\tau=0}^{t-1} x_\tau x_\tau^\top, \text{ with } \Sigma_0 = \lambda I.$$

The shape of the region BALL_t is defined through the feature covariance Σ_t .

Precisely, the uncertainty region, or confidence ball, is defined as:

$$\text{BALL}_t = \left\{ \mu \mid (\hat{\mu}_t - \mu)^\top \Sigma_t (\hat{\mu}_t - \mu) \leq \beta_t \right\}, \quad (5.2)$$

where β_t is a parameter of the algorithm.

Computation. Suppose that we have an efficient linear optimization oracle, i.e. that we can efficiently solve the problem:

$$\max_{x \in D} \nu \cdot x$$

for any ν . Even with this, Step 2 of LinUCB may not be computationally tractable. For example, suppose that D is provided to us as a polytope, then the above oracle can be efficiently computed using linear programming, while LinUCB is an NP-hard optimization. Here, we can actually use a wider confidence region, where we can keep track of ℓ_1 ball which contains BALL_t . See Section 5.4 for further reading.

5.2.2 Upper and Lower Bounds

Our main result here is that we have sublinear regret with only a polynomial dependence on the dimension d and, importantly, no dependence on the cardinality of the decision space D , i.e. on $|D|$.

Theorem 5.3. *Suppose that the expected costs are bounded, in magnitude, by 1, i.e. that $|\mu^* \cdot x| \leq 1$ for all $x \in D$; that $\|\mu^*\| \leq W$ and $\|x\| \leq B$ for all $x \in D$; and that the noise η_t is σ^2 sub-Gaussian¹. Set*

$$\lambda = \sigma^2 / W^2, \quad \beta_t := \sigma^2 \left(2 + 4d \log \left(1 + \frac{tB^2W^2}{d} \right) + 8 \log(4/\delta) \right).$$

We have that with probability greater than $1 - \delta$, that (simultaneously) for all $T \geq 0$,

$$R_T \leq c\sigma\sqrt{T} \left(d \log \left(1 + \frac{TB^2W^2}{d\sigma^2} \right) + \log(4/\delta) \right)$$

where c is an absolute constant. In other words, we have that R_T is $\tilde{O}(d\sqrt{T})$ with high probability.

The following shows that no algorithm can (uniformly) do better.

Theorem 5.4. *(Lower bound) There exists a distribution over linear bandit problems (i.e. a distribution over μ) with the rewards being bounded by 1, in magnitude, and $\sigma^2 \leq 1$, such that for every (randomized) algorithm, we have for $n \geq \max\{256, d^2/16\}$,*

$$\mathbb{E}_\mu \mathbb{E} R_T \geq \frac{1}{2500} d\sqrt{T}.$$

where the inner expectation is with respect to randomness in the problem and the algorithm.

We will only prove the upper bound (See Section bib:bandits).

¹Roughly, this assumes the tail probabilities of η_t decay no more slowly than a Gaussian distribution. See Definition A.2.

LinUCB and D -optimal design. Let us utilize the D -optimal design to improve the dependencies in Theorem 5.3. Let Σ_D denote the D -optimal design matrix from Theorem 3.2. Consider the coordinate transformation:

$$\tilde{x} = \Sigma_D^{-1/2} x, \quad \tilde{\mu}^* = \Sigma_D^{1/2} \mu^*.$$

Observe that $\tilde{\mu}^* \cdot \tilde{x} = \mu^* \cdot x$, so we still have a linear expected reward function in this new coordinate systems. Also, observe that $\|\tilde{x}\| = \|x\|_{\Sigma_D} \leq d$, which is a property of the D -optimal design, and that

$$\|\tilde{\mu}^*\| = \|\mu^*\|_{\Sigma_D} = \sqrt{(\mu^*)^\top \Sigma_D \mu^*} = \sqrt{\mathbb{E}_{x \sim \rho}[(\mu^* \cdot x)^2]} \leq 1,$$

where the last step uses our assumption that the rewards are bounded, in magnitude, by 1.

The following corollary shows that, without loss of generality, we can remove the dependencies on B and W from the previous theorem, due to that $B \leq \sqrt{d}$ and $W \leq 1$ when working under this coordinate transform.

Corollary 5.5. *Suppose that the expected costs are bounded, in magnitude, by 1, i.e. that $|\tilde{\mu}^* \cdot \tilde{x}| \leq 1$ for all $x \in D$ and that the noise η_t is σ^2 sub-Gaussian. Suppose linUCB is implemented in the $\tilde{x}$ coordinate system, as described above, with the following settings of the parameters.*

$$\lambda = \sigma^2, \quad \beta_t := \sigma^2 \left(2 + 4d \log(1+t) + 8 \log(4/\delta) \right).$$

We have that with probability greater than $1 - \delta$, that (simultaneously) for all $T \geq 0$,

$$R_T \leq c\sigma\sqrt{T} \left(d \log \left(1 + \frac{T}{\sigma^2} \right) + \log(4/\delta) \right)$$

where c is an absolute constant.

5.3 LinUCB Analysis

In establishing the upper bounds there are two main propositions from which the upper bounds follow. The first is in showing that the confidence region is appropriate.

Proposition 5.6. (Confidence) *Let $\delta > 0$. We have that*

$$\Pr(\forall t, \mu^* \in \text{BALL}_t) \geq 1 - \delta.$$

Section 5.3.2 is devoted to establishing this confidence bound. In essence, the proof seeks to understand the growth of the quantity $(\hat{\mu}_t - \mu^*)^\top \Sigma_t (\hat{\mu}_t - \mu^*)$.

The second main step in analyzing LinUCB is to show that as long as the aforementioned high-probability event holds, we have some control on the growth of the regret. Let us define

$$\text{regret}_t = \mu^* \cdot x^* - \mu^* \cdot x_t$$

which denotes the instantaneous regret.

The following bounds the sum of the squares of instantaneous regret.

Proposition 5.7. (Sum of Squares Regret Bound) *Suppose that $\|x\| \leq B$ for $x \in D$. Suppose β_t is increasing and that $\beta_t \geq 1$. For LinUCB, if $\mu^* \in \text{BALL}_t$ for all t , then*

$$\sum_{t=0}^{T-1} \text{regret}_t^2 \leq 8\beta_T d \log \left(1 + \frac{TB^2}{d\lambda} \right)$$

This is proven in Section 5.3.1. The idea of the proof involves a potential function argument on the log volume (i.e. the log determinant) of the “precision matrix” Σ_t (which tracks how accurate our estimates of μ^* are in each direction). The proof involves relating the growth of this volume to the regret.

Using these two results we are able to prove our upper bound as follows:

Proof:[Proof of Theorem 5.3] By Propositions 5.6 and 5.7 along with the Cauchy-Schwarz inequality, we have, with probability at least $1 - \delta$,

$$R_T = \sum_{t=0}^{T-1} \text{regret}_t \leq \sqrt{T \sum_{t=0}^{T-1} \text{regret}_t^2} \leq \sqrt{8T\beta_T d \log \left(1 + \frac{TB^2}{d\lambda}\right)}.$$

The remainder of the proof follows from using our chosen value of β_T and algebraic manipulations (that $2ab \leq a^2 + b^2$). $\blacksquare$

We now provide the proofs of these two propositions.

5.3.1 Regret Analysis

In this section, we prove Proposition 5.7, which says that the sum of the squares of the instantaneous regrets of the algorithm is small, assuming the evolving confidence balls always contain the true mean μ^* . An important observation is that on any round t in which $\mu^* \in \text{BALL}_t$, the instantaneous regret is at most the “width” of the ellipsoid in the direction of the chosen decision. Moreover, the algorithm’s choice of decisions forces the ellipsoids to shrink at a rate that ensures that the sum of the squares of the widths is small. We now formalize this.

Unless explicitly stated, all norms refer to the ℓ_2 norm.

Lemma 5.8. *Let $x \in D$. If $\mu \in \text{BALL}_t$ and $x \in D$. Then*

$$|(\mu - \hat{\mu}_t)^\top x| \leq \sqrt{\beta_t x^\top \Sigma_t^{-1} x}$$

Proof: By Cauchy-Schwarz, we have:

$$\begin{aligned} |(\mu - \hat{\mu}_t)^\top x| &= |(\mu - \hat{\mu}_t)^\top \Sigma_t^{1/2} \Sigma_t^{-1/2} x| = |(\Sigma_t^{1/2} (\mu - \hat{\mu}_t))^\top \Sigma_t^{-1/2} x| \\ &\leq \|\Sigma_t^{1/2} (\mu - \hat{\mu}_t)\| \|\Sigma_t^{-1/2} x\| = \|\Sigma_t^{1/2} (\mu - \hat{\mu}_t)\| \sqrt{x^\top \Sigma_t^{-1} x} \leq \sqrt{\beta_t x^\top \Sigma_t^{-1} x} \end{aligned}$$

where the last inequality holds since $\mu \in \text{BALL}_t$. $\blacksquare$

Define

$$w_t := \sqrt{x_t^\top \Sigma_t^{-1} x_t}$$

which we interpret as the “normalized width” at time t in the direction of the chosen decision. We now see that the width, $2\sqrt{\beta_t} w_t$, is an upper bound for the instantaneous regret.

Lemma 5.9. *Fix $t \leq T$. If $\mu^* \in \text{BALL}_t$ and $\beta_t \geq 1$, then*

$$\text{regret}_t \leq 2 \min(\sqrt{\beta_t} w_t, 1) \leq 2\sqrt{\beta_T} \min(w_t, 1)$$

Proof: Let $\tilde{\mu} \in \text{BALL}_t$ denote the vector which minimizes the dot product $\tilde{\mu}^\top x_t$. By the choice of x_t , we have

$$\tilde{\mu}^\top x_t = \max_{\mu \in \text{BALL}_t} \mu^\top x_t = \max_{x \in D} \max_{\mu \in \text{BALL}_t} \mu^\top x \geq (\mu^*)^\top x^*,$$

where the inequality used the hypothesis $\mu^* \in \text{BALL}_t$. Hence,

$$\begin{aligned} \text{regret}_t &= (\mu^*)^\top x^* - (\mu^*)^\top x_t \leq (\tilde{\mu} - \mu^*)^\top x_t \\ &= (\tilde{\mu} - \hat{\mu}_t)^\top x_t + (\hat{\mu}_t - \mu^*)^\top x_t \leq 2\sqrt{\beta_t} w_t \end{aligned}$$

where the last step follows from Lemma 5.8 since $\tilde{\mu}$ and μ^* are in BALL_t . Since $r_t \in [-1, 1]$, regret_t is always at most 2 and the first inequality follows. The final inequality is due to that β_t is increasing and larger than 1. ■

The following two lemmas prove useful in showing that we can treat the log determinant as a potential function, where can bound the sum of widths independently of the choices made by the algorithm.

Lemma 5.10. *We have:*

$$\det \Sigma_T = \det \Sigma_0 \prod_{t=0}^{T-1} (1 + w_t^2).$$

Proof: By the definition of Σ_{t+1} , we have

$$\begin{aligned} \det \Sigma_{t+1} &= \det(\Sigma_t + x_t x_t^\top) = \det(\Sigma_t^{1/2} (I + \Sigma_t^{-1/2} x_t x_t^\top \Sigma_t^{-1/2}) \Sigma_t^{1/2}) \\ &= \det(\Sigma_t) \det(I + \Sigma_t^{-1/2} x_t (\Sigma_t^{-1/2} x_t)^\top) = \det(\Sigma_t) \det(I + v_t v_t^\top), \end{aligned}$$

where $v_t := \Sigma_t^{-1/2} x_t$. Now observe that $v_t^\top v_t = w_t^2$ and

$$(I + v_t v_t^\top) v_t = v_t + v_t (v_t^\top v_t) = (1 + w_t^2) v_t$$

Hence $(1 + w_t^2)$ is an eigenvalue of $I + v_t v_t^\top$. Since $v_t v_t^\top$ is a rank one matrix, all other eigenvalues of $I + v_t v_t^\top$ equal 1. Hence, $\det(I + v_t v_t^\top)$ is $(1 + w_t^2)$, implying $\det \Sigma_{t+1} = (1 + w_t^2) \det \Sigma_t$. The result follows by induction. ■

Lemma 5.11. (*“Potential Function” Bound*) *For any sequence $x_0, \dots, x_{T-1}$ such that, for $t < T$, $\|x_t\|_2 \leq B$, we have:*

$$\log \left(\det \Sigma_{T-1} / \det \Sigma_0 \right) = \log \det \left(I + \frac{1}{\lambda} \sum_{t=0}^{T-1} x_t x_t^\top \right) \leq d \log \left(1 + \frac{TB^2}{d\lambda} \right).$$

Proof: Denote the eigenvalues of $\sum_{t=0}^{T-1} x_t x_t^\top$ as $\sigma_1, \dots, \sigma_d$, and note:

$$\sum_{i=1}^d \sigma_i = \text{Trace} \left(\sum_{t=0}^{T-1} x_t x_t^\top \right) = \sum_{t=0}^{T-1} \|x_t\|^2 \leq TB^2.$$

Using the AM-GM inequality,

$$\begin{aligned} \log \det \left(I + \frac{1}{\lambda} \sum_{t=0}^{T-1} x_t x_t^\top \right) &= \log \left(\prod_{i=1}^d (1 + \sigma_i / \lambda) \right) \\ &= d \log \left(\prod_{i=1}^d (1 + \sigma_i / \lambda) \right)^{1/d} \leq d \log \left(\frac{1}{d} \sum_{i=1}^d (1 + \sigma_i / \lambda) \right) \leq d \log \left(1 + \frac{TB^2}{d\lambda} \right), \end{aligned}$$

which concludes the proof. ■

Finally, we are ready to prove that if μ^* always stays within the evolving confidence region, then our regret is under control.

Proof:[Proof of Proposition 5.7] Assume that $\mu^* \in \text{BALL}_t$ for all t . We have that:

$$\begin{aligned} \sum_{t=0}^{T-1} \text{regret}_t^2 &\leq \sum_{t=0}^{T-1} 4\beta_t \min(w_t^2, 1) \leq 4\beta_T \sum_{t=0}^{T-1} \min(w_t^2, 1) \\ &\leq 8\beta_T \sum_{t=0}^{T-1} \ln(1 + w_t^2) \leq 8\beta_T \log \left(\det \Sigma_{T-1} / \det \Sigma_0 \right) = 8\beta_T d \log \left(1 + \frac{TB^2}{d\lambda} \right) \end{aligned}$$

where the first inequality follow from By Lemma 5.9; the second from that β_t is an increasing function of t ; the third uses that for $0 \leq y \leq 1$, $\ln(1 + y) \geq y/2$; the final two inequalities follow by Lemmas 5.10 and 5.11. ■

5.3.2 Confidence Analysis

Proof:[Proof of Proposition 5.6] Since $r_\tau = x_\tau \cdot \mu^* + \eta_\tau$, we have:

$$\begin{aligned} \hat{\mu}_t - \mu^* &= \Sigma_t^{-1} \sum_{\tau=0}^{t-1} r_\tau x_\tau - \mu^* = \Sigma_t^{-1} \sum_{\tau=0}^{t-1} x_\tau (x_\tau \cdot \mu^* + \eta_\tau) - \mu^* \\ &= \Sigma_t^{-1} \left(\sum_{\tau=0}^{t-1} x_\tau (x_\tau)^\top \right) \mu^* - \mu^* + \Sigma_t^{-1} \sum_{\tau=0}^{t-1} \eta_\tau x_\tau = \lambda \Sigma_t^{-1} \mu^* + \Sigma_t^{-1} \sum_{\tau=0}^{t-1} \eta_\tau x_\tau \end{aligned}$$

For any $0 < \delta_t < 1$, using Lemma A.9, it holds with probability at least $1 - \delta_t$,

$$\begin{aligned} \sqrt{(\hat{\mu}_t - \mu^*)^\top \Sigma_t (\hat{\mu}_t - \mu^*)} &= \|(\Sigma_t)^{1/2} (\hat{\mu}_t - \mu^*)\| \\ &\leq \left\| \lambda \Sigma_t^{-1/2} \mu^* \right\| + \left\| \Sigma_t^{-1/2} \sum_{\tau=0}^{t-1} \eta_\tau x_\tau \right\| \\ &\leq \sqrt{\lambda} \|\mu^*\| + \sqrt{2\sigma^2 \log(\det(\Sigma_t) \det(\Sigma_0)^{-1} / \delta_t)}. \end{aligned}$$

where we have also used the triangle inequality and that $\|\Sigma_t^{-1}\| \leq 1/\lambda$.

We seek to lower bound $\Pr(\forall t, \mu^* \in \text{BALL}_t)$. Note that at $t = 0$, by our choice of λ , we have that BALL_0 contains W^* , so $\Pr(\mu^* \notin \text{BALL}_0) = 0$. For $t \geq 1$, let us assign failure probability $\delta_t = (3/\pi^2)/t^2\delta$ for the t -th event, which, using the above, gives us an upper bound on the sum failure probability as

$$1 - \Pr(\forall t, \mu^* \in \text{BALL}_t) = \Pr(\exists t, \mu^* \notin \text{BALL}_t) \leq \sum_{t=1}^{\infty} \Pr(\mu^* \notin \text{BALL}_t) < \sum_{t=1}^{\infty} (1/t^2)(3/\pi^2)\delta = \delta/2.$$

This along with Lemma 5.11 completes the proof. ■

5.4 Bibliographic Remarks and Further Readings

The original multi-armed bandit model goes to back to [Robbins, 1952]. The linear bandit model was first introduced in [Abe and Long, 1999]. Our analysis of the LinUCB algorithm follows from the original proof in [Dani et al., 2008], with a simplified concentration analysis due to [Abbasi-Yadkori et al., 2011]. The first sublinear regret bound here was due to [Auer et al., 2002], which used a more complicated algorithm.

The lower bound we present is also due to [Dani et al., 2008], which also shows that LinUCB is minimax optimal.

Chapter 6

Strategic Exploration in Tabular MDPs

We now turn to how an agent acting in an unknown MDP can obtain a near-optimal reward over time. Compared with the previous setting with access to a generative model, we no longer have easy access to transitions at each state, but only have the ability to execute trajectories in the MDP. The main complexity this adds to the learning process is that the agent has to engage in exploration, that is, plan to reach new states where enough samples have not been seen yet, so that optimal behavior in those states can be learned.

We will work with finite-horizon MDPs with a fixed start state s_0 (see Section 1.2), and we will assume the agent learns in an episodic setting, as introduced in Chapter 2. Here, in every episode k , the learner acts for H step starting from a fixed starting state s_0 and, at the end of the H -length episode, the state is reset to s_0 . It is straightforward to extend this setting where the starting state is sampled from a distribution, i.e. $s_0 \sim \mu$.

The goal of the agent is to minimize her expected cumulative regret over K episodes:

$$\text{Regret} := \mathbb{E} \left[KV^*(s_0) - \sum_{k=0}^{K-1} \sum_{h=0}^{H-1} r(s_h^k, a_h^k) \right],$$

where the expectation is with respect to the randomness of the MDP environment and, possibly, any randomness of the agent's strategy.

This chapter considers tabular MDPs, where $\mathcal{S}$ and $\mathcal{A}$ are discrete. We will now present a (nearly) sublinear regret algorithm, UCB-Value Iteration. This chapter follows the proof in [Azar et al., 2017]. We first give a simplified analysis of UCB-VI that gives a regret bound in the order of $H^2 S \sqrt{AK}$, followed by a more refined analysis that gives a $H^2 \sqrt{SAK} + H^3 S^2 A$ regret bound.

6.1 On The Need for Strategic Exploration

Consider the chain MDP of length $H + 2$ shown in Figure 11.1. The starting state of interest is state s_0 . We consider a uniformly random policy $\pi_h(s)$ that at any time step h and any state s , selects an action from all A many actions uniform randomly. It is easy to see that starting from s_0 , the probability of such uniform random policy hitting the rewarding state s_{H+1} is exponentially small, i.e., $O(A^H)$, since it needs to select the right action that moves one step right at every time step h . This example demonstrates that we need to perform strategic exploration in order to avoid the exponential sample complexity.

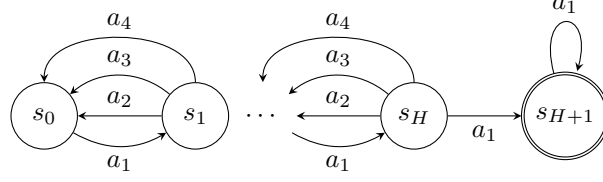


Figure 6.1: A deterministic, chain MDP of length $H + 2$. Rewards are 0 everywhere other than $r(s_{H+1}, a_1) = 1$. In this example we have $A = 4$.

Algorithm 5 UCBVI

Input: reward function r (assumed to be known), confidence parameters

- 1: **for** $k = 0 \dots K - 1$ **do**
 - 2: Compute $\hat{P}_h^k$ as the empirical estimates, for all h (Eq. 6.1)
 - 3: Compute reward bonus b_h^k for all h (Eq. 6.2)
 - 4: Run Value-Iteration on $\{\hat{P}_h^k, r + b_h^k\}_{h=0}^{H-1}$ (Eq. 6.3)
 - 5: Set π^k as the returned policy of VI.
 - 6: **end for**
-

6.2 The UCB-VI algorithm

If the learner then executes π^k in the underlying MDP to generate a single trajectory $\tau^k = \{s_h^k, a_h^k\}_{h=0}^{H-1}$ with $a_h = \pi_h^k(s_h^k)$ and $s_{h+1}^k \sim P_h(\cdot | s_h^k, a_h^k)$. We first define some notations below. Consider the very beginning of episode k . We use the history information up to the end of episode $k - 1$ (denoted as $\mathcal{H}_{<k}$) to form some statistics. Specifically, we define:

$$N_h^k(s, a, s') = \sum_{i=0}^{k-1} \mathbf{1}\{(s_h^i, a_h^i, s_{h+1}^i) = (s, a, s')\},$$

$$N_h^k(s, a) = \sum_{i=0}^{k-1} \mathbf{1}\{(s_h^i, a_h^i) = (s, a)\}, \forall h, s, a.$$

Namely, we maintain counts of how many times s, a, s' and s, a are visited at time step h from the beginning of the learning process to the end of the episode $k - 1$. We use these statistics to form an empirical model:

$$\hat{P}_h^k(s' | s, a) = \frac{N_h^k(s, a, s')}{N_h^k(s, a)}, \forall h, s, a, s'. \quad (6.1)$$

We will also use the counts to define a *reward bonus*, denoted as $b_h(s, a)$ for all h, s, a . Denote $L := \ln(SAHK/\delta)$ (δ as usual represents the failure probability which we will define later). We define reward bonus as follows:

$$b_h^k(s, a) = 2H \sqrt{\frac{L}{N_h^k(s, a)}}. \quad (6.2)$$

With reward bonus and the empirical model, the learner uses *Value Iteration* on the empirical transition $\hat{P}_h^k$ and the combined reward $r_h + b_h^k$. Starting at H (note that H is a fictitious extra step as an episode terminates at $H - 1$), we

perform dynamic programming all the way to $h = 0$:

$$\begin{aligned}\widehat{V}_H^k(s) &= 0, \forall s, \\ \widehat{Q}_h^k(s, a) &= \min \left\{ r_h(s, a) + b_h^k(s, a) + \widehat{P}_h^k(\cdot|s, a) \cdot \widehat{V}_{h+1}^k, H \right\}, \\ \widehat{V}_h^k(s) &= \max_a \widehat{Q}_h^k(s, a), \pi_h^k(s) = \operatorname{argmax}_a \widehat{Q}_h^k(s, a), \forall h, s, a.\end{aligned}\tag{6.3}$$

Note that when using $\widehat{V}_{h+1}^k$ to compute $\widehat{Q}_h^k$, we truncate the value by H . This is because we know that due to the assumption that $r(s, a) \in [0, 1]$, no policy's Q value will ever be larger than H .

Denote $\pi^k = \{\pi_0^k, \dots, \pi_{H-1}^k\}$. Learner then executes π^k in the MDP to get a new trajectory τ^k .

UCBVI repeats the above procedure for K episodes.

6.3 Analysis

We will prove the following theorem.

Theorem 6.1 (Regret Bound of UCBVI). *UCBVI achieves the following regret bound:*

$$\text{Regret} := \mathbb{E} \left[\sum_{k=0}^{K-1} \left(V^*(s_0) - V^{\pi^k}(s_0) \right) \right] \leq 10H^2 S \sqrt{AK \cdot \ln(SAH^2 K^2)} = \tilde{O} \left(H^2 S \sqrt{AK} \right)$$

Remark While the above regret is sub-optimal, the algorithm we presented here indeed achieves a sharper bound in the leading term $\tilde{O}(H^2 \sqrt{SAK})$ [Azar et al., 2017], which gives the tight dependency bound on S, A, K . In Section 6.4, we will present a refined analysis that indeed achieves $H^2 \sqrt{SAK}$ regret. The dependency on H is not tight and tightening the dependency on H requires modifications to the reward bonus (use Bernstein inequality rather than Hoeffding's inequality for reward bonus design).

We prove the above theorem in this section.

We start with bounding the error from the learned model $\widehat{P}_h^k$.

Lemma 6.2 (State-action wise model error). *Fix $\delta \in (0, 1)$. For all $k \in [0, \dots, K-1], s \in \mathcal{S}, a \in \mathcal{A}, h \in [0, \dots, H-1]$, with probability at least $1 - \delta$, we have that for any $f : \mathcal{S} \mapsto [0, H]$:*

$$\left| \left(\widehat{P}_h^k(\cdot|s, a) - P_h^*(\cdot|s, a) \right)^\top f \right| \leq 8H \sqrt{\frac{S \ln(SAHK/\delta)}{N_h^k(s, a)}}.$$

Remark The proof of the above lemma requires a careful argument using Martingale difference sequence mainly because $N_h^k(s, a)$ itself is a random quantity. We give a proof the above lemma at the end of this section (Sec. 6.3.1). Using the above lemma will result a sub-optimal regret bound in terms of the dependence on S . In Section 6.4, we will give a different approach to bound $\left| \left(\widehat{P}_h^k(\cdot|s, a) - P_h^*(\cdot|s, a) \right)^\top f \right|$ which leads to an improved regret bound.

The following lemma is still about model error, but this time we consider an average model error with respect to V^* — a deterministic quantity.

Lemma 6.3 (State-action wise average model error under V^*). *Fix $\delta \in (0, 1)$. For all $k \in [1, \dots, K-1]$, $s \in \mathcal{S}$, $a \in \mathcal{A}$, $h \in [0, \dots, H-1]$, and consider $V_h^* : \mathcal{S} \rightarrow [0, H]$, with probability at least $1 - \delta$, we have:*

$$\left| \widehat{P}_h^k(\cdot|s, a) \cdot V_{h+1}^* - P_h^*(\cdot|s, a) \cdot V_{h+1}^* \right| \leq 2H \sqrt{\frac{\ln(SAHN/\delta)}{N_h^k(s, a)}}.$$

Note that we can use Lemma 6.2 to bound $\left| \widehat{P}_h^k(\cdot|s, a) \cdot V_{h+1}^* - P_h^*(\cdot|s, a) \cdot V_{h+1}^* \right|$. However, the above lemma shows that by leveraging the fact that V^* is a deterministic quantity (i.e., independent of the data collected during learning), we can get a tighter upper bound which does not scale polynomially with respect to S .

Proof: Consider a fixed tuple s, a, k, h first. By the definition of $\widehat{P}_h^k$, we have:

$$\widehat{P}_h^k(\cdot|s, a) \cdot V_{h+1}^* = \frac{1}{N_h^k(s, a)} \sum_{i=0}^{k-1} \mathbf{1}\{(s_h^i, a_h^i) = (s, a)\} V_{h+1}^*(s_{h+1}^i).$$

Now denote $\mathcal{H}_{h,i}$ as the entire history from $t = 0$ to iteration $t = i$ where in iteration i , $\mathcal{H}_{h,i}$ includes history from time step 0 up to and including time step h . We define random variables: $X_i = \mathbf{1}\{(s_h^i, a_h^i) = (s, a)\} V_{h+1}^*(s_{h+1}^i) - \mathbb{E}[\mathbf{1}\{(s_h^i, a_h^i) = (s, a)\} V_{h+1}^*(s_{h+1}^i) | \mathcal{H}_{h,i}]$ for $i = 0, \dots, K-1$. These random variables have the following properties. First $|X_i| \leq H$ if $\mathbf{1}\{(s_h^i, a_h^i) = (s, a)\} = 1$, else $|X_i| = 0$, which implies that $\sum_{i=0}^k |X_i|^2 \leq N_h^k(s, a)$. Also we have $\mathbb{E}[X_i | \mathcal{H}_{h,i}] = 0$ for all i . Thus, $\{X_i\}_{i \geq 0}$ is a Martingale difference sequence. Using Azuma-Hoeffding's inequality, we get that for any fixed k , with probability at least $1 - \delta$,

$$\left| \sum_{i=0}^{k-1} X_i \right| = \left| \sum_{i=0}^{k-1} \mathbf{1}\{(s_h^i, a_h^i) = (s, a)\} V_{h+1}^*(s_{h+1}^i) - N_h^k(s, a) \mathbb{E}_{s' \sim P_h^*(s, a)} V^*(s') \right| \leq 2H \sqrt{N_h^k(s, a) \ln(1/\delta)}.$$

Divide $N_h^k(s, a)$ on both sides and use the definition of $\widehat{P}_h^k(\cdot|s, a)^\top V_{h+1}^*$, we get:

$$\left| \widehat{P}_h^k(\cdot|s, a)^\top V_{h+1}^* - P_h^*(\cdot|s, a)^\top V_{h+1}^* \right| \leq 2H \sqrt{\ln(1/\delta) / N_h^k(s, a)}.$$

With a union bound over all $s \in \mathcal{S}, a \in \mathcal{A}, k \in [N], h \in [H]$, we conclude the proof. $\blacksquare$

We denote the two inequalities in Lemma 6.2 and Lemma 6.3 as event $\mathcal{E}_{model}$. Note that the failure probability of $\mathcal{E}_{model}$ is at most 2δ . Below we condition on $\mathcal{E}_{model}$ being true (we deal with failure event at the very end).

Now we study the effect of reward bonus. Similar to the idea in multi-armed bandits, we want to pick a policy π^k , such that the value of π^k in under the combined reward $r_h + b_h^k$ and the empirical model $\widehat{P}_h^k$ is optimistic, i.e., we want $\widehat{V}_0^k(s_0) \geq V_0^*(s_0)$ for all s_0 . The following lemma shows that via reward bonus, we are able to achieve this optimism.

Lemma 6.4 (Optimism). *Assume $\mathcal{E}_{model}$ is true. For all episode k , we have:*

$$\widehat{V}_0^k(s_0) \geq V_0^*(s_0), \forall s_0 \in \mathcal{S};$$

where $\widehat{V}_h^k$ is computed based on VI in Eq. 6.3.

Proof: We prove via induction. At the additional time step H we have $\widehat{V}_H^k(s) = V_H^*(s) = 0$ for all s .

Starting at $h+1$, and assuming we have $\widehat{V}_{h+1}^k(s) \geq V_{h+1}^*(s)$ for all s , we move to h below.

Consider any $s, a \in \mathcal{S} \times \mathcal{A}$. First, if $Q_h^k(s, a) = H$, then we have $Q_h^k(s, a) \geq Q_h^*(s, a)$.

$$\begin{aligned}\hat{Q}_h^k(s, a) - Q_h^*(s, a) &= b_h^k(s, a) + \hat{P}_h^k(\cdot|s, a) \cdot \hat{V}_{h+1}^k - P_h^*(\cdot|s, a) \cdot V_{h+1}^* \\ &\geq b_h^k(s, a) + \hat{P}_h^k(\cdot|s, a) \cdot V_{h+1}^* - P_h^*(\cdot|s, a) \cdot V_{h+1}^* \\ &= b_h^k(s, a) + \left(\hat{P}_h^k(\cdot|s, a) - P_h^*(\cdot|s, a) \right) \cdot V_{h+1}^* \\ &\geq b_h^k(s, a) - 2H \sqrt{\frac{\ln(SAHK/\delta)}{N_h^k(s, a)}} \geq 0.\end{aligned}$$

where the first inequality is from the inductive hypothesis, and the last inequality uses Lemma 6.3 and the definition of bonus.

From $\hat{Q}_h^k$, one can finish the proof by showing $\hat{V}_h^k(s) \geq V_h^*(s), \forall s$. ■

Before we prove our final result, one more technical lemma will be helpful:

Lemma 6.5. Consider arbitrary K sequence of trajectories $\tau^k = \{s_h^k, a_h^k\}_{h=0}^{H-1}$ for $k = 0, \dots, K-1$. We have

$$\sum_{k=0}^{K-1} \sum_{h=0}^{H-1} \frac{1}{\sqrt{N_h^k(s_h^k, a_h^k)}} \leq 2H\sqrt{SAK}.$$

Proof: We swap the order of the two summation above:

$$\begin{aligned}\sum_{k=0}^{K-1} \sum_{h=0}^{H-1} \frac{1}{\sqrt{N_h^k(s_h^k, a_h^k)}} &= \sum_{h=0}^{H-1} \sum_{k=0}^{K-1} \frac{1}{\sqrt{N_h^k(s_h^k, a_h^k)}} = \sum_{h=0}^{H-1} \sum_{s, a \in \mathcal{S} \times \mathcal{A}} \sum_{i=1}^{N_h^K(s, a)} \frac{1}{\sqrt{i}} \\ &\leq 2 \sum_{h=0}^{H-1} \sum_{s, a \in \mathcal{S} \times \mathcal{A}} \sqrt{N_h^K(s, a)} \leq \sum_{h=0}^{H-1} \sqrt{SA \sum_{s, a} N_h^K(s, a)} = H\sqrt{SAK},\end{aligned}$$

where in the first inequality we use the fact that $\sum_{i=1}^N 1/\sqrt{i} \leq 2\sqrt{N}$, and in the second inequality we use CS inequality. ■

The proof of our main theorem now follows:

Proof:[Proof of Theorem 6.1]

Let us consider episode k and denote $\mathcal{H}_{<k}$ as the history up to the end of episode $k-1$. We consider bounding $V^* - V^{\pi^k}$. Using optimism and the simulation lemma, we can get the following result:

$$V^*(s_0) - V^{\pi^k}(s_0) \leq \hat{V}_0^k(s_0) - V_0^{\pi^k}(s_0) \leq \sum_{h=0}^{H-1} \mathbb{E}_{s_h, a_h \sim d_h^{\pi^k}} \left[b_h^k(s_h, a_h) + \left(\hat{P}_h^k(\cdot|s_h, a_h) - P^*(\cdot|s_h, a_h) \right) \cdot \hat{V}_{h+1}^{\pi^k} \right] \quad (6.4)$$

We prove the above two inequalities in the lecture. We leave the proof of the above inequality (Eq 6.4) as an exercise for readers. Note that this is slightly different from the usual simulation lemma, as here we truncate $\hat{V}$ by H during VI.

Under $\mathcal{E}_{model}$, we can bound $\left(\hat{P}_h^k(\cdot|s_h, a_h) - P^*(\cdot|s_h, a_h) \right) \cdot \hat{V}_{h+1}^{\pi^k}$ (recall Lemma 6.2) with a Holder's inequality:

$$\begin{aligned}\left| \left(\hat{P}_h^k(\cdot|s_h, a_h) - P^*(\cdot|s_h, a_h) \right) \cdot \hat{V}_{h+1}^{\pi^k} \right| &\leq \left\| \hat{P}_h^k(\cdot|s_h, a_h) - P^*(\cdot|s_h, a_h) \right\|_1 \left\| \hat{V}_{h+1}^{\pi^k} \right\|_\infty \\ &\leq 8H \sqrt{\frac{SL}{N_h^k(s, a)}},\end{aligned}$$

where recall $L := \ln(SAKH/\delta)$. Hence, for the per-episode regret $V^*(s_0) - V^{\pi^k}(s_0)$, we obtain:

$$\begin{aligned} V^*(s_0) - V^{\pi^k}(s_0) &\leq \sum_{h=0}^{H-1} \mathbb{E}_{s_h, a_h \sim d_h^{\pi^k}} \left[b_h^k(s_h, a_h) + 8H \sqrt{SL/N_h^k(s_h, a_h)} \right] \\ &\leq \sum_{h=0}^{H-1} \mathbb{E}_{s_h, a_h \sim d_h^{\pi^k}} \left[10H \sqrt{SL/N_h^k(s_h, a_h)} \right] \\ &= 10H \sqrt{SL} \cdot \mathbb{E} \left[\sum_{h=0}^{H-1} \frac{1}{\sqrt{N_h^k(s_h^k, a_h^k)}} \middle| \mathcal{H}_{<k} \right], \end{aligned}$$

where in the last term the expectation is taken with respect to the trajectory $\{s_h^k, a_h^k\}$ (which is generated from π^k) while conditioning on all history $\mathcal{H}_{<k}$ up to and including the end of episode $k-1$.

Now we sum all episodes together and take the failure event into consideration:

$$\begin{aligned} \mathbb{E} \left[\sum_{k=0}^{K-1} V^*(s_0) - V^{\pi^k}(s_0) \right] &= \mathbb{E} \left[\mathbf{1}\{\mathcal{E}_{model}\} \left(\sum_{k=0}^{K-1} V^*(s_0) - V^{\pi^k}(s_0) \right) \right] + \mathbb{E} \left[\mathbf{1}\{\bar{\mathcal{E}}_{model}\} \left(\sum_{k=0}^{K-1} V^*(s_0) - V^{\pi^k}(s_0) \right) \right] \\ &\leq \mathbb{E} \left[\mathbf{1}\{\mathcal{E}_{model}\} \left(\sum_{k=0}^{K-1} V^*(s_0) - V^{\pi^k}(s_0) \right) \right] + 2\delta KH \quad (6.5) \\ &\leq 10H \sqrt{S \ln(SAHK/\delta)} \mathbb{E} \left[\sum_{k=0}^{K-1} \sum_{h=0}^{H-1} \frac{1}{\sqrt{N_h^k(s_h^k, a_h^k)}} \right] + 2\delta KH, \end{aligned}$$

where the last inequality uses the law of total expectation.

We can bound the double summation term above using lemma 6.5. We can conclude that:

$$\mathbb{E} \left[\sum_{k=0}^{K-1} V^*(s_0) - V^{\pi^k}(s_0) \right] \leq 10H^2 S \sqrt{AK \ln(SAHK/\delta)} + 2\delta KH.$$

Now set $\delta = 1/KH$, we get:

$$\mathbb{E} \left[\sum_{k=0}^{K-1} V^*(s_0) - V^{\pi^k}(s_0) \right] \leq 10H^2 S \sqrt{AK \ln(SAH^2 K^2)} + 2 = O \left(H^2 S \sqrt{AK \ln(SAH^2 K^2)} \right).$$

This concludes the proof of Theorem 6.1. ■

6.3.1 Proof of Lemma 6.2

Here we include a proof of Lemma 6.2. The proof consists of two steps, first we use Martingale concentration to bound $|(\hat{P}_h^k(\cdot|s, a) - P_h^*(\cdot|s, a))^\top f|$ for all s, a, h, k , but for a fixed $f : \mathcal{S} \mapsto [0, H]$. Note that there are infinitely many such f that maps from $\mathcal{S}$ to $[0, H]$. Thus, the second step is to use a covering argument (i.e., ϵ -net on all $\{f : \mathcal{S} \mapsto [0, H]\}$) to show that $|(\hat{P}_h^k(\cdot|s, a) - P_h^*(\cdot|s, a))^\top f|$ is bounded for all s, a, h, k , and all $f : \mathcal{S} \mapsto [0, H]$.

Proof:[Proof of Lemma 6.2] We consider a fixed tuple s, a, k, h, f first. Recall the definition of $\hat{P}_h^k(s, a)$, and we have: $\hat{P}_h^k(\cdot|s, a)^\top f = \sum_{i=0}^{k-1} \mathbf{1}\{s_h^i, a_h^i = s, a\} f(s_{h+1}^i) / N_h^k(s, a)$.

Define $\mathcal{H}_{h,i}$ as the history starting from the beginning of iteration 0 all the way up to and include time step h at iteration i . Define random variables X_i as $X_i = \mathbf{1}\{s_h^i, a_h^i = s, a\}f(s_{h+1}^i) - \mathbf{1}\{s_h^i, a_h^i = s, a\}\mathbb{E}_{s' \sim P_h^*(s,a)}f(s')$. We now show that $\{X_i\}$ is a Martingale difference sequence. First we have $\mathbb{E}[X_i | \mathcal{H}_{h,i}] = 0$, since $\mathbf{1}\{s_h^i, a_h^i = s, a\}$ is a deterministic quantity given $\mathcal{H}_{h,i}$. Second, we have $|X_i| = 0$ for $(s_h^i, a_h^i) \neq (s, a)$, and $|X_i| \leq H$ when $(s_h^i, a_h^i) = (s, a)$. Thus, we have shown that it is a Martingale difference sequence.

Now we can apply Azuma-Hoeffding's inequality (Theorem A.5). With probability at least $1 - \delta$, we have:

$$\left| \sum_{i=0}^{k-1} X_i \right| = \left| \sum_{i=0}^{k-1} \mathbf{1}\{(s_h^i, a_h^i) = (s, a)\}f(s_{h+1}^i) - N_h^k(s, a)\mathbb{E}_{s' \sim P_h^*(s,a)}f(s') \right| \leq 2H\sqrt{N_h^k(s, a) \ln(1/\delta)}.$$

Now apply union bound over all $s \in \mathcal{S}, a \in \mathcal{A}, h \in [0, \dots, H-1], k \in [0, \dots, K-1]$, we have that with probability at least $1 - \delta$, for all s, a, h, k , we have:

$$\left| \sum_{i=0}^{k-1} \mathbf{1}\{(s_h^i, a_h^i) = (s, a)\}f(s_{h+1}^i) - N_h^k(s, a)\mathbb{E}_{s' \sim P_h^*(s,a)}f(s') \right| \leq 2H\sqrt{N_h^k(s, a) \ln(SAKH/\delta)}. \quad (6.6)$$

Note that the above result only holds for the fixed f that we specified at the beginning of the proof. We are left to perform a covering argument over all functions that map from $\mathcal{S}$ to $[0, H]$. We use a standard ϵ -net argument for that. We abuse notation a bit and denote f as a vector in $[0, H]^S$.

Note that $\|f\|_2 \leq H\sqrt{S}$ for any $f : \mathcal{S} \mapsto [0, H]$. A standard ϵ -net argument shows that we can construct an ϵ -net $\mathcal{N}_\epsilon$ over $[0, H]^S$ with $|\mathcal{N}_\epsilon| \leq (1 + 2H\sqrt{S}/\epsilon)^S$, such that for any $f \in [0, H]^S$, there exists a point $f' \in \mathcal{N}_\epsilon$, such that $\|f - f'\|_2 \leq \epsilon$. Note that the ϵ -net $\mathcal{N}_\epsilon$ is independent of the training data during the learning process.

Now using Eq. 6.6 and a union bound over all f in $\mathcal{N}_\epsilon$, we have that with probability at least $1 - \delta$, for all $s, a, k, h, f \in \mathcal{N}_\epsilon$, we have:

$$\left| \frac{\sum_{i=0}^{k-1} \mathbf{1}\{(s_h^i, a_h^i) = (s, a)\}f(s_{h+1}^i)}{N_h^k(s, a)} - \mathbb{E}_{s' \sim P_h^*(s,a)}f(s') \right| \leq 2H\sqrt{\frac{S \ln(SAKH(1 + 2H\sqrt{S}/\epsilon)/\delta)}{N_h^k(s, a)}}.$$

Finally, for any $f \in [0, H]^S$, denote its closest point in $\mathcal{N}_\epsilon$ as f' , we have:

$$\begin{aligned} & \left| \frac{\sum_{i=0}^{k-1} \mathbf{1}\{(s_h^i, a_h^i) = (s, a)\}f(s_{h+1}^i)}{N_h^k(s, a)} - \mathbb{E}_{s' \sim P_h^*(s,a)}f(s') \right| \\ & \leq \left| \frac{\sum_{i=0}^{k-1} \mathbf{1}\{(s_h^i, a_h^i) = (s, a)\}f'(s_{h+1}^i)}{N_h^k(s, a)} - \mathbb{E}_{s' \sim P_h^*(s,a)}f'(s') \right| + \left| \frac{\sum_{i=0}^{k-1} \mathbf{1}\{(s_h^i, a_h^i) = (s, a)\}(f(s_{h+1}^i) - f'(s_{h+1}^i))}{N_h^k(s, a)} \right| \\ & \quad + \left| \mathbb{E}_{s' \sim P_h^*(s,a)}(f(s') - f'(s')) \right| \\ & \leq 2H\sqrt{\frac{S \ln(SAKH(1 + 2H\sqrt{S}/\epsilon)/\delta)}{N_h^k(s, a)}} + 2\epsilon, \end{aligned}$$

where in the last inequality we use the fact that if $\|f - f'\|_2 \leq \epsilon$, then $|f(s) - f'(s)| \leq \epsilon, \forall s \in \mathcal{S}$.

Now we can set $\epsilon = 1/K$ and use the fact that $N_h^k(s, a) \leq K$, we have:

$$\begin{aligned} & \left| \frac{\sum_{i=0}^{k-1} \mathbf{1}\{(s_h^i, a_h^i) = (s, a)\}f(s_{h+1}^i)}{N_h^k(s, a)} - \mathbb{E}_{s' \sim P_h^*(s,a)}f(s') \right| \leq 2H\sqrt{\frac{S \ln(SAKH(1 + 2HK\sqrt{S}/\delta))}{N_h^k(s, a)}} + \frac{2}{K} \\ & \leq 4H\sqrt{\frac{S \ln(4H^2S^2K^2A/\delta)}{N_h^k(s, a)}} \leq 8H\sqrt{\frac{S \ln(HSKA/\delta)}{N_h^k(s, a)}}, \end{aligned}$$

where the last inequality uses the simple inequality $\ln(4H^2S^2K^2A) \leq 4\ln(HSKA)$. This concludes the proof. $\blacksquare$

6.4 An Improved Regret Bound

In this section, we show that UCB-VI also enjoys the regret bound $\tilde{O}\left(H^2\sqrt{SAK} + H^3S^2A\right)$.

Theorem 6.6 (Improved Regret Bound for UCB-VI). *The total expected regret is bounded as follows:*

$$\begin{aligned} \mathbb{E} \left[KV^*(s_0) - \sum_{k=0}^{K-1} V^{\pi^k}(s_0) \right] \\ \leq 8eH^2\sqrt{SAK} \cdot \sqrt{\ln(SAH^2K^2)} + 2eH^3S^2A \cdot \ln(K) \ln(SAH^2K^2) = \tilde{O}\left(H^2\sqrt{SAK} + H^3S^2A\right). \end{aligned}$$

Note that comparing to the regret bound we had in the previous section, this new regret improves the S dependence in the leading order term (i.e., the term that contains $\sqrt{K}$), but at the cost of introducing a lower order term H^3S^2A .

We first give an improved version of the model error presented in Lemma 6.2.

Lemma 6.7. *With probability at least $1 - \delta$, for all h, k, s, a , and all $f : \mathcal{S} \mapsto [0, H]$, we have:*

$$\left| \left(\hat{P}_h^k(\cdot|s, a) - P_h^*(\cdot|s, a) \right)^\top f \right| \leq \frac{\mathbb{E}_{s' \sim P_h^*(\cdot|s, a)} f(s')}{H} + \frac{3H^2SL}{N_h^k(s, a)}.$$

Below we prove this lemma. Recall the definition of $\hat{P}_h^k(s'|s, a)$ in Eq. 6.1. We can use Azuma-Bernstein's inequality to bound $|\hat{P}_h^k(s'|s, a) - P_h^*(s'|s, a)|$.

Lemma 6.8. *With probability at least $1 - \delta$, for all s, a, k, h , we have:*

$$\left| \hat{P}_h^k(s'|s, a) - P_h^*(s'|s, a) \right| \leq \sqrt{\frac{2P_h^*(s'|s, a)L}{N_h^k(s, a)}} + \frac{2L}{N_h^k(s, a)},$$

where $L := \ln(SAKH/\delta)$.

Proof: Consider a fixed tuple (s, a, h, k) . Define $\mathcal{H}_{h,i}$ as the history starting from the beginning of iteration 0 to time step h at iteration i (including step h , i.e., up to s_h^i, a_h^i). Define random variables $\{X_i\}_{i \geq 0}$ as $X_i = \mathbf{1}\{s_h^i, a_h^i, s_{h+1}^i = s, a, s'\} - \mathbf{1}\{s_h^i, a_h^i = s, a\}P_h^*(s'|s, a)$. We now show that $\{X_i\}$ is a Martingale difference sequence. Note that $\mathbb{E}[X_i|\mathcal{H}_{h,i}] = 0$. We have $|X_i| \leq 1$. To use Azuma-Bernstein's inequality, we note that $\mathbb{E}[X_i^2|\mathcal{H}_{h,i}]$ is bounded as:

$$\mathbb{E}[X_i^2|\mathcal{H}_{h,i}] = \mathbf{1}\{s_h^i, a_h^i = s, a\}P_h^*(s'|s, a)(1 - P_h^*(s'|s, a)),$$

where we use the fact that the variance of a Bernoulli with parameter p is $p(1-p)$. This means that $\sum_{i=0}^{k-1} \mathbb{E}[X_i^2|\mathcal{H}_{h,i}] = P_h^*(s'|s, a)(1 - P_h^*(s'|s, a))N_h^k(s, a)$. Now apply Bernstein's inequality on the martingale difference sequence $\{X_i\}$, we have:

$$\begin{aligned} \left| \sum_{i=0}^{k-1} X_i \right| &\leq \sqrt{\frac{2P_h^*(s'|s, a)(1 - P_h^*(s'|s, a)) \ln(1/\delta)}{N_h^k(s, a)}} + \frac{2 \ln(1/\delta)}{N_h^k(s, a)} \\ &\leq \sqrt{\frac{2P_h^*(s'|s, a) \ln(1/\delta)}{N_h^k(s, a)}} + \frac{2 \ln(1/\delta)}{N_h^k(s, a)} \end{aligned}$$

Recall the definition of $L := \ln(SAHK/\delta)$. Apply a union bound over all $s \in \mathcal{S}, a \in \mathcal{A}, h \in [0, \dots, H-1], k \in [0, \dots, K-1]$, we conclude the proof. $\blacksquare$

Now we are ready to prove Lemma 6.7.

Proof:[Proof of Lemma 6.7] Let us condition on the event in Lemma 6.8 being true, of which the probability is at least $1 - \delta$.

Take any function $f : \mathcal{S} \mapsto [0, H]$, we have:

$$\begin{aligned}
& \left| \left(\widehat{P}_h^k(\cdot|s, a) - P_h^*(\cdot|s, a) \right)^\top f \right| \\
& \leq \sum_{s' \in \mathcal{S}} \left| \widehat{P}_h^k(s'|s, a) - P_h^*(s'|s, a) \right| f(s') \leq \sum_{s' \in \mathcal{S}} \sqrt{\frac{2LP_h^*(s'|s, a)f^2(s')}{N_h^k(s, a)}} + \sum_{s' \in \mathcal{S}} \frac{2Lf(s')}{N_h^k(s, a)} \\
& \leq \sum_{s' \in \mathcal{S}} \sqrt{\frac{2LP_h^*(s'|s, a)f^2(s')}{N_h^k(s, a)}} + \frac{2SHL}{N_h^k(s, a)} \leq \sqrt{S} \sqrt{\frac{\sum_{s' \in \mathcal{S}} 2LP_h^*(s'|s, a)f^2(s')}{N_h^k(s, a)}} + \frac{2SHL}{N_h^k(s, a)} \\
& = \sqrt{\frac{2H^2SL}{N_h^k(s, a)} \cdot \frac{\sum_{s' \in \mathcal{S}} P_h^*(s'|s, a)f^2(s')}{H^2}} + \frac{2SHL}{N_h^k(s, a)} \leq \frac{H^2SL}{N_h^k(s, a)} + \frac{\mathbb{E}_{s' \sim P_h^*(\cdot|s, a)} f^2(s')}{H^2} + \frac{2SHL}{N_h^k(s, a)} \\
& \leq \frac{\mathbb{E}_{s' \sim P_h^*(\cdot|s, a)} f(s')}{H} + \frac{H^2SL}{N_h^k(s, a)} + \frac{2SHL}{N_h^k(s, a)},
\end{aligned}$$

where the third inequality uses that condition that $\|f\|_\infty \leq H$, the fourth inequality uses Cauchy-Schwarz inequality, the fifth inequality uses the inequality that $ab \leq (a^2 + b^2)/2$ for any real numbers a and b , and the last inequality uses the condition that $\|f\|_\infty \leq H$ again. This concludes the proof. $\blacksquare$

So far in Lemma 6.7, we have not specify what function f we will use. The following lemma instantiate f to be $\widehat{V}_{h+1}^k - V_{h+1}^*$. Recall the bonus definition, $b_h^k(s, a) := 2H\sqrt{L/N_h^k(s, a)}$ where $L := \ln(SAHK/\delta)$.

Lemma 6.9. Assume the events in Lemma 6.7 and Lemma 6.3 are true. For all s, a, k, h , set $f := \widehat{V}_{h+1}^k - V_{h+1}^*$, we have:

$$\left| \left(\widehat{P}_h^k(\cdot|s, a) - P_h^*(\cdot|s, a) \right)^\top f \right| \leq \xi_h^k(s, a) + \frac{e}{H} \mathbb{E} \left[\sum_{\tau=h+1}^{H-1} (\xi_\tau^k(s_\tau, a_\tau) + b_\tau^k(s_\tau, a_\tau)) \mid s_h = s, a_h = a, \pi^k \right]$$

where $\xi_\tau^i(s, a) = \frac{3H^2SL}{N_\tau^i(s, a)}$, for any $\tau \in [0, \dots, H-1], i \in [0, \dots, K-1]$, and the expectation is with respect to the randomness from the event of following policy π^k from (s, a) starting from time step h .

Proof: Using Lemma 6.7 and the fact that $f := \widehat{V}_{h+1}^k - V_{h+1}^*$, we have:

$$\left| \left(\widehat{P}_h^k(\cdot|s, a) - P_h^*(\cdot|s, a) \right)^\top f \right| \leq \frac{\mathbb{E}_{s' \sim P_h^*(\cdot|s, a)} (\widehat{V}_{h+1}^k(s') - V_{h+1}^*(s'))}{H} + \xi_h^k(s, a). \quad (6.7)$$

Now consider $\widehat{V}_{h+1}^k(s') - V_{h+1}^\star(s')$, we have:

$$\begin{aligned}
\widehat{V}_{h+1}^k(s') - V_{h+1}^\star(s') &\leq \widehat{V}_{h+1}^k(s') - (r(s', \pi^k(s')) + P_{h+1}^\star(\cdot|s', \pi^k(s'))^\top V_{h+2}^\star) \\
&\leq r(s', \pi^k(s')) + \widehat{P}_{h+1}^k(\cdot|s', \pi^k(s'))^\top \widehat{V}_{h+2}^k - (r(s', \pi^k(s')) + P_{h+1}^\star(\cdot|s', \pi^k(s'))^\top V_{h+2}^\star) \\
&= \widehat{P}_{h+1}^k(\cdot|s', \pi^k(s'))^\top \widehat{V}_{h+2}^k - P_{h+1}^\star(\cdot|s', \pi^k(s'))^\top \widehat{V}_{h+2}^k + P_{h+1}^\star(\cdot|s', \pi^k(s'))^\top (\widehat{V}_{h+2}^k - V_{h+2}^\star) \\
&= \left(\widehat{P}_{h+1}^k(\cdot|s', \pi^k(s')) - P_{h+1}^\star(\cdot|s', \pi^k(s')) \right)^\top (\widehat{V}_{h+2}^k - V_{h+2}^\star) \\
&\quad + \left(\widehat{P}_{h+1}^k(\cdot|s', \pi^k(s')) - P_{h+1}^\star(\cdot|s', \pi^k(s')) \right)^\top V_{h+2}^\star \\
&\quad + P_{h+1}^\star(\cdot|s', \pi^k(s'))^\top (\widehat{V}_{h+2}^k - V_{h+2}^\star)
\end{aligned}$$

where the first inequality uses the fact that $V_{h+1}^\star(s') = \max_{a'} (r(s', a') + P_{h+1}^\star(\cdot|s', a')^\top V_{h+2}^\star)$, the second inequality uses the definition of $\widehat{V}_{h+1}^k$ and π^k . Note that the second term in the RHS of the above inequality can be upper bounded exactly by the bonus using Lemma 6.3, and the first term can be further upper bounded using the same operation that we had in Lemma 6.7, i.e.,

$$\left(\widehat{P}_{h+1}^k(\cdot|s', \pi^k(s')) - P_{h+1}^\star(\cdot|s', \pi^k(s')) \right)^\top (\widehat{V}_{h+2}^k - V_{h+2}^\star) \leq \frac{P_{h+1}^\star(\cdot|s', \pi^k(s'))^\top (\widehat{V}_{h+2}^k - V_{h+2}^\star)}{H} + \xi_{h+1}^k(s', \pi^k(s')).$$

Combine the above inequalities together, we arrive at:

$$\widehat{V}_{h+1}^k(s') - V_{h+1}^\star(s') \leq b_{h+1}^k(s', \pi^k(s')) + \xi_{h+1}^k(s', \pi^k(s')) + \left(1 + \frac{1}{H} \right) \mathbb{E}_{s'' \sim P_{h+1}^\star(s', \pi^k(s'))} (\widehat{V}_{h+2}^k(s'') - V_{h+2}^\star(s'')).$$

We can recursively apply the same operation on $\widehat{V}_{h+2}^k(s'') - V_{h+2}^\star(s'')$ till step H , we have:

$$\begin{aligned}
\widehat{V}_{h+1}^k(s') - V_{h+1}^\star(s') &\leq \mathbb{E} \left[\sum_{\tau=h+1}^{H-1} \left(1 + \frac{1}{H} \right)^{\tau-h-1} (\xi_\tau^k(s_\tau, a_\tau) + b_\tau^k(s_\tau, a_\tau)) \mid s_{h+1} = s', \pi^k \right] \\
&\leq e \mathbb{E} \left[\sum_{\tau=h+1}^{H-1} (\xi_\tau^k(s_\tau, a_\tau) + b_\tau^k(s_\tau, a_\tau)) \mid s_{h+1} = s', \pi^k \right],
\end{aligned}$$

where the last inequality uses the fact that $(1 + 1/H)^x \leq e$ for any $x \in \mathbb{N}^+$. Substitute the above inequality into Eq. 6.7, we get:

$$\left| \left(\widehat{P}_h^k(\cdot|s, a) - P_h^\star(\cdot|s, a) \right)^\top f \right| \leq \xi_h^k(s, a) + \frac{e}{H} \mathbb{E} \left[\sum_{\tau=h+1}^{H-1} (\xi_\tau^k(s_\tau, a_\tau) + b_\tau^k(s_\tau, a_\tau)) \mid s_h = s, a_h = a, \pi^k \right]$$

■

Now we are ready to derive a refined upper bound for per-episode regret.

Lemma 6.10. Assume the events in Lemma 6.9 and Lemma 6.4 (optimism) hold. For all $k \in [0, K-1]$, we have:

$$V^\star(s_0) - V^{\pi^k}(s_0) \leq e \sum_{h=0}^{H-1} \mathbb{E}_{s_h, a_h \sim d_h^{\pi^k}} [2b_h^k(s_h, a_h) + \xi_h^k(s_h, a_h)].$$

Proof: Via optimism in Lemma 6.4, we have:

$$\begin{aligned}
V^*(s_0) - V^{\pi^k}(s_0) &\leq \widehat{V}_0^k(s_0) - V_0^{\pi^k}(s_0) \leq \sum_{h=0}^{H-1} \mathbb{E}_{s_h, a_h \sim d_h^{\pi^k}} \left[b_h^k(s_h, a_h) + \left(\widehat{P}_h^k(\cdot | s_h, a_h) - P^*(\cdot | s_h, a_h) \right) \cdot \widehat{V}_{h+1}^k \right] \\
&= \sum_{h=0}^{H-1} \mathbb{E}_{s_h, a_h \sim d_h^{\pi^k}} \left[b_h^k(s_h, a_h) + \left(\widehat{P}_h^k(\cdot | s_h, a_h) - P^*(\cdot | s_h, a_h) \right)^\top \left(\widehat{V}_{h+1}^k - V_{h+1}^* \right) + \left(\widehat{P}_h^k(\cdot | s_h, a_h) - P^*(\cdot | s_h, a_h) \right)^\top V_{h+1}^* \right] \\
&\leq \sum_{h=0}^{H-1} \mathbb{E}_{s_h, a_h \sim d_h^{\pi^k}} \left[2b_h^k(s_h, a_h) + \left(\widehat{P}_h^k(\cdot | s_h, a_h) - P^*(\cdot | s_h, a_h) \right)^\top \left(\widehat{V}_{h+1}^k - V_{h+1}^* \right) \right] \\
&\leq \sum_{h=0}^{H-1} \mathbb{E}_{s_h, a_h \sim d_h^{\pi^k}} \left[2b_h^k(s_h, a_h) + \xi_h^k(s_h, a_h) + \frac{e}{H} \mathbb{E} \left[\sum_{\tau=h+1}^{H-1} (\xi_\tau^k(s_\tau, a_\tau) + b_\tau^k(s_\tau, a_\tau)) \mid \pi^k \right] \right] \\
&\leq e \sum_{h=0}^{H-1} \mathbb{E}_{s_h, a_h \sim d_h^{\pi^k}} [2b_h^k(s_h, a_h) + \xi_h^k(s_h, a_h)]
\end{aligned}$$

where the first inequality uses optimism, and the fourth inequality uses Lemma 6.9, and the last inequality rearranges by grouping same terms together. $\blacksquare$

To conclude the final regret bound, let us denote the event $\mathcal{E}_{model}$ as the joint event in Lemma 6.3 and Lemma 6.8, which holds with probability at least $1 - 2\delta$.

Proof:[Proof of Theorem 6.6] Following the derivation we have in Eq. 6.5, we decompose the expected regret bound as follows:

$$\begin{aligned}
\mathbb{E} \left[\sum_{k=0}^{K-1} V^*(s_0) - V^{\pi^k}(s_0) \right] &\leq \mathbb{E} \left[\mathbf{1}\{\mathcal{E}_{model}\} \left(\sum_{k=0}^{K-1} V^*(s_0) - V^{\pi^k}(s_0) \right) \right] + 2\delta KH \\
&\leq e \mathbb{E} \left[\sum_{k=0}^{K-1} \sum_{h=0}^{H-1} [2b_h^k(s_h^k, a_h^k) + \xi_h^k(s_h^k, a_h^k)] \right] + 2\delta KH,
\end{aligned}$$

where the last inequality uses Lemma 6.10. Using Lemma 6.5, we have that:

$$\sum_{k=0}^{K-1} \sum_{h=0}^{H-1} 2b_h^k(s_h^k, a_h^k) \leq 4H\sqrt{L} \sum_k \sum_h \frac{1}{\sqrt{N_h^k(s_h^k, a_h^k)}} \leq 8H^2\sqrt{SAKL}.$$

For $\sum_k \sum_h \xi_h^k(s_h^k, a_h^k)$, we have:

$$\begin{aligned}
\sum_{k=0}^{K-1} \sum_{h=0}^{H-1} \xi_h^k(s_h^k, a_h^k) &= 2H^2SL \sum_{k=0}^{K-1} \sum_{h=0}^{H-1} \frac{1}{N_h^k(s_h^k, a_h^k)} = 2H^2SL \sum_{h=0}^{H-1} \sum_{s,a} \sum_{i=1}^{N_h^K(s,a)} \frac{1}{i} \\
&\leq 2H^2SALHS \ln(K) = 2H^3S^2AL \ln(K),
\end{aligned}$$

where in the last inequality, we use the fact that $N_h^K(s, a) \leq K$ for all s, a, h , and the inequality $\sum_{i=1}^t 1/i \leq \ln(t)$. Putting things together, and setting $\delta = 1/(KH)$, we conclude the proof. $\blacksquare$

6.5 Phased Q -learning

to be added

6.6 Bibliographic Remarks and Further Readings

The first provably correct PAC algorithm for reinforcement learning (which finds a near optimal policy) was due to Kearns and Singh [2002], which provided the E^3 algorithm; it achieves polynomial sample complexity in tabular MDPs. Brafman and Tennenholtz [2002] presents the Rmax algorithm which provides a refined PAC analysis over E^3 . Both are model based approaches [Kakade, 2003] improves the sample complexity to be $O(S^2 A)$. Both E^3 and Rmax uses the concept of absorbing MDPs to achieve optimism and balance exploration and exploitation.

Jaksch et al. [2010] provides the first $O(\sqrt{T})$ regret bound, where T is the number of timesteps in the MDP (T is proportional to K in our setting); this dependence on T is optimal. Subsequently, Azar et al. [2017], Dann et al. [2017] provide algorithms that, asymptotically, achieve minimax regret bound in tabular MDPs. By this, we mean that for sufficiently large T (for $T \geq \Omega(|\mathcal{S}|^2)$), the results in Azar et al. [2017], Dann et al. [2017] obtain optimal dependencies on $|\mathcal{S}|$ and $|\mathcal{A}|$. The requirement that $T \geq \Omega(|\mathcal{S}|^2)$ before these bounds hold is essentially the requirement that non-trivial model accuracy is required. It is an open question to remove this dependence.

Lower bounds are provided in [Dann and Brunskill, 2015, Osband and Van Roy, 2016, Azar et al., 2017].

Further exploration strategies. Refs and discussion for Q -learning, reward free, and thompson sampling to be added...

Chapter 7

Linearly Parameterized MDPs

In this chapter, we consider learning and exploration in linearly parameterized MDPs—the linear MDP. Linear MDP generalizes tabular MDPs into MDPs with potentially infinitely many state and action pairs.

This chapter largely follows the model and analysis first provided in [Jin et al., 2020].

7.1 Setting

We consider episodic finite horizon MDP with horizon H , $\mathcal{M} = \{\mathcal{S}, \mathcal{A}, \{r_h\}_h, \{P_h\}_h, H, s_0\}$, where s_0 is a fixed initial state, $r_h : \mathcal{S} \times \mathcal{A} \mapsto [0, 1]$ and $P_h : \mathcal{S} \times \mathcal{A} \mapsto \Delta(\mathcal{S})$ are time-dependent reward function and transition kernel. Note that for time-dependent finite horizon MDP, the optimal policy will be time-dependent as well. For simplicity, we overload notations a bit and denote $\pi = \{\pi_0, \dots, \pi_{H-1}\}$, where each $\pi_h : \mathcal{S} \mapsto \mathcal{A}$. We also denote $V^\pi := V_0^\pi(s_0)$, i.e., the expected total reward of π starting at $h = 0$ and s_0 .

We define the learning protocol below. Learning happens in an episodic setting. Every episode k , learner first proposes a policy π^k based on all the history information up to the end of episode $k - 1$. The learner then executes π^k in the underlying MDP to generate a single trajectory $\tau^k = \{s_h^k, a_h^k\}_{h=0}^{H-1}$ with $a_h = \pi_h^k(s_h^k)$ and $s_{h+1}^k \sim P_h(\cdot | s_h^k, a_h^k)$. The goal of the learner is to minimize the following cumulative regret over N episodes:

$$\text{Regret} := \mathbb{E} \left[\sum_{k=0}^{K-1} (V^* - V^{\pi^k}) \right],$$

where the expectation is with respect to the randomness of the MDP environment and potentially the randomness of the learner (i.e., the learner might make decisions in a randomized fashion).

7.1.1 Low-Rank MDPs and Linear MDPs

Note that here we do not assume $\mathcal{S}$ and $\mathcal{A}$ are finite anymore. Indeed in this note, both of them could be continuous. Without any further structural assumption, the lower bounds we saw in the Generalization Lecture forbid us to get a polynomially regret bound.

The structural assumption we make in this note is a linear structure in both reward and the transition.

Definition 7.1 (Linear MDPs). Consider transition $\{P_h\}$ and $\{r_h\}_h$. A linear MDP has the following structures on r_h

and P_h :

$$r_h(s, a) = \theta_h^* \cdot \phi(s, a), \quad P_h(\cdot | s, a) = \mu_h^* \phi(s, a), \forall h$$

where ϕ is a known state-action feature map $\phi : \mathcal{S} \times \mathcal{A} \mapsto \mathbb{R}^d$, and $\mu_h^* \in \mathbb{R}^{|\mathcal{S}| \times d}$. Here ϕ, θ_h^* are **known** to the learner, while μ^* is unknown. We further assume the following norm bound on the parameters: (1) $\sup_{s,a} \|\phi(s, a)\|_2 \leq 1$, (2) $\|v^\top \mu_h^*\|_2 \leq \sqrt{d}$ for any v such that $\|v\|_\infty \leq 1$, and all h , and (3) $\|\theta_h^*\|_2 \leq W$ for all h . We assume $r_h(s, a) \in [0, 1]$ for all h and s, a .

The model essentially says that the transition matrix $P_h \in \mathbb{R}^{|\mathcal{S}| \times |\mathcal{S}| \times |\mathcal{A}|}$ has rank at most d , and $P_h = \mu_h^* \Phi$, where $\Phi \in \mathbb{R}^{d \times |\mathcal{S}| \times |\mathcal{A}|}$ and each column of Φ corresponds to $\phi(s, a)$ for a pair $s, a \in \mathcal{S} \times \mathcal{A}$.

Linear Algebra Notations For real-valued matrix A , we denote $\|A\|_2 = \sup_{x: \|x\|_2=1} \|Ax\|_2$ which denotes the maximum singular value of A . We denote $\|A\|_F$ as the Frobenius norm $\|A\|_F^2 = \sum_{i,j} A_{i,j}^2$ where $A_{i,j}$ denotes the i, j 'th entry of A . For any Positive Definite matrix Λ , we denote $x^\top \Lambda x = \|x\|_\Lambda^2$. We denote $\det(A)$ as the determinant of the matrix A . For a PD matrix Λ , we note that $\det(\Lambda) = \prod_{i=1}^d \sigma_i$ where σ_i is the eigenvalues of Λ . For notation simplicity, during inequality derivation, we will use $\lesssim, \approx$ to suppress all absolute constants. We will use $\tilde{O}$ to suppress all absolute constants and log terms.

7.2 Planning in Linear MDPs

We first study how to do value iteration in linear MDP if μ is given.

We start from $Q_{H-1}^*(s, a) = \theta_{H-1}^* \cdot \phi(s, a)$, and $\pi_{H-1}^*(s) = \operatorname{argmax}_a Q_{H-1}^*(s, a) = \operatorname{argmax}_a \theta_{H-1}^* \cdot \phi(s, a)$, and $V_{H-1}^*(s) = \operatorname{argmax}_a Q_{H-1}^*(s, a)$.

Now we do dynamic programming from $h+1$ to h :

$$Q_h^*(s, a) = \theta_h^* \cdot \phi(s, a) + \mathbb{E}_{s' \sim P_h(\cdot | s, a)} V_{h+1}^*(s') = \theta_h^* \cdot \phi(s, a) + P_h(\cdot | s, a) \cdot V_{h+1}^* = \theta_h^* \cdot \phi(s, a) + (\mu_h^* \phi(s, a))^\top V_{h+1}^* \quad (7.1)$$

$$= \phi(s, a) \cdot (\theta_h^* + (\mu_h^*)^\top V_{h+1}^*) = \phi(s, a) \cdot w_h, \quad (7.2)$$

where we denote $w_h := \theta_h^* + (\mu_h^*)^\top V_{h+1}^*$. Namely we see that $Q_h^*(s, a)$ is a linear function with respect to $\phi(s, a)$! We can continue by defining $\pi_h^*(s) = \operatorname{argmax}_a Q_h^*(s, a)$ and $V_h^*(s) = \max_a Q_h^*(s, a)$.

At the end, we get a sequence of linear Q^* , i.e., $Q_h^*(s, a) = w_h \cdot \phi(s, a)$, and the optimal policy is also simple, $\pi_h^*(s) = \operatorname{argmax}_a w_h \cdot \phi(s, a)$, for all $h = 0, \dots, H-1$.

One key property of linear MDP is that a Bellman Backup of any function $f : \mathcal{S} \mapsto \mathbb{R}$ is a linear function with respect to $\phi(s, a)$. We summarize the key property in the following claim.

Claim 7.2. Consider any arbitrary function $f : \mathcal{S} \mapsto [0, H]$. At any time step $h \in [0, \dots, H-1]$, there must exist a $w \in \mathbb{R}^d$, such that, for all $s, a \in \mathcal{S} \times \mathcal{A}$:

$$r_h(s, a) + P_h(\cdot | s, a) \cdot f = w^\top \phi(s, a).$$

The proof of the above claim is essentially the Eq. 7.1.

7.3 Learning Transition using Ridge Linear Regression

In this section, we consider the following simple question: given a dataset of state-action-next state tuples, how can we learn the transition P_h for all h ?

Note that $\mu^* \in \mathbb{R}^{|\mathcal{S}| \times d}$. Hence explicitly writing down and storing the parameterization μ^* takes time at least $|\mathcal{S}|$. We show that we can represent the model in a non-parametric way.

We consider a particular episode n . Similar to Tabular-UCBVI, we learn a model at the very beginning of the episode n using all data from the previous episodes (episode 1 to the end of the episode $n-1$). We denote such dataset as:

$$\mathcal{D}_h^n = \{s_h^i, a_h^i, s_{h+1}^i\}_{i=0}^{n-1}.$$

We maintain the following statistics using $\mathcal{D}_h^n$:

$$\Lambda_h^n = \sum_{i=0}^{n-1} \phi(s_h^i, a_h^i) \phi(s_h^i, a_h^i)^\top + \lambda I,$$

where $\lambda \in \mathbb{R}^+$ (it will be set to 1 eventually, but we keep it here for generality).

To get intuition of Λ_h^n , think about the tabular setting where $\phi(s, a)$ is a one-hot vector (zeros everywhere except that the entry corresponding to (s, a) is one). Then Λ_h^n is a diagonal matrix and the diagonal entry contains $N^n(s, a)$ —the number of times (s, a) has been visited.

We consider the following multi-variate linear regression problem. Denote $\delta(s)$ as a one-hot vector that has zero everywhere except that the entry corresponding to s is one. Denote $\epsilon_h^i = P(\cdot | s_h^i, a_h^i) - \delta(s_{h+1}^i)$. Conditioned on history $\mathcal{H}_h^i$ (history $\mathcal{H}_h^i$ denotes all information from the very beginning of the learning process up to and including (s_h^i, a_h^i)), we have:

$$\mathbb{E}[\epsilon_h^i | \mathcal{H}_h^i] = 0,$$

simply because s_{h+1}^i is sampled from $P_h(\cdot | s_h^i, a_h^i)$ conditioned on (s_h^i, a_h^i) . Also note that $\|\epsilon_h^i\|_1 \leq 2$ for all h, i .

Since $\mu_h^* \phi(s_h^i, a_h^i) = P_h(\cdot | s_h^i, a_h^i)$, and $\delta(s_{h+1}^i)$ is an unbiased estimate of $P_h(\cdot | s_h^i, a_h^i)$ conditioned on s_h^i, a_h^i , it is reasonable to learn μ^* via regression from $\phi(s_h^i, a_h^i)$ to $\delta(s_{h+1}^i)$. This leads us to the following ridge linear regression:

$$\hat{\mu}_h^n = \operatorname{argmin}_{\mu \in \mathbb{R}^{|\mathcal{S}| \times d}} \sum_{i=0}^{n-1} \|\mu \phi(s_h^i, a_h^i) - \delta(s_{h+1}^i)\|_2^2 + \lambda \|\mu\|_F^2.$$

Ridge linear regression has the following closed-form solution:

$$\hat{\mu}_h^n = \sum_{i=0}^{n-1} \delta(s_{h+1}^i) \phi(s_h^i, a_h^i)^\top (\Lambda_h^n)^{-1} \quad (7.3)$$

Note that $\hat{\mu}_h^n \in \mathbb{R}^{|\mathcal{S}| \times d}$, so we never want to explicitly store it. Note that we will always use $\hat{\mu}_h^n$ together with a specific s, a pair and a value function V (think about value iteration case), i.e., we care about $\hat{P}_h^n(\cdot | s, a) \cdot V := (\hat{\mu}_h^n \phi(s, a)) \cdot V$, which can be re-written as:

$$\hat{P}_h^n(\cdot | s, a) \cdot V := (\hat{\mu}_h^n \phi(s, a)) \cdot V = \phi(s, a)^\top \sum_{i=0}^{n-1} (\Lambda_h^n)^{-1} \phi(s_h^i, a_h^i) V(s_{h+1}^i),$$

where we use the fact that $\delta(s)^\top V = V(s)$. Thus the operator $\hat{P}_h^n(\cdot|s, a) \cdot V$ simply requires storing all data and can be computed via simple linear algebra and the computation complexity is simply $\text{poly}(d, n)$ —no poly dependency on $|\mathcal{S}|$.

Let us calculate the difference between $\hat{\mu}_h^n$ and μ_h^* .

Lemma 7.3 (Difference between $\hat{\mu}_h$ and μ_h^*). *For all n and h , we must have:*

$$\hat{\mu}_h^n - \mu_h^* = -\lambda \mu_h^* (\Lambda_h^n)^{-1} + \sum_{i=1}^{n-1} \epsilon_h^i \phi(s_h^i, a_h^i)^\top (\Lambda_h^n)^{-1}.$$

Proof: We start from the closed-form solution of $\hat{\mu}_h^n$:

$$\begin{aligned} \hat{\mu}_h^n &= \sum_{i=0}^{n-1} \delta(s_{h+1}^i) \phi(s_h^i, a_h^i)^\top (\Lambda_h^n)^{-1} = \sum_{i=0}^{n-1} (P(\cdot|s_h^i, a_h^i) + \epsilon_h^n) \phi(s_h^i, a_h^i)^\top (\Lambda_h^n)^{-1} \\ &= \sum_{i=0}^{n-1} (\mu_h^* \phi(s_h^i, a_h^i) + \epsilon_h^i) \phi(s_h^i, a_h^i)^\top (\Lambda_h^n)^{-1} = \sum_{i=0}^{n-1} \mu_h^* \phi(s_h^i, a_h^i) \phi(s_h^i, a_h^i)^\top (\Lambda_h^n)^{-1} + \sum_{i=0}^{n-1} \epsilon_h^i \phi(s_h^i, a_h^i)^\top (\Lambda_h^n)^{-1} \\ &= \sum_{i=0}^{n-1} \mu_h^* \phi(s_h^i, a_h^i) \phi(s_h^i, a_h^i)^\top (\Lambda_h^n)^{-1} + \sum_{i=0}^{n-1} \epsilon_h^i \phi(s_h^i, a_h^i)^\top (\Lambda_h^n)^{-1} \\ &= \mu_h^* (\Lambda_h^n - \lambda I) (\Lambda_h^n)^{-1} + \sum_{i=0}^{n-1} \epsilon_h^i \phi(s_h^i, a_h^i)^\top (\Lambda_h^n)^{-1} = \mu_h^* - \lambda \mu_h^* (\Lambda_h^n)^{-1} + \sum_{i=0}^{n-1} \epsilon_h^i \phi(s_h^i, a_h^i)^\top (\Lambda_h^n)^{-1}. \end{aligned}$$

Rearrange terms, we conclude the proof. ■

Lemma 7.4. *Fix $V : \mathcal{S} \mapsto [0, H]$. For all n and $s, a \in \mathcal{S} \times \mathcal{A}$, and h , with probability at least $1 - \delta$, we have:*

$$\left\| \sum_{i=0}^{n-1} \phi(s_h^i, a_h^i) (V^\top \epsilon_h^i) \right\|_{(\Lambda_h^n)^{-1}} \leq 3H \sqrt{\ln \frac{H \det(\Lambda_h^n)^{1/2} \det(\lambda I)^{-1/2}}{\delta}}.$$

Proof: We first check the noise terms $\{V^\top \epsilon_h^i\}_{h,i}$. Since V is independent of the data (it's a pre-fixed function), and by linear property of expectation, we have:

$$\mathbb{E} [V^\top \epsilon_h^i | \mathcal{H}_h^i] = 0, \quad |V^\top \epsilon_h^i| \leq \|V\|_\infty \|\epsilon_h^i\|_1 \leq 2H, \forall h, i.$$

Hence, this is a Martingale difference sequence. Using the Self-Normalized vector-valued Martingale Bound (Lemma A.9), we have that for all n , with probability at least $1 - \delta$:

$$\left\| \sum_{i=0}^{n-1} \phi(s_h^i, a_h^i) (V^\top \epsilon_h^i) \right\|_{(\Lambda_h^n)^{-1}} \leq 3H \sqrt{\ln \frac{\det(\Lambda_h^n)^{1/2} \det(\lambda I)^{-1/2}}{\delta}}.$$

Apply union bound over all $h \in [H]$, we get that with probability at least $1 - \delta$, for all n, h :

$$\left\| \sum_{i=0}^{n-1} \phi(s_h^i, a_h^i) (V^\top \epsilon_h^i) \right\|_{(\Lambda_h^n)^{-1}} \leq 3H \sqrt{\ln \frac{H \det(\Lambda_h^n)^{1/2} \det(\lambda I)^{-1/2}}{\delta}}. \quad (7.4)$$

■

7.4 Uniform Convergence via Covering

Now we take a detour first and consider how to achieve a uniform convergence result over a function class $\mathcal{F}$ that contains infinitely many functions. Previously we know how to get uniform convergence if $\mathcal{F}$ is finite—we simply do a union bound. However, when $\mathcal{F}$ contains infinitely many functions, we cannot simply apply a union bound. We will use the covering argument here.

Consider the following ball with radius R : $\Theta = \{\theta \in \mathbb{R}^d : \|\theta\|_2 \leq R \in \mathbb{R}^+\}$. Fix an ϵ . An ϵ -net $\mathcal{N}_\epsilon \subset \Theta$ is a set such that for any $\theta \in \Theta$, there exists a $\theta' \in \mathcal{N}_\epsilon$, such that $\|\theta - \theta'\|_2 \leq \epsilon$. We call the smallest ϵ -net as ϵ -cover. Abuse notations a bit, we simply denote $\mathcal{N}_\epsilon$ as the ϵ -cover.

The ϵ -covering number is the size of ϵ -cover $\mathcal{N}_\epsilon$. We define the covering dimension as $\ln(|\mathcal{N}_\epsilon|)$

Lemma 7.5. *The ϵ -covering number of the ball $\Theta = \{\theta \in \mathbb{R}^d : \|\theta\|_2 \leq R \in \mathbb{R}^+\}$ is upper bounded by $(1 + 2R/\epsilon)^d$.*

We can extend the above definition to a function class. Specifically, we look at the following function. For a triple of (w, β, Λ) where $w \in \mathbb{R}^d$ and $\|w\|_2 \leq L$, $\beta \in [0, B]$, and Λ such that $\sigma_{\min}(\Lambda) \geq \lambda$, we define $f_{w, \beta, \Lambda} : \mathcal{S} \mapsto [0, R]$ as follows:

$$f_{w, \beta, \Lambda}(s) = \min \left\{ \max_a \left(w^\top \phi(s, a) + \beta \sqrt{\phi(s, a)^\top \Lambda^{-1} \phi(s, a)} \right), H \right\}, \forall s \in \mathcal{S}. \quad (7.5)$$

We denote the function class $\mathcal{F}$ as:

$$\mathcal{F} = \{f_{w, \beta, \Lambda} : \|w\|_2 \leq L, \beta \in [0, B], \sigma_{\min}(\Lambda) \geq \lambda\}. \quad (7.6)$$

Note that $\mathcal{F}$ contains infinitely many functions as the parameters are continuous. However we will show that it has finite covering number that scales exponentially with respect to the number of parameters in (w, β, Λ) .

Why do we look at $\mathcal{F}$? As we will see later in this chapter $\mathcal{F}$ contains all possible $\hat{Q}_h$ functions one could encounter during the learning process.

Lemma 7.6 (ϵ -covering dimension of $\mathcal{F}$). *Consider $\mathcal{F}$ defined in Eq. 7.6. Denote its ϵ -cover as $\mathcal{N}_\epsilon$ with the ℓ_∞ norm as the distance metric, i.e., $d(f_1, f_2) = \|f_1 - f_2\|_\infty$ for any $f_1, f_2 \in \mathcal{F}$. We have that:*

$$\ln(|\mathcal{N}_\epsilon|) \leq d \ln(1 + 6L/\epsilon) + \ln(1 + 6B/(\sqrt{\lambda}\epsilon)) + d^2 \ln(1 + 18B^2\sqrt{d}/(\lambda\epsilon^2)).$$

Note that the ϵ -covering dimension scales quadratically with respect to d .

Proof: We start from building a net over the parameter space (w, β, Λ) , and then we convert the net over parameter space to an ϵ -net over $\mathcal{F}$ under the ℓ_∞ distance metric.

We pick two functions that corresponding to parameters (w, β, Λ) and $(\hat{w}, \hat{\beta}, \hat{\Lambda})$.

$$\begin{aligned}
|f(s) - \hat{f}(s)| &\leq \left| \max_a \left(w^\top \phi(s, a) + \beta \sqrt{\phi(s, a)^\top \Lambda^{-1} \phi(s, a)} \right) - \max_a \left(\hat{w}^\top \phi(s, a) + \hat{\beta} \sqrt{\phi(s, a)^\top \hat{\Lambda}^{-1} \phi(s, a)} \right) \right| \\
&\leq \max_a \left| \left(w^\top \phi(s, a) + \beta \sqrt{\phi(s, a)^\top \Lambda^{-1} \phi(s, a)} \right) - \left(\hat{w}^\top \phi(s, a) + \hat{\beta} \sqrt{\phi(s, a)^\top \hat{\Lambda}^{-1} \phi(s, a)} \right) \right| \\
&\leq \max_a |(w - \hat{w})^\top \phi(s, a)| + \max_a \left| (\beta - \hat{\beta}) \sqrt{\phi(s, a)^\top \Lambda^{-1} \phi(s, a)} \right| \\
&\quad + \max_a \left| \hat{\beta} \left(\sqrt{\phi(s, a)^\top \Lambda^{-1} \phi(s, a)} - \sqrt{\phi(s, a)^\top \hat{\Lambda}^{-1} \phi(s, a)} \right) \right| \\
&\leq \|w - \hat{w}\|_2 + |\beta - \hat{\beta}|/\sqrt{\lambda} + B \sqrt{|\phi(s, a)^\top (\Lambda^{-1} - \hat{\Lambda}^{-1}) \phi(s, a)|} \\
&\leq \|w - \hat{w}\|_2 + |\beta - \hat{\beta}|/\sqrt{\lambda} + B \sqrt{\|\Lambda^{-1} - \hat{\Lambda}^{-1}\|_F}
\end{aligned}$$

Note that Λ^{-1} is a PD matrix with $\sigma_{\max}(\Lambda^{-1}) \leq 1/\lambda$.

Now we consider the $\epsilon/3$ -Net $\mathcal{N}_{\epsilon/3, w}$ over $\{w : \|w\|_2 \leq L\}$, $\sqrt{\lambda}\epsilon/3$ -net $\mathcal{N}_{\sqrt{\lambda}\epsilon/3, \beta}$ over interval $[0, B]$ for β , and $\epsilon^2/(9B^2)$ -net $\mathcal{N}_{\epsilon^2/(9B^2), \Lambda}$ over $\{\Lambda : \|\Lambda\|_F \leq \sqrt{d}/\lambda\}$. The product of these three nets provide a ϵ -cover for $\mathcal{F}$, which means that that size of the ϵ -net $\mathcal{N}_\epsilon$ for $\mathcal{F}$ is upper bounded as:

$$\begin{aligned}
\ln |\mathcal{N}_\epsilon| &\leq \ln |\mathcal{N}_{\epsilon/3, w}| + \ln |\mathcal{N}_{\sqrt{\lambda}\epsilon/3, \beta}| + \ln |\mathcal{N}_{\epsilon^2/(9B^2), \Lambda}| \\
&\leq d \ln(1 + 6L/\epsilon) + \ln(1 + 6B/(\sqrt{\lambda}\epsilon)) + d^2 \ln(1 + 18B^2\sqrt{d}/(\lambda\epsilon^2)).
\end{aligned}$$

■

Remark Covering gives a way to represent the complexity of function class (or hypothesis class). Relating to VC, covering number is upper bound roughly by $\exp(d)$ with d being the VC-dimension. However, there are cases where VC-dimension is infinite, but covering number is finite.

Now we can build a uniform convergence argument for all $f \in \mathcal{F}$.

Lemma 7.7 (Uniform Convergence Results). *Set $\lambda = 1$. Fix $\delta \in (0, 1)$. For all n, h , all s, a , and all $f \in \mathcal{F}$, with probability at least $1 - \delta$, we have:*

$$\left| \left(\hat{P}_h^n(\cdot | s, a) - P(\cdot | s, a) \right) \cdot f \right| \lesssim H \|\phi(s, a)\|_{(\Lambda_h^n)^{-1}} \left(\sqrt{d \ln(1 + 6L\sqrt{N})} + d \sqrt{\ln(1 + 18B^2\sqrt{d}N)} + \sqrt{\ln \frac{H}{\delta}} \right).$$

Proof: Recall Lemma 7.4, we have with probability at least $1 - \delta$, for all n, h , for a pre-fixed V (independent of the random process):

$$\left\| \sum_{i=1}^{n-1} \phi(s_h^i, a_h^i) (V^\top \epsilon_h^i) \right\|_{(\Lambda_h^n)^{-1}}^2 \leq 9H^2 \ln \frac{H \det(\Lambda_h^n)^{1/2} \det(\lambda I)^{-1/2}}{\delta} \leq 9H^2 \left(\ln \frac{H}{\delta} + d \ln(1 + N) \right)$$

where we have used the fact that $\|\phi\|_2 \leq 1$, $\lambda = 1$, and $\|\Lambda_h^n\|_2 \leq N + 1$.

Denote the ϵ -cover of $\mathcal{F}$ as $\mathcal{N}_\epsilon$. With an application of a union bound over all functions in $\mathcal{N}_\epsilon$, we have that with probability at least $1 - \delta$, for all $V \in \mathcal{N}_\epsilon$, all n, h , we have:

$$\left\| \sum_{i=1}^{n-1} \phi(s_h^i, a_h^i) (V^\top \epsilon_h^i) \right\|_{(\Lambda_h^n)^{-1}}^2 \leq 9H^2 \left(\ln \frac{H}{\delta} + \ln(|\mathcal{N}_\epsilon|) + d \ln(1 + N) \right).$$

Recall Lemma 7.6, substitute the expression of $\ln |\mathcal{N}_\epsilon|$ into the above inequality, we get:

$$\left\| \sum_{i=1}^{n-1} \phi(s_h^i, a_h^i) (V^\top \epsilon_h^i) \right\|_{(\Lambda_h^n)^{-1}}^2 \leq 9H^2 \left(\ln \frac{H}{\delta} + d \ln(1 + 6L/\epsilon) + d^2 \ln(1 + 18B^2 \sqrt{d}/\epsilon^2) + d \ln(1 + N) \right).$$

Now consider an arbitrary $f \in \mathcal{F}$. By the definition of ϵ -cover, we know that for f , there exists a $V \in \mathcal{N}_\epsilon$, such that $\|f - V\|_\infty \leq \epsilon$. Thus, we have:

$$\begin{aligned} \left\| \sum_{i=1}^{n-1} \phi(s_h^i, a_h^i) (f^\top \epsilon_h^i) \right\|_{(\Lambda_h^n)^{-1}}^2 &\leq 2 \left\| \sum_{i=1}^{n-1} \phi(s_h^i, a_h^i) (V^\top \epsilon_h^i) \right\|_{(\Lambda_h^n)^{-1}}^2 + 2 \left\| \sum_{i=1}^{n-1} \phi(s_h^i, a_h^i) ((V - f)^\top \epsilon_h^i) \right\|_{(\Lambda_h^n)^{-1}}^2 \\ &\leq 2 \left\| \sum_{i=1}^{n-1} \phi(s_h^i, a_h^i) (V^\top \epsilon_h^i) \right\|_{(\Lambda_h^n)^{-1}}^2 + 8\epsilon^2 N \\ &\leq 9H^2 \left(\ln \frac{H}{\delta} + d \ln(1 + 6L/\epsilon) + d^2 \ln(1 + 18B^2 \sqrt{d}/\epsilon^2) + d \ln(1 + N) \right) + 8\epsilon^2 N, \end{aligned}$$

where in the second inequality we use the fact that $\left\| \sum_{i=1}^{n-1} \phi(s_h^i, a_h^i) (V - f)^\top \epsilon_h^i \right\|_{(\Lambda_h^n)^{-1}}^2 \leq 4\epsilon^2 N$, which is from

$$\left\| \sum_{i=1}^{n-1} \phi(s_h^i, a_h^i) (V - f)^\top \epsilon_h^i \right\|_{(\Lambda_h^n)^{-1}}^2 \leq \left\| \sum_{i=1}^{n-1} \phi(s_h^i, a_h^i) \right\|_{(\Lambda_h^n)^{-1}}^2 \leq \frac{4\epsilon^2}{\lambda} \left\| \sum_{i=1}^{n-1} \phi(s_h^i, a_h^i) \right\|_2^2 \leq 4\epsilon^2 N.$$

Set $\epsilon = 1/\sqrt{N}$, we get:

$$\begin{aligned} \left\| \sum_{i=1}^{n-1} \phi(s_h^i, a_h^i) (f^\top \epsilon_h^i) \right\|_{(\Lambda_h^n)^{-1}}^2 &\leq 9H^2 \left(\ln \frac{H}{\delta} + d \ln(1 + 6L\sqrt{N}) + d^2 \ln(1 + 18B^2 \sqrt{d}N) + d \ln(1 + N) \right) + 8 \\ &\lesssim H^2 \left(\ln \frac{H}{\delta} + d \ln(1 + 6L\sqrt{N}) + d^2 \ln(1 + 18B^2 \sqrt{d}N) \right), \end{aligned}$$

where we recall $\lesssim$ ignores absolute constants.

Now recall that we can express $(\hat{P}_h^n(\cdot|s, a) - P(\cdot|s, a)) \cdot f = \phi(s, a)^\top (\hat{\mu}_h^n - \mu_h^*)^\top f$. Recall Lemma 7.3, we have:

$$\begin{aligned} |(\hat{\mu}_h^n \phi(s, a) - \mu_h^* \phi(s, a)) \cdot f| &\leq \left| \lambda \phi(s, a)^\top (\Lambda_h^n)^{-1} (\mu_h^*)^\top f \right| + \left| \sum_{i=1}^{n-1} \phi(s, a)^\top (\Lambda_h^n)^{-1} \phi(s_h^i, a_h^i) (\epsilon_h^i)^\top f \right| \\ &\lesssim H \sqrt{d} \|\phi(s, a)\|_{(\Lambda_h^n)^{-1}} + \|\phi(s, a)\|_{(\Lambda_h^n)^{-1}} \sqrt{\left(H^2 \left(\ln \frac{H}{\delta} + d \ln(1 + 6L\sqrt{N}) + d^2 \ln(1 + 18B^2 \sqrt{d}N) \right) \right)} \\ &\approx H \|\phi(s, a)\|_{(\Lambda_h^n)^{-1}} \left(\sqrt{\ln \frac{H}{\delta}} + \sqrt{d \ln(1 + 6L\sqrt{N})} + d \sqrt{\ln(1 + 18B^2 \sqrt{d}N)} \right). \end{aligned}$$

■

7.5 Algorithm

Our algorithm, Upper Confidence Bound Value Iteration (UCB-VI) will use reward bonus to ensure optimism. Specifically, we will use the following reward bonus, which is motivated from the reward bonus used in linear bandit:

$$b_h^n(s, a) = \beta \sqrt{\phi(s, a)^\top (\Lambda_h^n)^{-1} \phi(s, a)}, \quad (7.7)$$

where β contains poly of H and d , and other constants and log terms. Again to gain intuition, please think about what this bonus would look like when we specialize linear MDP to tabular MDP.

Algorithm 6 UCBVI for Linear MDPs

- 1: **Input: parameters** β, λ
 - 2: **for** $n = 1 \dots N$ **do**
 - 3: Compute $\hat{P}_h^n$ for all h (Eq. 7.3)
 - 4: Compute reward bonus b_h^n for all h (Eq. 7.7)
 - 5: Run Value-Iteration on $\{\hat{P}_h^n, r_h + b_h^n\}_{h=0}^{H-1}$ (Eq. 7.8)
 - 6: Set π^n as the returned policy of VI.
 - 7: **end for**
-

With the above setup, now we describe the algorithm. Every episode n , we learn the model $\hat{\mu}_h^n$ via ridge linear regression. We then form the quadratic reward bonus as shown in Eq. 7.7. With that, we can perform the following truncated Value Iteration (always truncate the Q function at H):

$$\begin{aligned} \hat{V}_H^n(s) &= 0, \forall s, \\ \hat{Q}_h^n(s, a) &= \theta^* \cdot \phi(s, a) + \beta \sqrt{\phi(s, a)^\top (\Lambda_h^n)^{-1} \phi(s, a)} + \phi(s, a)^\top (\hat{\mu}_h^n)^\top \hat{V}_{h+1}^n \\ &= \beta \sqrt{\phi(s, a)^\top (\Lambda_h^n)^{-1} \phi(s, a)} + (\theta^* + (\hat{\mu}_h^n)^\top \hat{V}_{h+1}^n)^\top \phi(s, a), \\ \hat{V}_h^n(s) &= \min_a \{ \hat{Q}_h^n(s, a), H \}, \quad \pi_h^n(s) = \operatorname{argmax}_a \hat{Q}_h^n(s, a). \end{aligned} \quad (7.8)$$

Note that above $\hat{Q}_h^n$ contains two components: a quadratic component and a linear component. And $\hat{V}_h^n$ has the format of $f_{w, \beta, \Lambda}$ defined in Eq. 7.5.

The following lemma bounds the norm of linear weights in $\hat{Q}_h^n$.

Lemma 7.8. Assume $\beta \in [0, B]$. For all n, h , we have $\hat{V}_h^n$ is in the form of Eq. 7.5, and $\hat{V}_h^n$ falls into the following class:

$$\mathcal{V} = \{f_{w, \beta, \Lambda} : \|w\|_2 \leq W + \frac{HN}{\lambda}, \beta \in [0, B], \sigma_{\min}(\Lambda) \geq \lambda\}. \quad (7.9)$$

Proof: We just need to show that $\theta^* + (\hat{\mu}_h^n)^\top \hat{V}_{h+1}^n$ has its ℓ_2 norm bounded. This is easy to show as we always have $\|\hat{V}_{h+1}^n\|_\infty \leq H$ as we do truncation at Value Iteration:

$$\left\| \theta^* + (\hat{\mu}_h^n)^\top \hat{V}_{h+1}^n \right\|_2 \leq W + \left\| (\hat{\mu}_h^n)^\top \hat{V}_{h+1}^n \right\|_2.$$

Now we use the closed-form of $\hat{\mu}_h^n$ from Eq. 7.3:

$$\left\| (\hat{\mu}_h^n)^\top \hat{V}_{h+1}^n \right\|_2 = \left\| \sum_{i=1}^{n-1} \hat{V}_{h+1}^n(s_{h+1}^i) \phi(s_h^i, a_h^i)^\top (\Lambda_h^n)^{-1} \right\|_2 \leq H \left\| (\Lambda_h^n)^{-1} \sum_{i=0}^{n-1} \phi(s_h^i, a_h^i) \right\|_2 \leq \frac{Hn}{\lambda},$$

where we use the fact that $\|\hat{V}_{h+1}^n\|_\infty \leq H$, $\sigma_{\max}(\Lambda^{-1}) \leq 1/\lambda$, and $\sup_{s,a} \|\phi(s, a)\|_2 \leq 1$. ■

7.6 Analysis of UCBVI for Linear MDPs

In this section, we prove the following regret bound for UCBVI.

Theorem 7.9 (Regret Bound). *Set $\beta = \tilde{O}(Hd)$, $\lambda = 1$. UCBVI (Algorithm 6) achieves the following regret bound:*

$$\mathbb{E} \left[NV^* - \sum_{i=0}^N V^{\pi^n} \right] \leq \tilde{O} \left(H^2 \sqrt{d^3 N} \right)$$

The main steps of the proof are similar to the main steps of UCBVI in tabular MDPs. We first prove optimism via induction, and then we use optimism to upper bound per-episode regret. Finally we use simulation lemma to decompose the per-episode regret.

In this section, to make notation simple, we set $\lambda = 1$ directly.

7.6.1 Proving Optimism

Proving optimism requires us to first bound model error which we have built in the uniform convergence result shown in Lemma 7.7, namely, the bound we get for $(\hat{P}_h^n(\cdot|s, a) - P(\cdot|s, a)) \cdot f$ for all $f \in \mathcal{V}$. Recall Lemma 7.7 but this time replacing $\mathcal{F}$ by $\mathcal{V}$ defined in Eq. 7.9. With probability at least $1 - \delta$, for all n, h, s, a and for all $f \in \mathcal{V}$,

$$\begin{aligned} \left| (\hat{P}_h^n(\cdot|s, a) - P(\cdot|s, a)) \cdot f \right| &\leq H \|\phi(s, a)\|_{(\Lambda_h^n)^{-1}} \left(\sqrt{\ln \frac{H}{\delta}} + \sqrt{d \ln(1 + 6(W + HN)\sqrt{N})} + d\sqrt{\ln(1 + 18B^2\sqrt{dN})} \right) \\ &\lesssim Hd \|\phi(s, a)\|_{(\Lambda_h^n)^{-1}} \left(\sqrt{\ln \frac{H}{\delta}} + \sqrt{\ln(WN + HN^2)} + \sqrt{\ln(B^2dN)} \right) \\ &\lesssim \|\phi(s, a)\|_{(\Lambda_h^n)^{-1}} Hd \underbrace{\left(\sqrt{\ln \frac{H}{\delta}} + \sqrt{\ln(W + H)} + \sqrt{\ln B} + \sqrt{\ln d} + \sqrt{\ln N} \right)}_{:=\beta}. \end{aligned}$$

Denote the above inequality as event $\mathcal{E}_{model}$. Below we are going to condition on $\mathcal{E}_{model}$ being hold. Note that here for notation simplicity, we denote

$$\beta = Hd \left(\sqrt{\ln \frac{H}{\delta}} + \sqrt{\ln(W + H)} + \sqrt{\ln B} + \sqrt{\ln d} + \sqrt{\ln N} \right) = \tilde{O}(Hd).$$

remark Note that in the definition of $\mathcal{V}$ (Eq. 7.9), we have $\beta \in [0, B]$. And in the above formulation of β , note that B appears inside a log term. So we need to set B such that $\beta \leq B$ and we can get the correct B by solving the inequality $\beta \leq B$ for B .

Lemma 7.10 (Optimism). *Assume event $\mathcal{E}_{model}$ is true. for all n and h ,*

$$\hat{V}_h^n(s) \geq V_h^*(s), \forall s.$$

Proof: We consider a fixed episode n . We prove via induction. Assume that $\widehat{V}_{h+1}^n(s) \geq V_{h+1}^*(s)$ for all s . For time step h , we have:

$$\begin{aligned} \widehat{Q}_h^n(s, a) - Q_h^*(s, a) &= \theta^* \cdot \phi(s, a) + \beta \sqrt{\phi(s, a)^\top (\Lambda_h^n)^{-1} \phi(s, a)} + \phi(s, a)^\top (\widehat{\mu}_h^n)^\top \widehat{V}_{h+1}^n - \theta^* \cdot \phi(s, a) - \phi(s, a)^\top (\mu_h^*)^\top V_{h+1}^* \\ &\geq \beta \sqrt{\phi(s, a)^\top (\Lambda_h^n)^{-1} \phi(s, a)} + \phi(s, a)^\top (\widehat{\mu}_h^n - \mu_h^*)^\top \widehat{V}_{h+1}^n, \end{aligned}$$

where in the last inequality we use the inductive hypothesis that $\widehat{V}_{h+1}^n(s) \geq V_{h+1}^*(s)$, and $\mu_h^* \phi(s, a)$ is a valid distribution (note that $\widehat{\mu}_h^n \phi(s, a)$ is not necessarily a valid distribution). We need to show that the bonus is big enough to offset the model error $\phi(s, a)^\top (\widehat{\mu}_h^n - \mu_h^*)^\top \widehat{V}_{h+1}^n$. Since we have event $\mathcal{E}_{model}$ being true, we have that:

$$\left| (\widehat{P}_h^n(\cdot|s, a) - P(\cdot|s, a)) \cdot \widehat{V}_{h+1}^n \right| \lesssim \beta \|\phi(s, a)\|_{(\Lambda_h^n)^{-1}},$$

as by the construction of $\mathcal{V}$, we know that $\widehat{V}_{h+1}^n \in \mathcal{V}$.

This concludes the proof. ■

7.6.2 Regret Decomposition

Now we can upper bound the per-episode regret as follows:

$$V^* - V^{\pi_n} \leq \widehat{V}_0^n(s_0) - V_0^{\pi_n}(s_0).$$

We can further bound the RHS of the above inequality using simulation lemma. Recall Eq. 6.4 that we derived in the note for tabular MDP (Chapter 6):

$$\widehat{V}_0^n(s_0) - V_0^{\pi_n}(s_0) \leq \sum_{h=0}^{H-1} \mathbb{E}_{s, a \sim d_h^{\pi_n}} \left[b_h^n(s, a) + \left(\widehat{P}_h^n(\cdot|s, a) - P(\cdot|s, a) \right) \cdot \widehat{V}_{h+1}^n \right].$$

(recall that the simulation lemma holds for any MDPs—it's not specialized to tabular).

In the event $\mathcal{E}_{model}$, we already know that for any s, a, h, n , we have $\left(\widehat{P}_h^n(\cdot|s, a) - P(\cdot|s, a) \right) \cdot \widehat{V}_{h+1}^n \lesssim \beta \|\phi(s, a)\|_{(\Lambda_h^n)^{-1}} = b_h^n(s, a)$. Hence, under $\mathcal{E}_{model}$, we have:

$$\widehat{V}_0^n(s_0) - V_0^{\pi_n}(s_0) \leq \sum_{h=0}^{H-1} \mathbb{E}_{s, a \sim d_h^{\pi_n}} [2b_h^n(s, a)] \lesssim \sum_{h=0}^{H-1} \mathbb{E}_{s, a \sim d_h^{\pi_n}} [b_h^n(s, a)].$$

Sum over all episodes, we have the following statement.

Lemma 7.11 (Regret Bound). *Assume the event $\mathcal{E}_{model}$ holds. We have:*

$$\sum_{n=0}^{N-1} (V_0^*(s_0) - V_0^{\pi_n}(s_0)) \leq \sum_{n=0}^{N-1} \sum_{h=0}^{H-1} \mathbb{E}_{s_h^n, a_h^n \sim d_h^{\pi_n}} [b_h^n(s_h^n, a_h^n)]$$

7.6.3 Concluding the Final Regret Bound

We first consider the following elliptical potential argument, which is similar to what we have seen in the linear bandit lecture.

Lemma 7.12 (Elliptical Potential). *Consider an arbitrary sequence of state action pairs s_h^i, a_h^i . Assume $\sup_{s,a} \|\phi(s, a)\|_2 \leq 1$. Denote $\Lambda_h^n = I + \sum_{i=0}^{n-1} \phi(s_h^i, a_h^i) \phi(s_h^i, a_h^i)^\top$. We have:*

$$\sum_{i=0}^{N-1} \phi(s_h^i, a_h^i) (\Lambda_h^i)^{-1} \phi(s_h^i, a_h^i) \leq 2 \ln \left(\frac{\det(\Lambda_h^{N+1})}{\det(I)} \right) \lesssim 2d \ln(N).$$

Proof: By the Lemma 3.7 and 3.8 in the linear bandit lecture note,

$$\begin{aligned} \sum_{i=1}^N \phi(s_h^i, a_h^i) (\Lambda_h^i)^{-1} \phi(s_h^i, a_h^i) &\leq 2 \sum_{i=1}^N \ln(1 + \phi(s_h^i, a_h^i) (\Lambda_h^i)^{-1} \phi(s_h^i, a_h^i)) \\ &\leq 2 \ln \left(\frac{\det(\Lambda_h^{N+1})}{\det(I)} \right) \\ &\leq 2d \ln(1 + \frac{N+1}{d\lambda}) \lesssim 2d \ln(N). \end{aligned}$$

where the first inequality uses that for $0 \leq y \leq 1$, $\ln(1+y) \geq y/2$. ■

Now we use Lemma 7.11 together with the above inequality to conclude the proof.

Proof:[Proof of main Theorem 7.9]

We split the expected regret based on the event $\mathcal{E}_{model}$.

$$\begin{aligned} \mathbb{E} \left[NV^* - \sum_{n=1}^N V^{\pi_n} \right] &= \mathbb{E} \left[\mathbf{1}_{\{\mathcal{E}_{model} \text{ holds}\}} \left(NV^* - \sum_{n=1}^N V^{\pi_n} \right) \right] \\ &\quad + \mathbb{E} \left[\mathbf{1}_{\{\mathcal{E}_{model} \text{ doesn't hold}\}} \left(NV^* - \sum_{n=1}^N V^{\pi_n} \right) \right] \\ &\leq \mathbb{E} \left[\mathbf{1}_{\{\mathcal{E}_{model} \text{ holds}\}} \left(NV^* - \sum_{n=1}^N V^{\pi_n} \right) \right] + \delta NH \\ &\lesssim \mathbb{E} \left[\sum_{n=1}^N \sum_{h=0}^{H-1} b_h^n(s_h^n, a_h^n) \right] + \delta NH. \end{aligned}$$

Note that:

$$\begin{aligned} \sum_{n=1}^N \sum_{h=0}^{H-1} b_h^n(s_h^n, a_h^n) &= \beta \sum_{n=1}^N \sum_{h=0}^{H-1} \sqrt{\phi(s_h^n, a_h^n)^\top (\Lambda_h^n)^{-1} \phi(s_h^n, a_h^n)} \\ &= \beta \sum_{h=0}^{H-1} \sum_{n=1}^N \sqrt{\phi(s_h^n, a_h^n)^\top (\Lambda_h^n)^{-1} \phi(s_h^n, a_h^n)} \\ &\leq \beta \sum_{h=0}^{H-1} \sqrt{N \sum_{n=1}^N \phi(s_h^n, a_h^n) (\Lambda_h^n)^{-1} \phi(s_h^n, a_h^n)} \lesssim \beta H \sqrt{Nd \ln(N)}. \end{aligned}$$

Recall that $\beta = \tilde{O}(Hd)$. This concludes the proof. ■

7.7 Bibliographic Remarks and Further Readings

There are number of ways to linearly parameterize an MDP such that it permits for efficient reinforcement learning (both statistically and computationally). The first observation that such assumptions lead to statistically efficient algorithms was due to [Jiang et al., 2017] due to that these models have low Bellman rank (as we shall see in Chapter 8). The first statistically and computationally efficient algorithm for a linearly parameterized MDP model was due to [Yang and Wang, 2019a,b]. Subsequently, [Jin et al., 2020] provided a computationally and statistically efficient algorithm for simplified version of this model, which is the model we consider here. The model of [Modi et al., 2020b, Jia et al., 2020, Ayoub et al., 2020b, Zhou et al., 2020] provides another linearly parameterized model, which can viewed as parameterizing $P(s'|s, a)$ as a linear combination of feature functions $\phi(s, a, s')$. One notable aspect of the model we choose to present here, where $P_h(\cdot|s, a) = \mu_h^* \phi(s, a)$, is that this model has a number of free parameters that is $|\mathcal{S}| \cdot d$ (note that μ is unknown and is of size $|\mathcal{S}| \cdot d$), and yet the statistical complexity does not depend on $|\mathcal{S}|$. Notably, this implies that accurate model estimation request $O(|\mathcal{S}|)$ samples, while the regret for reinforcement learning is only polynomial in d . The linearly parameterized models of [Modi et al., 2020b, Jia et al., 2020, Ayoub et al., 2020b, Zhou et al., 2020] are parameterized by $O(d)$ parameters, and, while $O(d)$ free parameters suggests lower model capacity (where accurate model based estimation requires only polynomial in d samples), these models are incomparable to the linearly parameterized models presented in this chapter;

It is worth observing that all of these models permit statistically efficient estimation due to that they have bounded Bellman rank [Jiang et al., 2017] (and bounded Witness rank [Sun et al., 2019a]), a point which we return to in the next Chapter.

The specific linear model we consider here was originally introduced by [Jin et al., 2020]. The non-parametric model-based algorithm we study here was first introduced by [Lykouris et al., 2019] (but under the context of adversarial attacks).

The analysis we present here does not easily extend to infinite dimensional feature ϕ (e.g., RBF kernel); here, [Agarwal et al., 2020a] provide an algorithm and an analysis that extends to infinite dimensional ϕ , i.e. where we have a Reproducing Kernel Hilbert Space (RKHS) and the regret is based on the concept of Information Gain.

Chapter 8

Generalization with Bounded Bilinear Rank

Our previous lecture on linear bandits and linear MDPs gave examples of statistically efficient reinforcement learning, where we can obtain a sublinear regret with no dependence on the number of states or actions in the MDP. Obtaining sample size results which are independent of the size of the state space (and possibly the action space) is essentially a question of *generalization*, which is the focus of this chapter. In particular, we now present a more general underlying assumption which permit sample efficient reinforcement learning in various cases.

Recall our lower bounds from Chapter ?? which suggest that assumptions which would permit sample efficient RL may be subtle due to that: (i) agnostic RL is not possible and (ii) any assumption which permits sample efficient requires stronger, and possibly more subtle, conditions than simply assuming Q^* is a linear in some known features. This chapter introduces the Bellman rank condition along with a generalization, the Bilinear rank condition, which permit sample efficient RL.

This chapter follows the ideas in the original introduction of the Bellman rank [Jiang et al., 2017], along with a simplified analysis and a generalized concept, the Bilinear class, provided by [Du et al., 2021].

Setting and notation and setting. This chapter focuses on PAC RL (see Section 1.4), where our goal is to find a near optimal policy. We work in the episodic setting where we can obtain trajectories under some fixed s_0 (see Section 1.4), and we work in the finite horizon setting. We use the notation $a_{0:h} \sim d^\pi$ to denote the sampling of actions a_0 to a_h from trajectories under π . We also overload notation where we write $s_h, a_h \sim d_h^\pi$ for the sampled state-action pair at timestep h under π , and $s_h \sim d_h^\pi$ as the sampled state at time h under π .

8.1 Hypothesis Classes

We assume access to a hypothesis class $\mathcal{H} = \mathcal{H}_0 \times \dots \times \mathcal{H}_{H-1}$, which can be abstract sets that permit for both *model-based* and *value-based* hypotheses. The only restriction we make is that for all $f \in \mathcal{H}$, we have an associated state-action value function $Q_{h,f}$ and a value function $V_{h,f}$. Furthermore, we assume the hypothesis class is constrained so that $V_{h,f}(s) = \max_a Q_{h,f}(s, a)$ for all $f \in \mathcal{H}$, $h \in [H]$, and $s \in \mathcal{S}$, which is always possible as we can remove hypothesis for which this is not true. We let $\pi_{h,f}$ be the greedy policy with respect to $Q_{h,f}$, i.e., $\pi_{h,f}(s) = \operatorname{argmax}_{a \in \mathcal{A}} Q_{h,f}(s, a)$, and π_f as the sequence of time-dependent policies $\{\pi_{h,f}\}_{h=0}^{H-1}$.

The following provide some examples of value based and model based hypothesis classes:

- An example of *value-based hypothesis class* $\mathcal{H}$ is an explicit set of state-action value Q and value functions V i.e.

$$\mathcal{H}_h \subset \{(Q_h, V_h) \mid Q_h \text{ is a function from } \mathcal{S} \times \mathcal{A} \mapsto \mathbb{R} \text{ and } V_h \text{ is a function from } \mathcal{S} \mapsto \mathbb{R}\}.$$

Note that in this case, for any hypothesis $f := ((Q_0, V_0), (Q_1, V_1), \dots, (Q_{H-1}, V_{H-1})) \in \mathcal{H}$, we can take the associated $Q_{h,f} = Q_h$ and associated $V_{h,f} = V_h$.

- Another example of *value-based hypothesis class* $\mathcal{H}$ is when $\mathcal{H}$ is just a set of state-action value Q functions i.e.

$$\mathcal{H}_h \subset \{Q_h \mid Q_h \text{ is a function from } \mathcal{S} \times \mathcal{A} \mapsto \mathbb{R}\}.$$

In this case, for any hypothesis $f := (Q_0, Q_1, \dots, Q_{H-1}) \in \mathcal{H}$, we can take the associated $Q_{h,f} = Q_h$ and the associated $V_{h,f}$ function to be greedy with respect to the $Q_{h,f}$ function i.e. $V_{h,f}(\cdot) = \max_{a \in \mathcal{A}} Q_{h,f}(\cdot, a)$.

- An example of *model-based hypothesis class* is when $\mathcal{H}_h$ is a set of transition models and reward functions, i.e.

$$\mathcal{H}_h \subset \{(P_h, r_h) \mid P_h \text{ is a function from } \mathcal{S} \times \mathcal{A} \mapsto \Delta(\mathcal{S}) \text{ and } r_h \text{ is a function from } \mathcal{S} \times \mathcal{A} \mapsto \mathbb{R}\}.$$

In this case, for any hypothesis $f := ((P_0, r_0), (P_1, r_1), \dots, (P_{H-1}, r_{H-1})) \in \mathcal{H}$, we can take the associated $Q_{h,f}$ and $V_{h,f}$ functions to be the optimal value functions corresponding to the transition models $\{P_h\}_{h=0}^{H-1}$ and reward functions $\{r_h\}_{h=0}^{H-1}$.

8.2 The Bellman Rank

We now make assumptions on the Bellman residual error. For the “Q-version” of the Bellman rank, we assume that the Bellman residual error, $Q - \mathcal{T}Q$, has a simple parametric form. In particular,

Definition 8.1 (*Q-Bellman Rank*). For a given MDP, a hypothesis class $\mathcal{H}$ has a *Q-Bellman rank* of dimension d if for all $h \in [H]$, there exist functions $W_h : \mathcal{H} \rightarrow \mathbb{R}^d$ and $X_h : \mathcal{H} \rightarrow \mathbb{R}^d$ with $\sup_{f \in \mathcal{H}} \|W_h(f)\|_2 \leq B_W$ and $\sup_{f \in \mathcal{H}} \|X_h(f)\|_2 \leq B_X$, such that for all $f, g \in \mathcal{H}$

$$\mathbb{E}_{a_0:h \sim d^{\pi_f}} [Q_{h,g}(s_h, a_h) - r(s_h, a_h) - V_{h+1,g}(s_{h+1})] = \langle W_h(g) - W_h(f^*), X_h(f) \rangle.$$

Importantly, the functions W_h and X_h need not be known to the learner.

Instead, of assuming this parametric assumption on the Bellman residual error of the state-action value, we consider a variant which is stated in terms of the values. Note that we can write the Bellman optimality equations (see Equation 1.7) in terms of the value functions as follows:

$$V_h(s) - \max_a (r_h(s, a) + \mathbb{E}_{s' \sim P_h(\cdot|s,a)} [V_{h+1}(s')]) = 0. \quad (8.1)$$

This motivates the following definition:

Definition 8.2 (*V-Bellman Rank*). For a given MDP, a hypothesis class $\mathcal{H}$ has a *V-Bellman rank* of dimension d if for all $h \in [H]$, there exist functions $W_h : \mathcal{H} \rightarrow \mathbb{R}^d$ and $X_h : \mathcal{H} \rightarrow \mathbb{R}^d$ with $\sup_{f \in \mathcal{H}} \|W_h(f)\|_2 \leq B_W$ and $\sup_{f \in \mathcal{H}} \|X_h(f)\|_2 \leq B_X$, such that for all $f, g \in \mathcal{H}$:

$$\mathbb{E}_{a_0:h-1 \sim d^{\pi_f}, a_h = \pi_g(s_h)} [V_{h,g}(s_h) - r(s_h, a_h) - V_{h+1,g}(s_{h+1})] = \langle W_h(g) - W_h(f^*), X_h(f) \rangle.$$

Note that the action a_h is chosen by π_g , and recall that for every g , there corresponds a $Q_{h,g}$, where both $V_{h,g}$ and π_g are the corresponding greedy value function and policy.

Let us remark on a few subtle differences in these two definitions, in terms their usage of functions $V_{h,f}$ vs $Q_{h,f}$. Roughly speaking, the former definition corresponds to an assumption on the Bellman residual error of the state-action value function, while the latter definition corresponds to an assumption on the Bellman residual error of the value functions. A more subtle distinction is that, for the Q -version, the expectation is with respect to $a_{0:h} \sim \pi_f$, while for the V -version, the expectation is with respect to $a_{0:h-1} \sim \pi_f$ and $a_h \sim \pi_g$. The use of π_g is subtly different than the expectation being taken directly on the Bellman residual error of $V_{h,g}$ due to that π_g may not correspond to the true argmax in Equation 8.1. However, as we shall see, we can find estimates of both of the relevant quantities in the Q and the V versions.

Realizability. We say that $\mathcal{H}$ is *realizable* for an MDP $\mathcal{M}$ if, for all $h \in [H]$, there exists a hypothesis $f^* \in \mathcal{H}$ such that $Q_h^*(s, a) = Q_{h,f^*}(s, a)$, where Q_h^* is the optimal state-action value at time step h in the ground truth MDP $\mathcal{M}$. For instance, for the model-based perspective, the realizability assumption is implied if the ground truth transitions $\{P_h\}_{h=0}^{H-1}$ and reward function $\{r_h\}_{h=0}^{H-1}$ belong to our hypothesis class $\mathcal{H}$.

8.3 Examples

8.3.1 Examples that have small Q -Bellman rank

Contextual Linear Bandits

Let us start with the $H = 1$ case and return to the linear bandit model from Chapter 5. Let us consider a contextual model where there is a distribution over states $s_0 \sim \mu$ and where we assume:

$$\mathbb{E}[r(s, a)] = w^* \cdot \phi(s, a).$$

Here, our hypothesis of state-action values is $\mathcal{H} = \{\theta^\top \phi(s, a) : \|\theta\|_2 \leq W\}$.

Take two hypothesis f, g where we denote $Q_{0,f}(s, a) := \theta^\top \phi(s, a)$, $Q_{0,g}(s, a) = w^\top \phi(s, a)$ (we take $h = 0$ due to that $H = 1$). We have:

$$\begin{aligned} \mathbb{E}_{a \sim d_0^{\pi_f}} [w^\top \phi(s, a) - r(s, a)] &= \mathbb{E}_{s \sim \mu, a = \pi_f(s)} [(w - w^*)^\top \phi(s, a)] \\ &= \langle w - w^*, \mathbb{E}_{s \sim \mu, a = \pi_f(s)} [\phi(s, a)] \rangle. \end{aligned}$$

Namely, we can set $W_0(g) = w$, and $X_0(f) = \mathbb{E}_{s \sim \mu, a = \pi_f(s)} [\phi(s, a)]$. Thus, we have the following:

Proposition 8.3. *In the contextual linear bandit model, we have that $\mathcal{H}$, as defined above, has a Q -Bellman rank of d .*

Linear Bellman Completion

Let us now consider the linear Bellman completion setting (Definition 3.1). We show that linear Bellman completion models have Q -Bellman rank equal to d . In this setting, since $Q_h^*(s, a) = (\theta_h^*)^\top \phi(s, a)$, $\forall s, a, h$, we use linear functions for our hypothesis class, i.e., $\mathcal{H}_h = \{\theta^\top \phi(s, a) : \|\theta\|_2 \leq W, \|\theta^\top \phi(\cdot, \cdot)\|_\infty \leq H\}$ for some $W \in \mathbb{R}^+$ where $\|\theta_h^*\|_2 \leq W$; as in Section 8.1, the set of values are those that are greedy with respect to these Q functions.

Recall from Definition 3.1, given θ , we use $\mathcal{T}_h(\theta) \in \mathbb{R}^d$ to represent the linear function resulting from the Bellman backup, i.e.,

$$\forall s, a : \mathcal{T}_h(\theta)^\top \phi(s, a) = r(s, a) + \mathbb{E}_{s' \sim P_h(\cdot | s, a)} [\max_{a'} \theta^\top \phi(s', a')].$$

We further assume feature is normalized, i.e., $\sup_{s,a} \|\phi(s,a)\|_2 \leq 1$, and $\sup_{h,\theta \in \mathcal{H}_h} \|\mathcal{T}_h(\theta)\|_2 \leq W$. Note that this condition holds in linear MDP with $W = 2H\sqrt{d}$.

Take two hypothesis f, g where we denote $Q_{h,f}(s,a) := \theta_h^\top \phi(s,a)$, $Q_{h,g}(s,a) = w_h^\top \phi(s,a)$, $\forall h$. Then for any time step $h \in [0, H-2]$, we have:

$$\begin{aligned} & \mathbb{E}_{s_h, a_h \sim d_h^{\pi_f}, s_{h+1} \sim P_h(\cdot | s_h, a_h)} \left[w_h^\top \phi(s_h, a_h) - r(s_h, a_h) - \max_{a'} w_{h+1}^\top \phi(s_{h+1}, a') \right] \\ &= \mathbb{E}_{s_h, a_h \sim d_h^{\pi_f}} \left[w_h^\top \phi(s_h, a_h) \mathcal{T}_h(w_{h+1})^\top \phi(s_h, a_h) \right] \\ &= \left\langle w_h - \mathcal{T}_h(w_{h+1}), \mathbb{E}_{s_h, a_h \sim d_h^{\pi_f}} [\phi(s_h, a_h)] \right\rangle. \end{aligned}$$

Taking $W_h(g) := w_h - \mathcal{T}_h(w_{h+1})$ (and note that $W_h(f^*) = 0$ due to the Bellman optimality condition) and $X_h(f) := \mathbb{E}_{s_h, a_h \sim d_h^{\pi_f}} [\phi(s_h, a_h)]$, we have shown the following:

Proposition 8.4. *Under the linear Bellman completion assumption, we have that $\mathcal{H}$, as defined above, has a Q -Bellman rank of d . Further more, we have $\sup_{g \in \mathcal{H}} \|W_h(g)\|_2 \leq B_W := 2W$, and $\sup_{f \in \mathcal{H}} \|X_h(f)\|_2 \leq B_X := 1$.*

There are a number of natural models which satisfy the Bellman completeness property, including contextual linear bandits, linear MDPs and Linear Quadratic Regulators (we will cover LQRs in Chapter 15).

Linear Q^* and V^*

Let us now suppose that Q_h^* and V_h^* are functions which are in the span of some given features $\phi : \mathcal{S} \times \mathcal{A} \mapsto \mathbb{R}^d$, $\psi : \mathcal{S} \mapsto \mathbb{R}^d$, i.e. that, for all $h \in [0, \dots, H-1]$, there exist $\theta_h^* \in \mathbb{R}^d$ and $w_h^* \in \mathbb{R}^d$, such that for all s, a , $Q_h^*(s, a) = (\theta_h^*)^\top \phi(s, a)$ and $V_h^*(s) = (w_h^*)^\top \psi(s)$.

Our hypothesis class $\mathcal{H}_h$ is defined as $\mathcal{H}_h = \{(\theta, w) : \|\theta\|_2 \leq W_1, \|w\|_2 \leq W_2, \forall s, \max_a \theta^\top \phi(s, a) = w^\top \psi(s)\}$. Take any f, g where we denote $Q_{h,g}(s, a) = \theta_h^\top \phi(s, a)$, $V_{h,g}(s) = w_h^\top \psi(s)$, we have that:

$$\begin{aligned} & \mathbb{E}_{s,a \sim d_h^{\pi_f}} [\theta_h^\top \phi(s, a) - r(s, a) - \mathbb{E}_{s' \sim P_h(\cdot | s, a)} [w_h^\top \psi(s')]] \\ &= \mathbb{E}_{s,a \sim d_h^{\pi_f}} [\theta_h^\top \phi(s, a) - Q_h^*(s, a) + \mathbb{E}_{s' \sim P_h(\cdot | s, a)} [V_h^*(s')] - \mathbb{E}_{s' \sim P_h(\cdot | s, a)} [w_h^\top \psi(s')]] \\ &= \mathbb{E}_{s,a \sim d_h^{\pi_f}} [(\theta_h - \theta_h^*)^\top \phi(s, a) + (w_h - w_h^*)^\top \mathbb{E}_{s' \sim P_h(\cdot | s, a)} [\psi(s')]] \\ &= \left\langle \begin{bmatrix} \theta_h - \theta_h^* \\ w_h - w_h^* \end{bmatrix}, \mathbb{E}_{s,a \sim d_h^{\pi_f}} \begin{bmatrix} \phi(s, a) \\ \mathbb{E}_{s' \sim P_h(\cdot | s, a)} [\psi(s')] \end{bmatrix} \right\rangle. \end{aligned}$$

Taking $W_h(g) := \begin{bmatrix} \theta_h \\ w_h \end{bmatrix}$, and $X_h(f) := \begin{bmatrix} \phi(s, a) \\ \mathbb{E}_{s' \sim P_h(\cdot | s, a)} [\psi(s')] \end{bmatrix}$, we have shown that:

Proposition 8.5. *Under the above linear Q^* and V^* assumption, our hypothesis class $\mathcal{H}$, as defined above, has a Q -Bellman rank of $2d$.*

The linear Q^* assumption vs. the linear Q^*/V^* assumption. As we saw in Chapter ??, simply assuming that Q^* is linear is not sufficient for sample efficient RL. Thus it may seem surprising that assuming both Q^* and V^* are linear (in different features) permits a sample efficient algorithm (as we shall see). However, it is worthwhile to observe how this is a substantially strong assumption. In particular, note that our hypothesis class must enforce the constraint that $\max_a \theta^\top \phi(s, a) = w^\top \psi(s)$ for all $(\theta, w) \in \mathcal{H}$; this *eliminates* functions from the set of all Q which are linear in ϕ . Furthermore, note that this pruning is done without collecting any data.

Q^* -state abstraction

The Q^* -state abstraction mode is defined as follows. There exists some abstraction function $\xi : \mathcal{S} \mapsto \mathcal{Z}$ where $|\mathcal{Z}| < |\mathcal{S}|$, such that for all h , we have:

$$\forall s, s', \xi(s) = \xi(s') \Rightarrow Q_h^*(s, a) = Q_h^*(s', a), \forall a \in \mathcal{A}, V_h^*(s) = V_h^*(s').$$

We show that this example has low Q -bellman rank by showing that it is a special case of the linear Q^* & V^* model. Specifically, let us define $\phi : \mathcal{S} \times \mathcal{A} \mapsto \mathbb{R}^{|\mathcal{Z}||\mathcal{A}|}$ and $\psi : \mathcal{S} \mapsto \mathbb{R}^{|\mathcal{Z}|}$ as follows:

$$\phi(s, a)_{z, a'} = \mathbf{1}\{\xi(s) = z, a = a'\}, \quad \psi(s)_z = \mathbf{1}\{\xi(s) = z\}.$$

With this feature setup, we have:

$$\begin{aligned} Q_h^*(s, a) &= (w_h^*)^\top \phi(s, a), \text{ where } (w_h^*)_{z, a} = Q^*(s, a) \text{ with } \xi(s) = z, \\ V_h^*(s) &= (\theta_h^*)^\top \psi(s), \text{ where } (\theta_h^*)_z = V^*(s) \text{ with } \xi(s) = z. \end{aligned}$$

Thus, similar to linear Q^*/V^* model above, we can set $\mathcal{H}$ such that $\mathcal{H}_h := \{(\theta, w) : \|\theta\|_2 \leq W_1, \|w\|_2 \leq W_2, \forall s, \max_s \theta^\top \phi(s, a) = w^\top \psi(s)\}$. we demonstrate that this model has Q -Bellman rank at most $|\mathcal{Z}||\mathcal{A}| + |\mathcal{Z}|$.

Proposition 8.6. *Under the above Q^* -state abstraction assumption, our hypothesis class $\mathcal{H}$ has a Q -Bellman rank of $|\mathcal{Z}||\mathcal{A}| + |\mathcal{Z}|$.*

Low-occupancy measure

An MDP has **low (state-action) occupancy measure** if the following is true. For all h , there exist two functions $\beta_h : \mathcal{H} \mapsto \mathbb{R}^d$, and $\phi_h : \mathcal{S} \times \mathcal{A} \mapsto \mathbb{R}^d$, such that for any $f \in \mathcal{H}$, we have:

$$d_h^{\pi_f}(s, a) = \langle \beta_h(f), \phi_h(s, a) \rangle.$$

We show that this model has Q -Bellman rank at most d . Consider any f, g and h , we have:

$$\begin{aligned} &\mathbb{E}_{s, a \sim d_h^{\pi_f}} [Q_{h, g}(s) - r(s, a) - \mathbb{E}_{s' \sim P_h(\cdot|s, a)} [V_{h+1, g}(s')]] \\ &= \beta_h(f)^\top \left(\sum_{s, a} \phi_h(s, a) [Q_{h, g}(s, a) - r(s, a) - \mathbb{E}_{s' \sim P_h(\cdot|s, a)} [V_{h+1, g}(s')]] \right), \end{aligned}$$

which shows that $W_h(g) := \sum_{s, a} \phi_h(s, a) [Q_{h, g}(s, a) - r(s, a) - \mathbb{E}_{s' \sim P_h(\cdot|s, a)} [V_{h+1, g}(s')]]$ (note that $W_h(f^*) = 0$ due to the Bellman optimality condition), and $X_h(f) := \beta_h(f)$. Note that in this model we can use any hypothesis class $\mathcal{H}$ as long as it contains f^* .

Proposition 8.7. *Any hypothesis class $\mathcal{H}$, together with the MDP that has a low-occupancy measure, has Q -Bellman rank d .*

Note that if the state distribution has low-occupancy, i.e.

$$d_h^{\pi_f}(s) = \beta_h(f)^\top \phi_h(s)$$

for some $\phi_h : \mathcal{S} \mapsto \mathbb{R}^d$, then it satisfies the V -Bellman rank condition, with rank d . In this latter case, we refer to the model as having **low (state) occupancy measure**.

8.3.2 Examples that have small V -Bellman rank

Contextual Bandit

Contextual bandit is a finite horizon MDP with $H = 1$ and finite number of actions. The contexts are states $s_0 \sim \mu$. Here our hypothesis class $\mathcal{H} = \mathcal{H}_0 = \{f : \mathcal{S} \times \mathcal{A} \mapsto [0, 1]\}$ with $r \in \mathcal{H}$. We show that the V -Bellman rank is at most A . We use the notation $V_1(s) = 0, \forall s$ (recall that $H = 1$ again). Consider any function pair $f, g \in \mathcal{H}$, note that since $H = 1$, the state distribution at $h = 0$ is independent of the policy π_f . Thus we have:

$$\mathbb{E}_{s \sim \mu} [V_{0,g}(s) - r(s, \pi_{0,g}(s))] = \sum_{a \in \mathcal{A}} \mathbb{E}_{s \sim \mu} [\mathbf{1}\{\pi_g(s) = a\} [Q_{0,g}(s, a) - r(s, a)]] .$$

This means that we can write the Bellman error of g averaged over μ as an inner product between two vectors — an all one vector, and a vector whose a -th entry is $\mathbb{E}_{s \sim \mu} [\mathbf{1}\{\pi_g(s) = a\} [Q_{0,g}(s, a) - r(s, a)]]$, i.e., we have $X_0(f) = \mathbf{1}_A$ where $\mathbf{1}_A$ represents a A -dim vector with all one, and $W_0(g) \in \mathbb{R}^A$ where the a -th entry of $W_0(g)$ is $\mathbb{E}_{s \sim \mu} [\mathbf{1}\{\pi_g(s) = a\} [Q_{0,g}(s, a) - r(s, a)]]$. Note that $W_0(f^*) = 0$ where f^* is the ground truth reward function here.

Proposition 8.8. *The hypothesis class $\mathcal{H}$ and the contextual bandit model together has a V -Bellman rank A .*

Feature Selection in Low-rank MDPs (Representation Learning)

We consider low-rank MDP where for all h , there exist $\mu^* : \mathcal{S} \mapsto \mathbb{R}^d, \phi^* : \mathcal{S} \times \mathcal{A} \mapsto \mathbb{R}^d$, such that for all s, s', a , we have $P_h(s'|s, a) = \mu^*(s')^\top \phi^*(s, a), r(s, a) = (\theta^*)^\top \phi^*(s, a)$. We denote $\Phi \subset \mathcal{S} \times \mathcal{A} \mapsto \mathbb{R}^d$ as our feature class such that $\phi^* \in \Phi$. We assume μ^*, ϕ^*, θ^* are normalized, i.e., $\|\phi^*(s, a)\|_2 \leq 1, \forall s, a, h$, and $\|v^\top \mu^*\|_2 \leq \sqrt{d}$, for all v such that $\|v\|_\infty \leq 1, \|\theta^*\|_2 \leq \sqrt{d}$.

We denote the hypothesis class $\mathcal{H}_h := \{w^\top \phi(s, a) : \|w\|_2 \leq W, \phi \in \Phi\}$ where we can set $W = 4H\sqrt{d}$ since we can verify that for $Q_h^*(s, a)$, it is linear with respect to ϕ^* , i.e., $Q_h^*(s, a) = (w_h^*)^\top \phi(s, a)$ for some w_h^* with $\|w_h^*\|_2 \leq 2H\sqrt{d}$. Consider a pair of hypothesis (f, g) where we denote $Q_{h,g} := w_h^\top \phi$, and recall $V_{h,g}(\cdot) = \max_a Q_{h,g}(\cdot, a)$, and $\pi_{h,g}(s) := \arg\max_a Q_{h,g}(s, a)$.

$$\mathbb{E}_{s \sim d_h^{\pi_f}} [V_{h,g}(s) - r(s, \pi_{h,g}(s)) - \mathbb{E}_{s' \sim P_h(\cdot|s, \pi_{h,g}(s))} [V_{h+1,g}(s')]] \quad (8.2)$$

$$\begin{aligned} &= \mathbb{E}_{\tilde{s}, \tilde{a} \sim d_{h-1}^{\pi_f}} \mathbb{E}_{s \sim P_{h-1}(\cdot|\tilde{s}, \tilde{a})} [V_{h,g}(s) - r(s, \pi_{h,g}(s)) - \mathbb{E}_{s' \sim P_h(\cdot|s, \pi_{h,g}(s))} [V_{h+1,g}(s')]] \\ &= \mathbb{E}_{\tilde{s}, \tilde{a} \sim d_{h-1}^{\pi_f}} \int_{\mathcal{S}} \mu^*(s)^\top \phi^*(\tilde{s}, \tilde{a}) [V_{h,g}(s) - r(s, \pi_{h,g}(s)) - \mathbb{E}_{s' \sim P_h(\cdot|s, \pi_{h,g}(s))} [V_{h+1,g}(s')]] d(s) \\ &= \left\langle \int_{\mathcal{S}} \mu^*(s) [V_{h,g}(s) - r(s, \pi_{h,g}(s)) - \mathbb{E}_{s' \sim P_h(\cdot|s, \pi_{h,g}(s))} [V_{h+1,g}(s')]] , \quad \mathbb{E}_{\tilde{s}, \tilde{a} \sim d_{h-1}^{\pi_f}} [\phi^*(\tilde{s}, \tilde{a})] \right\rangle. \quad (8.3) \end{aligned}$$

Taking $W_h(g) := \int_{\mathcal{S}} \mu^*(s) [V_{h,g}(s) - r(s, \pi_{h,g}(s)) - \mathbb{E}_{s' \sim P_h(\cdot|s, \pi_{h,g}(s))} [V_{h+1,g}(s')]]$ (note that $W_h(f^*) = 0$ due to the Bellman optimality condition) and $X_h(f) := \mathbb{E}_{\tilde{s}, \tilde{a} \sim d_{h-1}^{\pi_f}} [\phi^*(\tilde{s}, \tilde{a})]$, we have shown that:

Proposition 8.9. *The hypothesis class $\mathcal{H}$ and the low-rank MDP defined above together has V -Bellman rank d . Moreover, we have $\|X_h(f)\|_2 \leq B_X = 1$, and $\|W_h(g)\|_2 \leq B_W = 2H\sqrt{d}$.*

Block MDPs. The Block MDP has a small size discrete latent state space $\mathcal{Z}$, an emission distribution $v(\cdot; z) \in \Delta(\mathcal{S}), \forall z$, a decoder $\xi : \mathcal{S} \mapsto \mathcal{Z}$ which maps a state s to its corresponding latent state z , and a transition in latent state space $T_h(z'|z, a), \forall h$. To show that Low-rank MDP model captures it, we define $\phi^*(s, a) \in \{0, 1\}^{|\mathcal{Z}||\mathcal{A}|}$, where $\phi^*(s, a)$ is a one hot encoding vector which contains zero everywhere, except one in the entry corresponding to the latent state and action pair $(\xi(s), a)$. The corresponding $\mu^*(s') \in \mathbb{R}^{|\mathcal{Z}||\mathcal{A}|}$ is defined such that its entry corresponding to (z, a) is equal to $\sum_{z'} T(z'|z, a) v(s'; z')$. This shows that the Block MDP model is a special case of the Low-rank MDP model where the rank is the number of unique latent states.

Feature Selection in Sparse Linear MDP. A sparse linear MDP is a linear MDP where ϕ^* is known (but μ^* is unknown), and there is a subset $\mathcal{K}^* \in [d]$ with $|\mathcal{K}^*| = s$ and $s < d$, such that $P_h(s'|s, a) = \sum_{i \in \mathcal{K}^*} \mu_i^*(s') \phi_i^*(s, a)$, where we use x_i to denote the i -th element of the vector x . We can re-write the transition as $P_h(s'|s, a) = \mu_{\mathcal{K}^*}^*(s')^\top \phi_{\mathcal{K}^*}^*(s, a)$, where we denote $x_{\mathcal{K}}$ as the subvector of x that is indexed by elements in $\mathcal{K}$. Thus, this shows that a sparse linear MDP is a special case of low-rank MDP where the rank is s instead of d .

We consider the following hypothesis class. Define a representation class Φ with $|\Phi| = \binom{d}{s}$. Each $\phi \in \Phi$ corresponds to a s -size subset $\mathcal{K} \in [d]$. Given $\mathcal{K}$, we have $\phi(s, a) \in \mathbb{R}^s$, and $\phi(s, a) = \phi_{\mathcal{K}^*}^*(s, a), \forall s, a$. By the assumption that $\|\phi^*(s, a)\|_2 \leq 1$, we must have $\|\phi(s, a)\|_2 \leq 1$ for all $s, a, \phi \in \Phi$. We define $\mathcal{H}_h = \{w^\top \phi : \|w\|_2 \leq W, w \in \mathbb{R}^s, \phi \in \Phi_h, |w^\top \phi(\cdot, \cdot)| \leq H\}$. We set $W = 2H\sqrt{d}$ such that $Q_h^* \in \mathcal{H}_h$ (note that Q_h^* is a linear function with respect to $\phi_{\mathcal{K}^*}^*$ due to the fact that the MDP is a linear MDP with respect to feature $\phi_{\mathcal{K}^*}^*$).

8.4 Bilinear Classes

The Bilinear class model generalizes both the Q -Bellman rank and V -Bellman rank. We define the Bilinear class model as follows.

Definition 8.10 (Bilinear Class). Consider an MDP and a hypothesis class $\mathcal{H}$, a discrepancy function $\ell_f : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \times [H] \times \mathcal{H}$ (defined for each $f \in \mathcal{H}$), and a set of (non-stationary) estimation policy $\Pi_{est} = \{\pi_{est}(f) : f \in \mathcal{H}\}$. We say that $\{\mathcal{H}, \ell_f, \Pi_{est}, \mathcal{M}\}$ is a Bilinear class of rank d if $\mathcal{H}$ is realizable in $\mathcal{M}$, and if there exists $W_h : \mathcal{H} \mapsto \mathbb{R}^d$ and $X_h : \mathcal{H} \mapsto \mathbb{R}^d$, such that the following two properties hold for all $f \in \mathcal{H}$ and $h \in [H]$:

1. We have

$$\left| \mathbb{E}_{s, a \sim d_h^{\pi_f}, s' \sim P_h(\cdot|s, a)} [Q_{h,f}(s, a) - r(s, a) - V_{h+1,f}(s')] \right| \leq |\langle W_h(f) - W_h(f^*), X_h(f) \rangle|, \quad (8.4)$$

2. The policy $\pi_{est}(f)$ and the discrepancy ℓ_f can be used for estimation in the following sense: for any $g \in \mathcal{H}$, we have:

$$\left| \mathbb{E}_{s \sim d_h^{\pi_f}, a \sim \pi_{est,h}(s; f), s' \sim P_h(\cdot|s, a)} [\ell_f(s, a, s', h, g)] \right| = |\langle W_h(g) - W_h(f^*), X_h(f) \rangle|, \quad (8.5)$$

We suppress the h dependence on $\pi_{est,h}$ when clear from context. Typically $\pi_{est}(f)$ will be either the uniform distribution over $\mathcal{A}$ or π_f itself. In the latter case, we refer to the estimation strategy as being on-policy.

The first condition in the above definition assumes that the average Bellman error of f under π_f 's state-action distribution can be upper bounded by the value from the bilinear formulation. The second condition assumes that we have a discrepancy function that allows us to evaluate the value of the bilinear formulation. Note that the second condition permits data reuse, i.e., given a set of samples where $s \sim d_h^{\pi_f}, a \sim \pi_{est,h}(s; f), s' \sim P_h(\cdot|s, a)$, we can use the dataset to evaluate the loss for all $g \in \mathcal{H}$ simultaneously. This data reuse is the key to generalization. Also the second condition ensures that for all f , we must have $\mathbb{E}_{s \sim d_h^{\pi_f}, a \sim \pi_{est,h}(s; f), s' \sim P_h(\cdot|s, a)} [\ell_f(s, a, s', h, f^*)] = 0$.

8.4.1 Examples

Let now see how the Bilinear classes naturally generalize the Bellman rank.

Q-Bellman rank and V-Bellman rank

It is straightforward to see that the Bilinear class captures the Q -Bellman rank and V -Bellman rank models. To capture Q -Bellman rank, in Bilinear class, we will set $\ell_f := \ell$ for all f , with ℓ being defined in Eq. 8.6, and set

$\pi_{est,h}(f) = \pi_{f,h}, \forall f$. To capture V -Bellman rank, we will set $\ell_f := \ell$ for all f with ℓ being defined in Eq. 8.7, and $\pi_{est}(f) := \text{Unif}_{\mathcal{A}}$, i.e., uniform distribution over $\mathcal{A}$. More precisely, we have:

$$Q\text{-Bellman rank} : \ell(s, a, s', h, g) := Q_{h,g}(s, a) - r(s, a) - V_{h+1,g}(s'), \quad (8.6)$$

$$V\text{-Bellman rank} : \ell(s, a, s', h, g) := \frac{\mathbf{1}\{a = \pi_{h,g}(s)\}}{1/A} (Q_{h,g}(s, a) - r(s, a) - V_{h+1,g}(s')). \quad (8.7)$$

In the V -Bellman rank case, the importance weighting factor $\mathbf{1}\{a = \pi_{h,g}(s)\}/(1/A)$ ensures that given any s, g ,

$$\mathbb{E}_{a \sim \text{Unif}_{\mathcal{A}}, s' \sim P_h(\cdot|s,a)} \ell(s, a, s', g) = V_{h,g}(s) - r(s, \pi_{h,g}(s)) - \mathbb{E}_{s' \sim P_h(\cdot|s, \pi_{h,g}(s))} V_{h+1,g}(s'),$$

i.e., $\ell(s, a, s', g)$ is an unbiased estimate of the V -Bellman error of g at state s , given that $a \sim \text{Unif}_{\mathcal{A}}, s' \sim P_h(\cdot|s, a)$.

Q -Bellman and V -Bellman rank already assume the existence of two maps.

Note that by our assumption that $\|Q_{h,g}\|_{\infty} \leq H$, we have $\|\ell\|_{\infty} \leq H$ for the Q -Bellman rank case, and $\|\ell\|_{\infty} \leq AH$ for the V -Bellman rank case.

Linear mixture model

The additional f -dependent discrepancy function allows Bilinear classes to capture the following *linear mixture model*, which cannot be captured by Q or V version of the Bellman rank.

A linear mixture MDP model has a feature $\phi : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \mapsto \mathbb{R}^d$, such that $P_h(s'|s, a) = (\theta_h^*)^{\top} \phi(s, a, s')$ where we assume $\|\theta_h^*\|_2 \leq W$. We assume reward r is known.

Here, our hypothesis class $\mathcal{H}$ is a model based one, where $\mathcal{H}_h = \{(P_h, r) : P_h(s'|s, a) := \theta^{\top} \phi(s, a, s'), \|\theta\|_2 \leq W\}$. We assume that the reward function is known, and the associated Q and V function with any hypothesis is the optimal state-action value and optimal value for the associated model (as discussed in Section 8.1).

We design the discrepancy function ℓ_f as follows. Denote $g \in \mathcal{H}$ as $g = \{\theta_h^{\top} \psi(\cdot, \cdot, \cdot)\}_{h=0}^{H-1}$,

$$\ell_f(s, a, s', h, g) := \sum_{\tilde{s}} \theta_h^{\top} \phi(s, a, \tilde{s}) V_{h+1,f}(\tilde{s}) - V_{h+1,f}(s').$$

Below we show that it is a Bilinear class model. For any $f, g \in \mathcal{H}$, set $\pi_{est}(f) = \pi_f$ (i.e., on-policy), we have:

$$\begin{aligned} \mathbb{E}_{s, a \sim d_h^{\pi_f}, s' \sim P_h(\cdot|s,a)} \ell_f(s, a, s', h, g) &= \mathbb{E}_{s, a \sim d_h^{\pi_f}} \left[\sum_{\tilde{s}} \theta_h^{\top} \phi(s, a, \tilde{s}) V_{h+1,f}(\tilde{s}) - \sum_{\tilde{s}} (\theta_h^*)^{\top} \phi(s, a, \tilde{s}) V_{h+1,f}(s') \right] \\ &= (\theta_h - \theta_h^*)^{\top} \left(\mathbb{E}_{s, a \sim d_h^{\pi_f}} \left[\sum_{s'} \phi(s, a, s') \right] \right). \end{aligned}$$

Thus we have shown that $W_h(g) = \theta_h$, and $X_h(f) = \mathbb{E}_{s, a \sim d_h^{\pi_f}} [\sum_{s'} \phi(s, a, s')]$.

Proposition 8.11. *The hypothesis class $\mathcal{H}$ together with the linear mixture model is a Bilinear class.*

8.5 PAC-RL with Bounded Bilinear Rank

We now focus on the sample complexity of finding an ϵ -near optimal policy. We work in the episodic setting where we can obtain trajectories under some fixed initial state s_0 (i.e., $\mu := \delta(s_0)$). Note that generalizing to any distribution μ is straightforward.

8.5.1 Algorithm

Algorithm 7 BLin-UCB

- 1: **Input:** number of iteration T , batch size m , confidence radius R , estimation policy π_{est} , loss ℓ
- 2: **for** $t = 0 \dots T - 1$ **do**
- 3: Set f_t as the solution of the following program:

$$f_t = \operatorname{argmax}_{g \in \mathcal{H}} V_{0,g}(s_0),$$

$$\sum_{i=0}^{t-1} (\mathbb{E}_{\mathcal{D}_{i,h}} \ell_{f_i}(s, a, s', g))^2 \leq R^2, \forall h \in [0, \dots, H-1]$$

- 4: $\forall h$, create batch data $\mathcal{D}_{t,h} = \{s^i, a^i, (s')^i\}_{i=1}^m$,
 where $s^i \sim d_h^{\pi_{f_t}}, a^i \sim \pi_{est,h}(s^i; f_t), (s')^i \sim P_h(\cdot | s^i, a^i)$ (see text for discussion).
 - 5: **end for**
 - 6: **return** $\arg \max_{\pi \in \{\pi_{f_0}, \dots, \pi_{f_{T-1}}\}} V^\pi$.
-

We present the Bilinear-UCB algorithm in Algorithm 7. Algorithm 7 uses the estimation policy $\pi_{est,h}(\cdot; f_t)$ to generate action a_t when collecting the batch $\mathcal{D}_{t,h}$ at iteration t for time step h .

Note that the samples required by Bilinear-UCB algorithm can be obtained in episodic sampling model. We simply follow π until time step h , and then use $\pi_{est}(\cdot; f_t)$ to sample the action at the timestep.

While the algorithm returns the best policy found up to time T , i.e. $\max_{t \in [T]} V^{\pi_{f_t}}$, it is also straight forward to replace this step using a sample based estimate of the best policy seen so far (with a negligible increase in the sample size).

Intuitively, BLin-UCB uses optimism for exploration. Our constraints are designed such that they eliminate functions that are not f^* . Also the threshold R will be set such that f^* is always a feasible solution with high probability. Thus, the argmax procedure in the objective function ensures that it returns an optimistic estimate, i.e., $V_{0,f_t}(s_0) \geq V_0^*(s_0)$.

8.5.2 Sample Complexity

We begin by stating the following assumption about the uniform convergence property of our hypothesis class $\mathcal{H}$.

Assumption 8.12 (Uniform Convergence). We assume that there exists a function $\varepsilon_{gen}(m, \mathcal{H}, \delta)$ such that for any distribution $\nu \in \Delta(\mathcal{S} \times \mathcal{A} \times \mathcal{S})$ and $f \in \mathcal{H}$, and any $\delta \in (0, 1)$, with probability at least $1 - \delta$ over the m i.i.d samples $\mathcal{D} := \{s_i, a_i, s'_i\}_{i=1}^m \sim \nu$, we have:

$$\sup_{g \in \mathcal{H}} |\mathbb{E}_{\mathcal{D}}[\ell_f(s, a, s', h, g)] - \mathbb{E}_{\nu}[\ell_f(s, a, s', h, g)]| \leq \varepsilon_{gen}(m, \mathcal{H}, \delta).$$

The above assumption is the standard uniform convergence result and typically we have $\varepsilon_{gen}(m, \mathcal{H}, \delta) \rightarrow 0, m \rightarrow \infty$. Below we give three examples, one for finite hypothesis class $\mathcal{H}$, one for infinite hypothesis class under the linear Bellman completion setting, and one for the infinite hypothesis class under the problem of feature selection in sparse linear MDPs.

Example 8.13 (Finite hypothesis class case). Let us consider the example where $\mathcal{H}$ is a discrete hypothesis class. In this case, for Q -Bellman rank loss, via Hoeffding's inequality and a union bound over $\mathcal{H}$, we have that with probability at least $1 - \delta$,

$$\sup_{g \in \mathcal{H}} |\mathbb{E}_{\mathcal{D}}[\ell_f(s, a, s', h, g)] - \mathbb{E}_{\nu}[\ell_f(s, a, s', h, g)]| \leq 2H \sqrt{\ln(|\mathcal{H}|/\delta)/m},$$

which means we can set

$$\varepsilon_{gen}(m, \mathcal{H}, \delta) := 2H \sqrt{\ln(|\mathcal{H}|/\delta)/m}.$$

For V -Bellman rank loss, due to the importance weighting, we have an additional A term, i.e., with probability at least $1 - \delta$,

$$\sup_{g \in \mathcal{H}} |\mathbb{E}_{\mathcal{D}}[\ell_f(s, a, s', h, g)] - \mathbb{E}_{\nu}[\ell_f(s, a, s', h, g)]| \leq 2HA \sqrt{\ln(|\mathcal{H}|/\delta)/m}.$$

Example 8.14 (Linear Bellman completion). In the linear Bellman completion model, our hypothesis class $\mathcal{H}_h = \{w^\top \phi(s, a) : \|w\|_2 \leq W, \|w^\top \phi(\cdot, \cdot)\|_\infty \leq H\}$ is not discrete. However, with the loss function associated with Q -Bellman rank, via a standard uniform convergence analysis using ϵ -net for linear functions in $\mathcal{H}$, we can show (details in Lemma A.12):

$$\sup_{g \in \mathcal{H}} |\mathbb{E}_{\mathcal{D}}[\ell_f(s, a, s', h, g)] - \mathbb{E}_{\nu}[\ell_f(s, a, s', h, g)]| \leq 3H \sqrt{(2d \ln(1 + 4Wm) + \ln(1/\delta))/m}.$$

Thus, we have

$$\varepsilon_{gen}(m, \mathcal{H}, \delta) := 3H \sqrt{\frac{2d \ln(1 + 4Wm) + \ln(1/\delta)}{m}}.$$

Example 8.15 (Feature selection for sparse linear MDP). To give another example of infinite hypothesis class case, we consider the following $\mathcal{H}$ with $\mathcal{H}_h = \{w^\top \phi : \|w\|_2 \leq 2H\sqrt{d}, w \in \mathbb{R}^s, \phi \in \Phi_h, \|w^\top \phi\|_\infty \leq H\}$ from the example of feature selection in sparse linear MDP. Recall Φ is discrete and $|\Phi| = \binom{d}{s}$. We focus on the loss function associated with V -Bellman rank. Denote $\mathcal{H}_{h,\phi} = \{w^\top \phi : \|w\|_2 \leq 2H\sqrt{d}, w \in \mathbb{R}^s, \|w^\top \phi\|_\infty \leq H\}$ for a $\phi \in \Phi$. Via a standard uniform convergence analysis using ϵ -net for linear functions in $\mathcal{H}_{h,\phi}$, we can show (details in Lemma A.12):

$$\mathbb{P} \left(\sup_{g \in \mathcal{H}_\phi} |\mathbb{E}_{\mathcal{D}}[\ell_f(s, a, s', h, g)] - \mathbb{E}_{\nu}[\ell_f(s, a, s', h, g)]| \geq 3HA \sqrt{(2s \ln(1 + 8AH\sqrt{dm}) + \ln(1/\delta))/m} \right) \leq 1 - \delta.$$

Applying a union bound over all $\phi \in \Phi$ and use the fact that $|\Phi| \leq s^d$, we have:

$$\sup_{g \in \mathcal{H}} |\mathbb{E}_{\mathcal{D}}[\ell_f(s, a, s', h, g)] - \mathbb{E}_{\nu}[\ell_f(s, a, s', h, g)]| \leq 3HA \sqrt{(2s \ln(1 + 8AH\sqrt{dm}) + s \ln(d/\delta))/m},$$

with probability at least $1 - \delta$. Thus we have

$$\varepsilon_{gen}(m, \mathcal{H}, \delta) := 3HA \sqrt{\frac{2s \ln(1 + 8AH\sqrt{dm}) + s \ln(d/\delta)}{m}}.$$

Note the polynomial dependence on s and the poly-log dependence on d .

With the above assumption, now we are ready to state the main theorem.

Theorem 8.16 (PAC RL for BLin-UCB). Fix $\delta \in (0, 1)$, batch size $m \in \mathbb{N}^+$. Suppose our hypothesis class $\mathcal{H}$ forms a Bilinear class of rank d ; that the class is realizable; and that the Bilinear-UCB algorithm has access to both $\mathcal{H}$ and the associated discrepancy function ℓ . Set the parameters as follows:

$$T = \left\lceil 2Hd \ln \left(4Hd \left(\frac{B_X^2 B_W^2}{\varepsilon_{gen}^2(m, \mathcal{H}, \delta/(TH))} + 1 \right) \right) \right\rceil, \quad R = \sqrt{T} \varepsilon_{gen}(m, \mathcal{H}, \delta/(TH)).$$

Let π be the policy returned by the Bilinear-UCB algorithm (Algorithm 7). With probability at least $1 - \delta$, we have:

$$V_0^*(s_0) - V_0^\pi(s_0) \leq 5\varepsilon_{gen} \sqrt{dH^3 \ln \left(4Hd \left(\frac{B_X^2 B_W^2}{\varepsilon_{gen}^2} + 1 \right) \right)}.$$

Note the total number of samples obtained is mTH .

Before proving the main theorem, we revisit the three examples above, i.e., finite hypothesis case, linear Bellman completion, and feature selection in sparse linear MDP, and derive the sample complexity bounds for these examples.

Finite hypothesis class case (Example 8.13 continued). Plugging in specific forms of $\varepsilon_{gen}(m, \mathcal{H}, \delta/(TH))$ in the above general theorem allows us to instantiate PAC bounds. Consider the Q -Bellman rank example with a discrete hypothesis class $\mathcal{H}$. In this case, as we show above, we have $\varepsilon_{gen}(m, \mathcal{H}, \delta) := 2H\sqrt{\ln(|\mathcal{H}|/\delta)/m}$. To have $V_0^*(s_0) - \max_{i \in [0, \dots, T-1]} V_0^{\pi_{f_i}}(s_0) \leq \epsilon$ for some $\epsilon \in (0, H)$, we just need to set m such that:

$$12H\sqrt{\frac{\ln(|\mathcal{H}|/\delta)}{m}}\sqrt{dH^3 \ln\left(4Hd\left(\frac{B_X^2 B_W^2 m}{4H^2 \ln(|\mathcal{H}|/\delta)} + 1\right)\right)} \leq \epsilon,$$

Using the inequality from Equation 8.9, it is suffice to set m as:

$$m = \frac{cdH^5 \ln(|\mathcal{H}|/\delta)}{\epsilon^2} \cdot \nu, \text{ where } \nu := \ln\left(\frac{dH^5}{\epsilon^2} (HdB_X^2 B_W^2 + 4Hd \ln(|\mathcal{H}|/\delta))\right),$$

and c is an absolute constant. Using the fact that $T = \left\lceil 2Hd \ln\left(4Hd\left(\frac{B_X^2 B_W^2}{\varepsilon_{gen}^2(m, \mathcal{H}, \delta/(TH))} + 1\right)\right) \right\rceil$, we can reach the following corollary.

Corollary 8.17 (PAC bound for finite hypothesis $\mathcal{H}$ under Q -Bellman rank). *Fix $\epsilon, \delta \in (0, 1)$. With probability at least $1 - \delta$, BLin-UCB learns a policy π such that $V_0^*(s_0) - V_0^\pi(s_0) \leq \epsilon$, using at most*

$$\frac{cH^6 d^2 \ln(|\mathcal{H}|/\delta)}{\epsilon^2} \cdot \ln\left(4Hd\left(\frac{B_X^2 B_W^2 dH^3 \nu}{\epsilon^2} + 1\right)\right) \cdot \nu$$

many trajectories.

Linear Bellman completion (Example 8.14, continued). Plugging in the specific form of $\varepsilon_{gen}(m, \mathcal{H}, \delta/(TH))$ and $B_X = 1$ and $B_W = W$, with some linear algebra, we get:

$$\left(5\varepsilon_{gen}\sqrt{dH^3 \ln\left(4Hd\left(\frac{B_X^2 B_W^2}{\varepsilon_{gen}^2} + 1\right)\right)}\right)^2 \leq c\frac{H^5 d^2}{m} \ln^2\left(\frac{H^3 d^3 W^4 m}{\delta}\right),$$

for some absolute constant c . Thus, to have $V_0^*(s_0) - \max_{i \in [0, \dots, T-1]} V_0^{\pi_{f_i}}(s_0) \leq \epsilon$ for some $\epsilon \in (0, H)$, it is suffice to set m such that:

$$c\frac{H^5 d^2}{m} \ln^2\left(\frac{H^3 d^3 W^4 m}{\delta}\right) \leq \epsilon^2.$$

Using the fact that for any positive scalars a, b , $m = 9a \ln^2(9ab)$ is a solution to the inequality $m \geq a \ln^2(bm)$, we can set m as:

$$m := \frac{c'H^5 d^2}{\epsilon^2} \cdot \nu, \text{ where } \nu := \ln^2\left(\frac{H^8 d^5 W^4}{\epsilon^2 \delta}\right).$$

Also plugging in $\varepsilon_{gen}(m, \mathcal{H}, \delta/(HT))$ into T , we get:

$$T \leq c''Hd \ln(Hd + H^6 d^3 W^2 \nu / \epsilon^2)$$

Thus, the total number of trajectories used can be upper bounded as mT . This leads to the following corollary.

Corollary 8.18 (PAC bound for linear Bellman completion). *Fix $\epsilon, \delta \in (0, 1)$. With probability at least $1 - \delta$, for the linear Bellman completion model, BLin-UCB finds a policy π such that $V_0^*(s_0) - V_0^\pi(s_0) \leq \epsilon$, using at most*

$$\frac{\tilde{c}H^6 d^3}{\epsilon^2} \cdot \nu',$$

where $\tilde{c}$ is an absolute constant, and ν' only contains log terms:

$$\nu' := \ln^2\left(\frac{H^8 d^5 W^4}{\epsilon^2 \delta}\right) \ln\left(Hd + \frac{H^6 d^3 W^2}{\epsilon^2} \ln^2\left(\frac{H^8 d^5 W^4}{\epsilon^2 \delta}\right)\right).$$

Feature Selection in Sparse linear MDP (Example 8.15, continued). Plugging in the specific form of $\varepsilon_{gen}(m, \mathcal{H}, \delta/(TH))$ and the fact that $B_X = 1$ and $B_W = 2H\sqrt{d}$, with some linear algebra, we get:

$$\left(5\varepsilon_{gen}\sqrt{dH^3\ln\left(4Hd\left(\frac{B_X^2B_W^2}{\varepsilon_{gen}^2}+1\right)\right)}\right)^2 \leq c\frac{H^5A^2s^2}{m}\ln^2\left(\frac{AH^3d^2m}{\delta}\right),$$

for some absolute constant c . Thus, to have $V_0^*(s_0) - \max_{i \in [0, \dots, T-1]} V_0^{\pi_{f_i}}(s_0) \leq \epsilon$ for some $\epsilon \in (0, H)$, it is suffice to set m such that:

$$c\frac{H^5A^2s^2}{m}\ln^2\left(\frac{AH^3d^2m}{\delta}\right) \leq \epsilon^2.$$

Using the fact that for any positive scalars a, b , $m = 9a\ln^2(9ab)$ is a solution to the inequality $m \geq a\ln^2(bm)$, we can set m as:

$$m = \frac{c'H^5A^2s^2}{\epsilon^2} \cdot \nu, \text{ where } \nu := \ln^2\left(\frac{H^8A^3s^2d^2}{\epsilon^2\delta}\right).$$

Also plugging in $\varepsilon_{gen}(m, \mathcal{H}, \delta/(HT))$ into T , we get:

$$T = c''Hs\ln(4Hs + 4H^8A^2s^2d^2\nu/\epsilon^2)$$

Note the total number of trajectories used here is mT . This leads to the following corollary.

Corollary 8.19 (PAC bound for feature selection in sparse linear MDP). *Fix $\delta, \epsilon \in (0, 1)$. With probability at least $1 - \delta$, for the sparse linear MDP model, BLin-UCB finds a policy π such that $V_0^*(s_0) - V_0^\pi(s_0) \leq \epsilon$, using total number of trajectories at most*

$$\frac{\tilde{c}H^6A^2s^3}{\epsilon^2} \cdot \nu',$$

with $\tilde{c}$ being an absolute constant, and ν' only containing log terms:

$$\nu' := \ln^2\left(\frac{H^8A^3s^2d^2}{\epsilon^2\delta}\right)\ln\left(4Hs + \frac{4H^8A^2s^2d^2}{\epsilon^2}\ln^2\left(\frac{H^8A^3s^2d^2}{\epsilon^2\delta}\right)\right).$$

Note the polynomial dependence on s instead of d (d only appears inside log-terms) in the above PAC bound, which indicates that BLin-UCB leverages the sparsity.

8.5.3 Analysis

For notational convenience, let us overload notation and write $\varepsilon_{gen} := \varepsilon_{gen}(m, \mathcal{H}, \delta/TH)$.

Lemma 8.20. *For all $h \in [0, \dots, H-1]$ and $t \in [0, \dots, T-1]$, with probability at least $1 - \delta$, we have:*

$$\sup_{g \in \mathcal{H}} \left| \mathbb{E}_{\mathcal{D}_{t,h}}[\ell_{f_t}(s, a, s', g)] - \mathbb{E}_{s \sim d_h^{\pi_{f_t}}, a \sim \pi_{est}(s), s' \sim P_h(\cdot|s,a)}[\ell_{f_t}(s, a, s', g)] \right| \leq \varepsilon_{gen}.$$

Proof: Note that the (s, a, s') triple from $\mathcal{D}_{t,h}$ is sampled as $s \sim d_h^{\pi_{f_t}}, a \sim \pi_{est}(s), s' \sim P_h(\cdot|s, a)$, in an i.i.d fashion. Thus, using assumption 8.12 and a union bound over all h and t concludes the proof. $\blacksquare$

Our next lemma is with regards to feasibility and optimism; we show that with high probability, f^* is always a feasible solution of our constrained optimization program, which implies the optimism property holds.

Lemma 8.21 (Optimism and Feasibility of f^*). *Assume the event in Lemma 8.20 holds and that $R := \sqrt{T}\varepsilon_{gen}$. We have that for all $t \in [T]$, f^* is a feasible solution of the constrained program in Alg. 7. Furthermore, we have $V_{0,f_t}(s_0) \geq V_0^*(s_0)$, for all $t \in [T]$.*

Proof: Based on the second condition in the Bilinear class (Eq. 8.5), we have that for f^* :

$$|\mathbb{E}_{s \sim d_h^{\pi_{f_t}}, a \sim \pi_{est}(s), s' \sim P_h(\cdot|s,a)} [\ell_{f_t}(s, a, s', f^*)]| = |\langle W_h(f^*) - W_h(f_t), X_h(f_t) \rangle| = 0, \forall t.$$

This and our assumption that the event in Lemma 8.20 holds imply that:

$$\begin{aligned} |\mathbb{E}_{\mathcal{D}_{t,h}} [\ell_{f_t}(s, a, s', f^*)]| &= |\mathbb{E}_{\mathcal{D}_{t,h}} [\ell_{f_t}(s, a, s', f^*)] - \mathbb{E}_{s \sim d_h^{\pi_{f_t}}, a \sim \pi_{est}(s), s' \sim P_h(\cdot|s,a)} [\ell_{f_t}(s, a, s', f^*)]| \\ &\leq \sup_{g \in \mathcal{H}} |\mathbb{E}_{\mathcal{D}_{t,h}} [\ell_{f_t}(s, a, s', g)] - \mathbb{E}_{s \sim d_h^{\pi_{f_t}}, a \sim \pi_{est}(s), s' \sim P_h(\cdot|s,a)} [\ell_{f_t}(s, a, s', g)]| \\ &\leq \varepsilon_{gen} \end{aligned}$$

This implies $\sum_{i=0}^{t-1} (\mathbb{E}_{\mathcal{D}_{i,h}} \ell_{f_i}(s, a, s', f^*))^2 \leq T\varepsilon_{gen}^2$, which completes the proof of the first claim.

For the second claim, the feasibility of f^* and the fact that f_t is the maximizer of the constrained program immediately implies the lemma. $\blacksquare$

Now we are ready to bound the per-episode regret.

Lemma 8.22. *Assume the event in Lemma 8.20 holds. For all $t \in [0, \dots, T-1]$, we have:*

$$V_0^*(s_0) - V_0^{\pi_{f_t}}(s_0) \leq \sum_{h=0}^{H-1} |\langle W_h(f_t) - W_h(f^*), X_h(f_t) \rangle|.$$

Proof: We upper bound the regret as follows. Using optimism, we have

$$\begin{aligned} V_0^*(s_0) - V_0^{\pi_{f_t}}(s_0) &\leq V_{0,f_t}(s_0) - \mathbb{E}_{s_0 \sim \mu} [V_0^{\pi_{f_t}}(s_0)] \\ &= Q_{0,f_t}(s_0, a_0) - \mathbb{E} \left[\sum_{h=0}^{H-1} r(s_h, a_h) | \pi_{f_t} \right] \\ &= \mathbb{E} \left[\sum_{h=0}^{H-1} Q_{h,f_t}(s_h, a_h) - r(s_h, a_h) + Q_{h+1,f_t}(s_{h+1}, a_{h+1}) | \pi_{f_t} \right] \\ &\leq \sum_{h=0}^{H-1} \left| \mathbb{E}_{s_h, a_h \sim d_h^{\pi_{f_t}}} [Q_{h,f_t}(s_h, a_h) - r(s_h, a_h) - \mathbb{E}_{s_{h+1} \sim P_h(\cdot|s,a)} [V_{h+1,f_t}(s_{h+1})]] \right| \\ &\leq \sum_{h=0}^{H-1} |\langle W_h(f_t) - W_h(f^*), X_h(f_t) \rangle|, \end{aligned}$$

where the first equality uses that $a_0 = \pi_{0,f_t}(s_0)$, and the fact that $\pi_{0,f_t}(s) = \operatorname{argmax}_a Q_{0,f_t}(s, a)$, the second equality uses telescoping with the definition $Q_{H,f_0}(s, a) = 0$ for all s, a , and the last equality uses the first condition in Bilinear class (Eq. 8.4) $\blacksquare$

We are now ready to prove the main theorem. The proof uses the same elliptical potential argument from Chapter 5.

Proof:[Proof of Theorem 8.16] Define $\Sigma_{t,h} := \lambda I + \sum_{\tau=0}^{t-1} X_h(f_\tau) X_h^\top(f_\tau)$ (we will set λ later). First let us show that there exists an iteration t , such that for all $h \in [H]$,

$$\|X_h(f_t)\|_{\Sigma_{t,h}^{-1}}^2 \leq \exp \left(\frac{Hd}{T} \ln \left(1 + \frac{TB_X^2}{d\lambda} \right) \right) - 1, \quad \|W_h(f_t) - W_h(f^*)\|_{\Sigma_{t,h}}^2 \leq 4\lambda B_W^2 + 4T\varepsilon_{gen}^2. \quad (8.8)$$

We now prove the first claim above. Using Lemma 5.11 (from our linear bandit analysis),

$$\sum_{t=0}^{T-1} \ln \left(1 + \|X_h(f_t)\|_{\Sigma_{t,h}^{-1}}^2 \right) \leq \ln \frac{\det(\Sigma_{T,h})}{\det(\lambda I)} \leq d \ln \left(1 + \frac{TB_X^2}{d\lambda} \right),$$

which holds for all h . After multiplying by $1/T$ and summing over h , leads to:

$$\frac{1}{T} \sum_{t=0}^{T-1} \sum_{h=0}^{H-1} \ln \left(1 + \|X_h(f_t)\|_{\Sigma_{t,h}^{-1}}^2 \right) \leq \frac{Hd}{T} \ln \left(1 + \frac{TB_X^2}{d\lambda} \right).$$

By the intermediate value theorem, there exists a t such that $\sum_{h=0}^{H-1} \ln \left(1 + \|X_h(f_t)\|_{\Sigma_{t,h}^{-1}}^2 \right) \leq \frac{Hd}{T} \ln \left(1 + TB_X^2/(d\lambda) \right)$.

This further implies that for any h in this iteration t , we have $\ln \left(1 + \|X_h(f_t)\|_{\Sigma_{t,h}^{-1}}^2 \right) \leq \frac{Hd}{T} \ln \left(1 + TB_X^2/(d\lambda) \right)$. Rearranging terms proves the first claim.

Now let us prove our claim on the second term. Using the definition of $\pi_{est}(s)$ and that f_t is a feasible solution, we have that

$$\begin{aligned} \sum_{i=0}^{t-1} \left(\mathbb{E}_{s \sim d_h^{\pi_{f_i}}, a \sim \pi_{est}(s), s' \sim P_h(\cdot|s,a)} [\ell_{f_i}(s, a, s', f_t)] \right)^2 &\leq 2 \sum_{i=0}^{t-1} \left(\mathbb{E}_{\mathcal{D}_{i,h}} [\ell_{f_i}(s, a, s', f_t)] \right)^2 \\ &+ 2 \sum_{i=0}^{t-1} \left(\mathbb{E}_{s \sim d_h^{\pi_{f_t}}, a \sim \pi_{est}(s), s' \sim P_h(\cdot|s,a)} [\ell_{f_i}(s, a, s', f_t)] - \mathbb{E}_{\mathcal{D}_{i,h}} [\ell_{f_i}(s, a, s', f_t)] \right)^2 \leq 2T\varepsilon_{gen}^2 + 2T\varepsilon_{gen}^2 = 4T\varepsilon_{gen}^2, \end{aligned}$$

Using this and the second condition in the definition of Bilinear class (Eq. 8.5), we have:

$$\begin{aligned} \|W_h(f_t) - W_h(f^*)\|_{\Sigma_{t,h}}^2 &= \sum_{i=0}^{t-1} \left((W_h(f_t) - W_h(f^*))^\top X_h(f_i) \right)^2 + \lambda \|W_h(f_t) - W_h(f^*)\|_2^2 \\ &= \sum_{i=0}^{t-1} \left(\mathbb{E}_{s \sim d_h^{\pi_{f_i}}, a \sim \pi_{est}(s), s' \sim P_h(\cdot|s,a)} [\ell_{f_i}(s, a, s', f_t)] \right)^2 + \lambda \|W_h(f_t) - W_h(f^*)\|_2^2 \leq 4T\varepsilon_{gen}^2 + 4\lambda B_W^2, \end{aligned}$$

which proves our claimed bound on the second term.

Now consider the iteration t where Eqs. 8.8 hold. Using Lemma 8.22, we have:

$$\begin{aligned} V_0^*(s_0) - V_0^{\pi_{f_t}}(s_0) &\leq \sum_{h=0}^{H-1} |(W_h(f_t) - W_h(f^*))^\top X_h(f_t)| \\ &\leq \sum_{h=0}^{H-1} \|W_h(f_t) - W_h(f^*)\|_{\Sigma_{t,h}} \|X_h(f_t)\|_{\Sigma_{t,h}^{-1}} \\ &\leq H \sqrt{4\lambda B_W^2 + 4T\varepsilon_{gen}^2} \sqrt{\exp \left(\frac{Hd}{T} \ln \left(1 + \frac{TB_X^2}{d\lambda} \right) \right) - 1} \end{aligned}$$

Now let us set $1/\lambda = B_W^2/\varepsilon_{gen}^2 + 1/B_X^2$. Using this setting of λ and that $T = \left\lceil 2Hd \ln \left(4Hd \left(\frac{B_X^2 B_W^2}{\varepsilon_{gen}^2} + 1 \right) \right) \right\rceil$,

$$\frac{Hd}{T} \ln \left(1 + \frac{TB_X^2}{d\lambda} \right) \leq \frac{Hd}{T} \ln \left(1 + \frac{T}{d} \cdot \left(\frac{B_X^2 B_W^2}{\varepsilon_{gen}^2} + 1 \right) \right) \leq \frac{Hd}{T} \ln \left(2T \left(\frac{B_X^2 B_W^2}{\varepsilon_{gen}^2} + 1 \right) \right) \leq 1,$$

where the last step follows since we can show that, for any positive a, b ,

$$\text{If } T = 2a \ln(2ab) \text{ then } a \ln(bT)/T \leq 1. \quad (8.9)$$

This implies that:

$$\begin{aligned}
V_0^*(s_0) - V_0^{\pi_{f_t}}(s_0) &\leq H \sqrt{2(4\lambda B_W^2 + 4T\varepsilon_{gen}^2)} \\
&\leq H \sqrt{8\varepsilon_{gen}^2 + 16\varepsilon_{gen}^2 H d \ln \left(4Hd \left(\frac{B_X^2 B_W^2}{\varepsilon_{gen}^2} + 1 \right) \right)} \\
&\leq 5\varepsilon_{gen} \sqrt{dH^3 \ln \left(4Hd \left(\frac{B_X^2 B_W^2}{\varepsilon_{gen}^2} + 1 \right) \right)}.
\end{aligned}$$

where the second inequality uses that $\lambda \leq \varepsilon_{gen}^2/B_W^2$, due to our choice of λ . This concludes the proof. $\blacksquare$

8.6 The Eluder Dimension

To be added...

8.7 Bibliographic Remarks and Further Readings

The Bellman rank was originally proposed by [Jiang et al., 2017]; this original version was the V -version presented here. Both [Du et al., 2021, Jin et al., 2021] noted that there are two natural versions of the Bellman rank, which we refer to as the Q and V versions. While we focus on PAC RL (i.e. finding a good policy), the work in [Dong et al., 2020] showed how to obtain $O(\sqrt{T})$ regret bound for the case of the V -Bellman rank assumption; this algorithm utilized a careful sampling procedure (along with an “elimination-based” algorithm); the challenge here is that, for the V version, we do not have an “on-policy” estimation procedure, unlike for the Q version. With regards to regret bounds for the Q -version, a more direct algorithm is possible which obtains a $O(\sqrt{T})$ regret; this regret bound for the Q -version was provided by [Jin et al., 2021], which essentially follows along similar lines as the original argument provided in [Zanette et al., 2020] ([Zanette et al., 2020] provided a $O(\sqrt{T})$ regret for the linear Bellman completion case).

The bilinear class approach was proposed in [Du et al., 2021]. The algorithm and proof is inspired by [Jiang et al., 2017], though it provides a more simplified analysis to obtain PAC RL bounds, using a more direct elliptical function argument (which this chapter follows).

Another generalization is provided by the Eluder dimension. The Eluder dimension was first proposed in [Russo and Van Roy, 2013] in a bandit setting. It was extended to RL in [Osband and Roy, 2014, Wang et al., 2020, Ayoub et al., 2020a, Jin et al., 2021]. The Bellman eluder dimension and bounded bilinear rank are two different concepts. We focus on the latter one here due to that it captures nearly all the now standard parametric models which permit efficient RL, and it is relatively straightforward to verify. The Bellman eluder dimension can be thought of as specifying a more relaxed condition for when the algorithm terminates. Du et al. [2021] also shows that the Bilinear class can also capture Witness rank [Sun et al., 2019a] thus capturing examples such as Factored MDPs [Kearns and Koller, 1999], where Bellman rank and Bellman Eluder dimension [Jin et al., 2021] could be exponentially larger than the bilinear rank.

We also add a few remarks on further readings for important special cases. The problem of feature selection in low-rank MDP was first studied in [Jiang et al., 2017], and [Agarwal et al., 2020b, Uehara et al., 2021] later introduced two oracle-efficient model-based algorithms, and [Modi et al., 2021] introduces a model-free algorithm. The sparse linear MDP model was proposed in [Hao et al., 2021] though the algorithm from [Hao et al., 2021] does not perform strategic exploration. The linear mixture model was introduced and studied in [Modi et al., 2020a, Ayoub et al., 2020a,

Zhou et al., 2021b,a]. The linear Q^* and V^* model and the low-occupancy model were first introduced by Du et al. [2021].

Chapter 9

Deterministic MDPs with Linearly Parameterized Q^*

To be added

Part III

Policy Optimization

Chapter 10

Policy Gradient Methods and Non-Convex Optimization

For a distribution ρ over states, define:

$$V^\pi(\rho) := \mathbb{E}_{s_0 \sim \rho}[V^\pi(s_0)],$$

where we slightly overload notation. Consider a class of parametric policies $\{\pi_\theta | \theta \in \Theta \subset \mathbb{R}^d\}$. The optimization problem of interest is:

$$\max_{\theta \in \Theta} V^{\pi_\theta}(\rho).$$

We drop the MDP subscript in this chapter.

One immediate issue is that if the policy class $\{\pi_\theta\}$ consists of deterministic policies then π_θ will, in general, not be differentiable. This motivates us to consider policy classes that are stochastic, which permit differentiability.

Example 10.1 (Softmax policies). It is instructive to explicitly consider a “tabular” policy representation, given by the *softmax policy*:

$$\pi_\theta(a|s) = \frac{\exp(\theta_{s,a})}{\sum_{a'} \exp(\theta_{s,a'})}, \quad (10.1)$$

where the parameter space is $\Theta = \mathbb{R}^{|\mathcal{S}||\mathcal{A}|}$. Note that (the closure of) the set of softmax policies contains all stationary and deterministic policies.

Example 10.2 (Log-linear policies). For any state, action pair s, a , suppose we have a feature mapping $\phi_{s,a} \in \mathbb{R}^d$. Let us consider the policy class

$$\pi_\theta(a|s) = \frac{\exp(\theta \cdot \phi_{s,a})}{\sum_{a' \in \mathcal{A}} \exp(\theta \cdot \phi_{s,a'})}$$

with $\theta \in \mathbb{R}^d$.

Example 10.3 (Neural softmax policies). Here we may be interested in working with the policy class

$$\pi_\theta(a|s) = \frac{\exp(f_\theta(s, a))}{\sum_{a' \in \mathcal{A}} \exp(f_\theta(s, a'))}$$

where the scalar function $f_\theta(s, a)$ may be parameterized by a neural network, with $\theta \in \mathbb{R}^d$.

10.1 Policy Gradient Expressions and the Likelihood Ratio Method

Let τ denote a trajectory, whose unconditional distribution $\Pr_\mu^\pi(\tau)$ under π with starting distribution μ , is

$$\Pr_\mu^\pi(\tau) = \mu(s_0)\pi(a_0|s_0)P(s_1|s_0, a_0)\pi(a_1|s_1) \cdots \quad (10.2)$$

We drop the μ subscript when it is clear from context.

It is convenient to define the discounted total reward of a trajectory as:

$$R(\tau) := \sum_{t=0}^{\infty} \gamma^t r(s_t, a_t)$$

where s_t, a_t are the state-action pairs in τ . Observe that:

$$V^{\pi_\theta}(\mu) = \mathbb{E}_{\tau \sim \Pr_\mu^{\pi_\theta}}[R(\tau)].$$

Theorem 10.4. (Policy gradients) The following are expressions for $\nabla_\theta V^{\pi_\theta}(\mu)$:

- *REINFORCE:*

$$\nabla V^{\pi_\theta}(\mu) = \mathbb{E}_{\tau \sim \Pr_\mu^{\pi_\theta}} \left[R(\tau) \sum_{t=0}^{\infty} \nabla \log \pi_\theta(a_t|s_t) \right]$$

- *Action value expression:*

$$\begin{aligned} \nabla V^{\pi_\theta}(\mu) &= \mathbb{E}_{\tau \sim \Pr_\mu^{\pi_\theta}} \left[\sum_{t=0}^{\infty} \gamma^t Q^{\pi_\theta}(s_t, a_t) \nabla \log \pi_\theta(a_t|s_t) \right] \\ &= \frac{1}{1-\gamma} \mathbb{E}_{s \sim d^{\pi_\theta}} \mathbb{E}_{a \sim \pi_\theta(\cdot|s)} \left[Q^{\pi_\theta}(s, a) \nabla \log \pi_\theta(a|s) \right] \end{aligned}$$

- *Advantage expression:*

$$\nabla V^{\pi_\theta}(\mu) = \frac{1}{1-\gamma} \mathbb{E}_{s \sim d^{\pi_\theta}} \mathbb{E}_{a \sim \pi_\theta(\cdot|s)} \left[A^{\pi_\theta}(s, a) \nabla \log \pi_\theta(a|s) \right]$$

The alternative expressions are more helpful to use when we turn to Monte Carlo estimation.

Proof: We have:

$$\begin{aligned} \nabla V^{\pi_\theta}(\mu) &= \nabla \sum_{\tau} R(\tau) \Pr_\mu^{\pi_\theta}(\tau) \\ &= \sum_{\tau} R(\tau) \nabla \Pr_\mu^{\pi_\theta}(\tau) \\ &= \sum_{\tau} R(\tau) \Pr_\mu^{\pi_\theta}(\tau) \nabla \log \Pr_\mu^{\pi_\theta}(\tau) \\ &= \sum_{\tau} R(\tau) \Pr_\mu^{\pi_\theta}(\tau) \nabla \log (\mu(s_0)\pi_\theta(a_0|s_0)P(s_1|s_0, a_0)\pi_\theta(a_1|s_1) \cdots) \\ &= \sum_{\tau} R(\tau) \Pr_\mu^{\pi_\theta}(\tau) \left(\sum_{t=0}^{\infty} \nabla \log \pi_\theta(a_t|s_t) \right) \end{aligned}$$

which completes the proof of the first claim.

For the second claim, for any state s_0

$$\begin{aligned}
& \nabla V^{\pi_\theta}(s_0) \\
&= \nabla \sum_{a_0} \pi_\theta(a_0|s_0) Q^{\pi_\theta}(s_0, a_0) \\
&= \sum_{a_0} \left(\nabla \pi_\theta(a_0|s_0) \right) Q^{\pi_\theta}(s_0, a_0) + \sum_{a_0} \pi_\theta(a_0|s_0) \nabla Q^{\pi_\theta}(s_0, a_0) \\
&= \sum_{a_0} \pi_\theta(a_0|s_0) \left(\nabla \log \pi_\theta(a_0|s_0) \right) Q^{\pi_\theta}(s_0, a_0) \\
&\quad + \sum_{a_0} \pi_\theta(a_0|s_0) \nabla \left(r(s_0, a_0) + \gamma \sum_{s_1} P(s_1|s_0, a_0) V^{\pi_\theta}(s_1) \right) \\
&= \sum_{a_0} \pi_\theta(a_0|s_0) \left(\nabla \log \pi_\theta(a_0|s_0) \right) Q^{\pi_\theta}(s_0, a_0) + \gamma \sum_{a_0, s_1} \pi_\theta(a_0|s_0) P(s_1|s_0, a_0) \nabla V^{\pi_\theta}(s_1) \\
&= \mathbb{E}_{\tau \sim \text{Pr}_{s_0}^{\pi_\theta}} [Q^{\pi_\theta}(s_0, a_0) \nabla \log \pi_\theta(a_0|s_0)] + \gamma \mathbb{E}_{\tau \sim \text{Pr}_{s_0}^{\pi_\theta}} [\nabla V^{\pi_\theta}(s_1)].
\end{aligned}$$

By linearity of expectation,

$$\begin{aligned}
& \nabla V^{\pi_\theta}(\mu) \\
&= \mathbb{E}_{\tau \sim \text{Pr}_\mu^{\pi_\theta}} [Q^{\pi_\theta}(s_0, a_0) \nabla \log \pi_\theta(a_0|s_0)] + \gamma \mathbb{E}_{\tau \sim \text{Pr}_\mu^{\pi_\theta}} [\nabla V^{\pi_\theta}(s_1)] \\
&= \mathbb{E}_{\tau \sim \text{Pr}_\mu^{\pi_\theta}} [Q^{\pi_\theta}(s_0, a_0) \nabla \log \pi_\theta(a_0|s_0)] + \gamma \mathbb{E}_{\tau \sim \text{Pr}_\mu^{\pi_\theta}} [Q^{\pi_\theta}(s_1, a_1) \nabla \log \pi_\theta(a_1|s_1)] + \dots
\end{aligned}$$

where the last step follows from recursion. This completes the proof of the second claim.

The proof of the final claim is left as an exercise to the reader. ■

10.2 (Non-convex) Optimization

It is worth explicitly noting that $V^{\pi_\theta}(s)$ is non-concave in θ for the softmax parameterizations, so the standard tools of convex optimization are not applicable.

Lemma 10.5. (Non-convexity) *There is an MDP M (described in Figure 10.1) such that the optimization problem $V^{\pi_\theta}(s)$ is not concave for both the direct and softmax parameterizations.*

Proof: Recall the MDP in Figure 10.1. Note that since actions in terminal states s_3 , s_4 and s_5 do not change the expected reward, we only consider actions in states s_1 and s_2 . Let the "up/above" action as a_1 and "right" action as a_2 . Note that

$$V^\pi(s_1) = \pi(a_2|s_1)\pi(a_1|s_2) \cdot r$$

Now consider

$$\theta^{(1)} = (\log 1, \log 3, \log 3, \log 1), \quad \theta^{(2)} = (-\log 1, -\log 3, -\log 3, -\log 1)$$

where θ is written as a tuple $(\theta_{a_1, s_1}, \theta_{a_2, s_1}, \theta_{a_1, s_2}, \theta_{a_2, s_2})$. Then, for the softmax parameterization, we have:

$$\pi^{(1)}(a_2|s_1) = \frac{3}{4}; \quad \pi^{(1)}(a_1|s_2) = \frac{3}{4}; \quad V^{(1)}(s_1) = \frac{9}{16}r$$

and

$$\pi^{(2)}(a_2|s_1) = \frac{1}{4}; \quad \pi^{(2)}(a_1|s_2) = \frac{1}{4}; \quad V^{(2)}(s_1) = \frac{1}{16}r.$$

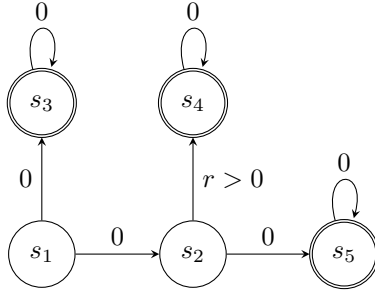


Figure 10.1: (Non-concavity example) A deterministic MDP corresponding to Lemma 10.5 where $V^{\pi_\theta}(s)$ is not concave. Numbers on arrows represent the rewards for each action.

Also, for $\theta^{(\text{mid})} = \frac{\theta^{(1)} + \theta^{(2)}}{2}$,

$$\pi^{(\text{mid})}(a_2|s_1) = \frac{1}{2}; \quad \pi^{(\text{mid})}(a_1|s_2) = \frac{1}{2}; \quad V^{(\text{mid})}(s_1) = \frac{1}{4}r.$$

This gives

$$V^{(1)}(s_1) + V^{(2)}(s_1) > 2V^{(\text{mid})}(s_1),$$

which shows that V^π is non-concave. ■

10.2.1 Gradient ascent and convergence to stationary points

Let us say a function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is β -smooth if

$$\|\nabla f(w) - \nabla f(w')\| \leq \beta \|w - w'\|, \quad (10.3)$$

where the norm $\|\cdot\|$ is the Euclidean norm. In other words, the derivatives of f do not change too quickly.

Gradient ascent, with a fixed stepsize η , follows the update rule:

$$\theta_{t+1} = \theta_t + \eta \nabla V^{\pi_{\theta_t}}(\mu).$$

It is convenient to use the shorthand notation:

$$\pi^{(t)} := \pi_{\theta_t}, \quad V^{(t)} := V^{\pi_{\theta_t}}$$

The next lemma is standard in non-convex optimization.

Lemma 10.6. (Convergence to Stationary Points) Assume that for all $\theta \in \Theta$, V^{π_θ} is β -smooth and bounded below by V_* . Suppose we use the constant stepsize $\eta = 1/\beta$. For all T , we have that

$$\min_{t \leq T} \|\nabla V^{(t)}(\mu)\|^2 \leq \frac{2\beta(V^*(\mu) - V^{(0)}(\mu))}{T}.$$

10.2.2 Monte Carlo estimation and stochastic gradient ascent

One difficulty is that even if we know the MDP M , computing the gradient may be computationally intensive. It turns out that we can obtain unbiased estimates of π with only simulation based access to our model, i.e. assuming we can obtain sampled trajectories $\tau \sim \text{Pr}_\mu^{\pi_\theta}$.

With respect to a trajectory τ , define:

$$\begin{aligned}\widehat{Q}^{\pi_\theta}(s_t, a_t) &:= \sum_{t'=t}^{\infty} \gamma^{t'-t} r(s_{t'}, a_{t'}) \\ \widehat{\nabla V}^{\pi_\theta}(\mu) &:= \sum_{t=0}^{\infty} \gamma^t \widehat{Q}^{\pi_\theta}(s_t, a_t) \nabla \log \pi_\theta(a_t | s_t)\end{aligned}$$

We now show this provides an unbiased estimated of the gradient:

Lemma 10.7. (Unbiased gradient estimate) We have :

$$\mathbb{E}_{\tau \sim \text{Pr}_\mu^{\pi_\theta}} [\widehat{\nabla V}^{\pi_\theta}(\mu)] = \nabla V^{\pi_\theta}(\mu)$$

Proof: Observe:

$$\begin{aligned}\mathbb{E}[\widehat{\nabla V}^{\pi_\theta}(\mu)] &= \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t \widehat{Q}^{\pi_\theta}(s_t, a_t) \nabla \log \pi_\theta(a_t | s_t) \right] \\ &\stackrel{(a)}{=} \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t \mathbb{E}[\widehat{Q}^{\pi_\theta}(s_t, a_t) | s_t, a_t] \nabla \log \pi_\theta(a_t | s_t) \right] \\ &\stackrel{(b)}{=} \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t Q^{\pi_\theta}(s_t, a_t) \nabla \log \pi_\theta(a_t | s_t) \right]\end{aligned}$$

where (a) follows from the tower property of the conditional expectations and (b) follows from that the Markov property implies $\mathbb{E}[Q^{\pi_\theta}(s_t, a_t) | s_t, a_t] = Q^{\pi_\theta}(s_t, a_t)$. $\blacksquare$

Hence, the following procedure is a stochastic gradient ascent algorithm:

1. initialize θ_0 .
2. For $t = 0, 1, \dots$
 - (a) Sample $\tau \sim \text{Pr}_\mu^{\pi_\theta}$.
 - (b) Update:

$$\theta_{t+1} = \theta_t + \eta_t \widehat{\nabla V}^{\pi_\theta}(\mu)$$

where η_t is the stepsize and $\widehat{\nabla V}^{\pi_\theta}(\mu)$ estimated with τ .

Note here that we are ignoring that τ is an infinite length sequence. It can be truncated appropriately so as to control the bias.

The following is standard result with regards to non-convex optimization. Again, with reasonably bounded variance, we will obtain a point θ_t with small gradient norm.

Lemma 10.8. (Stochastic Convergence to Stationary Points) Assume that for all $\theta \in \Theta$, V^{π_θ} is β -smooth and bounded below by V_* . Suppose the variance is bounded as follows:

$$\mathbb{E}[\|\widehat{\nabla V}^{\pi_\theta}(\mu) - \nabla V^{\pi_\theta}(\mu)\|^2] \leq \sigma^2$$

For $t \leq \beta(V^*(\mu) - V^{(0)}(\mu))/\sigma^2$, suppose we use a constant stepsize of $\eta_t = 1/\beta$, and thereafter, we use $\eta_t = \sqrt{2/(\beta T)}$. For all T , we have:

$$\min_{t \leq T} \mathbb{E}[\|\nabla V^{(t)}(\mu)\|^2] \leq \frac{2\beta(V^*(\mu) - V^{(0)}(\mu))}{T} + \sqrt{\frac{2\sigma^2}{T}}.$$

Baselines and stochastic gradient ascent

A significant practical issue is that the variance σ^2 is often large in practice. Here, a form of variance reduction is often critical in practice. A common method is as follows.

Let $f : \mathcal{S} \rightarrow \mathbb{R}$.

1. Construct f as an estimate of $V^{\pi_\theta}(\mu)$. This can be done using any previous data.
2. Sample a new trajectory τ , and define:

$$\begin{aligned}\widehat{Q}^{\pi_\theta}(s_t, a_t) &:= \sum_{t'=t}^{\infty} \gamma^{t'-t} r(s_{t'}, a_{t'}) \\ \widehat{\nabla V^{\pi_\theta}}(\mu) &:= \sum_{t=0}^{\infty} \gamma^t \left(\widehat{Q}^{\pi_\theta}(s_t, a_t) - f(s_t) \right) \nabla \log \pi_\theta(a_t | s_t)\end{aligned}$$

We often refer to $f(s)$ as a *baseline* at state s .

Lemma 10.9. (*Unbiased gradient estimate with Variance Reduction*) For any procedure used to construct the baseline function $f : \mathcal{S} \rightarrow \mathbb{R}$, if the samples used to construct f are independent of the trajectory τ , where $\widehat{Q}^{\pi_\theta}(s_t, a_t)$ is constructed using τ , then:

$$\mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t \left(\widehat{Q}^{\pi_\theta}(s_t, a_t) - f(s_t) \right) \nabla \log \pi_\theta(a_t | s_t) \right] = \nabla V^{\pi_\theta}(\mu)$$

where the expectation is with respect to both the random trajectory τ and the random function $f(\cdot)$.

Proof: For any function $g(s)$,

$$\mathbb{E} [\nabla \log \pi(a|s) g(s)] = \sum_a \nabla \pi(a|s) g(s) = g(s) \sum_a \nabla \pi(a|s) = g(s) \nabla \sum_a \pi(a|s) = g(s) \nabla 1 = 0$$

Using that $f(\cdot)$ is independent of τ , we have that for all t

$$\mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t f(s_t) \nabla \log \pi_\theta(a_t | s_t) \right] = 0$$

The result now follow from Lemma 10.7. ■

10.3 Bibliographic Remarks and Further Readings

The REINFORCE algorithm is due to [Williams, 1992], which is an example of the likelihood ratio method for gradient estimation [Glynn, 1990].

For standard optimization results in non-convex optimization (e.g. Lemma 10.6 and 10.8), we refer the reader to [Beck, 2017]. Our results for convergence rates for SGD to approximate stationary points follow from [Ghadimi and Lan, 2013].

Chapter 11

Optimality

We now seek to understand the global convergence properties of policy gradient methods, when given exact gradients. Here, we will largely limit ourselves to the (tabular) softmax policy class in Example 10.1.

Given our a starting distribution ρ over states, recall our objective is:

$$\max_{\theta \in \Theta} V^{\pi_\theta}(\rho).$$

where $\{\pi_\theta | \theta \in \Theta \subset \mathbb{R}^d\}$ is some class of parametric policies.

While we are interested in good performance under ρ , we will see how it is helpful to optimize under a different measure μ . Specifically, we consider optimizing $V^{\pi_\theta}(\mu)$, i.e.

$$\max_{\theta \in \Theta} V^{\pi_\theta}(\mu),$$

even though our ultimate goal is performance under $V^{\pi_\theta}(\rho)$.

We now consider the softmax policy parameterization (10.1). Here, we still have a non-concave optimization problem in general, as shown in Lemma 10.5, though we do show that global optimality can be reached under certain regularity conditions. From a practical perspective, the softmax parameterization of policies is preferable to the direct parameterization, since the parameters θ are unconstrained and standard unconstrained optimization algorithms can be employed. However, optimization over this policy class creates other challenges as we study in this section, as the optimal policy (which is deterministic) is attained by sending the parameters to infinity.

This chapter will study three algorithms for this problem, for the softmax policy class. The first performs direct policy gradient ascent on the objective without modification, while the second adds a log barrier regularizer to keep the parameters from becoming too large, as a means to ensure adequate exploration. Finally, we study the natural policy gradient algorithm and establish a global optimality convergence rate, with no dependence on the dimension-dependent factors.

The presentation in this chapter largely follows the results in [Agarwal et al., 2020d].

11.1 Vanishing Gradients and Saddle Points

To understand the necessity of optimizing under a distribution μ that is different from ρ , let us first give an informal argument that some condition on the state distribution of π , or equivalently μ , is necessary for stationarity to imply

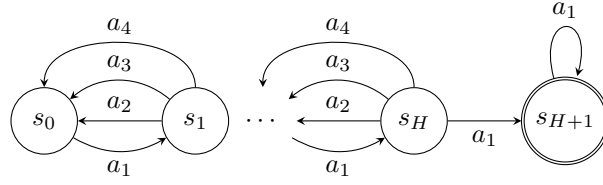


Figure 11.1: (Vanishing gradient example) A deterministic, chain MDP of length $H + 2$. We consider a policy where $\pi(a|s_i) = \theta_{s_i,a}$ for $i = 1, 2, \dots, H$. Rewards are 0 everywhere other than $r(s_{H+1}, a_1) = 1$. See Proposition 11.1.

optimality. For example, in a sparse-reward MDP (where the agent is only rewarded upon visiting some small set of states), a policy that does not visit *any* rewarding states will have zero gradient, even though it is arbitrarily suboptimal in terms of values. Below, we give a more quantitative version of this intuition, which demonstrates that even if π chooses all actions with reasonable probabilities (and hence the agent will visit all states if the MDP is connected), then there is an MDP where a large fraction of the policies π have vanishingly small gradients, and yet these policies are highly suboptimal in terms of their value.

Concretely, consider the chain MDP of length $H + 2$ shown in Figure 11.1. The starting state of interest is state s_0 and the discount factor $\gamma = H/(H + 1)$. Suppose we work with the direct parameterization, where $\pi_\theta(a|s) = \theta_{s,a}$ for $a = a_1, a_2, a_3$ and $\pi_\theta(a_4|s) = 1 - \theta_{s,a_1} - \theta_{s,a_2} - \theta_{s,a_3}$. Note we do not over-parameterize the policy. For this MDP and policy structure, if we were to initialize the probabilities over actions, say deterministically, then there is an MDP (obtained by permuting the actions) where all the probabilities for a_1 will be less than $1/4$.

The following result not only shows that the gradient is exponentially small in H , it also shows that many higher order derivatives, up to $O(H/\log H)$, are also exponentially small in H .

Proposition 11.1 (Vanishing gradients at suboptimal parameters). *Consider the chain MDP of Figure 11.1, with $H + 2$ states, $\gamma = H/(H + 1)$, and with the direct policy parameterization (with $3|S|$ parameters, as described in the text above). Suppose θ is such that $0 < \theta < 1$ (componentwise) and $\theta_{s,a_1} < 1/4$ (for all states s). For all $k \leq \frac{H}{40 \log(2H)} - 1$, we have $\|\nabla_\theta^k V^{\pi_\theta}(s_0)\| \leq (1/3)^{H/4}$, where $\nabla_\theta^k V^{\pi_\theta}(s_0)$ is a tensor of the k_{th} order derivatives of $V^{\pi_\theta}(s_0)$ and the norm is the operator norm of the tensor.¹ Furthermore, $V^*(s_0) - V^{\pi_\theta}(s_0) \geq (H + 1)/8 - (H + 1)^2/3^H$.*

We do not prove this lemma here (see Section 11.5). The lemma illustrates that lack of good exploration can indeed be detrimental in policy gradient algorithms, since the gradient can be small either due to π being near-optimal, or, simply because π does not visit advantageous states often enough. Furthermore, this lemma also suggests that varied results in the non-convex optimization literature, on escaping from saddle points, do not directly imply global convergence due to that the higher order derivatives are small.

While the chain MDP of Figure 11.1, is a common example where *sample* based estimates of gradients will be 0 under random exploration strategies; there is an exponentially small in H chance of hitting the goal state under a random exploration strategy. Note that this lemma is with regards to *exact* gradients. This suggests that even with exact computations (along with using exact higher order derivatives) we might expect numerical instabilities.

11.2 Policy Gradient Ascent

Let us now return to the softmax policy class, from Equation 10.1, where:

$$\pi_\theta(a|s) = \frac{\exp(\theta_{s,a})}{\sum_{a'} \exp(\theta_{s,a'})},$$

¹The operator norm of a k_{th} -order tensor $J \in \mathbb{R}^{d^{\otimes k}}$ is defined as $\sup_{u_1, \dots, u_k \in \mathbb{R}^d} : \|u_i\|_2=1 \langle J, u_1 \otimes \dots \otimes u_d \rangle$.

where the number of parameters in this policy class is $|\mathcal{S}||\mathcal{A}|$.

Observe that:

$$\frac{\partial \log \pi_\theta(a|s)}{\partial \theta_{s',a'}} = \mathbb{1}[s = s'] \left(\mathbb{1}[a = a'] - \pi_\theta(a'|s) \right)$$

where $\mathbb{1}[\mathcal{E}]$ is the indicator of $\mathcal{E}$ being true.

Lemma 11.2. *For the softmax policy class, we have:*

$$\frac{\partial V^{\pi_\theta}(\mu)}{\partial \theta_{s,a}} = \frac{1}{1-\gamma} d_\mu^{\pi_\theta}(s) \pi_\theta(a|s) A^{\pi_\theta}(s, a) \quad (11.1)$$

Proof: Using the advantage expression for the policy gradient (see Theorem 10.4),

$$\begin{aligned} \frac{\partial V^{\pi_\theta}(\mu)}{\partial \theta_{s',a'}} &= \frac{1}{1-\gamma} \mathbb{E}_{s \sim d_\mu^{\pi_\theta}} \mathbb{E}_{a \sim \pi_\theta(\cdot|s)} \left[A^{\pi_\theta}(s, a) \frac{\partial \log \pi_\theta(a|s)}{\partial \theta_{s',a'}} \right] \\ &= \frac{1}{1-\gamma} \mathbb{E}_{s \sim d_\mu^{\pi_\theta}} \mathbb{E}_{a \sim \pi_\theta(\cdot|s)} \left[A^{\pi_\theta}(s, a) \mathbb{1}[s = s'] \left(\mathbb{1}[a = a'] - \pi_\theta(a'|s) \right) \right] \\ &= \frac{1}{1-\gamma} d_\mu^{\pi_\theta}(s') \mathbb{E}_{a \sim \pi_\theta(\cdot|s')} \left[A^{\pi_\theta}(s', a) \left(\mathbb{1}[a = a'] - \pi_\theta(a'|s') \right) \right] \\ &= \frac{1}{1-\gamma} d_\mu^{\pi_\theta}(s') \left(\mathbb{E}_{a \sim \pi_\theta(\cdot|s')} \left[A^{\pi_\theta}(s', a) \mathbb{1}[a = a'] \right] - \pi_\theta(a'|s') \mathbb{E}_{a \sim \pi_\theta(\cdot|s')} \left[A^{\pi_\theta}(s', a) \right] \right) \\ &= \frac{1}{1-\gamma} d_\mu^{\pi_\theta}(s') \pi_\theta(a'|s') A^{\pi_\theta}(s', a') - 0, \end{aligned}$$

where the last step uses that for any policy $\sum_a \pi(a|s) A^\pi(s, a) = 0$. ■

The update rule for gradient ascent is:

$$\theta^{(t+1)} = \theta^{(t)} + \eta \nabla_\theta V^{(t)}(\mu). \quad (11.2)$$

Recall from Lemma 10.5 that, even for the case of the softmax policy class (which contains all stationary policies), our optimization problem is non-convex. Furthermore, due to the exponential scaling with the parameters θ in the softmax parameterization, *any* policy that is nearly deterministic will have gradients close to 0. Specifically, for any sequence of policies π^{θ_t} that becomes deterministic, $\|\nabla V^{\pi^{\theta_t}}\| \rightarrow 0$.

In spite of these difficulties, it turns out we have a positive result that gradient ascent asymptotically converges to the global optimum for the softmax parameterization.

Theorem 11.3 (Global convergence for softmax parameterization). *Assume we follow the gradient ascent update rule as specified in Equation (11.2) and that the distribution μ is strictly positive i.e. $\mu(s) > 0$ for all states s . Suppose $\eta \leq \frac{(1-\gamma)^3}{8}$, then we have that for all states s , $V^{(t)}(s) \rightarrow V^*(s)$ as $t \rightarrow \infty$.*

The proof is somewhat technical, and we do not provide a proof here (see Section 11.5).

A few remarks are in order. Theorem 11.3 assumed that optimization distribution μ was *strictly* positive, i.e. $\mu(s) > 0$ for all states s . We conjecture that any gradient ascent may not globally converge if this condition is not met. The concern is that if this condition is not met, then gradient ascent may not globally converge due to that $d_\mu^{\pi_\theta}(s)$ effectively scales down the learning rate for the parameters associated with state s (see Equation 11.1).

Furthermore, there is strong reason to believe that the convergence rate for this is algorithm (in the worst case) is exponentially slow in some of the relevant quantities, such as in terms of the size of state space. We now turn to a regularization based approach to ensure convergence at a polynomial rate in all relevant quantities.

11.3 Log Barrier Regularization

Due to the exponential scaling with the parameters θ , policies can rapidly become near deterministic, when optimizing under the softmax parameterization, which can result in slow convergence. Indeed a key challenge in the asymptotic analysis in the previous section was to handle the growth of the absolute values of parameters as they tend to infinity. Recall that the relative-entropy for distributions p and q is defined as:

$$\text{KL}(p, q) := \mathbb{E}_{x \sim p}[-\log q(x)/p(x)].$$

Denote the uniform distribution over a set $\mathcal{X}$ by $\text{Unif}_{\mathcal{X}}$, and define the following log barrier regularized objective as:

$$\begin{aligned} L_{\lambda}(\theta) &:= V^{\pi_{\theta}}(\mu) - \lambda \mathbb{E}_{s \sim \text{Unif}_{\mathcal{S}}} \left[\text{KL}(\text{Unif}_{\mathcal{A}}, \pi_{\theta}(\cdot|s)) \right] \\ &= V^{\pi_{\theta}}(\mu) + \frac{\lambda}{|\mathcal{S}||\mathcal{A}|} \sum_{s,a} \log \pi_{\theta}(a|s) + \lambda \log |\mathcal{A}|, \end{aligned} \quad (11.3)$$

where λ is a regularization parameter. The constant (i.e. the last term) is not relevant with regards to optimization. This regularizer is different from the more commonly utilized entropy regularizer, a point which we return to later.

The policy gradient ascent updates for $L_{\lambda}(\theta)$ are given by:

$$\theta^{(t+1)} = \theta^{(t)} + \eta \nabla_{\theta} L_{\lambda}(\theta^{(t)}). \quad (11.4)$$

We now see that any approximate first-order stationary points of the entropy-regularized objective is approximately globally optimal, provided the regularization is sufficiently small.

Theorem 11.4. (*Log barrier regularization*) Suppose θ is such that:

$$\|\nabla_{\theta} L_{\lambda}(\theta)\|_2 \leq \epsilon_{\text{opt}}$$

and $\epsilon_{\text{opt}} \leq \lambda/(2|\mathcal{S}||\mathcal{A}|)$. Then we have that for all starting state distributions ρ :

$$V^{\pi_{\theta}}(\rho) \geq V^{\star}(\rho) - \frac{2\lambda}{1-\gamma} \left\| \frac{d_{\rho}^{\pi^{\star}}}{\mu} \right\|_{\infty}.$$

We refer to $\left\| \frac{d_{\rho}^{\pi^{\star}}}{\mu} \right\|_{\infty}$ as the *distribution mismatch coefficient*. The above theorem shows the importance of having an appropriate measure $\mu(s)$ in order for the approximate first-order stationary points to be near optimal.

Proof: The proof consists of showing that $\max_a A^{\pi_{\theta}}(s, a) \leq 2\lambda/(\mu(s)|\mathcal{S}|)$ for all states. To see that this is sufficient, observe that by the performance difference lemma (Lemma 1.16),

$$\begin{aligned} V^{\star}(\rho) - V^{\pi_{\theta}}(\rho) &= \frac{1}{1-\gamma} \sum_{s,a} d_{\rho}^{\pi^{\star}}(s) \pi^{\star}(a|s) A^{\pi_{\theta}}(s, a) \\ &\leq \frac{1}{1-\gamma} \sum_s d_{\rho}^{\pi^{\star}}(s) \max_{a \in \mathcal{A}} A^{\pi_{\theta}}(s, a) \\ &\leq \frac{1}{1-\gamma} \sum_s 2d_{\rho}^{\pi^{\star}}(s) \lambda/(\mu(s)|\mathcal{S}|) \\ &\leq \frac{2\lambda}{1-\gamma} \max_s \left(\frac{d_{\rho}^{\pi^{\star}}(s)}{\mu(s)} \right). \end{aligned}$$

which would then complete the proof.

We now proceed to show that $\max_a A^{\pi_\theta}(s, a) \leq 2\lambda/(\mu(s)|\mathcal{S}|)$. For this, it suffices to bound $A^{\pi_\theta}(s, a)$ for any state-action pair s, a where $A^{\pi_\theta}(s, a) \geq 0$ else the claim is trivially true. Consider an (s, a) pair such that $A^{\pi_\theta}(s, a) > 0$. Using the policy gradient expression for the softmax parameterization (see Equation 11.1),

$$\frac{\partial L_\lambda(\theta)}{\partial \theta_{s,a}} = \frac{1}{1-\gamma} d_\mu^{\pi_\theta}(s) \pi_\theta(a|s) A^{\pi_\theta}(s, a) + \frac{\lambda}{|\mathcal{S}|} \left(\frac{1}{|\mathcal{A}|} - \pi_\theta(a|s) \right). \quad (11.5)$$

The gradient norm assumption $\|\nabla_\theta L_\lambda(\theta)\|_2 \leq \epsilon_{\text{opt}}$ implies that:

$$\begin{aligned} \epsilon_{\text{opt}} &\geq \frac{\partial L_\lambda(\theta)}{\partial \theta_{s,a}} = \frac{1}{1-\gamma} d_\mu^{\pi_\theta}(s) \pi_\theta(a|s) A^{\pi_\theta}(s, a) + \frac{\lambda}{|\mathcal{S}|} \left(\frac{1}{|\mathcal{A}|} - \pi_\theta(a|s) \right) \\ &\geq \frac{\lambda}{|\mathcal{S}|} \left(\frac{1}{|\mathcal{A}|} - \pi_\theta(a|s) \right), \end{aligned}$$

where we have used $A^{\pi_\theta}(s, a) \geq 0$. Rearranging and using our assumption $\epsilon_{\text{opt}} \leq \lambda/(2|\mathcal{S}||\mathcal{A}|)$,

$$\pi_\theta(a|s) \geq \frac{1}{|\mathcal{A}|} - \frac{\epsilon_{\text{opt}}|\mathcal{S}|}{\lambda} \geq \frac{1}{2|\mathcal{A}|}.$$

Solving for $A^{\pi_\theta}(s, a)$ in (11.5), we have:

$$\begin{aligned} A^{\pi_\theta}(s, a) &= \frac{1-\gamma}{d_\mu^{\pi_\theta}(s)} \left(\frac{1}{\pi_\theta(a|s)} \frac{\partial L_\lambda(\theta)}{\partial \theta_{s,a}} + \frac{\lambda}{|\mathcal{S}|} \left(1 - \frac{1}{\pi_\theta(a|s)|\mathcal{A}|} \right) \right) \\ &\leq \frac{1-\gamma}{d_\mu^{\pi_\theta}(s)} \left(2|\mathcal{A}|\epsilon_{\text{opt}} + \frac{\lambda}{|\mathcal{S}|} \right) \\ &\leq 2 \frac{1-\gamma}{d_\mu^{\pi_\theta}(s)} \frac{\lambda}{|\mathcal{S}|} \\ &\leq 2\lambda/(\mu(s)|\mathcal{S}|), \end{aligned}$$

where the penultimate step uses $\epsilon_{\text{opt}} \leq \lambda/(2|\mathcal{S}||\mathcal{A}|)$ and the final step uses $d_\mu^{\pi_\theta}(s) \geq (1-\gamma)\mu(s)$. This completes the proof. $\blacksquare$

The policy gradient ascent updates for $L_\lambda(\theta)$ are given by:

$$\theta^{(t+1)} = \theta^{(t)} + \eta \nabla_\theta L_\lambda(\theta^{(t)}). \quad (11.6)$$

By combining the above theorem with the convergence of gradient ascent to first order stationary points (Lemma 10.6), we obtain the following corollary.

Corollary 11.5. (*Iteration complexity with log barrier regularization*) Let $\beta_\lambda := \frac{8\gamma}{(1-\gamma)^3} + \frac{2\lambda}{|\mathcal{S}|}$. Starting from any initial $\theta^{(0)}$, consider the updates (11.6) with $\lambda = \frac{\epsilon(1-\gamma)}{2\left\|\frac{d_\rho^{\pi^*}}{\mu}\right\|_\infty}$ and $\eta = 1/\beta_\lambda$. Then for all starting state distributions ρ ,

we have

$$\min_{t \leq T} \left\{ V^*(\rho) - V^{(t)}(\rho) \right\} \leq \epsilon \quad \text{whenever} \quad T \geq \frac{320|\mathcal{S}|^2|\mathcal{A}|^2}{(1-\gamma)^6 \epsilon^2} \left\| \frac{d_\rho^{\pi^*}}{\mu} \right\|_\infty^2.$$

The corollary shows the importance of balancing how the regularization parameter λ is set relative to the desired accuracy ϵ , as well as the importance of the initial distribution μ to obtain global optimality.

Proof:[of Corollary 11.5] Let β_λ be the smoothness of $L_\lambda(\theta)$. A valid upper bound on β_λ is:

$$\beta_\lambda = \frac{8\gamma}{(1-\gamma)^3} + \frac{2\lambda}{|\mathcal{S}|},$$

where we leave the proof as an exercise to the reader.

Using Theorem 11.4, the desired optimality gap ϵ will follow if we set $\lambda = \frac{\epsilon(1-\gamma)}{2\left\|\frac{d\rho^*}{\mu}\right\|_\infty}$ and if $\|\nabla_\theta L_\lambda(\theta)\|_2 \leq \lambda/(2|\mathcal{S}||\mathcal{A}|)$. In order to complete the proof, we need to bound the iteration complexity of making the gradient sufficiently small.

By Lemma 10.6, after T iterations of gradient ascent with stepsize of $1/\beta_\lambda$, we have

$$\min_{t \leq T} \left\| \nabla_\theta L_\lambda(\theta^{(t)}) \right\|_2^2 \leq \frac{2\beta_\lambda(L_\lambda(\theta^*) - L_\lambda(\theta^{(0)}))}{T} \leq \frac{2\beta_\lambda}{(1-\gamma)T}, \quad (11.7)$$

where β_λ is an upper bound on the smoothness of $L_\lambda(\theta)$. We seek to ensure

$$\epsilon_{\text{opt}} \leq \sqrt{\frac{2\beta_\lambda}{(1-\gamma)T}} \leq \frac{\lambda}{2|\mathcal{S}||\mathcal{A}|}$$

Choosing $T \geq \frac{8\beta_\lambda |\mathcal{S}|^2 |\mathcal{A}|^2}{(1-\gamma)\lambda^2}$ satisfies the above inequality. Hence,

$$\begin{aligned} \frac{8\beta_\lambda |\mathcal{S}|^2 |\mathcal{A}|^2}{(1-\gamma)\lambda^2} &\leq \frac{64|\mathcal{S}|^2 |\mathcal{A}|^2}{(1-\gamma)^4 \lambda^2} + \frac{16|\mathcal{S}||\mathcal{A}|^2}{(1-\gamma)\lambda} \\ &\leq \frac{80|\mathcal{S}|^2 |\mathcal{A}|^2}{(1-\gamma)^4 \lambda^2} \\ &= \frac{320|\mathcal{S}|^2 |\mathcal{A}|^2}{(1-\gamma)^6 \epsilon^2} \left\| \frac{d\rho^*}{\mu} \right\|_\infty^2 \end{aligned}$$

where we have used that $\lambda < 1$. This completes the proof. $\blacksquare$

Entropy vs. log barrier regularization. A commonly considered regularizer is the entropy, where the regularizer would be:

$$\frac{1}{|\mathcal{S}|} \sum_s H(\pi_\theta(\cdot|s)) = \frac{1}{|\mathcal{S}|} \sum_s \sum_a -\pi_\theta(a|s) \log \pi_\theta(a|s).$$

Note the entropy is far less aggressive in penalizing small probabilities, in comparison to the log barrier, which is equivalent to the relative entropy. In particular, the entropy regularizer is always bounded between 0 and $\log |\mathcal{A}|$, while the relative entropy (against the uniform distribution over actions), is bounded between 0 and infinity, where it tends to infinity as probabilities tend to 0. Here, it can be shown that the convergence rate is asymptotically $O(1/\epsilon)$ (see Section 11.5) though it is unlikely that the convergence rate for this method is polynomial in other relevant quantities, including $|\mathcal{S}|$, $|\mathcal{A}|$, $1/(1-\gamma)$, and the distribution mismatch coefficient. The polynomial convergence rate using the log barrier (KL) regularizer crucially relies on the aggressive nature in which the relative entropy prevents small probabilities.

11.4 The Natural Policy Gradient

Observe that a policy constitutes a family of probability distributions $\{\pi_\theta(\cdot|s)|s \in \mathcal{S}\}$. We now consider a pre-conditioned gradient descent method based on this family of distributions. Recall that the Fisher information matrix

of a parameterized density $p_\theta(x)$ is defined as $\mathbb{E}_{x \sim p_\theta} [\nabla \log p_\theta(x) \nabla \log p_\theta(x)^\top]$. Now we let us define $\mathcal{F}_\rho^\theta$ as an (average) Fisher information matrix on the family of distributions $\{\pi_\theta(\cdot|s)|s \in \mathcal{S}\}$ as follows:

$$\mathcal{F}_\rho^\theta := \mathbb{E}_{s \sim d_\rho^{\pi_\theta}} \mathbb{E}_{a \sim \pi_\theta(\cdot|s)} [(\nabla \log \pi_\theta(a|s)) \nabla \log \pi_\theta(a|s)^\top].$$

Note that the average is under the state-action visitation frequencies. The NPG algorithm performs gradient updates in the geometry induced by this matrix as follows:

$$\theta^{(t+1)} = \theta^{(t)} + \eta F_\rho(\theta^{(t)})^\dagger \nabla_\theta V^{(t)}(\rho), \quad (11.8)$$

where $M^\dagger$ denotes the Moore-Penrose pseudoinverse of the matrix M .

Throughout this section, we restrict to using the initial state distribution $\rho \in \Delta(\mathcal{S})$ in our update rule in (11.8) (so our optimization measure μ and the performance measure ρ are identical). Also, we restrict attention to states $s \in \mathcal{S}$ reachable from ρ , since, without loss of generality, we can exclude states that are not reachable under this start state distribution².

For the softmax parameterization, this method takes a particularly convenient form; it can be viewed as a soft policy iteration update.

Lemma 11.6. (Softmax NPG as soft policy iteration) *For the softmax parameterization (10.1), the NPG updates (11.8) take the form:*

$$\theta^{(t+1)} = \theta^{(t)} + \frac{\eta}{1-\gamma} A^{(t)} + \eta v \quad \text{and} \quad \pi^{(t+1)}(a|s) = \pi^{(t)}(a|s) \frac{\exp(\eta A^{(t)}(s, a)/(1-\gamma))}{Z_t(s)},$$

where $Z_t(s) = \sum_{a \in \mathcal{A}} \pi^{(t)}(a|s) \exp(\eta A^{(t)}(s, a)/(1-\gamma))$. Here, v is only a state dependent offset (i.e. $v_{s,a} = c_s$ for some $c_s \in \mathbb{R}$ for each state s), and, owing to the normalization $Z_t(s)$, v has no effect on the update rule.

It is important to note that while the ascent direction was derived using the gradient $\nabla_\theta V^{(t)}(\rho)$, which depends on ρ , the NPG update rule actually has no dependence on the measure ρ . Furthermore, there is no dependence on the state distribution $d_\rho^{(\pi)}$, which is due to the pseudoinverse of the Fisher information cancelling out the effect of the state distribution in NPG.

Proof: By definition of the Moore-Penrose pseudoinverse, we have that $(\mathcal{F}_\rho^\theta)^\dagger \nabla V^{\pi_\theta}(\rho) = w_\star$ if and only if $w_\star$ is the minimum norm solution of:

$$\min_w \|\nabla V^{\pi_\theta}(\rho) - \mathcal{F}_\rho^\theta w\|^2.$$

Let us first evaluate $\mathcal{F}_\rho^\theta w$. For the softmax policy parameterization, Lemma 11.1 implies:

$$w^\top \nabla_\theta \log \pi_\theta(a|s) = w_{s,a} - \sum_{a' \in \mathcal{A}} w_{s,a'} \pi_\theta(a'|s) := w_{s,a} - \bar{w}_s$$

where $\bar{w}_s$ is not a function of a . This implies that:

$$\begin{aligned} \mathcal{F}_\rho^\theta w &= \mathbb{E}_{s \sim d_\rho^{\pi_\theta}} \mathbb{E}_{a \sim \pi_\theta(\cdot|s)} \left[\nabla \log \pi_\theta(a|s) \left(w^\top \nabla_\theta \log \pi_\theta(a|s) \right) \right] \\ &= \mathbb{E}_{s \sim d_\rho^{\pi_\theta}} \mathbb{E}_{a \sim \pi_\theta(\cdot|s)} \left[\nabla \log \pi_\theta(a|s) \left(w_{s,a} - \bar{w}_s \right) \right] = \mathbb{E}_{s \sim d_\rho^{\pi_\theta}} \mathbb{E}_{a \sim \pi_\theta(\cdot|s)} \left[w_{s,a} \nabla \log \pi_\theta(a|s) \right], \end{aligned}$$

where the last equality uses that $\bar{w}_s$ is not a function of s . Again using the functional form of derivative of the softmax policy parameterization, we have:

$$\left[\mathcal{F}_\rho^\theta w \right]_{s',a'} = d^{\pi_\theta}(s') \pi_\theta(a'|s') \left(w_{s',a'} - \bar{w}_{s'} \right).$$

²Specifically, we restrict the MDP to the set of states $\{s \in \mathcal{S} : \exists \pi \text{ such that } d_\rho^\pi(s) > 0\}$.

This implies:

$$\begin{aligned}\|\nabla V^{\pi_\theta}(\rho) - \mathcal{F}_\rho^\theta w\|^2 &= \sum_{s,a} \left(d^{\pi_\theta}(s) \pi_\theta(a|s) \left(\frac{1}{1-\gamma} A^{\pi_\theta}(s,a) - [\mathcal{F}_\rho^\theta w]_{s,a} \right) \right)^2 \\ &= \sum_{s,a} \left(d^{\pi_\theta}(s) \pi_\theta(a|s) \left(\frac{1}{1-\gamma} A^{\pi_\theta}(s,a) - w_{s,a} - \sum_{a' \in \mathcal{A}} w_{s,a'} \pi_\theta(a'|s) \right) \right)^2.\end{aligned}$$

Due to that $w = \frac{1}{1-\gamma} A^{\pi_\theta}$ leads to 0 error, the above implies that all 0 error solutions are of the form $w = \frac{1}{1-\gamma} A^{\pi_\theta} + v$, where v is only a state dependent offset (i.e. $v_{s,a} = c_s$ for some $c_s \in \mathbb{R}$ for each state s). The first claim follows due to that the minimum norm solution is one of these solutions. The proof of the second claim now follows by the definition of the NPG update rule, along with that v has no effect on the update rule due to the normalization constant $Z_t(s)$. ■

We now see that this algorithm enjoys a dimension free convergence rate.

Theorem 11.7 (Global convergence for NPG). *Suppose we run the NPG updates (11.8) using $\rho \in \Delta(\mathcal{S})$ and with $\theta^{(0)} = 0$. Fix $\eta > 0$. For all $T > 0$, we have:*

$$V^{(T)}(\rho) \geq V^*(\rho) - \frac{\log |\mathcal{A}|}{\eta T} - \frac{1}{(1-\gamma)^2 T}.$$

Note in the above the theorem that the NPG algorithm is directly applied to the performance measure $V^\pi(\rho)$, and the guarantees are also with respect to ρ . In particular, there is no distribution mismatch coefficient in the rate of convergence.

Now setting $\eta \geq (1-\gamma)^2 \log |\mathcal{A}|$, we see that NPG finds an ϵ -optimal policy in a number of iterations that is at most:

$$T \leq \frac{2}{(1-\gamma)^2 \epsilon},$$

which has no dependence on the number of states or actions, despite the non-concavity of the underlying optimization problem.

The proof strategy we take borrows ideas from the classical multiplicative weights algorithm (see Section 11.5).

First, the following improvement lemma is helpful:

Lemma 11.8 (Improvement lower bound for NPG). *For the iterates $\pi^{(t)}$ generated by the NPG updates (11.8), we have for all starting state distributions μ*

$$V^{(t+1)}(\mu) - V^{(t)}(\mu) \geq \frac{(1-\gamma)}{\eta} \mathbb{E}_{s \sim \mu} \log Z_t(s) \geq 0.$$

Proof: First, let us show that $\log Z_t(s) \geq 0$. To see this, observe:

$$\begin{aligned}\log Z_t(s) &= \log \sum_a \pi^{(t)}(a|s) \exp(\eta A^{(t)}(s,a)/(1-\gamma)) \\ &\geq \sum_a \pi^{(t)}(a|s) \log \exp(\eta A^{(t)}(s,a)/(1-\gamma)) = \frac{\eta}{1-\gamma} \sum_a \pi^{(t)}(a|s) A^{(t)}(s,a) = 0.\end{aligned}$$

where we have used Jensen's inequality on the concave function $\log x$ and that $\sum_a \pi^{(t)}(a|s) A^{(t)}(s,a) = 0$. By the

performance difference lemma,

$$\begin{aligned}
V^{(t+1)}(\mu) - V^{(t)}(\mu) &= \frac{1}{1-\gamma} \mathbb{E}_{s \sim d_\mu^{(t+1)}} \sum_a \pi^{(t+1)}(a|s) A^{(t)}(s, a) \\
&= \frac{1}{\eta} \mathbb{E}_{s \sim d_\mu^{(t+1)}} \sum_a \pi^{(t+1)}(a|s) \log \frac{\pi^{(t+1)}(a|s) Z_t(s)}{\pi^{(t)}(a|s)} \\
&= \frac{1}{\eta} \mathbb{E}_{s \sim d_\mu^{(t+1)}} \text{KL}(\pi_s^{(t+1)} || \pi_s^{(t)}) + \frac{1}{\eta} \mathbb{E}_{s \sim d_\mu^{(t+1)}} \log Z_t(s) \\
&\geq \frac{1}{\eta} \mathbb{E}_{s \sim d_\mu^{(t+1)}} \log Z_t(s) \geq \frac{1-\gamma}{\eta} \mathbb{E}_{s \sim \mu} \log Z_t(s),
\end{aligned}$$

where the last step uses that $d_\mu^{(t+1)} \geq (1-\gamma)\mu$ (by (1.12)) and that $\log Z_t(s) \geq 0$. ■

With this lemma, we now prove Theorem 11.7.

Proof:[of Theorem 11.7] Since ρ is fixed, we use d^* as shorthand for d_ρ^* ; we also use π_s as shorthand for the vector of $\pi(\cdot|s)$. By the performance difference lemma (Lemma 1.16),

$$\begin{aligned}
V^{\pi^*}(\rho) - V^{(t)}(\rho) &= \frac{1}{1-\gamma} \mathbb{E}_{s \sim d^*} \sum_a \pi^*(a|s) A^{(t)}(s, a) \\
&= \frac{1}{\eta} \mathbb{E}_{s \sim d^*} \sum_a \pi^*(a|s) \log \frac{\pi^{(t+1)}(a|s) Z_t(s)}{\pi^{(t)}(a|s)} \\
&= \frac{1}{\eta} \mathbb{E}_{s \sim d^*} \left(\text{KL}(\pi_s^* || \pi_s^{(t)}) - \text{KL}(\pi_s^* || \pi_s^{(t+1)}) + \sum_a \pi^*(a|s) \log Z_t(s) \right) \\
&= \frac{1}{\eta} \mathbb{E}_{s \sim d^*} \left(\text{KL}(\pi_s^* || \pi_s^{(t)}) - \text{KL}(\pi_s^* || \pi_s^{(t+1)}) + \log Z_t(s) \right),
\end{aligned}$$

where we have used the closed form of our updates from Lemma 11.6 in the second step.

By applying Lemma 11.8 with d^* as the starting state distribution, we have:

$$\frac{1}{\eta} \mathbb{E}_{s \sim d^*} \log Z_t(s) \leq \frac{1}{1-\gamma} \left(V^{(t+1)}(d^*) - V^{(t)}(d^*) \right)$$

which gives us a bound on $\mathbb{E}_{s \sim d^*} \log Z_t(s)$.

Using the above equation and that $V^{(t+1)}(\rho) \geq V^{(t)}(\rho)$ (as $V^{(t+1)}(s) \geq V^{(t)}(s)$ for all states s by Lemma 11.8), we have:

$$\begin{aligned}
V^{\pi^*}(\rho) - V^{(T-1)}(\rho) &\leq \frac{1}{T} \sum_{t=0}^{T-1} (V^{\pi^*}(\rho) - V^{(t)}(\rho)) \\
&= \frac{1}{\eta T} \sum_{t=0}^{T-1} \mathbb{E}_{s \sim d^*} (\text{KL}(\pi_s^* || \pi_s^{(t)}) - \text{KL}(\pi_s^* || \pi_s^{(t+1)})) + \frac{1}{\eta T} \sum_{t=0}^{T-1} \mathbb{E}_{s \sim d^*} \log Z_t(s) \\
&\leq \frac{\mathbb{E}_{s \sim d^*} \text{KL}(\pi_s^* || \pi^{(0)})}{\eta T} + \frac{1}{(1-\gamma)T} \sum_{t=0}^{T-1} (V^{(t+1)}(d^*) - V^{(t)}(d^*)) \\
&= \frac{\mathbb{E}_{s \sim d^*} \text{KL}(\pi_s^* || \pi^{(0)})}{\eta T} + \frac{V^{(T)}(d^*) - V^{(0)}(d^*)}{(1-\gamma)T} \\
&\leq \frac{\log |\mathcal{A}|}{\eta T} + \frac{1}{(1-\gamma)^2 T}.
\end{aligned}$$

The proof is completed using that $V^{(T)}(\rho) \geq V^{(T-1)}(\rho)$. ■

11.5 Bibliographic Remarks and Further Readings

The natural policy gradient method was originally presented in [Kakade, 2001]; a number of arguments for this method have been provided based on information geometry [Kakade, 2001, Bagnell and Schneider, 2003, Peters and Schaal, 2008].

The convergence rates in this chapter are largely derived from [Agarwal et al., 2020d]. The proof strategy for the NPG analysis has origins in the online regret framework in changing MDPs [Even-Dar et al., 2009], which would result in a worst rate in comparison to [Agarwal et al., 2020d]. This observation that the proof strategy from [Even-Dar et al., 2009] provided a convergence rate for the NPG was made in [Neu et al., 2017]. The faster NPG rate we present here is due to [Agarwal et al., 2020d]. The analysis of the MD-MPI algorithm [Geist et al., 2019] also implies a $O(1/T)$ rate for the NPG, though with worse dependencies on other parameters.

Building on ideas in [Agarwal et al., 2020d], [Mei et al., 2020] showed that, for the softmax policy class, both the gradient ascent and entropy regularized gradient ascent asymptotically converge at a $O(1/t)$; it is unlikely these methods have finite rate which are polynomial in other quantities (such as the $|S|$, $|A|$, $1/(1 - \gamma)$, and the distribution mismatch coefficient).

[Mnih et al., 2016] introduces the entropy regularizer (also see [Ahmed et al., 2019] for a more detailed empirical investigation).

Chapter 12

Function Approximation and the NPG

We now analyze the case of using parametric policy classes:

$$\Pi = \{\pi_\theta \mid \theta \in \mathbb{R}^d\},$$

where Π may not contain all stochastic policies (and it may not even contain an optimal policy). In contrast with the tabular results in the previous sections, the policy classes that we are often interested in are not fully expressive, e.g. $d \ll |\mathcal{S}||\mathcal{A}|$ (indeed $|\mathcal{S}|$ or $|\mathcal{A}|$ need not even be finite for the results in this section); in this sense, we are in the regime of function approximation.

We focus on obtaining *agnostic* results, where we seek to do as well as the best policy in this class (or as well as some other comparator policy). While we are interested in a solution to the (unconstrained) policy optimization problem

$$\max_{\theta \in \mathbb{R}^d} V^{\pi_\theta}(\rho),$$

(for a given initial distribution ρ), we will see that optimization with respect to a different distribution will be helpful, just as in the tabular case,

We will consider variants of the NPG update rule (11.8):

$$\theta \leftarrow \theta + \eta F_\rho(\theta)^\dagger \nabla_\theta V^\theta(\rho). \quad (12.1)$$

Our analysis will leverage a close connection between the NPG update rule (11.8) with the notion of *compatible function approximation*. We start by formalizing this connection. The compatible function approximation error also provides a measure of the expressivity of our parameterization, allowing us to quantify the relevant notion of approximation error for the NPG algorithm.

The main results in this chapter establish the effectiveness of NPG updates where there is error both due to statistical estimation (where we may not use exact gradients) and approximation (due to using a parameterized function class). We will see an precise estimation/approximation decomposition based on the compatible function approximation error.

The presentation in this chapter largely follows the results in [Agarwal et al., 2020d].

12.1 Compatible function approximation and the NPG

We now introduce the notion of *compatible function approximation*, which both provides some intuition with regards to policy gradient methods and it will help us later on with regards to characterizing function approximation.

Lemma 12.1 (Gradients and compatible function approximation). *Let w^* denote the following minimizer:*

$$w^* \in \mathbb{E}_{s \sim d_\mu^{\pi_\theta}} \mathbb{E}_{a \sim \pi_\theta(\cdot|s)} \left[\left(A^{\pi_\theta}(s, a) - w \cdot \nabla_\theta \log \pi_\theta(a|s) \right)^2 \right],$$

where the squared error above is referred to as the compatible function approximation. Denote the best linear predictor of $A^{\pi_\theta}(s, a)$ using $\nabla_\theta \log \pi_\theta(a|s)$ by $\hat{A}^{\pi_\theta}(s, a)$, i.e.

$$\hat{A}^{\pi_\theta}(s, a) := w^* \cdot \nabla_\theta \log \pi_\theta(a|s).$$

We have that:

$$\nabla_\theta V^{\pi_\theta}(\mu) = \frac{1}{1-\gamma} \mathbb{E}_{s \sim d_\mu^{\pi_\theta}} \mathbb{E}_{a \sim \pi_\theta(\cdot|s)} [\nabla_\theta \log \pi_\theta(a|s) \hat{A}^{\pi_\theta}(s, a)].$$

Proof: The first order optimality conditions for w^* imply

$$\mathbb{E}_{s \sim d_\mu^{\pi_\theta}} \mathbb{E}_{a \sim \pi_\theta(\cdot|s)} \left[\left(A^{\pi_\theta}(s, a) - w^* \cdot \nabla_\theta \log \pi_\theta(a|s) \right) \nabla_\theta \log \pi_\theta(a|s) \right] = 0 \quad (12.2)$$

Rearranging and using the definition of $\hat{A}^{\pi_\theta}(s, a)$,

$$\begin{aligned} \nabla_\theta V^{\pi_\theta}(\mu) &= \frac{1}{1-\gamma} \mathbb{E}_{s \sim d_\mu^{\pi_\theta}} \mathbb{E}_{a \sim \pi_\theta(\cdot|s)} [\nabla_\theta \log \pi_\theta(a|s) A^{\pi_\theta}(s, a)] \\ &= \frac{1}{1-\gamma} \mathbb{E}_{s \sim d_\mu^{\pi_\theta}} \mathbb{E}_{a \sim \pi_\theta(\cdot|s)} [\nabla_\theta \log \pi_\theta(a|s) \hat{A}^{\pi_\theta}(s, a)], \end{aligned}$$

which completes the proof. ■

The next lemma shows that the weight vector above is precisely the NPG ascent direction. Precisely,

Lemma 12.2. *We have that:*

$$F_\rho(\theta)^\dagger \nabla_\theta V^\theta(\rho) = \frac{1}{1-\gamma} w^*, \quad (12.3)$$

where w^* is a minimizer of the following regression problem:

$$w^* \in \operatorname{argmin}_w \mathbb{E}_{s \sim d_\rho^{\pi_\theta}, a \sim \pi_\theta(\cdot|s)} \left[(w^\top \nabla_\theta \log \pi_\theta(\cdot|s) - A^{\pi_\theta}(s, a))^2 \right].$$

Proof: The above is a straightforward consequence of the first order optimality conditions (see Equation 12.2). Specifically, Equation 12.2, along with the advantage expression for the policy gradient (see Theorem 10.4), imply that w^* must satisfy:

$$\nabla_\theta V^\theta(\rho) = \frac{1}{1-\gamma} F_\rho(\theta) w^*$$

which completes the proof. ■

This lemma implies that we might write the NPG update rule as:

$$\theta \leftarrow \theta + \frac{\eta}{1-\gamma} w^*.$$

where w^* is minimizer of the compatible function approximation error (which depends on θ).

The above regression problem can be viewed as “compatible” function approximation: we are approximating $A^{\pi_\theta}(s, a)$ using the $\nabla_\theta \log \pi_\theta(\cdot|s)$ as features. We also consider a variant of the above update rule, Q -NPG, where instead of using advantages in the above regression we use the Q -values. This viewpoint provides a methodology for approximate updates, where we can solve the relevant regression problems with samples.

12.2 Examples: NPG and Q-NPG

In practice, the most common policy classes are of the form:

$$\Pi = \left\{ \pi_\theta(a|s) = \frac{\exp(f_\theta(s, a))}{\sum_{a' \in \mathcal{A}} \exp(f_\theta(s, a'))} \mid \theta \in \mathbb{R}^d \right\}, \quad (12.4)$$

where f_θ is a differentiable function. For example, the tabular softmax policy class is one where $f_\theta(s, a) = \theta_{s,a}$. Typically, f_θ is either a linear function or a neural network. Let us consider the NPG algorithm, and a variant Q-NPG, in each of these two cases.

12.2.1 Log-linear Policy Classes and Soft Policy Iteration

For any state-action pair (s, a) , suppose we have a feature mapping $\phi_{s,a} \in \mathbb{R}^d$. Each policy in the log-linear policy class is of the form:

$$\pi_\theta(a|s) = \frac{\exp(\theta \cdot \phi_{s,a})}{\sum_{a' \in \mathcal{A}} \exp(\theta \cdot \phi_{s,a'})},$$

with $\theta \in \mathbb{R}^d$. Here, we can take $f_\theta(s, a) = \theta \cdot \phi_{s,a}$.

With regards to compatible function approximation for the log-linear policy class, we have:

$$\nabla_\theta \log \pi_\theta(a|s) = \bar{\phi}_{s,a}^\theta, \text{ where } \bar{\phi}_{s,a}^\theta = \phi_{s,a} - \mathbb{E}_{a' \sim \pi_\theta(\cdot|s)}[\phi_{s,a'}],$$

that is, $\bar{\phi}_{s,a}^\theta$ is the centered version of $\phi_{s,a}$. With some abuse of notation, we accordingly also define $\bar{\phi}^\pi$ for any policy π . Here, using (12.3), the NPG update rule (12.1) is equivalent to:

$$\text{NPG: } \theta \leftarrow \theta + \eta w_\star, \quad w_\star \in \operatorname{argmin}_w \mathbb{E}_{s \sim d_\rho^\pi, a \sim \pi_\theta(\cdot|s)} \left[(A^{\pi_\theta}(s, a) - w \cdot \bar{\phi}_{s,a}^\theta)^2 \right].$$

(We have rescaled the learning rate η in comparison to (12.1)). Note that we recompute $w_\star$ for every update of θ . Here, the compatible function approximation error measures the expressivity of our parameterization in how well linear functions of the parameterization can capture the policy's advantage function.

We also consider a variant of the NPG update rule (12.1), termed Q-NPG, where:

$$\text{Q-NPG: } \theta \leftarrow \theta + \eta w_\star, \quad w_\star \in \operatorname{argmin}_w \mathbb{E}_{s \sim d_\rho^\pi, a \sim \pi_\theta(\cdot|s)} \left[(Q^{\pi_\theta}(s, a) - w \cdot \phi_{s,a})^2 \right].$$

Note we do not center the features for Q-NPG; observe that $Q^\pi(s, a)$ is also not 0 in expectation under $\pi(\cdot|s)$, unlike the advantage function.

(NPG/Q-NPG and Soft-Policy Iteration) We now see how we can view both NPG and Q-NPG as an incremental (soft) version of policy iteration, just as in Lemma 11.6 for the tabular case. Rather than writing the update rule in terms of the parameter θ , we can write an equivalent update rule directly in terms of the (log-linear) policy π :

$$\text{NPG: } \pi(a|s) \leftarrow \pi(a|s) \exp(w_\star \cdot \phi_{s,a}) / Z_s, \quad w_\star \in \operatorname{argmin}_w \mathbb{E}_{s \sim d_\rho^\pi, a \sim \pi(\cdot|s)} \left[(A^\pi(s, a) - w \cdot \bar{\phi}_{s,a}^\pi)^2 \right],$$

where Z_s is normalization constant. While the policy update uses the original features ϕ instead of $\bar{\phi}^\pi$, whereas the quadratic error minimization is in terms of the centered features $\bar{\phi}^\pi$, this distinction is not relevant due to that we may also instead use $\bar{\phi}^\pi$ (in the policy update) which would result in an equivalent update; the normalization makes the update invariant to (constant) translations of the features. Similarly, an equivalent update for Q-NPG, where we update π directly rather than θ , is:

$$\text{Q-NPG: } \pi(a|s) \leftarrow \pi(a|s) \exp(w_\star \cdot \phi_{s,a}) / Z_s, \quad w_\star \in \operatorname{argmin}_w \mathbb{E}_{s \sim d_\rho^\pi, a \sim \pi(\cdot|s)} \left[(Q^\pi(s, a) - w \cdot \phi_{s,a})^2 \right].$$

(On the equivalence of NPG and Q -NPG) If it is the case that the compatible function approximation error is 0, then it is straightforward to verify that the NPG and Q -NPG are equivalent algorithms, in that their corresponding policy updates will be equivalent to each other.

12.2.2 Neural Policy Classes

Now suppose $f_\theta(s, a)$ is a neural network parameterized by $\theta \in \mathbb{R}^d$, where the policy class Π is of form in (12.4). Observe:

$$\nabla_\theta \log \pi_\theta(a|s) = g_\theta(s, a), \text{ where } g_\theta(s, a) = \nabla_\theta f_\theta(s, a) - \mathbb{E}_{a' \sim \pi_\theta(\cdot|s)}[\nabla_\theta f_\theta(s, a')],$$

and, using (12.3), the NPG update rule (12.1) is equivalent to:

$$\text{NPG: } \theta \leftarrow \theta + \eta w_\star, \quad w_\star \in \operatorname{argmin}_w \mathbb{E}_{s \sim d_\rho^{\pi_\theta}, a \sim \pi_\theta(\cdot|s)} \left[(A^{\pi_\theta}(s, a) - w \cdot g_\theta(s, a))^2 \right]$$

(Again, we have rescaled the learning rate η in comparison to (12.1)).

The Q -NPG variant of this update rule is:

$$Q\text{-NPG: } \theta \leftarrow \theta + \eta w_\star, \quad w_\star \in \operatorname{argmin}_w \mathbb{E}_{s \sim d_\rho^{\pi_\theta}, a \sim \pi_\theta(\cdot|s)} \left[(Q^{\pi_\theta}(s, a) - w \cdot \nabla_\theta f_\theta(s, a))^2 \right].$$

12.3 The NPG “Regret Lemma”

It is helpful for us to consider NPG more abstractly, as an update rule of the form

$$\theta^{(t+1)} = \theta^{(t)} + \eta w^{(t)}. \quad (12.5)$$

We will now provide a lemma where $w^{(t)}$ is an *arbitrary* (bounded) sequence, which will be helpful when specialized.

Recall a function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is said to be β -smooth if for all $x, x' \in \mathbb{R}^d$:

$$\|\nabla f(x) - \nabla f(x')\|_2 \leq \beta \|x - x'\|_2,$$

and, due to Taylor’s theorem, recall that this implies:

$$\left| f(x') - f(x) - \nabla f(x) \cdot (x' - x) \right| \leq \frac{\beta}{2} \|x' - x\|_2^2. \quad (12.6)$$

The following analysis of NPG draws close connections to the mirror-descent approach used in online learning (see Section 12.6), which motivates us to refer to it as a “regret lemma”.

Lemma 12.3. (NPG Regret Lemma) Fix a comparison policy $\tilde{\pi}$ and a state distribution p . Assume for all $s \in \mathcal{S}$ and $a \in \mathcal{A}$ that $\log \pi_\theta(a|s)$ is a β -smooth function of θ . Consider the update rule (12.5), where $\pi^{(0)}$ is the uniform distribution (for all states) and where the sequence of weights $w^{(0)}, \dots, w^{(T)}$, satisfies $\|w^{(t)}\|_2 \leq W$ (but is otherwise arbitrary). Define:

$$\text{err}_t = \mathbb{E}_{s \sim \tilde{d}} \mathbb{E}_{a \sim \tilde{\pi}(\cdot|s)} \left[A^{(t)}(s, a) - w^{(t)} \cdot \nabla_\theta \log \pi^{(t)}(a|s) \right].$$

Using $\eta = \sqrt{2 \log |\mathcal{A}| / (\beta W^2 T)}$, we have that:

$$\min_{t < T} \left\{ V^{\tilde{\pi}}(\rho) - V^{(t)}(\rho) \right\} \leq \frac{1}{1 - \gamma} \left(W \sqrt{\frac{2\beta \log |\mathcal{A}|}{T}} + \frac{1}{T} \sum_{t=0}^{T-1} \text{err}_t \right).$$

This lemma is the key tool in understanding the role of function approximation of various algorithms. We will consider one example in detail with regards to the log-linear policy class (from Example 10.2).

Note that when $\text{err}_t = 0$, as will be the case with the (tabular) softmax policy class with exact gradients, we obtain a convergence rate of $O(\sqrt{1/T})$ using a learning rate of $\eta = O(\sqrt{1/T})$. Note that this is slower than the faster rate of $O(1/T)$, provided in Theorem 11.7. Obtaining a bound that leads to a faster rate in the setting with errors requires more complicated dependencies on err_t than those stated above.

Proof: By smoothness (see (12.6)),

$$\begin{aligned} \log \frac{\pi^{(t+1)}(a|s)}{\pi^{(t)}(a|s)} &\geq \nabla_{\theta} \log \pi^{(t)}(a|s) \cdot (\theta^{(t+1)} - \theta^{(t)}) - \frac{\beta}{2} \|\theta^{(t+1)} - \theta^{(t)}\|_2^2 \\ &= \eta \nabla_{\theta} \log \pi^{(t)}(a|s) \cdot w^{(t)} - \eta^2 \frac{\beta}{2} \|w^{(t)}\|_2^2. \end{aligned}$$

We use $\tilde{d}$ as shorthand for $d_{\rho}^{\tilde{\pi}}$ (note ρ and $\tilde{\pi}$ are fixed); for any policy π , we also use π_s as shorthand for the distribution $\pi(\cdot|s)$. Using the performance difference lemma (Lemma 1.16),

$$\begin{aligned} &\mathbb{E}_{s \sim \tilde{d}} \left(\text{KL}(\tilde{\pi}_s \| \pi_s^{(t)}) - \text{KL}(\tilde{\pi}_s \| \pi_s^{(t+1)}) \right) \\ &= \mathbb{E}_{s \sim \tilde{d}} \mathbb{E}_{a \sim \tilde{\pi}(\cdot|s)} \left[\log \frac{\pi^{(t+1)}(a|s)}{\pi^{(t)}(a|s)} \right] \\ &\geq \eta \mathbb{E}_{s \sim \tilde{d}} \mathbb{E}_{a \sim \tilde{\pi}(\cdot|s)} \left[\nabla_{\theta} \log \pi^{(t)}(a|s) \cdot w^{(t)} \right] - \eta^2 \frac{\beta}{2} \|w^{(t)}\|_2^2 \quad (\text{using previous display}) \\ &= \eta \mathbb{E}_{s \sim \tilde{d}} \mathbb{E}_{a \sim \tilde{\pi}(\cdot|s)} \left[A^{(t)}(s, a) \right] - \eta^2 \frac{\beta}{2} \|w^{(t)}\|_2^2 \\ &\quad + \eta \mathbb{E}_{s \sim \tilde{d}} \mathbb{E}_{a \sim \tilde{\pi}(\cdot|s)} \left[\nabla_{\theta} \log \pi^{(t)}(a|s) \cdot w^{(t)} - A^{(t)}(s, a) \right] \\ &= (1 - \gamma) \eta \left(V^{\tilde{\pi}}(\rho) - V^{(t)}(\rho) \right) - \eta^2 \frac{\beta}{2} \|w^{(t)}\|_2^2 - \eta \text{err}_t \end{aligned}$$

Rearranging, we have:

$$V^{\tilde{\pi}}(\rho) - V^{(t)}(\rho) \leq \frac{1}{1 - \gamma} \left(\frac{1}{\eta} \mathbb{E}_{s \sim \tilde{d}} \left(\text{KL}(\tilde{\pi}_s \| \pi_s^{(t)}) - \text{KL}(\tilde{\pi}_s \| \pi_s^{(t+1)}) \right) + \frac{\eta \beta}{2} W^2 + \text{err}_t \right)$$

Proceeding,

$$\begin{aligned} \frac{1}{T} \sum_{t=0}^{T-1} (V^{\tilde{\pi}}(\rho) - V^{(t)}(\rho)) &\leq \frac{1}{\eta T(1 - \gamma)} \sum_{t=0}^{T-1} \mathbb{E}_{s \sim \tilde{d}} (\text{KL}(\tilde{\pi}_s \| \pi_s^{(t)}) - \text{KL}(\tilde{\pi}_s \| \pi_s^{(t+1)})) \\ &\quad + \frac{1}{T(1 - \gamma)} \sum_{t=0}^{T-1} \left(\frac{\eta \beta W^2}{2} + \text{err}_t \right) \\ &\leq \frac{\mathbb{E}_{s \sim \tilde{d}} \text{KL}(\tilde{\pi}_s \| \pi^{(0)})}{\eta T(1 - \gamma)} + \frac{\eta \beta W^2}{2(1 - \gamma)} + \frac{1}{T(1 - \gamma)} \sum_{t=0}^{T-1} \text{err}_t \\ &\leq \frac{\log |\mathcal{A}|}{\eta T(1 - \gamma)} + \frac{\eta \beta W^2}{2(1 - \gamma)} + \frac{1}{T(1 - \gamma)} \sum_{t=0}^{T-1} \text{err}_t, \end{aligned}$$

which completes the proof. ■

12.4 Q -NPG: Performance Bounds for Log-Linear Policies

For a state-action distribution ν , define:

$$L(w; \theta, \nu) := \mathbb{E}_{s,a \sim \nu} \left[\left(Q^{\pi_\theta}(s, a) - w \cdot \phi_{s,a} \right)^2 \right].$$

The iterates of the Q -NPG algorithm can be viewed as minimizing this loss under some (changing) distribution ν .

We now specify an approximate version of Q -NPG. It is helpful to consider a slightly more general version of the algorithm in the previous section, where instead of optimizing under a starting state distribution ρ , we have a different starting *state-action* distribution ν . The motivation for this is similar in spirit to our log barrier regularization: we seek to maintain exploration (and estimation) over the action space even if the current policy does not have coverage over the action space.

Analogous to the definition of the state-action visitation measure $d_{s_0}^\pi$ (see Equation 1.9), we define another state-action visitation measure, d_{s_0, a_0}^π , as follows:

$$d_{s_0, a_0}^\pi(s, a) := (1 - \gamma) \sum_{t=0}^{\infty} \gamma^t \Pr^\pi(s_t = s, a_t = a | s_0, a_0) \quad (12.7)$$

where $\Pr^\pi(s_t = s, a_t = a | s_0, a_0)$ is the probability that $s_t = s$ and $a_t = a$, after starting at state s_0 , taking action a_0 , and following π thereafter. Again, we overload notation and write:

$$d_\nu^\pi(s, a) = \mathbb{E}_{s_0, a_0 \sim \nu} [d_{s_0, a_0}^\pi]$$

where ν is a distribution over $\mathcal{S} \times \mathcal{A}$.

Q -NPG will be defined with respect to the *on-policy* state action measure starting with $s_0, a_0 \sim \nu$. As per our convention, we define

$$d^{(t)} := d_\nu^{\pi^{(t)}}.$$

The approximate version of this algorithm is:

$$\text{Approx. } Q\text{-NPG: } \theta^{(t+1)} = \theta^{(t)} + \eta w^{(t)}, \quad w^{(t)} \approx \operatorname{argmin}_{\|w\|_2 \leq W} L(w; \theta^{(t)}, d^{(t)}), \quad (12.8)$$

where the above update rule also permits us to constrain the norm of the update direction $w^{(t)}$ (alternatively, we could use ℓ_2 regularization as is also common in practice). The exact minimizer is denoted as:

$$w_\star^{(t)} \in \operatorname{argmin}_{\|w\|_2 \leq W} L(w; \theta^{(t)}, d^{(t)}).$$

Note that $w_\star^{(t)}$ depends on the current parameter $\theta^{(t)}$.

Our analysis will take into account both the *excess risk* (often also referred to as estimation error) and the *approximation error*. The standard approximation-estimation error decomposition is as follows:

$$L(w^{(t)}; \theta^{(t)}, d^{(t)}) = \underbrace{L(w^{(t)}; \theta^{(t)}, d^{(t)}) - L(w_\star^{(t)}; \theta^{(t)}, d^{(t)})}_{\text{Excess risk}} + \underbrace{L(w_\star^{(t)}; \theta^{(t)}, d^{(t)})}_{\text{Approximation error}}$$

Using a sample based approach, we would expect $\epsilon_{\text{stat}} = O(1/\sqrt{N})$ or better, where N is the number of samples used to estimate $w_\star^{(t)}$. In contrast, the approximation error is due to modeling error, and does not tend to 0 with more samples. We will see how these two errors have strikingly different impact on our final performance bound.

Note that we have already considered two cases where $\epsilon_{\text{approx}} = 0$. For the tabular softmax policy class, it is immediate that $\epsilon_{\text{approx}} = 0$. A more interesting example (where the state and action space could be infinite) is provided by

the linear parameterized MDP model from Chapter 7. Here, provided that we use the log-linear policy class (see Section 12.2.1) with features corresponding to the linear MDP features, it is straightforward to see that $\epsilon_{\text{approx}} = 0$ for this log-linear policy class. More generally, we will see the effect of model misspecification in our performance bounds.

We make the following assumption on these errors:

Assumption 12.4 (Approximation/estimation error bounds). Let $w^{(0)}, w^{(1)}, \dots, w^{(T-1)}$ be the sequence of iterates used by the Q -NPG algorithm. Suppose the following holds for all $t < T$:

1. (*Excess risk*) Suppose the estimation error is bounded as follows:

$$L(w^{(t)}; \theta^{(t)}, d^{(t)}) - L(w_{\star}^{(t)}; \theta^{(t)}, d^{(t)}) \leq \epsilon_{\text{stat}}$$

2. (*Approximation error*) Suppose the approximation error is bounded as follows:

$$L(w_{\star}^{(t)}; \theta^{(t)}, d^{(t)}) \leq \epsilon_{\text{approx}}.$$

We will also see how, with regards to our estimation error, we will need a far more mild notion of coverage. Here, with respect to any state-action distribution v , define:

$$\Sigma_v = \mathbb{E}_{s,a \sim v} [\phi_{s,a} \phi_{s,a}^{\top}].$$

We make a the following conditioning assumption:

Assumption 12.5 (Relative condition number). Fix a state distribution ρ (this will be what ultimately be the performance measure that we seek to optimize). Consider an arbitrary comparator policy π^* (not necessarily an optimal policy). With respect to π^* , define the state-action measure d^* as

$$d^*(s, a) = d_{\rho}^{\pi^*}(s) \cdot \text{Unif}_{\mathcal{A}}(a)$$

i.e. d^* samples states from the comparators state visitation measure, $d_{\rho}^{\pi^*}$ and actions from the uniform distribution. Define

$$\sup_{w \in \mathbb{R}^d} \frac{w^{\top} \Sigma_{d^*} w}{w^{\top} \Sigma_{\nu} w} = \kappa,$$

and assume that κ is finite.

We later discuss why it is reasonable to expect that κ is not a quantity related to the size of the state space.

The main result of this chapter shows how the approximation error, the excess risk, and the conditioning, determine the final performance.

Theorem 12.6. Fix a state distribution ρ ; a state-action distribution ν ; an arbitrary comparator policy π^* (not necessarily an optimal policy). Suppose Assumption 12.5 holds with respect to these choices and that $\|\phi_{s,a}\|_2 \leq B$ for all s, a . Suppose the Q -NPG update rule (in (12.8)) starts with $\theta^{(0)} = 0$, $\eta = \sqrt{2 \log |\mathcal{A}| / (B^2 W^2 T)}$, and the (random) sequence of iterates satisfies Assumption 12.4. We have that:

$$\mathbb{E} \left[\min_{t < T} \left\{ V^{\pi^*}(\rho) - V^{(t)}(\rho) \right\} \right] \leq \frac{BW}{1-\gamma} \sqrt{\frac{2 \log |\mathcal{A}|}{T}} + \sqrt{\frac{4|\mathcal{A}|}{(1-\gamma)^3} \left(\kappa \cdot \epsilon_{\text{stat}} + \left\| \frac{d^*}{\nu} \right\|_{\infty} \cdot \epsilon_{\text{approx}} \right)}.$$

Note when $\epsilon_{\text{approx}} = 0$, our convergence rate is $O(\sqrt{1/T})$ plus a term that depends on the excess risk; hence, provided we obtain enough samples, then ϵ_{stat} will also tend to 0, and we will be competitive with the comparison policy π^* .

The above also shows the striking difference between the effects of estimation error and approximation error. A few remarks are now in order.

Transfer learning, distribution shift, and the approximation error. In large scale problems, the worst case distribution mismatch factor $\left\| \frac{d^*}{\nu} \right\|_\infty$ is unlikely to be small. However, this factor is ultimately due to transfer learning. Our approximation error is with respect to the fitting distribution $d^{(t)}$, where we assume that $L(w_\star^{(t)}; \theta^{(t)}, d^{(t)}) \leq \epsilon_{\text{approx}}$. As the proof will show, the relevant notion of approximation error will be $L(w_\star^{(t)}; \theta^{(t)}, d^*)$, where d^* is the *fixed* comparators measure. In others words, to get a good performance bound we need to successfully have low transfer learning error to the fixed measure d^* . Furthermore, in many modern machine learning applications, this error is often is favorable, in that it is substantially better than worst case theory might suggest.

See Section 12.6 for further remarks on this point.

Dimension dependence in κ and the importance of ν . It is reasonable to think about κ as being dimension dependent (or worse), but it is not necessarily related to the size of the state space. For example, if $\|\phi_{s,a}\|_2 \leq B$, then $\kappa \leq \frac{B^2}{\sigma_{\min}(\mathbb{E}_{s,a \sim \nu} [\phi_{s,a} \phi_{s,a}^\top])}$ though this bound may be pessimistic. Here, we also see the importance of choice of ν in having a small (relative) condition number; in particular, this is the motivation for considering the generalization which allows for a starting state-action distribution ν vs. just a starting state distribution μ (as we did in the tabular case). Roughly speaking, we desire a ν which provides good coverage over the features. As the following lemma shows, there always exists a universal distribution ν , which can be constructed only with knowledge of the feature set (without knowledge of d^*), such that $\kappa \leq d$.

Lemma 12.7. (*$\kappa \leq d$ is always possible*) Let $\Phi = \{\phi(s,a) | (s,a) \in \mathcal{S} \times \mathcal{A}\} \subset \mathbb{R}^d$ and suppose Φ is a compact set. There always exists a state-action distribution ν , which is supported on at most d^2 state-action pairs and which can be constructed only with knowledge of Φ (without knowledge of the MDP or d^*), such that:

$$\kappa \leq d.$$

Proof: The distribution can be found through constructing the minimal volume ellipsoid containing Φ , i.e. the Löwner-John ellipsoid. **To be added...** ■

Direct policy optimization vs. approximate value function programming methods Part of the reason for the success of the direct policy optimization approaches is to due their more mild dependence on the approximation error. Here, our theoretical analysis has a dependence on a distribution mismatch coefficient, $\left\| \frac{d^*}{\nu} \right\|_\infty$, while approximate value function methods have even worse dependencies. See Chapter 4. As discussed earlier and as can be seen in the regret lemma (Lemma 12.3), the distribution mismatch coefficient is due to that the relevant error for NPG is a transfer error notion to a *fixed* comparator distribution, while approximate value function methods have more stringent conditions where the error has to be small under, essentially, the distribution of *any* other policy.

12.4.1 Analysis

Proof: (of Theorem 12.6) For the log-linear policy class, due to that the feature mapping ϕ satisfies $\|\phi_{s,a}\|_2 \leq B$, then it is not difficult to verify that $\log \pi_\theta(a|s)$ is a B^2 -smooth function. Using this and the NPG regret lemma (Lemma 12.3), we have:

$$\min_{t \leq T} \left\{ V^{\pi^*}(\rho) - V^{(t)}(\rho) \right\} \leq \frac{BW}{1-\gamma} \sqrt{\frac{2 \log |\mathcal{A}|}{T}} + \frac{1}{(1-\gamma)T} \sum_{t=0}^{T-1} \text{err}_t.$$

where we have used our setting of η .

We make the following decomposition of err_t :

$$\begin{aligned} \text{err}_t &= \mathbb{E}_{s \sim d_\rho^*, a \sim \pi^*(\cdot|s)} \left[A^{(t)}(s, a) - w_\star^{(t)} \cdot \nabla_\theta \log \pi^{(t)}(a|s) \right] \\ &\quad + \mathbb{E}_{s \sim d_\rho^*, a \sim \pi^*(\cdot|s)} \left[(w_\star^{(t)} - w^{(t)}) \cdot \nabla_\theta \log \pi^{(t)}(a|s) \right]. \end{aligned}$$

For the first term, using that $\nabla_\theta \log \pi_\theta(a|s) = \phi_{s,a} - \mathbb{E}_{a' \sim \pi_\theta(\cdot|s)} [\phi_{s,a'}]$ (see Section 12.2.1), we have:

$$\begin{aligned} &\mathbb{E}_{s \sim d_\rho^*, a \sim \pi^*(\cdot|s)} \left[A^{(t)}(s, a) - w_\star^{(t)} \cdot \nabla_\theta \log \pi^{(t)}(a|s) \right] \\ &= \mathbb{E}_{s \sim d_\rho^*, a \sim \pi^*(\cdot|s)} \left[Q^{(t)}(s, a) - w_\star^{(t)} \cdot \phi_{s,a} \right] - \mathbb{E}_{s \sim d_\rho^*, a' \sim \pi^{(t)}(\cdot|s)} \left[Q^{(t)}(s, a') - w_\star^{(t)} \cdot \phi_{s,a'} \right] \\ &\leq \sqrt{\mathbb{E}_{s \sim d_\rho^*, a \sim \pi^*(\cdot|s)} \left(Q^{(t)}(s, a) - w_\star^{(t)} \cdot \phi_{s,a} \right)^2} + \sqrt{\mathbb{E}_{s \sim d_\rho^*, a' \sim \pi^{(t)}(\cdot|s)} \left(Q^{(t)}(s, a') - w_\star^{(t)} \cdot \phi_{s,a'} \right)^2} \\ &\leq 2\sqrt{|\mathcal{A}| \mathbb{E}_{s \sim d_\rho^*, a \sim \text{Unif}_{\mathcal{A}}} \left[\left(Q^{(t)}(s, a) - w_\star^{(t)} \cdot \phi_{s,a} \right)^2 \right]} = 2\sqrt{|\mathcal{A}| L(w_\star^{(t)}; \theta^{(t)}, d^*)}. \end{aligned}$$

where we have used the definition of d^* and $L(w_\star^{(t)}; \theta^{(t)}, d^*)$ in the last step. Using following crude upper bound,

$$L(w_\star^{(t)}; \theta^{(t)}, d^*) \leq \left\| \frac{d^*}{d^{(t)}} \right\|_\infty L(w_\star^{(t)}; \theta^{(t)}, d^{(t)}) \leq \frac{1}{1-\gamma} \left\| \frac{d^*}{\nu} \right\|_\infty L(w_\star^{(t)}; \theta^{(t)}, d^{(t)}),$$

(where the last step uses the definition of $d^{(t)}$, see Equation 12.7), we have that:

$$\mathbb{E}_{s \sim d_\rho^*, a \sim \pi^*(\cdot|s)} \left[A^{(t)}(s, a) - w_\star^{(t)} \cdot \nabla_\theta \log \pi^{(t)}(a|s) \right] \leq 2\sqrt{\frac{|\mathcal{A}|}{1-\gamma} \left\| \frac{d^*}{\nu} \right\|_\infty L(w_\star^{(t)}; \theta^{(t)}, d^{(t)})}. \quad (12.9)$$

For the second term, let us now show that:

$$\begin{aligned} &\mathbb{E}_{s \sim d_\rho^*, a \sim \pi^*(\cdot|s)} \left[(w_\star^{(t)} - w^{(t)}) \cdot \nabla_\theta \log \pi^{(t)}(a|s) \right] \\ &\leq 2\sqrt{\frac{|\mathcal{A}| \kappa}{1-\gamma} \left(L(w^{(t)}; \theta^{(t)}, d^{(t)}) - L(w_\star^{(t)}; \theta^{(t)}, d^{(t)}) \right)} \end{aligned} \quad (12.10)$$

To see this, first observe that a similar argument to the above leads to:

$$\begin{aligned} &\mathbb{E}_{s \sim d_\rho^*, a \sim \pi^*(\cdot|s)} \left[(w_\star^{(t)} - w^{(t)}) \cdot \nabla_\theta \log \pi^{(t)}(a|s) \right] \\ &= \mathbb{E}_{s \sim d_\rho^*, a \sim \pi^*(\cdot|s)} \left[(w_\star^{(t)} - w^{(t)}) \cdot \phi_{s,a} \right] - \mathbb{E}_{s \sim d_\rho^*, a' \sim \pi^{(t)}(\cdot|s)} \left[(w_\star^{(t)} - w^{(t)}) \cdot \phi_{s,a'} \right] \\ &\leq 2\sqrt{|\mathcal{A}| \mathbb{E}_{s, a \sim d^*} \left[\left((w_\star^{(t)} - w^{(t)}) \cdot \phi_{s,a} \right)^2 \right]} = 2\sqrt{|\mathcal{A}| \cdot \|w_\star^{(t)} - w^{(t)}\|_{\Sigma_{d^*}}^2}, \end{aligned}$$

where we use the notation $\|x\|_M^2 := x^\top M x$ for a matrix M and a vector x . From the definition of κ ,

$$\|w_\star^{(t)} - w^{(t)}\|_{\Sigma_{d^*}}^2 \leq \kappa \|w_\star^{(t)} - w^{(t)}\|_{\Sigma_\nu}^2 \leq \frac{\kappa}{1-\gamma} \|w_\star^{(t)} - w^{(t)}\|_{\Sigma_{d^{(t)}}}^2$$

using that $(1-\gamma)\nu \leq d_\nu^{(t)}$ (see (12.7)). Due to that $w_\star^{(t)}$ minimizes $L(w; \theta^{(t)}, d^{(t)})$ over the set $\mathcal{W} := \{w : \|w\|_2 \leq W\}$, for any $w \in \mathcal{W}$ the first-order optimality conditions for $w_\star^{(t)}$ imply that:

$$(w - w_\star^{(t)}) \cdot \nabla L(w_\star^{(t)}; \theta^{(t)}, d^{(t)}) \geq 0.$$

Therefore, for any $w \in \mathcal{W}$,

$$\begin{aligned}
& L(w; \theta^{(t)}, d^{(t)}) - L(w_\star^{(t)}; \theta^{(t)}, d^{(t)}) \\
&= \mathbb{E}_{s,a \sim d^{(t)}} \left[\left(w \cdot \phi(s, a) - w_\star \cdot \phi(s, a) + w_\star \cdot \phi(s, a) - Q^{(t)}(s, a) \right)^2 \right] - L(w_\star^{(t)}; \theta^{(t)}, d^{(t)}) \\
&= \mathbb{E}_{s,a \sim d^{(t)}} \left[\left(w \cdot \phi(s, a) - w_\star \cdot \phi(s, a) \right)^2 \right] + 2(w - w_\star) \mathbb{E}_{s,a \sim d^{(t)}} \left[\phi(s, a) (w_\star \cdot \phi(s, a) - Q^{(t)}(s, a)) \right] \\
&= \|w - w_\star^{(t)}\|_{\Sigma_{d^{(t)}}}^2 + (w - w_\star^{(t)}) \cdot \nabla L(w_\star^{(t)}; \theta^{(t)}, d^{(t)}) \\
&\geq \|w - w_\star^{(t)}\|_{\Sigma_{d^{(t)}}}^2.
\end{aligned}$$

Noting that $w^{(t)} \in \mathcal{W}$ by construction in Algorithm 12.8 yields the claimed bound on the second term in (12.10).

Using the bounds on the first and second terms in (12.9) and (12.10), along with concavity of the square root function, we have that:

$$\text{err}_t \leq 2\sqrt{\frac{|\mathcal{A}|}{1-\gamma} \left\| \frac{d^\star}{\nu} \right\|_\infty} L(w_\star^{(t)}; \theta^{(t)}, d^{(t)}) + 2\sqrt{\frac{|\mathcal{A}|\kappa}{1-\gamma}} \left(L(w^{(t)}; \theta^{(t)}, d^{(t)}) - L(w_\star^{(t)}; \theta^{(t)}, d^{(t)}) \right).$$

The proof is completed by substitution and using our assumptions on ϵ_{stat} and ϵ_{bias} . ■

12.5 Q -NPG Sample Complexity

To be added...

12.6 Bibliographic Remarks and Further Readings

The notion of *compatible function approximation* was due to [Sutton et al., 1999], which also proved the claim in Lemma 12.1. The close connection of the NPG update rule to compatible function approximation (Lemma 12.3) was noted in [Kakade, 2001].

The regret lemma (Lemma 12.3) for the NPG analysis has origins in the online regret framework in changing MDPs [Even-Dar et al., 2009]. The convergence rates in this chapter are largely derived from [Agarwal et al., 2020d]. The Q-NPG algorithm for the log-linear policy classes is essentially the same algorithm as POLITEX [Abbasi-Yadkori et al., 2019], with a distinction that it is important to use a state action measure over the initial distribution. The analysis and error decomposition of Q-NPG is from [Agarwal et al., 2020d], which has a more general analysis of NPG with function approximation under the regret lemma. This more general approach also permits the analysis of neural policy classes, as shown in [Agarwal et al., 2020d]. Also, [Liu et al., 2019] provide an analysis of the TRPO algorithm [Schulman et al., 2015] (essentially the same as NPG) for neural network parameterizations in the somewhat restrictive linearized “neural tangent kernel” regime.

Chapter 13

CPI, TRPO, and More

In this chapter, we consider conservative policy iteration (CPI) and trust-region constrained policy optimization (TRPO). Both CPI and TRPO can be understood as making small incremental update to the policy by forcing that the new policy's state action distribution is not too far away from the current policy's. We will see that CPI achieves that by forming a new policy that is a mixture of the current policy and a local greedy policy, while TRPO forcing that by explicitly adding a KL constraint (over policies' induced trajectory distributions space) in the optimization procedure. We will show that TRPO gives an equivalent update procedure as Natural Policy Gradient.

Along the way, we discuss the benefit of incremental policy update, by contrasting it to another family of policy update procedure called *Approximate Policy Iteration* (API), which performs local greedy policy search and could potentially lead to abrupt policy change. We show that API in general fails to converge or make local improvement, unless under a much stronger concentrability ratio assumption.

The algorithm and analysis of CPI is adapted from the original one in [Kakade and Langford, 2002], and the we follow the presentation of TRPO from [Schulman et al., 2015], while making a connection to the NGP algorithm.

13.1 Conservative Policy Iteration

As the name suggests, we will now describe a more conservative version of the policy iteration algorithm, which shifts the next policy away from the current policy with a small step size to prevent drastic shifts in successive state distributions.

We consider the discounted MDP here $\{\mathcal{S}, \mathcal{A}, P, r, \gamma, \rho\}$ where ρ is the initial state distribution. Similar to Policy Gradient Methods, we assume that we have a restart distribution μ (i.e., the μ -restart setting). Throughout this section, for any policy π , we denote d_μ^π as the state-action visitation starting from $s_0 \sim \mu$ instead of ρ , and d^π the state-action visitation starting from the true initial state distribution ρ , i.e., $s_0 \sim \rho$. Similarly, we denote V_μ^π as expected discounted total reward of policy π starting at μ , while V^π as the expected discounted total reward of π with ρ as the initial state distribution. We assume $\mathcal{A}$ is discrete but $\mathcal{S}$ could be continuous.

CPI is based on the concept of *Reduction to Supervised Learning*. Specifically we will use the Approximate Greedy Policy Selector defined in Chapter 4 (Definition ??). We recall the definition of the ε -approximate Greedy Policy Selector $\mathcal{G}_\varepsilon(\pi, \Pi, \mu)$ below. Given a policy π , policy class Π , and a restart distribution μ , denote $\hat{\pi} = \mathcal{G}_\varepsilon(\pi, \Pi, \mu)$, we have that:

$$\mathbb{E}_{s \sim d_\mu^\pi} A^\pi(s, \hat{\pi}(s)) \geq \max_{\tilde{\pi} \in \Pi} \mathbb{E}_{s \sim d_\mu^\pi} A^\pi(s, \tilde{\pi}(s)) - \varepsilon.$$

Recall that in Chapter 4 we explained two approach to implement such approximate oracle: one with a reduction to classification oracle, and the other one with a reduction to regression oracle.

13.1.1 The CPI Algorithm

CPI, summarized in Alg. 8, will iteratively generate a sequence of policies π^t . Note we use $\pi_\alpha = (1-\alpha)\pi + \alpha\pi'$ to refer to a randomized policy which at any state s , chooses an action according to π with probability $1 - \alpha$ and according to π' with probability α . The greedy policy π' is computed using the ε -approximate greedy policy selector $\mathcal{G}_\varepsilon(\pi^t, \Pi, \mu)$. The algorithm is terminate when there is no significant one-step improvement over π^t , i.e., $\mathbb{E}_{s \sim d_\mu^{\pi^t}} A^{\pi^t}(s, \pi'(s)) \leq \varepsilon$.

Algorithm 8 Conservative Policy Iteration (CPI)

Input: Initial policy $\pi^0 \in \Pi$, accuracy parameter ε .

```

1: for  $t = 0, 1, 2 \dots$  do
2:    $\pi' = \mathcal{G}_\varepsilon(\pi^t, \Pi, \mu)$ 
3:   if  $\mathbb{E}_{s \sim d_\mu^{\pi^t}} A^{\pi^t}(s, \pi'(s)) \leq \varepsilon$  then
4:     Return  $\pi^t$ 
5:   end if
6:   Update  $\pi^{t+1} = (1 - \alpha)\pi^t + \alpha\pi'$ 
7: end for
```

The main intuition behind the algorithm is that the stepsize α controls the difference between state distributions of π^t and π^{t+1} . Let us look into the performance difference lemma to get some intuition on this conservative update. From PDL, we have:

$$V_\mu^{\pi^{t+1}} - V_\mu^{\pi^t} = \frac{1}{1-\gamma} \mathbb{E}_{s \sim d_\mu^{\pi^{t+1}}} A^{\pi^t}(s, \pi^{t+1}(s)) = \frac{\alpha}{1-\gamma} \mathbb{E}_{s \sim d_\mu^{\pi^{t+1}}} A^{\pi^t}(s, \pi'(s)),$$

where the last equality we use the fact that $\pi^{t+1} = (1-\alpha)\pi^t + \alpha\pi'$ and $A^{\pi^t}(s, \pi^t(s)) = 0$ for all s . Thus, if we can search for a policy $\pi' \in \Pi$ that maximizes $\mathbb{E}_{s \sim d_\mu^{\pi^{t+1}}} A^{\pi^t}(s, \pi'(s))$ and makes $\mathbb{E}_{s \sim d_\mu^{\pi^{t+1}}} A^{\pi^t}(s, \pi'(s)) > 0$, then we can guarantee policy improvement. However, at episode t , we do not know the state distribution of π^{t+1} and all we have access to is $d_\mu^{\pi^t}$. Thus, we explicitly make the policy update procedure to be conservative such that $d_\mu^{\pi^t}$ and the new policy's distribution $d_\mu^{\pi^{t+1}}$ is guaranteed to be not that different. Thus we can hope that $\mathbb{E}_{s \sim d_\mu^{\pi^{t+1}}} A^{\pi^t}(s, \pi'(s))$ is close to $\mathbb{E}_{s \sim d_\mu^{\pi^t}} A^{\pi^t}(s, \pi'(s))$, and the latter is something that we can manipulate using the greedy policy selector.

Below we formalize the above intuition and show that with small enough α , we indeed can ensure monotonic policy improvement.

We start from the following lemma which shows that π^{t+1} and π^t are close to each other in terms of total variation distance at any state, and $d_\mu^{\pi^{t+1}}$ and $d_\mu^{\pi^t}$ are close as well.

Lemma 13.1 (Similar Policies imply similar state visitations). *Consider any t , we have that:*

$$\|\pi^{t+1}(\cdot|s) - \pi^t(\cdot|s)\|_1 \leq 2\alpha, \forall s;$$

Further, we have:

$$\|d_\mu^{\pi^{t+1}} - d_\mu^{\pi^t}\|_1 \leq \frac{2\alpha\gamma}{1-\gamma}.$$

Proof: The first claim in the above lemma comes from the definition of policy update:

$$\|\pi^{t+1}(\cdot|s) - \pi^t(\cdot|s)\|_1 = \alpha \|\pi^t(\cdot|s) - \pi'(\cdot|s)\|_1 \leq 2\alpha.$$

Denote $\mathbb{P}_h^\pi$ as the state distribution resulting from π at time step h with μ as the initial state distribution. We consider bounding $\|\mathbb{P}_h^{\pi^{t+1}} - \mathbb{P}_h^{\pi^t}\|_1$ with $h \geq 1$.

$$\begin{aligned} \mathbb{P}_h^{\pi^{t+1}}(s') - \mathbb{P}_h^{\pi^t}(s') &= \sum_{s,a} \left(\mathbb{P}_{h-1}^{\pi^{t+1}}(s) \pi^{t+1}(a|s) - \mathbb{P}_{h-1}^{\pi^t}(s) \pi^t(a|s) \right) P(s'|s, a) \\ &= \sum_{s,a} \left(\mathbb{P}_{h-1}^{\pi^{t+1}}(s) \pi^{t+1}(a|s) - \mathbb{P}_{h-1}^{\pi^{t+1}}(s) \pi^t(a|s) + \mathbb{P}_{h-1}^{\pi^{t+1}}(s) \pi^t(a|s) - \mathbb{P}_{h-1}^{\pi^t}(s) \pi^t(a|s) \right) P(s'|s, a) \\ &= \sum_s \mathbb{P}_{h-1}^{\pi^{t+1}}(s) \sum_a (\pi^{t+1}(a|s) - \pi^t(a|s)) P(s'|s, a) \\ &\quad + \sum_s \left(\mathbb{P}_{h-1}^{\pi^{t+1}}(s) - \mathbb{P}_{h-1}^{\pi^t}(s) \right) \sum_a \pi^t(a|s) P(s'|s, a). \end{aligned}$$

Take absolute value on both sides, we get:

$$\begin{aligned} \sum_{s'} \left| \mathbb{P}_h^{\pi^{t+1}}(s') - \mathbb{P}_h^{\pi^t}(s') \right| &\leq \sum_s \mathbb{P}_{h-1}^{\pi^{t+1}}(s) \sum_a |\pi^{t+1}(a|s) - \pi^t(a|s)| \sum_{s'} P(s'|s, a) \\ &\quad + \sum_s \left| \mathbb{P}_{h-1}^{\pi^{t+1}}(s) - \mathbb{P}_{h-1}^{\pi^t}(s) \right| \sum_{s'} \sum_a \pi^t(a|s) P(s'|s, a) \\ &\leq 2\alpha + \|\mathbb{P}_{h-1}^{\pi^{t+1}} - \mathbb{P}_{h-1}^{\pi^t}\|_1 \leq 4\alpha + \|\mathbb{P}_{h-2}^{\pi^{t+1}} - \mathbb{P}_{h-2}^{\pi^t}\|_1 = 2h\alpha. \end{aligned}$$

Now use the definition of d_μ^π , we have:

$$d_\mu^{\pi^{t+1}} - d_\mu^{\pi^t} = (1 - \gamma) \sum_{h=0}^{\infty} \gamma^h \left(\mathbb{P}_h^{\pi^{t+1}} - \mathbb{P}_h^{\pi^t} \right).$$

Add ℓ_1 norm on both sides, we get:

$$\left\| d_\mu^{\pi^{t+1}} - d_\mu^{\pi^t} \right\|_1 \leq (1 - \gamma) \sum_{h=0}^{\infty} \gamma^h 2h\alpha$$

It is not hard to verify that $\sum_{h=0}^{\infty} \gamma^h h = \frac{\gamma}{(1-\gamma)^2}$. Thus, we can conclude that:

$$\left\| d_\mu^{\pi^{t+1}} - d_\mu^{\pi^t} \right\|_1 \leq \frac{2\alpha\gamma}{1-\gamma}.$$

■

The above lemma states that if π^{t+1} and π^t are close in terms of total variation distance for every state, then the total variation distance between the resulting state visitations from π^{t+1} and π^t will be small up to a effective horizon $1/(1-\gamma)$ amplification.

The above lemma captures the key of the conservative policy update. Via the conservative policy update, we make sure that $d_\mu^{\pi^{t+1}}$ and $d_\mu^{\pi^t}$ are close to each other in terms of total variation distance. Now we use the above lemma to show a monotonic policy improvement.

Theorem 13.2 (Monotonic Improvement in CPI). *Consider any episode t . Denote $\mathbb{A} = \mathbb{E}_{s \sim d_\mu^{\pi^t}} A^{\pi^t}(s, \pi'(s))$. We have:*

$$V_\mu^{\pi^{t+1}} - V_\mu^{\pi^t} \geq \frac{\alpha}{1-\gamma} \left(\mathbb{A} - \frac{2\alpha\gamma}{(1-\gamma)^2} \right)$$

Set $\alpha = \frac{\mathbb{A}(1-\gamma)^2}{4\gamma}$, we get:

$$V_\mu^{\pi^{t+1}} - V_\mu^{\pi^t} \geq \frac{\mathbb{A}^2(1-\gamma)}{8\gamma}.$$

The above lemma shows that as long as we still have positive one-step improvement, i.e., $\mathbb{A} > 0$, then we guarantee that π^{t+1} is strictly better than π^t .

Proof: Via PDL, we have:

$$V_\mu^{\pi^{t+1}} - V_\mu^{\pi^t} = \frac{1}{1-\gamma} \mathbb{E}_{s \sim d_\mu^{\pi^{t+1}}} \alpha A^{\pi^t}(s, \pi'(s)).$$

Recall Lemma 13.1, we have:

$$\begin{aligned} (1-\gamma) \left(V_\mu^{\pi^{t+1}} - V_\mu^{\pi^t} \right) &= \mathbb{E}_{s \sim d_\mu^{\pi^t}} \alpha A^{\pi^t}(s, \pi'(s)) + \mathbb{E}_{s \sim d_\mu^{\pi^{t+1}}} \alpha A^{\pi^t}(s, \pi'(s)) - \mathbb{E}_{s \sim d_\mu^{\pi^t}} \alpha A^{\pi^t}(s, \pi'(s)) \\ &\geq \mathbb{E}_{s \sim d_\mu^{\pi^t}} \alpha A^{\pi^t}(s, \pi'(s)) - \alpha \sup_{s, a, \pi} |A^\pi(s, a)| \|d_\mu^{\pi^t} - d_\mu^{\pi^{t+1}}\|_1 \\ &\geq \mathbb{E}_{s \sim d_\mu^{\pi^t}} \alpha A^{\pi^t}(s, \pi'(s)) - \frac{\alpha}{1-\gamma} \|d_\mu^{\pi^t} - d_\mu^{\pi^{t+1}}\|_1 \\ &\geq \mathbb{E}_{s \sim d_\mu^{\pi^t}} \alpha A^{\pi^t}(s, \pi'(s)) - \frac{2\alpha^2\gamma}{(1-\gamma)^2} = \alpha \left(\mathbb{A} - \frac{2\alpha\gamma}{(1-\gamma)^2} \right) \end{aligned}$$

where the first inequality we use the fact that for any two distributions P_1 and P_2 and any function f , $|\mathbb{E}_{x \sim P_1} f(x) - \mathbb{E}_{x \sim P_2} f(x)| \leq \sup_x |f(x)| \|P_1 - P_2\|_1$, for the second inequality, we use the fact that $|A^\pi(s, a)| \leq 1/(1-\gamma)$ for any π, s, a , and the last inequality uses Lemma 13.1.

For the second part of the above theorem, note that we want to maximum the policy improvement as much as possible by choosing α . So we can pick α which maximizes $\alpha(\mathbb{A} - 2\alpha\gamma/(1-\gamma)^2)$. This gives us the α we claimed in the lemma. Plug in α back into $\alpha(\mathbb{A} - 2\alpha\gamma/(1-\gamma)^2)$, we conclude the second part of the theorem. ■

The above theorem indicates that with the right choice of α , we guarantee that the policy is making improvement as long as $\mathbb{A} > 0$. Recall the termination criteria in CPI where we terminate CPI when $\mathbb{A} \leq \varepsilon$. Putting these results together, we obtain the following overall convergence guarantee for the CPI algorithm.

Theorem 13.3 (Local optimality of CPI). *Algorithm 8 terminates in at most $8\gamma/\varepsilon^2$ steps and outputs a policy π^t satisfying $\max_{\pi \in \Pi} \mathbb{E}_{s \sim d_\mu^{\pi^t}} A^{\pi^t}(s, \pi(s)) \leq 2\varepsilon$.*

Proof: Note that our reward is bounded in $[0, 1]$ which means that $V_\mu^\pi \in [0, 1/(1-\gamma)]$. Note that we have shown in Theorem 13.2, every iteration t , we have policy improvement at least $\frac{\mathbb{A}^2(1-\gamma)}{8\gamma}$, where recall $\mathbb{A}$ at episode t is defined as $\mathbb{A} = \mathbb{E}_{s \sim d_\mu^{\pi^t}} A^{\pi^t}(s, \pi'(s))$. If the algorithm does not terminate at episode t , then we guarantee that:

$$V_\mu^{\pi^{t+1}} \geq V_\mu^{\pi^t} + \frac{\varepsilon^2(1-\gamma)}{8\gamma}$$

Since V_μ^π is upper bounded by $1/(1-\gamma)$, so we can at most make improvement $8\gamma/\varepsilon^2$ many iterations.

Finally, recall that ε -approximate greedy policy selector $\pi' = \mathcal{G}_\varepsilon(\pi^t, \Pi, \mu)$, we have:

$$\max_{\pi \in \Pi} \mathbb{E}_{s \sim d_\mu^{\pi^t}} A^{\pi^t}(s, \pi(s)) \leq \mathbb{E}_{s \sim d_\mu^{\pi^t}} A^{\pi^t}(s, \pi'(s)) + \varepsilon \leq 2\varepsilon$$

This concludes the proof. $\blacksquare$

Theorem 13.3 can be viewed as a local optimality guarantee in a sense. It shows that when CPI terminates, we cannot find a policy $\pi \in \Pi$ that achieves local improvement over the returned policy more than ε . However, this does not necessarily imply that the value of π is close to V^* . However, similar to the policy gradient analysis, we can turn this local guarantee into a global one when the restart distribution μ covers d^{π^*} . We formalize this intuition next.

Theorem 13.4 (Global optimality of CPI). *Upon termination, we have a policy π such that:*

$$V^* - V^\pi \leq \frac{2\varepsilon + \epsilon_\Pi}{(1-\gamma)^2} \left\| \frac{d^{\pi^*}}{\mu} \right\|_\infty,$$

where $\epsilon_\Pi := \mathbb{E}_{s \sim d_\mu^\pi} [\max_{a \in \mathcal{A}} A^\pi(s, a)] - \max_{\pi \in \Pi} \mathbb{E}_{s \sim d_\mu^\pi} [A^\pi(s, \pi(s))]$.

In other words, if our policy class is rich enough to approximate the policy $\max_{a \in \mathcal{A}} A^\pi(s, a)$ under d_μ^π , i.e., ϵ_Π is small, and μ covers d^{π^*} in a sense that $\left\| \frac{d^{\pi^*}}{\mu} \right\|_\infty \leq \infty$, CPI guarantees to find a near optimal policy.

Proof: By the performance difference lemma,

$$\begin{aligned} V^* - V^\pi &= \frac{1}{1-\gamma} \mathbb{E}_{s \sim d^{\pi^*}} A^\pi(s, \pi^*(s)) \\ &\leq \frac{1}{1-\gamma} \mathbb{E}_{s \sim d^{\pi^*}} \max_{a \in \mathcal{A}} A^\pi(s, a) \\ &\leq \frac{1}{1-\gamma} \left\| \frac{d^{\pi^*}}{d_\mu^\pi} \right\|_\infty \mathbb{E}_{s \sim d_\mu^\pi} \max_{a \in \mathcal{A}} A^\pi(s, a) \\ &\leq \frac{1}{(1-\gamma)^2} \left\| \frac{d^{\pi^*}}{\mu} \right\|_\infty \mathbb{E}_{s \sim d_\mu^\pi} \max_{a \in \mathcal{A}} A^\pi(s, a) \\ &= \frac{1}{(1-\gamma)^2} \left\| \frac{d^{\pi^*}}{\mu} \right\|_\infty \left[\max_{\hat{\pi} \in \Pi} \mathbb{E}_{s \sim d_\mu^{\hat{\pi}}} A^\pi(s, \hat{\pi}(s)) - \max_{\hat{\pi} \in \Pi} \mathbb{E}_{s \sim d_\mu^{\hat{\pi}}} A^\pi(s, \hat{\pi}(s)) + \mathbb{E}_{s \sim d_\mu^\pi} \max_{a \in \mathcal{A}} A^\pi(s, a) \right] \\ &\leq \frac{1}{(1-\gamma)^2} \left\| \frac{d^{\pi^*}}{\mu} \right\|_\infty (2\varepsilon + \epsilon_\Pi), \end{aligned}$$

where the second inequality holds due to the fact that $\max_a A^\pi(s, a) \geq 0$, the third inequality uses the fact that $d_\mu^\pi(s) \geq (1-\gamma)\mu(s)$ for any s and π , and the last inequality uses the definition ϵ_Π and Theorem 13.3. $\blacksquare$

It is informative to contrast CPI and policy gradient algorithms due to the similarity of their guarantees. Both provide local optimality guarantees. For CPI, the local optimality always holds, while for policy gradients it requires a smooth value function as a function of the policy parameters. If the distribution mismatch between an optimal policy and the output of the algorithm is not too large, then both algorithms further yield a near optimal policy. The similarities are not so surprising. Both algorithms operate by making local improvements to the current policy at each iteration, by inspecting its advantage function. The changes made to the policy are controlled using a stepsize parameter in both the approaches. It is the actual mechanism of the improvement which differs in the two cases. Policy gradients assume that the policy's reward is a differentiable function of the parameters, and hence make local improvements through gradient ascent. The differentiability is certainly an assumption and does not necessarily hold for all policy classes. An easy example is when the policy itself is not an easily differentiable function of its parameters. For instance, if the policy is parametrized by regression trees, then performing gradient updates can be challenging.

In CPI, on the other hand, the basic computational primitive required on the policy class is the ability to maximize the advantage function relative to the current policy. Notice that Algorithm 8 does not necessarily restrict to a policy class, such as a set of parametrized policies as in policy gradients. Indeed, due to the reduction to supervised learning approach (e.g., using the weighted classification oracle CO), we can parameterize policy class via decision tree, for

instance. This property makes CPI extremely attractive. Any policy class over which efficient supervised learning algorithms exist can be adapted to reinforcement learning with performance guarantees.

A second important difference between CPI and policy gradients is in the notion of locality. Policy gradient updates are local in the parameter space, and we hope that this makes small enough changes to the state distribution that the new policy is indeed an improvement on the older one (for instance, when we invoke the performance difference lemma between successive iterates). While this is always true in expectation for correctly chosen stepsizes based on properties of stochastic gradient ascent on smooth functions, the variance of the algorithm and lack of robustness to suboptimal stepsizes can make the algorithm somewhat finicky. Indeed, there are a host of techniques in the literature to both lower the variance (through control variates) and explicitly control the state distribution mismatch between successive iterates of policy gradients (through trust region techniques). On the other hand, CPI explicitly controls the amount of perturbation to the state distribution by carefully mixing policies in a manner which does not drastically alter the trajectories with high probability. Indeed, this insight is central to the proof of CPI, and has been instrumental in several follow-ups, both in the direct policy improvement as well as policy gradient literature.

13.2 Trust Region Methods and Covariant Policy Search

So far we have seen policy gradient methods and CPI which all uses a small step-size to ensure incremental update in policies. Another popular approach for incremental policy update is to explicitly forcing small change in policies' distribution via a trust region constraint. More specifically, let us go back to the general policy parameterization π_θ . At iteration t with the current policy π_{θ_t} , we are interested in the following local trust-region constrained optimization:

$$\begin{aligned} & \max_{\theta} \mathbb{E}_{s \sim d_{\mu}^{\pi_{\theta_t}}} \mathbb{E}_{a \sim \pi_{\theta}(\cdot|s)} A^{\pi_{\theta_t}}(s, a) \\ & \text{s.t., } KL\left(\Pr_{\mu}^{\pi_{\theta_t}} \parallel \Pr_{\mu}^{\pi_{\theta}}\right) \leq \delta, \end{aligned}$$

where recall $\Pr_{\mu}^{\pi}(\tau)$ is the trajectory distribution induced by π starting at $s_0 \sim \mu$, and $KL(P_1 \parallel P_2)$ are KL-divergence between two distribution P_1 and P_2 . Namely we explicitly perform local policy search with a constraint forcing the new policy not being too far away from $\Pr_{\mu}^{\pi_{\theta_t}}$ in terms of KL divergence.

As we are interested in small local update in parameters, we can perform sequential quadratic programming here, i.e., we can further linearize the objective function at θ_t and quadratize the KL constraint at θ_t to form a local quadratic programming:

$$\max_{\theta} \left\langle \mathbb{E}_{s \sim d_{\mu}^{\pi_{\theta_t}}} \mathbb{E}_{a \sim \pi_{\theta_t}(\cdot|s)} \nabla_{\theta} \ln \pi_{\theta_t}(a|s) A^{\pi_{\theta_t}}(s, a), \theta \right\rangle \quad (13.1)$$

$$\text{s.t., } \langle \nabla_{\theta} KL\left(\Pr_{\mu}^{\pi_{\theta_t}} \parallel \Pr_{\mu}^{\pi_{\theta}}\right) |_{\theta=\theta_t}, \theta - \theta_t \rangle + \frac{1}{2}(\theta - \theta_t)^{\top} \left(\nabla_{\theta}^2 KL\left(\Pr_{\mu}^{\pi_{\theta_t}} \parallel \Pr_{\mu}^{\pi_{\theta}}\right) |_{\theta=\theta_t} \right) (\theta - \theta_t) \leq \delta, \quad (13.2)$$

where we denote $\nabla^2 KL|_{\theta=\theta_t}$ as the Hessian of the KL constraint measured at θ_t . Note that KL divergence is not a valid metric as it is not symmetric. However, its local quadratic approximation can serve as a valid local distance metric, as we prove below that the Hessian $\nabla^2 KL|_{\theta=\theta_t}$ is a positive semi-definite matrix. Indeed, we will show that the Hessian of the KL constraint is exactly equal to the fisher information matrix, and the above quadratic programming exactly reveals a Natural Policy Gradient update. Hence Natural policy gradient can also be interpreted as performing sequential quadratic programming with KL constraint over policy's trajectory distributions.

To match to the practical algorithms in the literature (e.g., TRPO), below we focus on episode finite horizon setting again (i.e., an MDP with horizon H).

Claim 13.5. Consider a finite horizon MDP with horizon H . Consider any fixed θ_t . We have:

$$\begin{aligned}\nabla_{\theta} KL \left(\Pr_{\mu}^{\pi^{\theta_t}} || \Pr_{\mu}^{\pi^{\theta}} \right) |_{\theta=\theta_t} &= 0, \\ \nabla_{\theta}^2 KL \left(\Pr_{\mu}^{\pi^{\theta_t}} || \Pr_{\mu}^{\pi^{\theta}} \right) |_{\theta=\theta_t} &= H \mathbb{E}_{s, a \sim d^{\pi_{\theta_t}}} \nabla \ln \pi_{\theta_t}(a|s) (\nabla \ln \pi_{\theta_t}(a|s))^{\top}.\end{aligned}$$

Proof: We first recall the trajectory distribution in finite horizon setting.

$$\Pr_{\mu}^{\pi}(\tau) = \mu(s_0) \prod_{h=0}^{H-1} \pi(a_h|s_h) P(s_{h+1}|s_h, a_h).$$

We first prove that the gradient of KL is zero. First note that:

$$\begin{aligned}KL \left(\Pr_{\mu}^{\pi^{\theta_t}} || \Pr_{\mu}^{\pi^{\theta}} \right) &= \sum_{\tau} \Pr_{\mu}^{\pi^{\theta_t}}(\tau) \ln \frac{\Pr_{\mu}^{\pi^{\theta_t}}(\tau)}{\Pr_{\mu}^{\pi^{\theta}}(\tau)} \\ &= \sum_{\tau} \Pr_{\mu}^{\pi^{\theta_t}}(\tau) \left(\sum_{h=0}^{H-1} \ln \frac{\pi_{\theta_t}(a_h|s_h)}{\pi_{\theta}(a_h|s_h)} \right) = \sum_{h=0}^{H-1} \mathbb{E}_{s_h, a_h \sim \mathbb{P}_h^{\pi_{\theta_t}}} \ln \frac{\pi_{\theta_t}(a_h|s_h)}{\pi_{\theta}(a_h|s_h)}.\end{aligned}$$

Thus, for $\nabla_{\theta} KL \left(\Pr_{\mu}^{\pi^{\theta_t}} || \Pr_{\mu}^{\pi^{\theta}} \right)$, we have:

$$\begin{aligned}\nabla KL \left(\Pr_{\mu}^{\pi^{\theta_t}} || \Pr_{\mu}^{\pi^{\theta}} \right) |_{\theta=\theta_t} &= \sum_{h=0}^{H-1} \mathbb{E}_{s_h, a_h \sim \mathbb{P}_h^{\pi_{\theta_t}}} - \nabla \ln \pi_{\theta_t}(a_h|s_h) \\ &= - \sum_{h=0}^{H-1} \mathbb{E}_{s_h \sim \mathbb{P}_h^{\pi_{\theta_t}}} \mathbb{E}_{a_h \sim \pi_{\theta_t}(\cdot|s_h)} \nabla \ln \pi_{\theta_t}(a_h|s_h) = 0,\end{aligned}$$

where we have seen the last step when we argue the unbiased nature of policy gradient with an action independent baseline.

Now we move to the Hessian.

$$\begin{aligned}\nabla^2 KL \left(\Pr_{\mu}^{\pi^{\theta_t}} || \Pr_{\mu}^{\pi^{\theta}} \right) |_{\theta=\theta_t} &= - \sum_{h=0}^{H-1} \mathbb{E}_{s_h, a_h \sim \mathbb{P}_h^{\pi_{\theta_t}}} \nabla^2 \ln \pi_{\theta}(a_h|s_h) |_{\theta=\theta_t} \\ &= - \sum_{h=0}^{H-1} \mathbb{E}_{s_h, a_h \sim \mathbb{P}_h^{\pi_{\theta_t}}} \left(\nabla \left(\frac{\nabla \pi_{\theta}(a|s)}{\pi_{\theta}(a|s)} \right) \right) \\ &= - \sum_{h=0}^{H-1} \mathbb{E}_{s_h, a_h \sim \mathbb{P}_h^{\pi_{\theta_t}}} \left(\frac{\nabla^2 \pi_{\theta}(a_h|s_h)}{\pi_{\theta}(a_h|s_h)} - \frac{\nabla \pi_{\theta}(a_h|s_h) \nabla \pi_{\theta}(a_h|s_h)^{\top}}{\pi_{\theta}^2(a_h|s_h)} \right) \\ &= \sum_{h=0}^{H-1} \mathbb{E}_{s_h, a_h \sim \mathbb{P}_h^{\pi_{\theta_t}}} \nabla_{\theta} \ln \pi_{\theta}(a_h|s_h) \nabla_{\theta} \ln \pi_{\theta}(a_h|s_h)^{\top},\end{aligned}$$

where in the last equation we use the fact that $\mathbb{E}_{s_h, a_h \sim \mathbb{P}_h^{\pi_{\theta_t}}} \frac{\nabla^2 \pi_{\theta}(a_h|s_h)}{\pi_{\theta}(a_h|s_h)} = 0$. ■

The above claim shows that a second order taylor expansion of the KL constraint over trajectory distribution gives a local distance metric at θ_t :

$$(\theta - \theta_t) F_{\theta_t} (\theta - \theta_t),$$

where again $F_{\theta_t} := H\mathbb{E}_{s,a \sim d^{\pi_{\theta_t}}} \nabla \ln \pi_{\theta_t}(a|s) (\nabla \ln \pi_{\theta_t}(a|s))^\top$ is proportional to the fisher information matrix. Note that F_{θ_t} is a PSD matrix and thus $d(\theta, \theta_t) := (\theta - \theta_t)^\top F_{\theta_t} (\theta - \theta_t)$ is a valid distance metric. By sequential quadratic programming, we are using local geometry information of the trajectory distribution manifold induced by the parameterization θ , rather the naive Euclidean distance in the parameter space. Such a method is sometimes referred to as *Covariant Policy Search*, as the policy update procedure will be invariant to linear transformation of parameterization (See Section 13.3 for further discussion).

Now using the results from Claim 13.5, we can verify that the local policy optimization procedure in Eq. 13.2 exactly recovers the NPG update, where the step size is based on the trust region parameter δ . Denote $\Delta = \theta - \theta_t$, we have:

$$\begin{aligned} & \max_{\Delta} \langle \Delta, \nabla_{\theta} V^{\pi_{\theta_t}} \rangle, \\ & \text{s.t.}, \Delta^\top F_{\theta_t} \Delta \leq \delta, \end{aligned}$$

which gives the following update procedure:

$$\theta_{t+1} = \theta_t + \Delta = \theta_t + \sqrt{\frac{\delta}{(\nabla V^{\pi_{\theta_t}})^\top F_{\theta_t}^{-1} \nabla V^{\pi_{\theta_t}}}} \cdot F_{\theta_t}^{-1} \nabla V^{\pi_{\theta_t}},$$

where note that we use the self-normalized learning rate computed using the trust region parameter δ .

13.2.1 Proximal Policy Optimization

Here we consider an ℓ_∞ style trust region constraint:

$$\max_{\theta} \mathbb{E}_{s \sim d_{\mu}^{\pi_{\theta_t}}} \mathbb{E}_{a \sim \pi_{\theta}(\cdot|s)} A^{\pi_{\theta_t}}(s, a) \quad (13.3)$$

$$\text{s.t.}, \sup_s \|\pi_{\theta}(\cdot|s) - \pi_{\theta_t}(\cdot|s)\|_{tv} \leq \delta, \quad (13.4)$$

Namely, we restrict the new policy such that it is close to π_{θ_t} at every state s under the total variation distance. Recall the CPI's update, CPI indeed makes sure that the new policy will be close the old policy at every state. In other words, the new policy computed by CPI is a feasible solution of the constraint Eq. 13.4, but is not the optimal solution of the above constrained optimization program. Also one downside of the CPI algorithm is that one needs to keep all previous learned policies around, which requires large storage space when policies are parameterized by large deep neural networks.

Proximal Policy Optimization (PPO) aims to directly optimize the objective Eq. 13.3 using multiple steps of gradient updates, and approximating the constraints Eq. 13.4 via a clipping trick. We first rewrite the objective function using importance weighting:

$$\max_{\theta} \mathbb{E}_{s \sim d_{\mu}^{\pi_{\theta_t}}} \mathbb{E}_{a \sim \pi_{\theta_t}(\cdot|s)} \frac{\pi_{\theta}(a|s)}{\pi_{\theta_t}(a|s)} A^{\pi_{\theta_t}}(s, a), \quad (13.5)$$

where we can easily approximate the expectation $\mathbb{E}_{s \sim d_{\mu}^{\pi_{\theta_t}}} \mathbb{E}_{a \sim \pi_{\theta_t}(\cdot|s)}$ via finite samples $s \sim d^{\pi_{\theta_t}}, a \sim \pi_{\theta_t}(\cdot|s)$.

To make sure $\pi_{\theta}(a|s)$ and $\pi_{\theta_t}(a|s)$ are not that different, PPO modifies the objective function by clipping the density ratio $\pi_{\theta}(a|s)$ and $\pi_{\theta_t}(a|s)$:

$$L(\theta) := \mathbb{E}_{s \sim d_{\mu}^{\pi_{\theta_t}}} \mathbb{E}_{a \sim \pi_{\theta_t}(\cdot|s)} \min \left\{ \frac{\pi_{\theta}(a|s)}{\pi_{\theta_t}(a|s)} A^{\pi_{\theta_t}}(s, a), \text{clip} \left(\frac{\pi_{\theta}(a|s)}{\pi_{\theta_t}(a|s)}; 1 - \epsilon, 1 + \epsilon \right) A^{\pi_{\theta_t}}(s, a) \right\}, \quad (13.6)$$

where $\text{clip}(x; 1 - \epsilon, 1 + \epsilon) = \begin{cases} 1 - \epsilon & x \leq 1 - \epsilon \\ 1 + \epsilon & x \geq 1 + \epsilon \\ x & \text{else} \end{cases}$. The clipping operator ensures that for π_θ , at any state action pair

where $\pi^\theta(a|s)/\pi^{\theta_t}(a|s) \notin [1 - \epsilon, 1 + \epsilon]$, we get zero gradient, i.e., $\nabla_\theta \left[\text{clip} \left(\frac{\pi^\theta(a|s)}{\pi^{\theta_t}(\cdot|s)}; 1 - \epsilon, 1 + \epsilon \right) A^{\pi^{\theta_t}}(s, a) \right] = 0$. The outer min makes sure the objective function $L(\theta)$ is a lower bound of the original objective. PPO then proposes to collect a dataset (s, a) with $s \sim d_\mu^{\pi^{\theta_t}}$ and $a \sim \pi^{\theta_t}(\cdot|s)$, and then perform multiple steps of mini-batch stochastic gradient ascent on $L(\theta)$.

One of the key difference between PPO and other algorithms such as NPG is that PPO targets to optimize objective Eq. 13.5 via multiple steps of mini-batch stochastic gradient ascent with mini-batch data from $d_\mu^{\pi^{\theta_t}}$ and π^{θ_t} , while algorithm such as NPG indeed optimizes the first order Taylor expansion of Eq. 13.5 at θ_t , i.e.,

$$\max_{\theta} \left\langle (\theta - \theta_t), \mathbb{E}_{s, a \sim d_\mu^{\pi^{\theta_t}}} \nabla_\theta \ln \pi^{\theta_t}(a|s) A^{\pi^{\theta_t}}(s, a) \right\rangle,$$

upper to some trust region constraints (e.g., $\|\theta - \theta_t\|_2 \leq \delta$ in policy gradient, and $\|\theta - \theta_t\|_{F_{\theta_t}}^2 \leq \delta$ for NPG).

13.3 Bibliographic Remarks and Further Readings

The analysis of CPI is adapted from the original one in [Kakade and Langford, 2002]. There have been a few further interpretations of CPI. One interesting perspective is that CPI can be treated as a boosting algorithm [Scherrer and Geist, 2014].

More generally, CPI and NPG are part of family of *incremental* algorithms, including Policy Search by Dynamic Programming (PSDP) [Bagnell et al., 2004] and MD-MPI [Geist et al., 2019]. PSDP operates in a finite horizon setting and optimizes a sequence of time-dependent policies; from the last time step to the first time step, every iteration of, PSDP only updates the policy at the current time step while holding the future policies fixed — thus making incremental update on the policy. See [Scherrer, 2014] for more a detailed discussion and comparison of some of these approaches. Mirror Descent-Modified Policy Iteration (MD-MPI) algorithm [Geist et al., 2019] is a family of actor-critic style algorithms which is based on regularization and is incremental in nature; with negative entropy as the Bregman divergence (for the tabular case), MD-MPI recovers the NPG the tabular case (for the softmax parameterization).

Broadly speaking, these incremental algorithms can improve upon the stringent concentrability conditions for approximate value iteration methods, presented in Chapter 4. Scherrer [2014] provide a more detailed discussion of bounds which depend on these density ratios. As discussed in the last chapter, the density ratio for NPG can be interpreted as a factor due to transfer learning to a single, *fixed* distribution.

The interpretation of NPG as Covariant Policy Search is due to [Bagnell and Schneider, 2003], as the policy update procedure will be invariant to linear transformations of the parameterization; see [Bagnell and Schneider, 2003] for a more detailed discussion on this.

The TRPO algorithm is due to [Schulman et al., 2015]. The original TRPO analysis provides performance guarantees, largely relying on a reduction to the CPI guarantees. In this Chapter, we make a sharper connection of TRPO to NPG, which was subsequently observed by a number of researchers; this connection provides a sharper analysis for the generalization and approximation behavior of TRPO (e.g. via the results presented in Chapter 12). In practice, a popular variant is the Proximal Policy Optimization (PPO) algorithm [Schulman et al., 2017].

Part IV

Further Topics

Chapter 14

Imitation Learning

In this chapter, we study imitation learning. Unlike the Reinforcement Learning setting, in Imitation Learning, we do not have access to the ground truth reward function (or cost function), but instead, we have expert demonstrations. We often assume that the expert is a policy that approximately optimizes the underlying reward (or cost) functions. The goal is to leverage the expert demonstrations to learn a policy that performs as good as the expert.

We consider three settings of imitation learning: (1) pure offline where we only have expert demonstrations and no more real world interaction is allowed, (2) hybrid where we have expert demonstrations, and also is able to interact with the real world (e.g., have access to the ground truth transition dynamics); (3) interactive setting where we have an interactive expert and also have access to the underlying reward (cost) function.

14.1 Setting

We will focus on finite horizon MDPs $\mathcal{M} = \{\mathcal{S}, \mathcal{A}, r, \mu, P, H\}$ where r is the reward function but is unknown to the learner. We represent the expert as a closed-loop policy $\pi^* : \mathcal{S} \mapsto \Delta(\mathcal{A})$. For analysis simplicity, we assume the expert π^* is indeed the optimal policy of the original MDP $\mathcal{M}$ with respect to the ground truth reward r . Again, our goal is to learn a policy $\hat{\pi} : \mathcal{S} \mapsto \Delta(\mathcal{A})$ that performs as well as the expert, i.e., $V^{\hat{\pi}}$ needs to be close to V^* , where V^π denotes the expected total reward of policy π under the MDP $\mathcal{M}$. We denote d^π as the state-action visitation of policy π under $\mathcal{M}$.

We assume we have a pre-collected expert dataset in the format of $\{s_i^*, a_i^*\}_{i=1}^M$ where $s_i^*, a_i^* \sim d^{\pi^*}$.

14.2 Offline IL: Behavior Cloning

We study offline IL here. Specifically, we study the classic Behavior Cloning algorithm.

We consider a policy class $\Pi = \{\pi : \mathcal{S} \mapsto \Delta(\mathcal{A})\}$. We consider the following realizable assumption.

Assumption 14.1. We assume Π is rich enough such that $\pi^* \in \Pi$.

For analysis simplicity, we assume Π is discrete. But our sample complexity will only scale with respect to $\ln(|\Pi|)$.

Behavior cloning is one of the simplest Imitation Learning algorithm which only uses the expert data $\mathcal{D}^*$ and does not require any further interaction with the MDP. It computes a policy via a reduction to supervised learning. Specifically,

we consider a reduction to Maximum Likelihood Estimation (MLE):

$$\text{Behavior Cloning (BC): } \hat{\pi} = \operatorname{argmax}_{\pi \in \Pi} \sum_{i=1}^N \ln \pi(a_i^* | s_i^*). \quad (14.1)$$

Namely we try to find a policy from Π that has the maximum likelihood of fitting the training data. As this is a reduction to MLE, we can leverage the existing classic analysis of MLE (e.g., Chapter 7 in [Geer and van de Geer, 2000]) to analyze the performance of the learned policy.

Theorem 14.2 (MLE Guarantee). *Consider the MLE procedure (Eq. 14.1). With probability at least $1 - \delta$, we have:*

$$\mathbb{E}_{s \sim d^{\pi^*}} \|\hat{\pi}(\cdot | s) - \pi^*(\cdot | s)\|_{tv}^2 \leq \frac{2 \ln(|\Pi|/\delta)}{M}.$$

We refer readers to Section E, Theorem 21 in [Agarwal et al., 2020b] for detailed proof of the above MLE guarantee.

Now we can transfer the average divergence between $\hat{\pi}$ and π^* to their performance difference $V^{\hat{\pi}}$ and V^{π^*} .

One thing to note is that BC only ensures that the learned policy $\hat{\pi}$ is close to π^* under d^{π^*} . Outside of d^{π^*} 's support, we have no guarantee that $\hat{\pi}$ and π^* will be close to each other. The following theorem shows that a compounding error occurs when we study the performance of the learned policy $V^{\hat{\pi}}$.

Theorem 14.3 (Sample Complexity of BC). *With probability at least $1 - \delta$, BC returns a policy $\hat{\pi}$ such that:*

$$V^* - V^{\hat{\pi}} \leq \frac{3}{(1-\gamma)^2} \sqrt{\frac{\ln(|\Pi|/\delta)}{M}}.$$

Proof: We start with the performance difference lemma and the fact that $\mathbb{E}_{a \sim \pi(\cdot | s)} A^\pi(s, a) = 0$:

$$\begin{aligned} (1-\gamma) (V^* - V^{\hat{\pi}}) &= \mathbb{E}_{s \sim d^{\pi^*}} \mathbb{E}_{a \sim \pi^*(\cdot | s)} A^{\hat{\pi}}(s, a) \\ &= \mathbb{E}_{s \sim d^{\pi^*}} \mathbb{E}_{a \sim \pi^*(\cdot | s)} A^{\hat{\pi}}(s, a) - \mathbb{E}_{s \sim d^{\pi^*}} \mathbb{E}_{a \sim \hat{\pi}(\cdot | s)} A^{\hat{\pi}}(s, a) \\ &\leq \mathbb{E}_{s \sim d^{\pi^*}} \frac{1}{1-\gamma} \|\pi^*(\cdot | s) - \hat{\pi}(\cdot | s)\|_1 \\ &\leq \frac{1}{1-\gamma} \sqrt{\mathbb{E}_{s \sim d^{\pi^*}} \|\pi^*(\cdot | s) - \hat{\pi}(\cdot | s)\|_1^2} \\ &= \frac{1}{1-\gamma} \sqrt{4 \mathbb{E}_{s \sim d^{\pi^*}} \|\pi^*(\cdot | s) - \hat{\pi}(\cdot | s)\|_{tv}^2}. \end{aligned}$$

where we use the fact that $\sup_{s,a,\pi} |A^\pi(s, a)| \leq \frac{1}{1-\gamma}$, and the fact that $(\mathbb{E}[x])^2 \leq \mathbb{E}[x^2]$.

Using Theorem 14.2 and rearranging terms conclude the proof. ■

For Behavior cloning, from the supervised learning error the quadratic polynomial dependency on the effect horizon $1/(1-\gamma)$ is not avoidable in worst case [Ross and Bagnell, 2010]. This is often referred as the distribution shift issue in the literature. Note that $\hat{\pi}$ is trained under d^{π^*} , but during execution, $\hat{\pi}$ makes prediction on states that are generated by itself, i.e., $d^{\hat{\pi}}$, instead of the training distribution d^{π^*} .

14.3 The Hybrid Setting: Statistical Benefit and Algorithm

The question we want to answer here is that if we know the underlying MDP's transition P (but the reward is still unknown), can we improve Behavior Cloning? In other words:

what is the benefit of the known transition in addition to the expert demonstrations?

Instead of a quadratic dependency on the effective horizon, we should expect a linear dependency on the effective horizon. The key benefit of the known transition is that we can test our policy using the known transition to see how far away we are from the expert's demonstrations, and then use the known transition to plan to move closer to the expert demonstrations.

In this section, we consider a statistical efficient, but computationally intractable algorithm, which we use to demonstrate that informationally theoretically, by interacting with the underlying known transition, we can do better than Behavior Cloning. In the next section, we will introduce a popular and computationally efficient algorithm Maximum Entropy Inverse Reinforcement Learning (MaxEnt-IRL) which operates under this setting (i.e., expert demonstrations with a known transition).

We start from the same policy class Π as we have in the BC algorithm, and again we assume realizability (Assumption 14.1) and Π is discrete.

Since we know the transition P , information theoretically, for any policy π , we have d^π available (though it is computationally intractable to compute d^π for large scale MDPs). We have $(s_i^*, a_i^*)_{i=1}^M \sim d^{\pi^*}$.

Below we present an algorithm which we called Distribution Matching with Scheffé Tournament (DM-ST).

For any two policies π and π' , we denote $f_{\pi, \pi'}$ as the following witness function:

$$f_{\pi, \pi'} := \operatorname{argmax}_{f: \|f\|_\infty \leq 1} \left[\mathbb{E}_{s, a \sim d^\pi} f(s, a) - \mathbb{E}_{s, a \sim d^{\pi'}} f(s, a) \right].$$

We denote the set of witness functions as:

$$\mathcal{F} = \{f_{\pi, \pi'} : \pi, \pi' \in \Pi, \pi \neq \pi'\}.$$

Note that $|\mathcal{F}| \leq |\Pi|^2$.

DM-ST selects $\hat{\pi}$ using the following procedure:

$$\text{DM-ST: } \hat{\pi} \in \operatorname{argmin}_{\pi \in \Pi} \left[\max_{f \in \mathcal{F}} \left[\mathbb{E}_{s, a \sim d^\pi} f(s, a) - \frac{1}{M} \sum_{i=1}^M f(s_i^*, a_i^*) \right] \right].$$

The following theorem captures the sample complexity of DM-ST.

Theorem 14.4 (Sample Complexity of DM-ST). *With probability at least $1 - \delta$, DM-ST finds a policy $\hat{\pi}$ such that:*

$$V^* - V^{\hat{\pi}} \leq \frac{4}{1 - \gamma} \sqrt{\frac{2 \ln(|\Pi|) + \ln(\frac{1}{\delta})}{M}}.$$

Proof: The proof basically relies on a uniform convergence argument over $\mathcal{F}$ of which the size is $|\Pi|^2$. First we note that for all policy $\pi \in \Pi$:

$$\max_{f \in \mathcal{F}} [\mathbb{E}_{s, a \sim d^\pi} f(s, a) - \mathbb{E}_{s, a \sim d^*} f(s, a)] = \max_{f: \|f\|_\infty \leq 1} [\mathbb{E}_{s, a \sim d^\pi} f(s, a) - \mathbb{E}_{s, a \sim d^*} f(s, a)] = \|d^\pi - d^{\pi^*}\|_1,$$

where the first equality comes from the fact that $\mathcal{F}$ includes $\operatorname{argmax}_{f: \|f\|_\infty \leq 1} [\mathbb{E}_{s, a \sim d^\pi} f(s, a) - \mathbb{E}_{s, a \sim d^*} f(s, a)]$.

Via Hoeffding's inequality and a union bound over $\mathcal{F}$, we get that with probability at least $1 - \delta$, for all $f \in \mathcal{F}$:

$$\left| \frac{1}{M} \sum_{i=1}^M f(s_i^*, a_i^*) - \mathbb{E}_{s, a \sim d^*} f(s, a) \right| \leq 2 \sqrt{\frac{\ln(|\mathcal{F}|/\delta)}{M}} := \epsilon_{\text{stat}}.$$

Denote $\hat{f} := \arg \max_{f \in \mathcal{F}} [\mathbb{E}_{s,a \sim d^{\hat{\pi}}} f(s, a) - \mathbb{E}_{s,a \sim d^{\star}} f(s, a)]$, and $\tilde{f} := \arg \max_{f \in \mathcal{F}} \mathbb{E}_{s,a \sim d^{\hat{\pi}}} f(s, a) - \frac{1}{M} \sum_{i=1}^M f(s_i, a_i)$. Hence, for $\hat{\pi}$, we have:

$$\begin{aligned} \|d^{\hat{\pi}} - d^{\star}\|_1 &= \mathbb{E}_{s,a \sim d^{\hat{\pi}}} \hat{f}(s, a) - \mathbb{E}_{s,a \sim d^{\star}} \hat{f}(s, a) \leq \mathbb{E}_{s,a \sim d^{\hat{\pi}}} \hat{f}(s, a) - \frac{1}{M} \sum_{i=1}^M \hat{f}(s_i^{\star}, a_i^{\star}) + \epsilon_{stat} \\ &\leq \mathbb{E}_{s,a \sim d^{\hat{\pi}}} \tilde{f}(s, a) - \frac{1}{M} \sum_{i=1}^M \tilde{f}(s_i, a_i) + \epsilon_{stat} \\ &\leq \mathbb{E}_{s,a \sim d^{\pi^{\star}}} \tilde{f}(s, a) - \frac{1}{M} \sum_{i=1}^M \tilde{f}(s_i, a_i) + \epsilon_{stat} \\ &\leq \mathbb{E}_{s,a \sim d^{\pi^{\star}}} \tilde{f}(s, a) - \mathbb{E}_{s,a \sim d^{\star}} \tilde{f}(s, a) + 2\epsilon_{stat} = 2\epsilon_{stat}, \end{aligned}$$

where in the third inequality we use the optimality of $\hat{\pi}$.

Recall that $V^{\pi} = \mathbb{E}_{s,a \sim d^{\pi}} r(s, a) / (1 - \gamma)$, we have:

$$V^{\hat{\pi}} - V^{\star} = \frac{1}{1 - \gamma} (\mathbb{E}_{s,a \sim d^{\hat{\pi}}} r(s, a) - \mathbb{E}_{s,a \sim d^{\star}} r(s, a)) \leq \frac{\sup_{s,a} |r(s, a)|}{1 - \gamma} \|d^{\hat{\pi}} - d^{\star}\|_1 \leq \frac{2}{1 - \gamma} \epsilon_{stat}.$$

This concludes the proof. ■

Note that above theorem confirms the statistical benefit of having the access to a known transition: comparing to the classic Behavior Cloning algorithm, the approximation error of DM-ST only scales linearly with respect to horizon $1/(1 - \gamma)$ instead of quadratically.

14.3.1 Extension to Agnostic Setting

So far we focused on agnostic learning setting. What would happen if $\pi^{\star} \notin \Pi$? We can still run our DM-ST algorithm as is. We state the sample complexity of DM-ST in agnostic setting below.

Theorem 14.5 (Agnostic Guarantee of DM-ST). *Assume Π is finite, but $\pi^{\star} \notin \Pi$. With probability at least $1 - \delta$, DM-ST learns a policy $\hat{\pi}$ such that:*

$$V^{\star} - V^{\hat{\pi}} \leq \frac{1}{1 - \gamma} \|d^{\star} - d^{\hat{\pi}}\|_1 \leq \frac{3}{1 - \gamma} \min_{\pi \in \Pi} \|d^{\pi} - d^{\star}\|_1 + \tilde{O} \left(\frac{1}{1 - \gamma} \sqrt{\frac{\ln(|\Pi|) + \ln(1/\delta))}{M}} \right).$$

Proof: We first define some terms below. Denote $\tilde{\pi} := \arg \min_{\pi \in \Pi} \|d^{\pi} - d^{\star}\|_1$. Let us denote:

$$\begin{aligned} \tilde{f} &= \arg \max_{f \in \mathcal{F}} [\mathbb{E}_{s,a \sim d^{\tilde{\pi}}} f(s, a) - \mathbb{E}_{s,a \sim d^{\star}} f(s, a)], \\ \bar{f} &= \arg \max_{f \in \mathcal{F}} \left[\mathbb{E}_{s,a \sim d^{\tilde{\pi}}} f(s, a) - \frac{1}{M} \sum_{i=1}^M f(s_i^{\star}, a_i^{\star}) \right], \\ f' &= \arg \max_{f \in \mathcal{F}} \left[\mathbb{E}_{s,a \sim d^{\tilde{\pi}}} [f(s, a)] - \frac{1}{M} \sum_{i=1}^M f(s_i^{\star}, a_i^{\star}) \right]. \end{aligned}$$

Starting with a triangle inequality, we have:

$$\begin{aligned}
\|d^{\hat{\pi}} - d^{\pi^*}\|_1 &\leq \|d^{\hat{\pi}} - d^{\tilde{\pi}}\|_1 + \|d^{\tilde{\pi}} - d^{\pi^*}\|_1 \\
&= \mathbb{E}_{s,a \sim d^{\hat{\pi}}} [\tilde{f}(s,a)] - \mathbb{E}_{s,a \sim d^{\tilde{\pi}}} [\tilde{f}(s,a)] + \|d^{\tilde{\pi}} - d^{\pi^*}\|_1 \\
&= \mathbb{E}_{s,a \sim d^{\hat{\pi}}} [\tilde{f}(s,a)] - \frac{1}{M} \sum_{i=1}^M \tilde{f}(s_i^*, a_i^*) + \frac{1}{M} \sum_{i=1}^M \tilde{f}(s_i^*, a_i^*) - \mathbb{E}_{s,a \sim d^{\tilde{\pi}}} [\tilde{f}(s,a)] + \|d^{\tilde{\pi}} - d^{\pi^*}\|_1 \\
&\leq \mathbb{E}_{s,a \sim d^{\hat{\pi}}} [\tilde{f}(s,a)] - \frac{1}{M} \sum_{i=1}^M \tilde{f}(s_i^*, a_i^*) + \frac{1}{M} \sum_{i=1}^M \tilde{f}(s_i^*, a_i^*) - \mathbb{E}_{s,a \sim d^{\tilde{\pi}}} [\tilde{f}(s,a)] \\
&\quad + [\mathbb{E}_{s,a \sim d^{\tilde{\pi}}} \tilde{f}(s,a) - \mathbb{E}_{s,a \sim d^{\tilde{\pi}}} [\tilde{f}(s,a)]] + \|d^{\tilde{\pi}} - d^{\pi^*}\|_1 \\
&\leq \mathbb{E}_{s,a \sim d^{\tilde{\pi}}} [f'(s,a)] - \frac{1}{M} \sum_{i=1}^M f'(s_i^*, a_i^*) + 2\sqrt{\frac{\ln(|\mathcal{F}|/\delta)}{M}} + 2\|d^{\tilde{\pi}} - d^{\pi^*}\|_1 \\
&\leq \mathbb{E}_{s,a \sim d^{\tilde{\pi}}} [f'(s,a)] - \mathbb{E}_{s,a \sim d^{\pi^*}} [f'(s,a)] + 4\sqrt{\frac{\ln(|\mathcal{F}|/\delta)}{M}} + 2\|d^{\tilde{\pi}} - d^{\pi^*}\|_1 \\
&\leq 3\|d^{\pi^*} - d^{\tilde{\pi}}\|_1 + 4\sqrt{\frac{\ln(|\mathcal{F}|/\delta)}{M}},
\end{aligned}$$

where the first inequality uses the definition of $\tilde{f}$, the second inequality uses the fact that $\hat{\pi}$ is the minimizer of $\max_{f \in \mathcal{F}} \mathbb{E}_{s,a \sim d^{\hat{\pi}}} f(s,a) - \frac{1}{M} \sum_{i=1}^M f(s_i^*, a_i^*)$, along the way we also use Hoeffding's inequality where $\forall f \in \mathcal{F}$, $|\mathbb{E}_{s,a \sim d^{\hat{\pi}}} f(s,a) - \sum_{i=1}^M f(s_i^*, a_i^*)| \leq 2\sqrt{\ln(|\mathcal{F}|/\delta)/M}$, with probability at least $1 - \delta$. $\blacksquare$

As we can see, comparing to the realizable setting, here we have an extra term that is related to $\min_{\pi \in \Pi} \|d^{\pi} - d^{\pi^*}\|_1$. Note that the dependency on horizon also scales linearly in this case. In general, the constant 3 in front of $\min_{\pi \in \Pi} \|d^{\pi} - d^{\pi^*}\|_1$ is not avoidable in Scheffé estimator [Devroye and Lugosi, 2012].

14.4 Maximum Entropy Inverse Reinforcement Learning

Similar to Behavior cloning, we assume we have a dataset of state-action pairs from expert $\mathcal{D}^* = \{s_i^*, a_i^*\}_{i=1}^N$ where $s_i^*, a_i^* \sim d^{\pi^*}$. Different from Behavior cloning, here we assume that we have access to the underlying MDP's transition, i.e., we assume transition is known and we can do planning if we were given a cost function.

We assume that we are given a state-action feature mapping $\phi : \mathcal{S} \times \mathcal{A} \mapsto \mathbb{R}^d$ (this can be extended infinite dimensional feature space in RKHS, but we present finite dimensional setting for simplicity). We assume the true ground truth cost function as $c(s,a) := \theta^* \cdot \phi(s,a)$ with $\|\theta^*\|_2 \leq 1$ and θ^* being unknown.

The goal of the learner is to compute a policy $\pi : \mathcal{S} \mapsto \Delta(\mathcal{A})$ such that when measured under the true cost function, it performs as good as the expert, i.e., $\mathbb{E}_{s,a \sim d^{\pi}} \theta^* \cdot \phi(s,a) \approx \mathbb{E}_{s,a \sim d^{\pi^*}} \theta^* \cdot \phi(s,a)$.

We will focus on finite horizon MDP setting in this section. We denote $\rho^{\pi}(\tau)$ as the trajectory distribution induced by π and d^{π} as the average state-action distribution induced by π .

14.4.1 MaxEnt IRL: Formulation and The Principle of Maximum Entropy

MaxEnt IRL uses the principle of Maximum Entropy and poses the following policy optimization program:

$$\begin{aligned} \max_{\pi} & - \sum_{\tau} \rho^{\pi}(\tau) \ln \rho^{\pi}(\tau), \\ \text{s.t.}, & \mathbb{E}_{s,a \sim d^{\pi}} \phi(s, a) = \sum_{i=1}^N \phi(s_i^*, a_i^*)/N. \end{aligned}$$

Note that $\sum_{i=1}^N \phi(s_i^*, a_i^*)/N$ is an unbiased estimate of $\mathbb{E}_{s,a \sim d^{\pi^*}} \phi(s, a)$. MaxEnt-IRL searches for a policy that maximizes the entropy of its trajectory distribution subject to a moment matching constraint.

Note that there could be many policies that satisfy the moment matching constraint, i.e., $\mathbb{E}_{s,a \sim d^{\pi}} \phi(s, a) = \mathbb{E}_{s,a \sim d^{\pi^*}} \phi(s, a)$, and any feasible solution is guaranteed to achieve the same performance of the expert under the ground truth cost function $\theta^* \cdot \phi(s, a)$. The maximum entropy objective ensures that the solution is unique.

Using the Markov property, we notice that:

$$\operatorname{argmax}_{\pi} - \sum_{\tau} \rho^{\pi}(\tau) \ln \rho^{\pi}(\tau) = \operatorname{argmax}_{\pi} - \mathbb{E}_{s,a \sim d^{\pi}} \ln \pi(a|s).$$

This, we can rewrite the MaxEnt-IRL as follows:

$$\min_{\pi} \mathbb{E}_{s,a \sim d^{\pi}} \ln \pi(a|s), \tag{14.2}$$

$$\text{s.t.}, \mathbb{E}_{s,a \sim d^{\pi}} \phi(s, a) = \sum_{i=1}^N \phi(s_i^*, a_i^*)/N. \tag{14.3}$$

14.4.2 Algorithm

To better interpret the objective function, below we replace $\sum_i \phi(s_i^*, a_i^*)/N$ by its expectation $\mathbb{E}_{s,a \sim d^{\pi^*}} \phi(s, a)$. Note that we can use standard concentration inequalities to bound the difference between $\sum_i \phi(s_i^*, a_i^*)/N$ and $\mathbb{E}_{s,a \sim d^{\pi^*}} \phi(s, a)$.

Using Lagrange multipliers, we can rewrite the constrained optimization program in Eq. 14.2 as follows:

$$\min_{\pi} \mathbb{E}_{s,a \sim d^{\pi}} \ln \pi(a|s) + \max_{\theta} \mathbb{E}_{s,a \sim d^{\pi}} \theta^{\top} \phi(s, a) - \mathbb{E}_{s,a \sim d^{\pi^*}} \theta^{\top} \phi(s, a).$$

The above objective conveys a clear goal of our imitation learning problem: we are searching for π that minimizes the MMD between d^{π} and d^{π^*} with a (negative) entropy regularization on the trajectory distribution of policy π .

To derive an algorithm that optimizes the above objective, we first again swap the min max order via the minimax theorem:

$$\max_{\theta} \left(\min_{\pi} \mathbb{E}_{s,a \sim d^{\pi}} \theta^{\top} \phi(s, a) - \mathbb{E}_{s,a \sim d^{\pi^*}} \theta^{\top} \phi(s, a) + \mathbb{E}_{s,a \sim d^{\pi}} \ln \pi(a|s) \right).$$

The above objective proposes a natural algorithm where we perform projected gradient ascent on θ , while perform best response update on π , i.e., given θ , we solve the following planning problem:

$$\operatorname{argmin}_{\pi} \mathbb{E}_{s,a \sim d^{\pi}} \theta^{\top} \phi(s, a) + \mathbb{E}_{s,a \sim d^{\pi}} \ln \pi(a|s). \tag{14.4}$$

Note that the above objective can be understood as planning with cost function $\theta^\top \phi(s, a)$ with an additional negative entropy regularization on the trajectory distribution. On the other hand, given π , we can compute the gradient of θ , which is simply the difference between the expected features:

$$\mathbb{E}_{s,a \sim d^\pi} \phi(s, a) - \mathbb{E}_{s,a \sim d^{\pi^*}} \phi(s, a),$$

which gives the following gradient ascent update on θ :

$$\theta := \theta + \eta \left(\mathbb{E}_{s,a \sim d^\pi} \phi(s, a) - \mathbb{E}_{s,a \sim d^{\pi^*}} \phi(s, a) \right). \quad (14.5)$$

Algorithm of MaxEnt-IRL MaxEnt-IRL updates π and θ alternatively using Eq. 14.4 and Eq. 14.5, respectively. We summarize the algorithm in Alg. 9. Note that for the stochastic gradient of θ_t , we see that it is the empirical average feature difference between the current policy π_t and the expert policy π^* .

Algorithm 9 MaxEnt-IRL

Input: Expert data $\mathcal{D}^* = \{s_i^*, a_i^*\}_{i=1}^M$, MDP $\mathcal{M}$, parameters β, η, N .

- 1: Initialize θ_0 with $\|\theta_0\|_2 \leq 1$.
 - 2: **for** $t = 1, 2, \dots$, **do**
 - 3: Entropy-regularized Planning with cost $\theta_t^\top \phi(s, a)$: $\pi_t \in \operatorname{argmin}_\pi \mathbb{E}_{s,a \sim d^\pi} [\theta_t^\top \phi(s, a) + \beta \ln \pi(a|s)]$.
 - 4: Draw samples $\{s_i, a_i\}_{i=1}^N \sim d^{\pi_t}$ by executing π_t in $\mathcal{M}$.
 - 5: Stochastic Gradient Update: $\theta' = \theta_t + \eta \left[\frac{1}{N} \sum_{i=1}^N \phi(s_i, a_i) - \frac{1}{M} \sum_{i=1}^M \phi(s_i^*, a_i^*) \right]$.
 - 6: **end for**
-

Note that Alg. 9 uses an entropy-regularized planning oracle. Below we discuss how to implement such entropy-regularized planning oracle via dynamic programming.

14.4.3 Maximum Entropy RL: Implementing the Planning Oracle in Eq. 14.4

The planning oracle in Eq. 14.4 can be implemented in a value iteration fashion using Dynamic Programming. We denote $c(s, a) := \theta \cdot \phi(s, a)$.

We are interested in implementing the following planning objective:

$$\operatorname{argmin}_\pi \mathbb{E}_{s,a \sim d^\pi} [c(s, a) + \ln \pi(a|s)]$$

The subsection has its own independent interests beyond the framework of imitation learning. This maximum entropy regularized planning formulation is widely used in RL as well and it is well connected to the framework of RL as Inference.

As usually, we start from the last time step $H - 1$. For any policy π and any state-action (s, a) , we have the cost-to-go $Q_{H-1}^\pi(s, a)$:

$$Q_{H-1}^\pi(s, a) = c(s, a) + \ln \pi(a|s), \quad V_{H-1}^\pi(s) = \sum_a \pi(a|s) (c(s, a) + \ln \pi(a|s)).$$

We have:

$$V_{H-1}^*(s) = \min_{\rho \in \Delta(\mathcal{A})} \sum_a \rho(a) c(s, a) + \rho(a) \ln \rho(a). \quad (14.6)$$

Take gradient with respect to ρ , set it to zero and solve for ρ , we get:

$$\pi_{H-1}^*(a|s) \propto \exp(-c(s, a)), \forall s, a.$$

Substitute π_{H-1}^* back to the expression in Eq. 14.6, we get:

$$V_{H-1}^*(s) = -\ln \left(\sum_a \exp(-c(s, a)) \right),$$

i.e., we apply a softmin operator rather than the usual min operator (recall here we are minimizing cost).

With V_{h+1}^* , we can continue to h . Denote $Q^*(s, a)$ as follows:

$$Q_h^*(s, a) = r(s, a) + \mathbb{E}_{s' \sim P(\cdot|s, a)} V_{h+1}^*(s').$$

For V_h^* , we have:

$$V_h^*(s) = \min_{\rho \in \Delta(\mathcal{A})} \sum_a \rho(a) (c(s, a) + \ln \rho(a) + \mathbb{E}_{s' \sim P(\cdot|s, a)} V_{h+1}^*(s')).$$

Again we can show that the minimizer of the above program has the following form:

$$\pi_h^*(a|s) \propto \exp(-Q_h^*(s, a)). \quad (14.7)$$

Substitute π_h^* back to Q_h^* , we can show that:

$$V_h^*(s) = -\ln \left(\sum_a \exp(-Q_h^*(s, a)) \right),$$

where we see again that V_h^* is based on a softmin operator on Q^* .

Thus the *soft value iteration* can be summarized below:

Soft Value Iteration:

$$Q_{H-1}^*(s, a) = c(s, a), \quad \pi_h^*(a|s) \propto \exp(-Q_h^*(s, a)), \quad V_h^*(s) = -\ln \left(\sum_a \exp(-Q_h^*(s, a)) \right), \forall h.$$

We can continue the above procedure to $h = 0$, which gives us the optimal policies, all in the form of Eq. 14.7

14.5 Interactive Imitation Learning: AggreVaTe and Its Statistical Benefit over Offline IL Setting

We study the Interactive Imitation Learning setting where we have an expert policy that can be queried at any time during training, and we also have access to the ground truth reward signal.

We present AggreVaTe (Aggregate Values to Imitate) first and then analyze its sample complexity under the realizable setting.

Again we start with a realizable policy class Π that is discrete. We denote $\Delta(\Pi)$ as the convex hull of Π and each policy $\pi \in \Delta(\Pi)$ is a mixture policy represented by a distribution $\rho \in \mathbb{R}^{|\Pi|}$ with $\rho[i] \geq 0$ and $\|\rho\|_1 = 1$. With this parameterization, our decision space simply becomes $\Delta(|\Pi|)$, i.e., any point in $\Delta(|\Pi|)$ corresponds to a mixture policy. Notation wise, given $\rho \in \Delta(|\Pi|)$, we denote π_ρ as the corresponding mixture policy. We denote π_i as the i -th policy in Π . We denote $\rho[i]$ as the i -th element of the vector ρ .

Algorithm 10 AggreVaTe

Input: The interactive expert, regularization λ

- 1: Initialize ρ_0 to be a uniform distribution.
 - 2: **for** $t = 0, 2, \dots$, **do**
 - 3: Sample $s_t \sim d^{\pi_{\rho_t}}$
 - 4: Query expert to get $A^*(s_t, a)$ for all $a \in \mathcal{A}$
 - 5: Policy update: $\rho_{t+1} = \operatorname{argmax}_{\rho \in \Delta(|\Pi|)} \sum_{j=0}^t \sum_{i=1}^{|\Pi|} \rho[i] (\mathbb{E}_{a \sim \pi_i(\cdot|s_t)} A^*(s_t, a)) - \lambda \sum_{i=1}^{|\Pi|} \rho[i] \ln(\rho[i])$
 - 6: **end for**
-

AggreVaTe assumes an interactive expert from whom I can query for action feedback. Basically, given a state s , let us assume that expert returns us the advantages of all actions, i.e., *one query of expert oracle at any state s returns $A^*(s, a), \forall a \in \mathcal{A}$* .¹

The algorithm is summarized in Alg. 10.

To analyze the algorithm, let us introduce some additional notations. Let us denote $\ell_t(\rho) = \sum_{i=1}^{|\Pi|} \rho[i] \mathbb{E}_{a \sim \pi_i(\cdot|s)} A^*(s_t, a)$, which is dependent on state s_t generated at iteration t , and is a linear function with respect to decision variable ρ . AggreVaTe is essentially running the specific online learning algorithm, Follow-the-Regularized Leader (FTRL) (e.g., see [Shalev-Shwartz, 2011]) with Entropy regularization:

$$\rho_{t+1} = \operatorname{argmax}_{\rho \in \Delta(|\Pi|)} \sum_{i=0}^t \ell_t(\rho) + \lambda \operatorname{Entropy}(\rho).$$

Denote $c = \max_{s,a} A^*(s, a)$. This implies that $\sup_{\rho,t} \|\ell_t(\rho)\| \leq c$. FTRL with linear loss functions and entropy regularization gives the following deterministic regret guarantees ([Shalev-Shwartz, 2011]):

$$\max_{\rho} \sum_{t=0}^{T-1} \ell_t(\rho) - \sum_{t=0}^{T-1} \ell_t(\rho_t) \leq c \sqrt{\log(|\Pi|)T}. \quad (14.8)$$

We will analyze AggreVaTe's sample complexity using the above result.

Theorem 14.6 (Sample Complexity of AggreVaTe). *Denote $c = \sup_{s,a} |A^*(s, a)|$. Let us denote ϵ_{Π} and ϵ_{stat} as follows:*

$$\epsilon_{\Pi} := \max_{\rho \in \Delta(|\Pi|)} \frac{1}{M} \sum_{t=0}^{M-1} \ell_t(\rho^*), \quad \epsilon_{stat} := \sqrt{\frac{\ln(|\Pi|)/\delta}{M}} + 2\sqrt{\frac{\ln(1/\delta)}{M}}.$$

With probability at least $1 - \delta$, after M iterations (i.e., M calls of expert oracle), AggreVaTe finds a policy $\hat{\pi}$ such that:

$$V^* - V^{\hat{\pi}} \leq \frac{c}{1 - \gamma} \epsilon_{stat} - \frac{1}{1 - \gamma} \epsilon_{\Pi}.$$

Proof: At each iteration t , let us define $\tilde{\ell}_t(\rho_t)$ as follows:

$$\tilde{\ell}_t(\rho_t) = \mathbb{E}_{s \sim d^{\pi_{\rho_t}}} \sum_{i=1}^{|\Pi|} \rho_t[i] \mathbb{E}_{a \sim \pi_i(\cdot|s)} A^*(s, a)$$

¹Technically one cannot use one query to get $A^*(s, a)$ for all a . But one can use importance weighting to get an unbiased estimate of $A^*(s, a)$ for all a via just one expert roll-out. For analysis simplicity, we assume one expert query at s returns the whole vector $A^*(s, \cdot) \in \mathbb{R}^{|\mathcal{A}|}$.

Denote $\mathbb{E}_t[\ell_t(\rho_t)]$ as the conditional expectation of $\ell_t(\rho_t)$, conditioned on all history up and including the end of iteration $t - 1$. Thus, we have $\mathbb{E}_t[\ell_t](\rho_t) = \tilde{\ell}_t(\rho_t)$ as ρ_t only depends on the history up to the end of iteration $t - 1$. Also note that $|\ell_t(\rho)| \leq c$. Thus by Azuma-Hoeffding inequality (Theorem A.5), we get:

$$\left| \frac{1}{M} \sum_{t=0}^{M-1} \tilde{\ell}_t(\rho_t) - \frac{1}{T} \sum_{t=0}^{M-1} \ell_t(\rho_t) \right| \leq 2c\sqrt{\frac{\ln(1/\delta)}{M}},$$

with probability at least $1 - \delta$. Now use Eq. 14.8, and denote $\rho^* = \operatorname{argmin}_{\rho \in \Delta(\Pi)} \frac{1}{M} \sum_{t=0}^{M-1} \ell_t(\rho)$, i.e., the best minimizer that minimizes the average loss $\sum_{t=0}^{M-1} \ell_t(\rho)/M$. we get:

$$\frac{1}{M} \sum_{t=0}^{M-1} \tilde{\ell}_t(\rho_t) \geq \frac{1}{M} \sum_{t=0}^{M-1} \ell_t(\rho_t) - 2c\sqrt{\frac{\ln(1/\delta)}{M}} \geq \frac{1}{M} \sum_{t=0}^{M-1} \ell_t(\rho^*) - c\sqrt{\frac{\ln(|\Pi|)}{M}} - 2c\sqrt{\frac{\ln(1/\delta)}{M}},$$

which means that there must exist a $\hat{t} \in \{0, \dots, M-1\}$, such that:

$$\tilde{\ell}_{\hat{t}}(\rho_{\hat{t}}) \geq \frac{1}{M} \sum_{t=0}^{M-1} \ell_t(\rho^*) - c\sqrt{\frac{\ln(|\Pi|)}{M}} - 2c\sqrt{\frac{\ln(1/\delta)}{M}}.$$

Now use Performance difference lemma, we have:

$$\begin{aligned} (1 - \gamma)(-V^* + V^{\pi_{\rho_{\hat{t}}}}) &= \mathbb{E}_{s \sim d^{\pi_{\rho_{\hat{t}}}}} \sum_{i=1}^{|\Pi|} \rho_{\hat{t}}[i] \mathbb{E}_{a \sim \pi_i(\cdot|s)} A^*(s, a) = \ell_{\hat{t}}(\rho_{\hat{t}}) \\ &\geq \frac{1}{M} \sum_{t=0}^{M-1} \ell_t(\rho^*) - c\sqrt{\frac{\ln(|\Pi|)}{T}} - 2c\sqrt{\frac{\ln(1/\delta)}{T}}. \end{aligned}$$

Rearrange terms, we get:

$$V^* - V^{\pi_{\rho_{\hat{t}}}} \leq -\frac{1}{1 - \gamma} \left[\frac{1}{M} \sum_{t=0}^{M-1} \ell_t(\rho^*) \right] + \frac{c}{1 - \gamma} \left[\sqrt{\frac{\ln(|\Pi|)/\delta}{M}} + 2\sqrt{\frac{\ln(1/\delta)}{M}} \right],$$

which concludes the proof. ■

Remark We analyze the ϵ_{Π} by discussing realizable setting and non-realizable setting separately. When Π is realizable, i.e., $\pi^* \in \Pi$, by the definition of our loss function ℓ_t , we immediately have $\sum_{i=1}^M \ell_t(\rho^*) \geq 0$ since $A^*(s, \pi^*(s)) = 0$ for all $s \in \mathcal{S}$. Moreover, when π^* is not the globally optimal policy, it is possible that $\epsilon_{\Pi} := \sum_{i=1}^M \ell_t(\rho^*)/M > 0$, which implies that when $M \rightarrow \infty$ (i.e., $\epsilon_{stat} \rightarrow 0$), AggreVaTe indeed can learn a policy that outperforms the expert policy π^* . In general when $\pi^* \notin \Pi$, there might not exist a mixture policy ρ that achieves positive advantage against π^* under the M training samples $\{s_0, \dots, s_{M-1}\}$. In this case, $\epsilon_{\Pi} < 0$.

Does AggreVaTe avoids distribution shift? Under realizable setting, note that the bound explicitly depends on $\sup_{s,a} |A^*(s, a)|$ (i.e., it depends on $|\ell_t(\rho_t)|$ for all t). In worst case, it is possible that $\sup_{s,a} |A^*(s, a)| = \Theta(1/(1 - \gamma))$, which implies that AggreVaTe could suffer a quadratic horizon dependency, i.e., $1/(1 - \gamma)^2$. Note that DM-ST provably scales linearly with respect to $1/(1 - \gamma)$, but DM-ST requires a stronger assumption that the transition P is known.

When $\sup_{s,a} |A^*(s, a)| = o(1/(1 - \gamma))$, then AggreVaTe performs strictly better than BC. This is possible when the MDP is mixing quickly under π^* , or Q^* is L -Lipschitz continuous, i.e., $|Q^*(s, a) - Q^*(s, a')| \leq L\|a - a'\|$, with bounded range in action space, e.g., $\sup_{a,a'} \|a - a'\| \leq \beta \in \mathbb{R}^+$. In this case, if L and β are independent of $1/(1 - \gamma)$, then $\sup_{s,a} |A^*(s, a)| \leq L\beta$, which leads to a $\frac{\beta L}{1 - \gamma}$ dependency.

When $\pi^* \notin \Pi$, the agnostic result of AggreVaTe and the agnostic result of DM-ST is not directly comparable. Note that the model class error that DM-ST suffers is algorithmic-independent, i.e., it is $\min_{\pi \in \Pi} \|d^\pi - d^{\pi^*}\|_1$ and it only depends on π^* and Π , while the model class ϵ_Π in AggreVaTe is algorithmic path-dependent, i.e., in addition to π^* and Π , it depends the policies $\pi_1, \pi_2, \dots$ computed during the learning process. Another difference is that $\min_{\pi \in \Pi} \|d^\pi - d^{\pi^*}\|_1 \in [0, 2]$, while ϵ_Π indeed could scale linearly with respect to $1/(1 - \gamma)$, i.e., $\epsilon_\pi \in [-\frac{1}{1-\gamma}, 0]$ (assume π^* is the globally optimal policy for simplicity).

14.6 Bibliographic Remarks and Further Readings

Behavior cloning was used in autonomous driving back in 1989 [Pomerleau, 1989], and the distribution shift and compounding error issue was studied by Ross and Bagnell [2010]. Ross and Bagnell [2010], Ross et al. [2011] proposed using interactive experts to alleviate the distribution shift issue.

MaxEnt-IRL was proposed by Ziebart et al. [2008]. The original MaxEnt-IRL focuses on deterministic MDPs and derived distribution over trajectories. Later Ziebart et al. [2010] proposed the Principle of Maximum Causal Entropy framework which captures MDPs with stochastic transition. MaxEnt-IRL has been widely used in real applications (e.g., [Kitani et al., 2012, Ziebart et al., 2009]).

To the best of our knowledge, the algorithm Distribution Matching with Scheffé Tournament introduced in this chapter is new here and is the first to demonstrate the statistical benefit (in terms of sample complexity of the expert policy) of the hybrid setting over the pure offline setting with general function approximation.

Recently, there are approaches that extend the linear cost functions used in MaxEnt-IRL to deep neural networks which are treated as discriminators to distinguish between experts datasets and the datasets generated by the policies. Different distribution divergences have been proposed, for instance, JS-divergence [Ho and Ermon, 2016], general f-divergence which generalizes JS-divergence [Ke et al., 2019], and Integral Probability Metric (IPM) which includes Wasserstein distance [Sun et al., 2019b]. While these adversarial IL approaches are promising as they fall into the hybrid setting, it has been observed that empirically, adversarial IL sometimes cannot outperform simple algorithms such as Behavior cloning which operates in pure offline setting (e.g., see experiments results on common RL benchmarks from Brantley et al. [2019]).

Another important direction in IL is to combine expert demonstrations and reward functions together (i.e., perform Imitation together with RL). There are many works in this direction, including learning with pre-collected expert data [Hester et al., 2017, Rajeswaran et al., 2017, Cheng et al., 2018, Le et al., 2018, Sun et al., 2018], learning with an interactive expert [Daumé et al., 2009, Ross and Bagnell, 2014, Chang et al., 2015, Sun et al., 2017, Cheng et al., 2020, Cheng and Boots, 2018]. The algorithm AggreVaTe was originally introduced in [Ross and Bagnell, 2014]. A policy gradient and natural policy gradient version of AggreVaTe are introduced in [Sun et al., 2017]. The precursor of AggreVaTe is the algorithm Data Aggregation (DAgger) [Ross et al., 2011], which leverages interactive experts and a reduction to no-regret online learning, but without assuming access to the ground truth reward signals.

The maximum entropy RL formulation has been widely used in RL literature as well. For instance, Guided Policy Search (GPS) (and its variants) use maximum entropy RL formulation as its planning subroutine [Levine and Koltun, 2013, Levine and Abbeel, 2014]. We refer readers to the excellent survey from Levine [2018] for more details of Maximum Entropy RL formulation and its connections to the framework of RL as Probabilistic Inference.

Chapter 15

Linear Quadratic Regulators

This chapter will introduce some of the fundamentals of optimal control for the linear quadratic regulator model. This model is an MDP, with continuous states and actions. While the model itself is often inadequate as a global model, it can be quite effective as a locally linear model (provided our system does not deviate away from the regime where our linear model is reasonable approximation).

The basics of optimal control theory can be found in any number of standards text Anderson and Moore [1990], Evans [2005], Bertsekas [2017]. The treatment of Gauss-Newton and the NPG algorithm are due to Fazel et al. [2018].

15.1 The LQR Model

In the standard optimal control problem, a dynamical system is described as

$$x_{t+1} = f_t(x_t, u_t, w_t),$$

where f_t maps a state $x_t \in \mathbb{R}^d$, a control (the action) $u_t \in \mathbb{R}^k$, and a disturbance w_t , to the next state $x_{t+1} \in \mathbb{R}^d$, starting from an initial state x_0 . The objective is to find the control policy π which minimizes the long term cost,

$$\begin{aligned} \text{minimize} \quad & \mathbb{E}_\pi \left[\sum_{t=0}^H c_t(x_t, u_t) \right] \\ \text{such that} \quad & x_{t+1} = f_t(x_t, u_t, w_t) \quad t = 0, \dots, H. \end{aligned}$$

where H is the time horizon (which can be finite or infinite).

In practice, this is often solved by considering the linearized control (sub-)problem where the dynamics are approximated by

$$x_{t+1} = A_t x_t + B_t u_t + w_t,$$

with the matrices A_t and B_t are derivatives of the dynamics f and where the costs are approximated by a quadratic function in x_t and u_t .

This chapter focuses on an important special case: finite and infinite horizon problem referred to as the linear quadratic regulator (LQR) problem. We can view this model as being an local approximation to non-linear model. However, we will analyze these models under the assumption that they are globally valid.

Finite Horizon LQRs. The finite horizon LQR problem is given by

$$\begin{aligned} \text{minimize} \quad & \mathbb{E} \left[x_H^\top Q x_H + \sum_{t=0}^{H-1} (x_t^\top Q x_t + u_t^\top R u_t) \right] \\ \text{such that} \quad & x_{t+1} = A_t x_t + B_t u_t + w_t, \quad x_0 \sim \mathcal{D}, \quad w_t \sim N(0, \sigma^2 I), \end{aligned}$$

where initial state $x_0 \sim \mathcal{D}$ is assumed to be randomly distributed according to distribution $\mathcal{D}$; the disturbance $w_t \in \mathbb{R}^d$ follows the law of a multi-variate normal with covariance $\sigma^2 I$; the matrices $A_t \in \mathbb{R}^{d \times d}$ and $B_t \in \mathbb{R}^{d \times k}$ are referred to as system (or transition) matrices; $Q \in \mathbb{R}^{d \times d}$ and $R \in \mathbb{R}^{k \times k}$ are both positive definite matrices that parameterize the quadratic costs. Note that this model is a finite horizon MDP, where the $\mathcal{S} = \mathbb{R}^d$ and $\mathcal{A} = \mathbb{R}^k$.

Infinite Horizon LQRs. We also consider the infinite horizon LQR problem:

$$\begin{aligned} \text{minimize} \quad & \lim_{H \rightarrow \infty} \frac{1}{H} \mathbb{E} \left[\sum_{t=0}^H (x_t^\top Q x_t + u_t^\top R u_t) \right] \\ \text{such that} \quad & x_{t+1} = A x_t + B u_t + w_t, \quad x_0 \sim \mathcal{D}, \quad w_t \sim N(0, \sigma^2 I). \end{aligned}$$

Note that here we are assuming the dynamics are time homogenous. We will assume that the optimal objective function (i.e. the optimal average cost) is finite; this is referred to as the system being *controllable*. This is a special case of an MDP with an average reward objective.

Throughout this chapter, we assume A and B are such that the optimal cost is finite. Due to the geometric nature of the system dynamics (say for a controller which takes controls u_t that are linear in the state x_t), there may exist linear controllers with infinite costs. This instability of LQRs (at least for some A and B and some controllers) leads to that the theoretical analysis often makes various assumptions on A and B in order to guarantee some notion of stability. In practice, the finite horizon setting is more commonly used in practice, particularly due to that the LQR model is only a good local approximation of the system dynamics, where the infinite horizon model tends to be largely of theoretical interest. See Section 15.6.

The infinite horizon discounted case? The infinite horizon discounted case tends not to be studied for LQRs. This is largely due to that, for the undiscounted case (with the average cost objective), we have may infinite costs (due to the aforementioned geometric nature of the system dynamics); in such cases, discounting will not necessarily make the average cost finite.

15.2 Bellman Optimality: Value Iteration & The Algebraic Riccati Equations

A standard result in optimal control theory shows that the optimal control input can be written as a linear function in the state. As we shall see, this is a consequence of the Bellman equations.

15.2.1 Planning and Finite Horizon LQRs

Slightly abusing notation, it is convenient to define the value function and state-action value function with respect to the costs as follows: For a policy π , a state x , and $h \in \{0, \dots, H-1\}$, we define the value function $V_h^\pi : \mathbb{R}^d \rightarrow \mathbb{R}$ as

$$V_h^\pi(x) = \mathbb{E} \left[x_H^\top Q x_H + \sum_{t=h}^{H-1} (x_t^\top Q x_t + u_t^\top R u_t) \mid \pi, x_h = x \right],$$

where again expectation is with respect to the randomness of the trajectory, that is, the randomness in state transitions. Similarly, the state-action value (or Q-value) function $Q_h^\pi : \mathbb{R}^d \times \mathbb{R}^k \rightarrow \mathbb{R}$ is defined as

$$Q_h^\pi(x, u) = \mathbb{E} \left[x_H^\top Q x_H + \sum_{t=h}^{H-1} (x_t^\top Q x_t + u_t^\top R u_t) \mid \pi, x_h = x, u_h = u \right].$$

We define V^* and Q^* analogously.

The following theorem provides a characterization of the optimal policy, via the algebraic Riccati equations. These equations are simply the value iteration algorithm for the special case of LQRs.

Theorem 15.1. (*Value Iteration and the Riccati Equations*). *Suppose R is positive definite. The optimal policy is a linear controller specified by:*

$$\pi^*(x_t) = -K_t^* x_t$$

where

$$K_t^* = (B_t^\top P_{t+1} B_t + R)^{-1} B_t^\top P_{t+1} A_t.$$

Here, P_t can be computed iteratively, in a backwards manner, using the following algebraic Riccati equations, where for $t \in [H]$,

$$\begin{aligned} P_t &:= A_t^\top P_{t+1} A_t + Q - A_t^\top P_{t+1} B_t (B_t^\top P_{t+1} B_t + R)^{-1} B_t^\top P_{t+1} A_t \\ &= A_t^\top P_{t+1} A_t + Q - (K_{t+1}^*)^\top (B_t^\top P_{t+1} B_t + R) K_{t+1}^* \end{aligned}$$

and where $P_H = Q$. (The above equation is simply the value iteration algorithm).

Furthermore, for $t \in [H]$, we have that:

$$V_t^*(x) = x^\top P_t x + \sigma^2 \sum_{h=t+1}^H \text{Trace}(P_h).$$

We often refer to K_t^* as the *optimal control gain* matrices. It is straightforward to generalize the above when $\sigma \geq 0$.

We have assumed that R is strictly positive definite, to avoid have to working with the pseudo-inverse; the theorem is still true when $R = 0$, provided we use a pseudo-inverse.

Proof: By the Bellman optimality conditions (Theorem ?? for episodic MDPs), we know the optimal policy (among all possibly history dependent, non-stationary, and randomized policies), is given by a deterministic stationary policy which is only a function of x_t and t . We have that:

$$\begin{aligned} Q_{H-1}(x, u) &= \mathbb{E}[(A_{H-1}x + B_{H-1}u + w_{H-1})^\top Q(A_{H-1}x + B_{H-1}u + w_{H-1})] + x^\top Q x + u^\top R u \\ &= (A_{H-1}x + B_{H-1}u)^\top Q(A_{H-1}x + B_{H-1}u) + \sigma^2 \text{Trace}(Q) + x^\top Q x + u^\top R u \end{aligned}$$

due to that $x_H = A_{H-1}x + B_{H-1}u + w_{H-1}$, and $\mathbb{E}[w_{H-1}^\top Q w_{H-1}] = \text{Trace}(\sigma^2 Q)$. Due to that this is a quadratic function of u , we can immediately derive that the optimal control is given by:

$$\pi_{H-1}^*(x) = -(B_{H-1}^\top Q B_{H-1} + R)^{-1} B_{H-1}^\top Q A_{H-1} x = -K_{H-1}^* x,$$

where the last step uses that $P_H := Q$.

For notational convenience, let $K = K_{H-1}^*$, $A = A_{H-1}$, and $B = B_{H-1}$. Using the optimal control at x , i.e.

$u = -K_{H-1}^* x$, we have:

$$\begin{aligned}
V_{H-1}^*(x) &= Q_{H-1}(x, -K_{H-1}^* x) \\
&= x^\top \left((A - BK)^\top Q (A - BK) + Q + K^\top R K \right) x + \sigma^2 \text{Trace}(Q) \\
&= x^\top \left(AQA + Q - 2K^\top B^\top Q A + K^\top (B^\top Q B + R) K \right) x + \sigma^2 \text{Trace}(Q) \\
&= x^\top \left(AQA + Q - 2K^\top (B^\top Q B + R) K + K^\top (B^\top Q B + R) K \right) x + \sigma^2 \text{Trace}(Q) \\
&= x^\top \left(AQA + Q - K^\top (B^\top Q B + R) K \right) x + \sigma^2 \text{Trace}(Q) \\
&= x^\top P_{H-1} x + \sigma^2 \text{Trace}(Q).
\end{aligned}$$

where the fourth step uses our expression for $K = K_{H-1}^*$. This proves our claim for $t = H - 1$.

This implies that:

$$\begin{aligned}
Q_{H-2}^*(x, u) &= \mathbb{E}[V_{H-1}^*(A_{H-2}x + B_{H-2}u + w_{H-2})] + x^\top Qx + u^\top Ru \\
&= (A_{H-2}x + B_{H-2}u)^\top P_{H-1} (A_{H-2}x + B_{H-2}u) + \sigma^2 \text{Trace}(P_{H-1}) + \sigma^2 \text{Trace}(Q) + x^\top Qx + u^\top Ru.
\end{aligned}$$

The remainder of the proof follows from a recursive argument, which can be verified along identical lines to the $t = H - 1$ case. $\blacksquare$

15.2.2 Planning and Infinite Horizon LQRs

Theorem 15.2. *Suppose that the optimal cost is finite and that R is positive definite. Let P be a solution to the following algebraic Riccati equation:*

$$P = A^T P A + Q - A^T P B (B^T P B + R)^{-1} B^T P A. \quad (15.1)$$

(Note that P is a positive definite matrix). We have that the optimal policy is:

$$\pi^*(x) = -K^* x$$

where the optimal control gain is:

$$K^* = -(B^T P B + R)^{-1} B^T P A. \quad (15.2)$$

We have that P is unique and that the optimal average cost is $\sigma^2 \text{Trace}(P)$.

As before, P parameterizes the optimal value function. We do not prove this theorem here though it follows along similar lines to the previous proof, via a limiting argument.

To find P , we can again run the recursion:

$$P \leftarrow Q + A^T P A - A^T P B (R + B^T P B)^{-1} B^T P A$$

starting with $P = Q$, which can be shown to converge to the unique positive semidefinite solution of the Riccati equation (since one can show the fixed-point iteration is contractive). Again, this approach is simply value iteration.

15.3 Convex Programs to find P and K^*

For the infinite horizon LQR problem, the optimization may be formulated as a convex program. In particular, the LQR problem can also be expressed as a semidefinite program (SDP) with variable P , for the infinite horizon case. We now present this primal program along with the dual program.

Note that these programs are the analogues of the linear programs from Section ?? for MDPs. While specifying these linear programs for an LQR (as an LQR is an MDP) would result in infinite dimensional linear programs, the special structure of the LQR implies these primal and dual programs have a more compact formulation when specified as an SDP.

15.3.1 The Primal for Infinite Horizon LQR

The primal optimization problem is given as:

$$\begin{aligned} & \text{maximize} && \sigma^2 \text{Trace}(P) \\ & \text{subject to} && \begin{bmatrix} A^T P A + Q - P & A^T P B \\ B^T P A & B^T P B + R \end{bmatrix} \succeq 0, \quad P \succeq 0, \end{aligned}$$

where the optimization variable is P . This SDP has a unique solution, P^* , which satisfies the algebraic Riccati equations (Equation 15.1); the optimal average cost of the infinite horizon LQR is $\sigma^2 \text{Trace}(P^*)$; and the optimal policy is given by Equation 15.2.

The SDP can be derived by relaxing the equality in the Riccati equation to an inequality, then using the Schur complement lemma to rewrite the resulting Riccati inequality as linear matrix inequality. In particular, we can consider the relaxation where P must satisfy:

$$A^T P A + Q - A^T P B (B^T P B + R)^{-1} B^T P A \succeq P.$$

That the solution to this relaxed optimization problem leads to the optimal P^* is due to the Bellman optimality conditions.

Now the Schur complement lemma for positive semi-definiteness is as follows: define

$$X = \begin{bmatrix} D & E \\ E^T & F \end{bmatrix}.$$

(for matrices D , E , and F of appropriate size, with D and F being square symmetric matrices). We have that X is PSD if and only if

$$D - E^T F^{-1} E \succeq 0.$$

This shows that the constraint set is equivalent to the above relaxation.

15.3.2 The Dual

The dual optimization problem is given as:

$$\begin{aligned} & \text{minimize} && \text{Trace} \left(\Sigma \cdot \begin{bmatrix} Q & 0 \\ 0 & R \end{bmatrix} \right) \\ & \text{subject to} && \Sigma_{xx} = (A \ B) \Sigma (A \ B)^T + \sigma^2 I, \quad \Sigma \succeq 0, \end{aligned}$$

where the optimization variable is a symmetric matrix Σ , which is a $(d+k) \times (d+k)$ matrix with the block structure:

$$\Sigma = \begin{bmatrix} \Sigma_{xx} & \Sigma_{xu} \\ \Sigma_{ux} & \Sigma_{uu} \end{bmatrix}.$$

The interpretation of Σ is that it is the covariance matrix of the stationary distribution. This analogous to the state-visitation measure for an MDP.

This SDP has a unique solution, say Σ^* . The optimal gain matrix is then given by:

$$K^* = -\Sigma_{ux}^* (\Sigma_{xx}^*)^{-1}.$$

15.4 Policy Iteration, Gauss Newton, and NPG

Note that the noise variance σ^2 does not impact the optimal policy, in either the discounted case or in the infinite horizon case.

Here, when we examine local search methods, it is more convenient to work with case where $\sigma = 0$. In this case, we can work with cumulative cost rather the average cost. Precisely, when $\sigma = 0$, the infinite horizon LQR problem takes the form:

$$\min_K C(K), \text{ where } C(K) = \mathbb{E}_{x_0 \sim \mathcal{D}} \left[\sum_{t=0}^H (x_t^\top Q x_t + u_t^\top R u_t) \right],$$

where the dynamics evolves as

$$x_{t+1} = (A - BK)x_t.$$

Note that we have directly parameterized our policy as a linear policy in terms of the gain matrix K , due to that we know the optimal policy is linear in the state. Again, we assume that $C(K^*)$ is finite; this assumption is referred to as the system being *controllable*.

We now examine local search based approaches, where we will see a close connection to policy iteration. Again, we have a non-convex optimization problem:

Lemma 15.3. (Non-convexity) *If $d \geq 3$, there exists an LQR optimization problem, $\min_K C(K)$, which is not convex or quasi-convex.*

Regardless, we will see that gradient based approaches are effective. For local search based approaches, the importance of (some) randomization, either in x_0 or noise through having a disturbance, is analogous to our use of having a wide-coverage distribution μ (for MDPs).

15.4.1 Gradient Expressions

Gradient descent on $C(K)$, with a fixed stepsize η , follows the update rule:

$$K \leftarrow K - \eta \nabla C(K).$$

It is helpful to explicitly write out the functional form of the gradient. Define P_K as the solution to:

$$P_K = Q + K^\top R K + (A - BK)^\top P_K (A - BK).$$

and, under this definition, it follows that $C(K)$ can be written as:

$$C(K) = \mathbb{E}_{x_0 \sim \mathcal{D}} x_0^\top P_K x_0.$$

Also, define Σ_K as the (un-normalized) state correlation matrix, i.e.

$$\Sigma_K = \mathbb{E}_{x_0 \sim \mathcal{D}} \sum_{t=0}^{\infty} x_t x_t^\top.$$

Lemma 15.4. (Policy Gradient Expression) *The policy gradient is:*

$$\nabla C(K) = 2 \left((R + B^\top P_K B) K - B^\top P_K A \right) \Sigma_K$$

For convenience, define E_K to be

$$E_K = \left((R + B^\top P_K B) K - B^\top P_K A \right),$$

as a result the gradient can be written as $\nabla C(K) = 2E_K \Sigma_K$.

Proof: Observe:

$$\begin{aligned} C_K(x_0) &= x_0^\top P_K x_0 = x_0^\top (Q + K^\top R K) x_0 + x_0^\top (A - BK)^\top P_K (A - BK) x_0 \\ &= x_0^\top (Q + K^\top R K) x_0 + C_K((A - BK)x_0). \end{aligned}$$

Let ∇ denote the gradient with respect to K ; note that $\nabla C_K((A - BK)x_0)$ has two terms as function of K , one with respect to K in the subscript and one with respect to the input $(A - BK)x_0$. This implies

$$\begin{aligned} \nabla C_K(x_0) &= 2RKx_0x_0^\top - 2B^\top P_K(A - BK)x_0x_0^\top + \nabla C_K(x_1)|_{x_1=(A-BK)x_0} \\ &= 2((R + B^\top P_K B)K - B^\top P_K A) \sum_{t=0}^{\infty} x_t x_t^\top \end{aligned}$$

where we have used recursion and that $x_1 = (A - BK)x_0$. Taking expectations completes the proof. $\blacksquare$

The natural policy gradient. Let us now motivate a version of the natural gradient. The natural policy gradient follows the update:

$$\theta \leftarrow \theta - \eta F(\theta)^{-1} \nabla C(\theta), \text{ where } F(\theta) = \mathbb{E} \left[\sum_{t=0}^{\infty} \nabla \log \pi_\theta(u_t | x_t) \nabla \log \pi_\theta(u_t | x_t)^\top \right],$$

where $F(\theta)$ is the Fisher information matrix. A natural special case is using a linear policy with additive Gaussian noise, i.e.

$$\pi_K(x, u) = \mathcal{N}(Kx, \sigma^2 I) \quad (15.3)$$

where $K \in \mathbb{R}^{k \times d}$ and σ^2 is the noise variance. In this case, the natural policy gradient of K (when σ is considered fixed) takes the form:

$$K \leftarrow K - \eta \nabla C(\pi_K) \Sigma_K^{-1} \quad (15.4)$$

Note a subtlety here is that $C(\pi_K)$ is the randomized policy.

To see this, one can verify that the Fisher matrix of size $kd \times kd$, which is indexed as $[G_K]_{(i,j),(i',j')}$ where $i, i' \in \{1, \dots, k\}$ and $j, j' \in \{1, \dots, d\}$, has a block diagonal form where the only non-zeros blocks are $[G_K]_{(i,\cdot),(i,\cdot)} = \Sigma_K$ (this is the block corresponding to the i -th coordinate of the action, as i ranges from 1 to k). This form holds more generally, for any diagonal noise.

15.4.2 Convergence Rates

We consider three exact rules, where we assume access to having exact gradients. As before, we can also estimate these gradients through simulation. For gradient descent, the update is

$$K_{n+1} = K_n - \eta \nabla C(K_n). \quad (15.5)$$

For natural policy gradient descent, the direction is defined so that it is consistent with the stochastic case, as per Equation 15.4, in the exact case the update is:

$$K_{n+1} = K_n - \eta \nabla C(K_n) \Sigma_{K_n}^{-1} \quad (15.6)$$

One show that the Gauss-Newton update is:

$$K_{n+1} = K_n - \eta (R + B^\top P_{K_n} B)^{-1} \nabla C(K_n) \Sigma_{K_n}^{-1}. \quad (15.7)$$

(Gauss-Newton is non-linear optimization approach which uses a certain Hessian approximation. It can be show that this leads to the above update rule.) Interestingly, for the case when $\eta = 1$, the Gauss-Newton method is equivalent to the policy iteration algorithm, which optimizes a one-step deviation from the current policy.

The Gauss-Newton method requires the most complex oracle to implement: it requires access to $\nabla C(K)$, Σ_K , and $R + B^\top P_K B$; as we shall see, it also enjoys the strongest convergence rate guarantee. At the other extreme, gradient descent requires oracle access to only $\nabla C(K)$ and has the slowest convergence rate. The natural policy gradient sits in between, requiring oracle access to $\nabla C(K)$ and Σ_K , and having a convergence rate between the other two methods.

In this theorem, $\|M\|_2$ denotes the spectral norm of a matrix M .

Theorem 15.5. (*Global Convergence of Gradient Methods*) Suppose $C(K_0)$ is finite and, for μ defined as

$$\mu := \sigma_{\min}(\mathbb{E}_{x_0 \sim \mathcal{D}} x_0 x_0^\top),$$

suppose $\mu > 0$.

- *Gauss-Newton case:* Suppose $\eta = 1$, the Gauss-Newton algorithm (Equation 15.7) enjoys the following performance bound:

$$C(K_N) - C(K^*) \leq \epsilon, \text{ for } N \geq \frac{\|\Sigma_{K^*}\|_2}{\mu} \log \frac{C(K_0) - C(K^*)}{\epsilon}.$$

- *Natural policy gradient case:* For a stepsize $\eta = 1/(\|R\|_2 + \frac{\|B\|_2^2 C(K_0)}{\mu})$, natural policy gradient descent (Equation 15.6) enjoys the following performance bound:

$$C(K_N) - C(K^*) \leq \epsilon, \text{ for } N \geq \frac{\|\Sigma_{K^*}\|_2}{\mu} \left(\frac{\|R\|_2}{\sigma_{\min}(R)} + \frac{\|B\|_2^2 C(K_0)}{\mu \sigma_{\min}(R)} \right) \log \frac{C(K_0) - C(K^*)}{\epsilon}.$$

- *Gradient descent case:* For any starting policy K_0 , there exists a (constant) stepsize η (which could be a function of K_0), such that:

$$C(K_N) \rightarrow C(K^*), \text{ as } N \rightarrow \infty.$$

15.4.3 Gauss-Newton Analysis

We only provide a proof for the Gauss-Newton case (see 15.6 for further readings).

We overload notation and let K denote the policy $\pi(x) = Kx$. For the infinite horizon cost function, define:

$$\begin{aligned} V_K(x) &:= \sum_{t=0}^{\infty} (x_t^\top Q x_t + u_t^\top R u_t) \\ &= x^\top P_K x, \end{aligned}$$

and

$$Q_K(x, u) := x^\top Q x + u^\top R u + V_K(Ax + Bu),$$

and

$$A_K(x, u) = Q_K(x, u) - V_K(x).$$

The next lemma is identical to the performance difference lemma.

Lemma 15.6. (*Cost difference lemma*) Suppose K and K' have finite costs. Let $\{x'_t\}$ and $\{u'_t\}$ be state and action sequences generated by K' , i.e. starting with $x'_0 = x$ and using $u'_t = -K'x'_t$. It holds that:

$$V_{K'}(x) - V_K(x) = \sum_t A_K(x'_t, u'_t).$$

Also, for any x , the advantage is:

$$A_K(x, K'x) = 2x^\top (K' - K)^\top E_K x + x^\top (K' - K)^\top (R + B^\top P_K B)(K' - K)x. \quad (15.8)$$

Proof: Let c'_t be the cost sequence generated by K' . Telescoping the sum appropriately:

$$\begin{aligned} V_{K'}(x) - V_K(x) &= \sum_{t=0} c'_t - V_K(x) = \sum_{t=0} (c'_t + V_K(x'_t) - V_K(x'_t)) - V_K(x) \\ &= \sum_{t=0} (c'_t + V_K(x'_{t+1}) - V_K(x'_t)) = \sum_{t=0} A_K(x'_t, u'_t) \end{aligned}$$

which completes the first claim (the third equality uses the fact that $x = x_0 = x'_0$).

For the second claim, observe that:

$$V_K(x) = x^\top (Q + K^\top R K) x + x^\top (A - BK)^\top P_K (A - BK) x$$

And, for $u = K'x$,

$$\begin{aligned} A_K(x, u) &= Q_K(x, u) - V_K(x) \\ &= x^\top (Q + (K')^\top R K') x + x^\top (A - BK')^\top P_K (A - BK') x - V_K(x) \\ &= x^\top (Q + (K' - K + K)^\top R (K' - K + K)) x + \\ &\quad x^\top (A - BK - B(K' - K))^\top P_K (A - BK - B(K' - K)) x - V_K(x) \\ &= 2x^\top (K' - K)^\top ((R + B^\top P_K B)K - B^\top P_K A) x + \\ &\quad x^\top (K' - K)^\top (R + B^\top P_K B)(K' - K) x, \end{aligned}$$

which completes the proof. ■

We have the following corollary, which can be viewed analogously to a smoothness lemma.

Corollary 15.7. (*“Almost” smoothness*) $C(K)$ satisfies:

$$C(K') - C(K) = -2\text{Trace}(\Sigma_{K'}(K - K')^\top E_K) + \text{Trace}(\Sigma_{K'}(K - K')^\top (R + B^\top P_K B)(K - K'))$$

To see why this is related to smoothness (recall the definition of a smooth function in Equation 12.6), suppose K' is sufficiently close to K so that:

$$\Sigma_{K'} \approx \Sigma_K + O(\|K - K'\|) \quad (15.9)$$

and the leading order term $2\text{Trace}(\Sigma_{K'}(K' - K)^\top E_K)$ would then behave as $\text{Trace}((K' - K)^\top \nabla C(K))$.

Proof: The claim immediately results from Lemma 15.6, by using Equation 15.8 and taking an expectation. ■

We now use this cost difference lemma to show that $C(K)$ is gradient dominated.

Lemma 15.8. (*Gradient domination*) Let K^* be an optimal policy. Suppose K has finite cost and $\mu > 0$. It holds that:

$$\begin{aligned} C(K) - C(K^*) &\leq \|\Sigma_{K^*}\| \text{Trace}(E_K^\top (R + B^\top P_K B)^{-1} E_K) \\ &\leq \frac{\|\Sigma_{K^*}\|}{\mu^2 \sigma_{\min}(R)} \text{Trace}(\nabla C(K)^\top \nabla C(K)) \end{aligned}$$

Proof: From Equation 15.8 and by completing the square,

$$\begin{aligned} A_K(x, K'x) &= Q_K(x, K'x) - V_K(x) \\ &= 2\text{Trace}(xx^\top (K' - K)^\top E_K) + \text{Trace}(xx^\top (K' - K)^\top (R + B^\top P_K B)(K' - K)) \\ &= \text{Trace}(xx^\top (K' - K + (R + B^\top P_K B)^{-1} E_K)^\top (R + B^\top P_K B)(K' - K + (R + B^\top P_K B)^{-1} E_K)) \\ &\quad - \text{Trace}(xx^\top E_K^\top (R + B^\top P_K B)^{-1} E_K) \\ &\geq -\text{Trace}(xx^\top E_K^\top (R + B^\top P_K B)^{-1} E_K) \end{aligned} \quad (15.10)$$

with equality when $K' = K - (R + B^\top P_K B)^{-1} E_K$.

Let x_t^* and u_t^* be the sequence generated under K^* . Using this and Lemma 15.6,

$$\begin{aligned} C(K) - C(K^*) &= -\mathbb{E} \sum_t A_K(x_t^*, u_t^*) \leq \mathbb{E} \sum_t \text{Trace}(x_t^* (x_t^*)^\top E_K^\top (R + B^\top P_K B)^{-1} E_K) \\ &= \text{Trace}(\Sigma_{K^*} E_K^\top (R + B^\top P_K B)^{-1} E_K) \leq \|\Sigma_{K^*}\| \text{Trace}(E_K^\top (R + B^\top P_K B)^{-1} E_K) \end{aligned}$$

which completes the proof of first inequality. For the second inequality, observe:

$$\begin{aligned} \text{Trace}(E_K^\top (R + B^\top P_K B)^{-1} E_K) &\leq \|(R + B^\top P_K B)^{-1}\| \text{Trace}(E_K^\top E_K) \leq \frac{1}{\sigma_{\min}(R)} \text{Trace}(E_K^\top E_K) \\ &= \frac{1}{\sigma_{\min}(R)} \text{Trace}(\Sigma_K^{-1} \nabla C(K)^\top \nabla C(K) \Sigma_K^{-1}) \leq \frac{1}{\sigma_{\min}(\Sigma_K)^2 \sigma_{\min}(R)} \text{Trace}(\nabla C(K)^\top \nabla C(K)) \\ &\leq \frac{1}{\mu^2 \sigma_{\min}(R)} \text{Trace}(\nabla C(K)^\top \nabla C(K)) \end{aligned}$$

which completes the proof of the upper bound. Here the last step is because $\Sigma_K \succeq \mathbb{E}[x_0 x_0^\top]$. ■

The next lemma bounds the one step progress of Gauss-Newton.

Lemma 15.9. (*Gauss-Newton Contraction*) Suppose that:

$$K' = K - \eta(R + B^\top P_K B)^{-1} \nabla C(K) \Sigma_K^{-1},$$

If $\eta \leq 1$, then

$$C(K') - C(K^*) \leq \left(1 - \frac{\eta \mu}{\|\Sigma_{K^*}\|}\right) (C(K) - C(K^*))$$

Proof: Observe that for PSD matrices A and B , we have that $\text{Trace}(AB) \geq \sigma_{\min}(A) \text{Trace}(B)$. Also, observe $K' = K - \eta(R + B^\top P_K B)^{-1} E_K$. Using Lemma 15.7 and the condition on η ,

$$\begin{aligned} C(K') - C(K) &= -2\eta \text{Trace}(\Sigma_{K'} E_K^\top (R + B^\top P_K B)^{-1} E_K) + \eta^2 \text{Trace}(\Sigma_{K'} E_K^\top (R + B^\top P_K B)^{-1} E_K) \\ &\leq -\eta \text{Trace}(\Sigma_{K'} E_K^\top (R + B^\top P_K B)^{-1} E_K) \\ &\leq -\eta \sigma_{\min}(\Sigma_{K'}) \text{Trace}(E_K^\top (R + B^\top P_K B)^{-1} E_K) \\ &\leq -\eta \mu \text{Trace}(E_K^\top (R + B^\top P_K B)^{-1} E_K) \\ &\leq -\eta \frac{\mu}{\|\Sigma_{K^*}\|} (C(K) - C(K^*)), \end{aligned}$$

where the last step uses Lemma 15.8. ■

With this lemma, the proof of the convergence rate of the Gauss Newton algorithm is immediate.

Proof: (of Theorem 15.5, Gauss-Newton case) The theorem is due to that $\eta = 1$ leads to a contraction of $1 - \frac{\mu}{\|\Sigma_{K^*}\|}$ at every step. ■

15.5 System Level Synthesis for Linear Dynamical Systems

We demonstrate another parameterization of controllers which admits convexity. Specifically in this section, we consider finite horizon setting, i.e.,

$$x_{t+1} = Ax_t + Bu_t + w_t, \quad x_0 \sim \mathcal{D}, w_t \sim \mathcal{N}(0, \sigma^2 I), \quad t \in [0, 1, \dots, H-1].$$

Instead of focusing on quadratic cost function, we consider general convex cost function $c(x_t, u_t)$, and the goal is to optimize:

$$\mathbb{E}_\pi \left[\sum_{t=0}^{H-1} c_t(x_t, u_t) \right],$$

where π is a time-dependent policy.

While we relax the assumption on the cost function from quadratic cost to general convex cost, we still focus on linearly parameterized policy, i.e., we are still interested in searching for a sequence of time-dependent linear controllers $\{-K_t^*\}_{t=0}^{H-1}$ with $u_t = -K_t^* x_t$, such that it minimizes the above objective function. We again assume the system is controllable, i.e., the expected total cost under $\{-K_t^*\}_{t=0}^{H-1}$ is finite.

Note that due to the fact that now the cost function is not quadratic anymore, the Riccati equation we studied before does not hold, and it is not necessarily true that the value function of any linear controllers will be a quadratic function.

We present another parameterization of linear controllers which admit convexity in the objective function with respect to parameterization (recall that the objective function is non-convex with respect to linear controllers $-K_t$). With the new parameterization, we will see that due to the convexity, we can directly apply gradient descent to find the globally optimal solution.

Consider any fixed time-dependent linear controllers $\{-K_t\}_{t=0}^{H-1}$. We start by rolling out the linear system under the controllers. Note that during the rollout, once we observe x_{t+1} , we can compute w_t as $w_t = x_{t+1} - Ax_t - Bu_t$. With $x_0 \sim \mu$, we have that at time step t :

$$\begin{aligned} u_t &= -K_t x_t = -K_t (Ax_{t-1} - BK_{t-1}x_{t-1} + w_{t-1}) \\ &= -K_t w_{t-1} - K_t (A - BK_{t-1})x_{t-1} \\ &= -K_t w_{t-1} - K_t (A - BK_{t-1})(Ax_{t-2} - BK_{t-2}x_{t-2} + w_{t-2}) \\ &= \underbrace{-K_t}_{:=M_{t-1;t}} w_{t-1} - \underbrace{K_t(A - BK_{t-1})}_{:=M_{t-2;t}} w_{t-2} - \underbrace{K_t(A - BK_{t-1})(A - BK_{t-2})}_{:=M_{t-3;t}} x_{t-2} \\ &= -K_t \left(\prod_{\tau=1}^t (A - BK_{t-\tau}) \right) x_0 - \sum_{\tau=0}^{t-1} K_t \prod_{h=1}^{t-1-\tau} (A - BK_{t-h}) w_\tau \end{aligned}$$

Note that we can equivalently write u_t using x_0 and the noises $w_0, \dots, w_{t-1}$. Hence for time step t , let us do the following re-parameterization. Denote

$$M_t := -K_t \left(\prod_{\tau=1}^t (A - BK_{t-\tau}) \right), \quad M_{\tau;t} := -K_t \prod_{h=1}^{t-1-\tau} (A - BK_{t-h}), \quad \text{where } \tau = [0, \dots, t-1].$$

Now we can express the control u_t using $M_{\tau;t}$ for $\tau \in [0, \dots, t-1]$ and M_t as follows:

$$u_t = M_t x_0 + \sum_{\tau=0}^{t-1} M_{\tau;t} w_\tau,$$

which is equal to $u_t = -K_t x_t$. Note that above we only reasoned time step t . We can repeat the same calculation for all $t = 0 \rightarrow H-1$.

The above calculation basically proves the following claim.

Claim 15.10. *For any linear controllers $\pi := \{-K_0, \dots, -K_{H-1}\}$, there exists a parameterization, $\tilde{\pi} := \{\{M_t, M_{0;t}, \dots, M_{t-1;t}\}\}_{t=0}^{H-1}$, such that when execute π and $\tilde{\pi}$ under any initialization x_0 , and any sequence of noises $\{w_t\}_{t=0}^{H-1}$, they generate exactly the same state-control trajectory.*

We can execute $\tilde{\pi} := \{\{M_t, M_{0;t}, \dots, M_{t-1;t}\}\}_{t=0}^{H-1}$ in the following way. Given any x_0 , we execute $u_0 = M_0 x_0$ and observe x_1 ; at time step t with the observed x_t , we calculate $w_{t-1} = x_t - Ax_{t-1} - Bu_{t-1}$, and execute the control $u_t = M_t x_0 + \sum_{\tau=0}^{t-1} M_{\tau;t} w_\tau$. We repeat until we execute the last control u_{H-1} and reach x_H .

What is the benefit of the above parameterization? Note that for any t , this is clearly over-parameterized: the simple linear controllers $-K_t$ has $d \times k$ many parameters, while the new controller $\{M_t; \{M_{\tau;t}\}_{\tau=0}^{t-1}\}$ has $t \times d \times k$ many parameters. The benefit of the above parameterization is that the objective function now is convex! The following claim formally shows the convexity.

Claim 15.11. *Given $\tilde{\pi} := \{\{M_t, M_{0;t}, \dots, M_{t-1;t}\}\}_{t=0}^{H-1}$ and denote its expected total cost as $J(\tilde{\pi}) := \mathbb{E} \left[\sum_{t=0}^{H-1} c(x_t, u_t) \right]$, where the expectation is with respect to the noise $w_t \sim \mathcal{N}(0, \sigma^2 I)$ and the initial state $x_0 \sim \mu$, and c is any convex function with respect to x, u . We have that $J(\tilde{\pi})$ is convex with respect to parameters $\{\{M_t, M_{0;t}, \dots, M_{t-1;t}\}\}_{t=0}^{H-1}$.*

Proof: We consider a fixed x_0 and a fixed sequence of noises $\{w_t\}_{t=0}^{H-1}$. Recall that for u_t , we can write it as:

$$u_t = M_t x_0 + \sum_{\tau=0}^{t-1} M_{\tau;t} w_\tau,$$

which is clearly linear with respect to the parameterization $\{M_t; M_{0;t}, \dots, M_{t-1;t}\}$.

For x_t , we can show by induction that it is linear with respect to $\{M_\tau; M_{0;\tau}, \dots, M_{\tau-1;\tau}\}_{\tau=0}^{t-1}$. Note that it is clearly true for x_0 which is independent of any parameterizations. Assume this claim holds for x_t with $t \geq 0$, we now check x_{t+1} . We have:

$$x_{t+1} = Ax_t + Bu_t = Ax_t + BM_t x_0 + \sum_{\tau=0}^{t-1} BM_{\tau;t} w_\tau$$

Note that by inductive hypothesis x_t is linear with respect to $\{M_\tau; M_{0;\tau}, \dots, M_{\tau-1;\tau}\}_{\tau=0}^{t-1}$, and the part $BM_t x_0 + \sum_{\tau=0}^{t-1} BM_{\tau;t} w_\tau$ is clearly linear with respect to $\{M_t; M_{0;t}, \dots, M_{t-1;t}\}$. Together, this implies that x_{t+1} is linear with respect to $\{M_\tau; M_{0;\tau}, \dots, M_{\tau-1;\tau}\}_{\tau=0}^t$, which concludes that for any x_t with $t = 0, \dots, H-1$, we have x_t is linear with respect to $\{M_\tau; M_{0;\tau}, \dots, M_{\tau-1;\tau}\}_{\tau=0}^{H-1}$.

Note that the trajectory total cost is $\sum_{t=0}^{H-1} c(x_t, u_t)$. Since c_t is convex with respect to x_t and u_t , and x_t and u_t are linear with respect to $\{M_\tau; M_{0;\tau}, \dots, M_{\tau-1;\tau}\}_{\tau=0}^{H-1}$, we have that $\sum_{t=0}^{H-1} c_t(x_t, u_t)$ is convex with respect to $\{M_\tau; M_{0;\tau}, \dots, M_{\tau-1;\tau}\}_{\tau=0}^{H-1}$.

In last step we can simply take expectation with respect to x_0 and $w_1, \dots, w_{H-1}$. By linearity of expectation, we conclude the proof. $\blacksquare$

The above immediately suggests that (sub) gradient-based algorithms such as projected gradient descent on parameters $\{M_t, M_{0;t}, \dots, M_{t-1;t}\}_{t=0}^{H-1}$ can converge to the globally optimal solutions for any convex function c_t . Recall that the best linear controllers $\{-K_t^*\}_{t=0}^{H-1}$ has its own corresponding parameterization $\{M_t^*, M_{0;t}^*, \dots, M_{t-1;t}^*\}_{t=0}^{H-1}$. Thus gradient-based optimization (with some care of the boundness of the parameters $\{M_t^*, M_{0;t}^*, \dots, M_{t-1;t}^*\}_{t=0}^{H-1}$) can find a solution that at least as good as the best linear controllers $\{-K_t^*\}_{t=0}^{H-1}$.

Remark The above claims easily extend to time-dependent transition and cost function, i.e., A_t, B_t, c_t are time-dependent. One can also extend it to episodic online control setting with adversarial noises w_t and adversarial cost functions using no-regret online convex programming algorithms [Shalev-Shwartz, 2011]. In episodic online control, every episode k , an adversary determines a sequence of bounded noises $w_0^k, \dots, w_{H-1}^k$, and cost function $c^k(x, u)$; the learner proposes a sequence of controllers $\tilde{\pi}^k = \{M_t^k, M_{0;t}^k, \dots, M_{t-1;t}^k\}_{t=0}^{H-1}$ and executes them (learner does

not know the cost function c^k until the end of the episode, and the noises are revealed in a way when she observes x_t^k and calculates w_{t-1}^k as $x_t^k - Ax_{t-1}^k - Bu_{t-1}^k$; at the end of the episode, learner observes c^k and suffers total cost $\sum_{t=0}^{H-1} c^k(x_t^k, u_t^k)$. The goal of the episodic online control is to be no-regret with respect to the best linear controllers in hindsight:

$$\sum_{k=0}^{K-1} \sum_{t=0}^{H-1} c^k(x_t^k, u_t^k) - \min_{\{-K_t^*\}_{t=0}^{H-1}} \sum_{k=0}^{K-1} J^k(\{-K_t^*\}_{t=0}^{H-1}) = o(K),$$

where we denote $J^k(\{-K_t^*\}_{t=0}^{H-1})$ as the total cost of executing $\{-K_t^*\}_{t=0}^{H-1}$ in episodic k under c^k and noises $\{w_t^k\}$, i.e., $\sum_{t=0}^{H-1} c(x_t, u_t)$ with $u_t = -K_t^* x_t$ and $x_{t+1} = Ax_t + Bu_t + w_t^k$, for all t .

15.6 Bibliographic Remarks and Further Readings

The basics of optimal control theory can be found in any number of standards text [Anderson and Moore, 1990, Evans, 2005, Bertsekas, 2017]. The primal and dual formulations for the continuous time LQR are derived in Balakrishnan and Vandenberghe [2003], while the dual formulation for the discrete time LQR, that we use here, is derived in Cohen et al. [2019].

The treatment of Gauss-Newton, NPG, and PG algorithm are due to Fazel et al. [2018]. We have only provided the proof for the Gauss-Newton case based; the proofs of convergence rates for NPG and PG can be found in Fazel et al. [2018].

For many applications, the finite horizon LQR model is widely used as a model of locally linear dynamics, e.g. [Ahn et al., 2007, Todorov and Li, 2005, Tedrake, 2009, Perez et al., 2012]. The issue of instabilities are largely due to model misspecification and in the accuracy of the Taylor’s expansion; it is less evident how the infinite horizon LQR model captures these issues. In contrast, for MDPs, practice tends to deal with stationary (and discounted) MDPs, due that stationary policies are more convenient to represent and learn; here, the non-stationary, finite horizon MDP model tends to be more of theoretical interest, due to that it is straightforward (and often more effective) to simply incorporate temporal information into the state. Roughly, if our policy is parametric, then practical representational constraints leads us to use stationary policies (incorporating temporal information into the state), while if we tend to use non-parametric policies (e.g. through some rollout based procedure, say with “on the fly” computations like in model predictive control, e.g. [Williams et al., 2017]), then it is often more effective to work with finite horizon, non-stationary models.

Sample Complexity and Regret for LQRs. We have not treated the sample complexity of learning in an LQR (see [Dean et al., 2017, Simchowitz et al., 2018, Mania et al., 2019] for rates). Here, the basic analysis follows from the online regression approach, which was developed in the study of linear bandits [Dani et al., 2008, Abbasi-Yadkori et al., 2011]; in particular, the self-normalized bound for vector-valued martingales [Abbasi-Yadkori et al., 2011] (see Theorem A.9) provides a direct means to obtain sharp confidence intervals for estimating the system matrices A and B from data from a single trajectory (e.g. see [Simchowitz and Foster, 2020]).

Another family of work provides regret analyses of online LQR problems [Abbasi-Yadkori and Szepesvári, 2011, Dean et al., 2018, Mania et al., 2019, Cohen et al., 2019, Simchowitz and Foster, 2020]. Here, naive random search suffices for sample efficient learning of LQRs [Simchowitz and Foster, 2020]. For the learning and control of more complex nonlinear dynamical systems, one would expect this is insufficient, where strategic exploration is required for sample efficient learning; just as in the case for MDPs (e.g. the UCB-VI algorithm).

Convex Parameterization of Linear Controllers The convex parameterization in Section 15.5 is based on [Agarwal et al., 2019] which is equivalent to the System Level Synthesis (SLS) parameterization [Wang et al., 2019]. Agar-

wal et al. [2019] uses SLS parameterization in the infinite horizon online control setting and leverages a reduction to online learning with memory. Note that in episodic online control setting, we can just use classic no-regret online learner such as projected gradient descent [Zinkevich, 2003]. For partial observable linear systems, the classic Youla parameterization [Youla et al., 1976] introduces a convex parameterization. We also refer readers to [Simchowitz et al., 2020] for more detailed discussion about Youla parameterization and the generalization of Youla parameterization. Moreover, using the performance difference lemma, the exact optimal control policy for adversarial noise with full observations can be exactly characterized Foster and Simchowitz [2020], Goel and Hassibi [2020] and yields a form reminiscent of the SLS parametrization Wang et al. [2019].

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Appendix A

Concentration

Lemma A.1. (Hoeffding's inequality) Suppose $X_1, X_2, \dots, X_n$ are a sequence of independent, identically distributed (i.i.d.) random variables with mean μ . Let $\bar{X}_n = n^{-1} \sum_{i=1}^n X_i$. Suppose that $X_i \in [b_-, b_+]$ with probability 1, then

$$P(\bar{X}_n \geq \mu + \epsilon) \leq e^{-2n\epsilon^2 / (b_+ - b_-)^2}.$$

Similarly,

$$P(\bar{X}_n \leq \mu - \epsilon) \leq e^{-2n\epsilon^2 / (b_+ - b_-)^2}.$$

The Chernoff bound implies that with probability $1 - \delta$:

$$\bar{X}_n - EX \leq (b_+ - b_-) \sqrt{\ln(1/\delta) / (2n)}.$$

Now we introduce the definition of subGaussian random variables.

Definition A.2. [Sub-Gaussian Random Variable] A random variable X is σ -subGaussian if for all $\lambda \in \mathbb{R}$, it holds that $\mathbb{E}[\exp(\lambda X)] \leq \exp(\lambda^2 \sigma^2 / 2)$.

One can show that a Gaussian random variable with zero mean and standard deviation σ is a σ -subGaussian random variable.

The following theorem shows that the tails of a σ -subGaussian random variable decay approximately as fast as that of a Gaussian variable with zero mean and standard deviation σ .

Theorem A.3. If X is a σ -subGaussian, then for any $\epsilon > 0$, we have $\Pr(X \geq \epsilon) \leq \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right)$.

The following lemma shows that the sum of independent sub-Gaussian variables is still sub-Gaussian.

Lemma A.4. Suppose that X_1 and X_2 are independent and σ_1 and σ_2 subGaussian, respectively. Then for any $c \in \mathbb{R}$, we have cX being $|c|\sigma$ -subGaussian. We also have $X_1 + X_2$ being $\sqrt{\sigma_1^2 + \sigma_2^2}$ -subGaussian.

Theorem A.5 (Hoeffding-Azuma Inequality). Suppose $X_1, \dots, X_T$ is a martingale difference sequence where each X_t is a σ_t sub-Gaussian, Then, for all $\epsilon > 0$ and all positive integer N ,

$$P\left(\sum_{i=1}^N X_i \geq \epsilon\right) \leq \exp\left(\frac{-\epsilon^2}{2 \sum_{i=1}^N \sigma_i^2}\right).$$

Lemma A.6. (Bernstein's inequality) Suppose $X_1, \dots, X_n$ are independent random variables. Let $\bar{X}_n = n^{-1} \sum_{i=1}^n X_i$, $\mu = \mathbb{E}\bar{X}_n$, and $\text{Var}(X_i)$ denote the variance of X_i . If $X_i - EX_i \leq b$ for all i , then

$$P(\bar{X}_n \geq \mu + \epsilon) \leq \exp \left[-\frac{n^2 \epsilon^2}{2 \sum_{i=1}^n \text{Var}(X_i) + 2nb\epsilon/3} \right].$$

If all the variances are equal, the Bernstein inequality implies that, with probability at least $1 - \delta$,

$$\bar{X}_n - EX \leq \sqrt{2\text{Var}(X) \ln(1/\delta)/n} + \frac{2b \ln(1/\delta)}{3n}.$$

Lemma A.7 (Bernstein's Inequality for Martingales). Suppose $X_1, X_2 \dots$ is a martingale difference sequence where $|X_i| \leq M \in \mathbb{R}^+$ almost surely. Then for all positive t and $n \in \mathbb{N}^+$, we have:

$$P\left(\sum_{i=1}^n X_i \geq t\right) \leq \exp\left(-\frac{t^2/2}{\sum_{i=1}^n \mathbb{E}[X_i^2 | X_{<i}] + Mt/3}\right).$$

The following concentration bound is a simple application of the McDiarmid's inequality [McDiarmid, 1989] (e.g. see [Hsu et al., 2008] for proof).

Proposition A.8. (Concentration for Discrete Distributions) Let z be a discrete random variable that takes values in $\{1, \dots, d\}$, distributed according to q . We write q as a vector where $\vec{q} = [\Pr(z = j)]_{j=1}^d$. Assume we have N iid samples, and that our empirical estimate of $\vec{q}$ is $[\hat{q}]_j = \sum_{i=1}^N \mathbf{1}[z_i = j]/N$.

We have that $\forall \epsilon > 0$:

$$\Pr\left(\|\hat{q} - \vec{q}\|_2 \geq 1/\sqrt{N} + \epsilon\right) \leq e^{-N\epsilon^2}.$$

which implies that:

$$\Pr\left(\|\hat{q} - \vec{q}\|_1 \geq \sqrt{d}(1/\sqrt{N} + \epsilon)\right) \leq e^{-N\epsilon^2}.$$

Lemma A.9 (Self-Normalized Bound for Vector-Valued Martingales; [Abbasi-Yadkori et al., 2011]). Let $\{\varepsilon_i\}_{i=1}^\infty$ be a real-valued stochastic process with corresponding filtration $\{\mathcal{F}_i\}_{i=1}^\infty$ such that ε_i is $\mathcal{F}_i$ measurable, $\mathbb{E}[\varepsilon_i | \mathcal{F}_{i-1}] = 0$, and ε_i is conditionally σ -sub-Gaussian with $\sigma \in \mathbb{R}^+$. Let $\{X_i\}_{i=1}^\infty$ be a stochastic process with $X_i \in \mathcal{H}$ (some Hilbert space) and X_i being $\mathcal{F}_i$ measurable. Assume that a linear operator $\Sigma : \mathcal{H} \rightarrow \mathcal{H}$ is positive definite, i.e., $x^\top \Sigma x > 0$ for any $x \in \mathcal{H}$. For any t , define the linear operator $\Sigma_t = \Sigma_0 + \sum_{i=1}^t X_i X_i^\top$ (here $xx^\top$ denotes outer-product in $\mathcal{H}$). With probability at least $1 - \delta$, we have for all $t \geq 1$:

$$\left\| \sum_{i=1}^t X_i \varepsilon_i \right\|_{\Sigma_t^{-1}}^2 \leq \sigma^2 \log \left(\frac{\det(\Sigma_t) \det(\Sigma_0)^{-1}}{\delta^2} \right).$$

Theorem A.10 (OLS with a fixed design). Consider a fixed dataset $\mathcal{D} = \{x_i, y_i\}_{i=1}^N$ where $x_i \in \mathbb{R}^d$, and $y_i = (\theta^*)^\top x_i + \epsilon_i$, where $\{\epsilon_i\}_{i=1}^N$ are independent σ -sub-Gaussian random variables, and $\mathbb{E}[\epsilon_i] = 0$. Denote $\Lambda = \sum_{i=1}^N x_i x_i^\top$, and we further assume that Λ is full rank such that Λ^{-1} exists. Denote $\hat{\theta}$ as the least squares solution, i.e., $\hat{\theta} = \Lambda^{-1} \sum_{i=1}^N x_i y_i$. Pick $\delta \in (0, 1)$, with probability at least $1 - \delta$, we have:

$$\left\| \hat{\theta} - \theta^* \right\|_{\Lambda}^2 \leq \sigma^2 (d + \sqrt{d \ln(1/\delta)} + 2 \ln(1/\delta)).$$

The proof of the above theorem is standard (e.g. the proof can be easily adapted from Lemma 6 in [Hsu et al., 2012]).

Lemma A.11 (Least Squares Generalization Bound). Given a dataset $\mathcal{D} = \{x_i, y_i\}_{i=1}^n$ where $x_i \in \mathcal{X}$ and $x_i, y_i \sim \nu$,

$y_i = f^*(x_i) + \epsilon_i$, where $|y_i| \leq Y$, $\max_x |f^*(x)| \leq Y$ and $|\epsilon_i| \leq \sigma$ for all i , and $\{\epsilon_i\}$ are independent from each other. Given a function class $\mathcal{F} : \mathcal{X} \mapsto [0, Y]$, we assume approximate realizable, i.e., $\min_{f \in \mathcal{F}} \mathbb{E}_{x \sim \nu} |f^*(x) - f(x)|^2 \leq \epsilon_{\text{approx}}$. Denote $\hat{f}$ as the least square solution, i.e., $\hat{f} = \operatorname{argmin}_{f \in \mathcal{F}} \sum_{i=1}^n (f(x_i) - y_i)^2$. With probability at least $1 - \delta$, we have:

$$\mathbb{E}_{x \sim \nu} \left(\hat{f}(x) - f^*(x) \right)^2 \leq \frac{22Y^2 \ln(|\mathcal{F}|/\delta)}{n} + 20\epsilon_{\text{approx}}.$$

Proof:

For notation simplicity, let us denote $f^*(x) := \mathbb{E}[y|x]$. Let us consider a fixed function $f \in \mathcal{F}$.

Denote random variables $z_i^f := (f(x_i) - y_i)^2 - (f^*(x_i) - y_i)^2$. We now show that $\mathbb{E}[z_i^f] = \|f - f^*\|_{2,\nu}^2$, and $\mathbb{E}[(z_i^f)^2] \leq 4Y^2 \|f - f^*\|_{2,\nu}^2$:

$$\mathbb{E}[z_i^f] = \mathbb{E}[(f(x_i) - f^*(x_i))(f(x_i) + f^*(x_i) - 2y_i)] = \mathbb{E}[(f(x_i) - f^*(x_i))^2] := \|f - f^*\|_{2,\nu}^2$$

where the last equality uses the fact that $\mathbb{E}[y_i|x_i] = f^*(x_i)$. For the second moment, we have:

$$\mathbb{E}[(z_i^f)^2] = \mathbb{E}[(f(x_i) - f^*(x_i))^2(f(x_i) + f^*(x_i) - 2y_i)^2] \leq 4Y^2 \mathbb{E}[(f(x_i) - f^*(x_i))^2] := 4Y^2 \|f - f^*\|_{2,\nu}^2,$$

where the inequality uses the conditions that $\|f^*\|_\infty \leq Y$, $\|f\|_\infty \leq Y$, $|y_i| \leq Y$. Now we can bound the deviation from the empirical mean $\sum_{i=1}^n z_i^f/n$ to its expectation $\|f - f^*\|_{2,\nu}^2$ using Bernstein's inequality. Together with a union bound over all $f \in \mathcal{F}$, with probability at least $1 - \delta$, we have

$$\forall f \in \mathcal{F} : \|f - f^*\|_{2,\nu}^2 - \sum_{i=1}^n z_i^f/n \leq \sqrt{\frac{8Y^2 \|f - f^*\|_{2,\nu}^2 \ln(|\mathcal{F}|/\delta)}{n}} + \frac{4Y^2 \ln(|\mathcal{F}|/\delta)}{n}. \quad (\text{A.1})$$

Below we first bound $\sum_{i=1}^n z_i^{\hat{f}}/n$ where recall $\hat{f}$ is the least squares solution.

Denote $\tilde{f} := \operatorname{argmin}_{f \in \mathcal{F}} \|f - f^*\|_{2,\nu}^2$. Note that $\tilde{f}$ is independent of the dataset by its definition. Our previous calculation implies that $\mathbb{E}[z_i^{\tilde{f}}] = \|\tilde{f} - f^*\|_{2,\nu}^2$, and $\mathbb{E}[(z_i^{\tilde{f}})^2] \leq 4Y^2 \|\tilde{f} - f^*\|_{2,\nu}^2$. Directly applying Bernstein's inequality to bound $\sum_{i=1}^n z_i^{\tilde{f}}/n - \|\tilde{f} - f^*\|_{2,\nu}^2$, we have that with probability at least $1 - \delta$:

$$\sum_{i=1}^n z_i^{\tilde{f}}/n - \|\tilde{f} - f^*\|_{2,\nu}^2 \leq \sqrt{\frac{8Y^2 \|\tilde{f} - f^*\|_{2,\nu}^2 \ln(1/\delta)}{n}} + \frac{4Y^2 \ln(1/\delta)}{3n}.$$

Using this inequality, we discuss two cases. First when $\sum_{i=1}^n z_i^{\tilde{f}}/n \geq Y^2 \ln(1/\delta)/n$. In this case, the above inequality implies that:

$$\sum_{i=1}^n z_i^{\tilde{f}}/n - \|\tilde{f} - f^*\|_{2,\nu}^2 \leq \sqrt{8\|\tilde{f} - f^*\|_{2,\nu}^2 \cdot \sum_{i=1}^n z_i^{\tilde{f}}/n} + \frac{4Y^2 \ln(1/\delta)}{3n}.$$

Solve for $\sum_{i=1}^n z_i^{\tilde{f}}/n$, we get:

$$\sum_{i=1}^n z_i^{\tilde{f}}/n \leq 10\|\tilde{f} - f^*\|_{2,\nu}^2 + \frac{8Y^2 \ln(1/\delta)}{3n}.$$

For the other case where $\sum_{i=1}^n z_i^{\tilde{f}}/n \leq Y^2 \ln(1/\delta)/n$, we immediately have:

$$\sum_{i=1}^n z_i^{\tilde{f}}/n \leq Y^2 \ln(1/\delta)/n \leq 10\|\tilde{f} - f^*\|_{2,\nu}^2 + \frac{8Y^2 \ln(1/\delta)}{3n}.$$

Thus $\sum_{i=1}^n z_i^{\tilde{f}}/n \leq 10\|\tilde{f} - f^*\|_{2,\nu}^2 + \frac{8Y^2 \ln(1/\delta)}{3n}$ regardless.

Since $\|\tilde{f} - f^*\|_{2,\nu}^2 \leq \epsilon_{approx}$ by the assumption, we can conclude that:

$$\sum_{i=1}^n z_i^{\tilde{f}}/n \leq 10\epsilon_{approx} + \frac{8Y^2 \ln(1/\delta)}{3n}.$$

Note that by definition of $\hat{f}$, we have $\sum_{i=1}^n z_i^{\hat{f}} \leq \sum_{i=1}^n z_i^{\tilde{f}}$, which implies that:

$$\sum_{i=1}^n z_i^{\hat{f}}/n \leq 10\epsilon_{approx} + \frac{8Y^2 \ln(1/\delta)}{3n},$$

which concludes the upper bound of $\sum_{i=1}^n z_i^{\hat{f}}/n$.

Going back to Eq. A.1, by setting $f = \hat{f}$ there, and use the upper bound on $\sum_{i=1}^n z_i^{\hat{f}}/n$, we have that:

$$\|\hat{f} - f^*\|_{2,\nu}^2 \leq \sqrt{\frac{8Y^2 \|\hat{f} - f^*\|_{2,\nu}^2 \ln(|\mathcal{F}|/\delta)}{n}} + \frac{4Y^2 \ln(|\mathcal{F}|/\delta)}{n} + \frac{8Y^2 \ln(|\mathcal{F}|/\delta)}{3n} + 10\epsilon_{approx}.$$

Solve for $\|\hat{f} - f^*\|_{2,\nu}^2$ from the above inequality, we get:

$$\|\hat{f} - f^*\|_{2,\nu}^2 \leq \frac{22Y^2 \ln(|\mathcal{F}|/\delta)}{n} + 20\epsilon_{approx}.$$

This concludes the proof. ■

Lemma A.12 (Uniform convergence for Bellman error under linear function hypothesis class). *Given a feature $\phi : \mathcal{S} \times \mathcal{A} \mapsto \mathbb{R}^d$ with $\|\phi(s, a)\|_2 \leq 1, \forall s, a$. Define $\mathcal{H}_h = \{w^\top \phi(s, a) : \|w\|_2 \leq W, w \in \mathbb{R}^d, \|w^\top \phi(\cdot, \cdot)\|_\infty \leq H\}$. Given $g \in \mathcal{H}$, denote $\ell(s, a, s', g)$ as follows:*

$$\ell(s, a, s', h, g) = \frac{\mathbf{1}\{\pi_{h,g}(s) = a\}}{1/A} \left(w_h^\top \phi(s, a) - r(s, a) - \max_{a'} w_{h+1}^\top \phi(s', a') \right).$$

For any distribution $\nu \in \Delta(\mathcal{S})$, with probability $1 - \delta$ over the randomness of the m i.i.d triples $\{s_i, a_i, s'_i\}_{i=1}^m$ with $s_i \sim \mu, a_i \sim \text{Unif}_{\mathcal{A}}, s'_i \sim P_h(\cdot|s_i, a_i)$, we have:

$$\max_{g \in \mathcal{H}} \left| \mathbb{E}_{s \sim \nu, a \sim \text{Unif}_{\mathcal{A}}, s' \sim P_h(\cdot|s, a)} [\ell(s, a, s', h, g)] - \sum_{i=1}^m \ell(s_i, a_i, s'_i, h, g)/m \right| \leq 3HA \sqrt{\frac{2d \ln(1 + 4AWm) + \ln(1/\delta)}{m}}.$$

Similarly, when $\ell(s, a, s', g)$ is defined as:

$$\ell(s, a, s', h, g) = w_h^\top \phi(s, a) - r(s, a) - \max_{a'} w_{h+1}^\top \phi(s', a'),$$

for any distribution $\nu \in \Delta(\mathcal{S} \times \mathcal{A})$, with probability $1 - \delta$ over randomness of the m i.i.d triples $\{s_i, a_i, s'_i\}_{i=1}^m$ with $s_i, a_i \sim \nu, s'_i \sim P_h(\cdot|s_i, a_i)$, we have:

$$\max_{g \in \mathcal{H}} \left| \mathbb{E}_{s \sim \nu, a \sim \text{Unif}_{\mathcal{A}}, s' \sim P_h(\cdot|s, a)} [\ell(s, a, s', h, g)] - \sum_{i=1}^m \ell(s_i, a_i, s'_i, h, g)/m \right| \leq 3H \sqrt{\frac{2d \ln(1 + 4Wm) + \ln(1/\delta)}{m}}.$$

Proof: The proof uses the standard ϵ -net argument. Define $\mathcal{N}_\epsilon$ as the ϵ -net for $\mathcal{W} = \{w : w \in \mathbb{R}^d, \|w\|_2 \leq W, w^\top \phi(\cdot, \cdot) \in \mathcal{H}\}$. Standard ϵ -net argument shows that $|\mathcal{N}_\epsilon| \leq (1 + W/\epsilon)^d$. Denote $g' = \{(w'_h)^\top \phi\}_{h=0}^{H-1}$, we have:

$$\begin{aligned} |\ell(s, a, s', h, g) - \ell(s, a, s', h, g')| &\leq \frac{\mathbf{1}\{\pi_{h,g}(s) = a\}}{1/A} \left[|w_h^\top \phi(s, a) - (w'_h)^\top \phi(s, a)| + \max_{a'} |w_{h+1}^\top \phi(s', a') - (w'_{h+1})^\top \phi(s', a')| \right] \\ &\leq A \sup_{s,a} |w_h^\top \phi(s, a) - (w'_h)^\top \phi(s, a)| + A \sup_{s,a} |w_{h+1}^\top \phi(s, a) - (w'_{h+1})^\top \phi(s, a)| \\ &\leq A \|w_h - w'_h\|_2 + A \|w_{h+1} - w'_{h+1}\|_2. \end{aligned}$$

Thus, if $\|w_h - w'_h\|_2 \leq \epsilon$, $\|w_{h+1} - w'_{h+1}\|_2 \leq \epsilon$, we have $|\ell(s, a, s', h, g) - \ell(s, a, s', h, g')| \leq 2A\epsilon$.

Consider a $w_h \in \mathcal{N}_\epsilon$, $w_{h+1} \in \mathcal{N}_\epsilon$, via Hoeffding's inequality, we have that with probability at least $1 - \delta$:

$$|\mathbb{E}[\ell(s, a, s', h, (w_h, w_{h+1}))] - \bar{\mathbb{E}}[\ell(s, a, s', h, (w_h, w_{h+1}))]| \leq 2HA\sqrt{\ln(1/\delta)/m}.$$

where for notation simplicity, we denote $\bar{\mathbb{E}}$ as the empirical average over the m data triples. Apply a union bound over all $w_h \in \mathcal{N}_\epsilon$ and $w_{h+1} \in \mathcal{N}_\epsilon$, we have that with probability at least $1 - \delta$:

$$\forall w_h \in \mathcal{N}_\epsilon, w_{h+1} \in \mathcal{N}_\epsilon : |\mathbb{E}[\ell(s, a, s', h, (w_h, w_{h+1}))] - \bar{\mathbb{E}}[\ell(s, a, s', h, (w_h, w_{h+1}))]| \leq 2HA\sqrt{2 \ln(|\mathcal{N}_\epsilon|/\delta)/m}.$$

Thus for any $w_h \in \mathcal{W}$, $w_{h+1} \in \mathcal{W}$, we have:

$$|\mathbb{E}[\ell(s, a, s', h, (w_h, w_{h+1}))] - \bar{\mathbb{E}}[\ell(s, a, s', h, (w_h, w_{h+1}))]| \leq 2HA\sqrt{2 \ln(|\mathcal{N}_\epsilon|/\delta)/m} + 4A\epsilon.$$

Set $\epsilon = 1/(4Am)$, we have:

$$\begin{aligned} |\mathbb{E}[\ell(s, a, s', h, (w_h, w_{h+1}))] - \bar{\mathbb{E}}[\ell(s, a, s', h, (w_h, w_{h+1}))]| &\leq 2HA\sqrt{(2d \ln(1 + 4AWm) + \ln(1/\delta))/m} + 1/m \\ &\leq 3HA\sqrt{\frac{2d \ln(1 + 4AWm) + \ln(1/\delta)}{m}}. \end{aligned}$$

This proves the first claim in the lemma.

The second claim in the lemma can be proved in a similar way, noting that without the importance weighting term on the loss ℓ , we have:

$$|\ell(s, a, s', h, g) - \ell(s, a, s', h, g')| \leq \|w_h - w'_h\|_2 + \|w_{h+1} - w'_{h+1}\|_2.$$

■